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THE STACK IN THE FIELD

When Mathematics Meets the Physical Trades

First-Principles Applied Mathematics, Operational Shorthand, and Four-Register RPN Stack Traces Across Surveying, Electrical Trades, Embedded Systems, Cinema, RF, Chemistry, Brewing, Clinical Dosing & Finance

StackCalc
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First Edition

THE STACK IN THE FIELD: When Mathematics Meets the Physical Trades
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For the surveyors, electricians, systems programmers, focus pullers, RF engineers, chemists, bakers, brewmasters, clinical nurses, and analysts
who put mathematics to work in the physical world.

"When you can measure what you are speaking about,
and express it in numbers, you know something about it."

— **WILLIAM THOMSON (LORD KELVIN),** ***Electrical Units of Measurement*** **(1883)**

Prologue: When Mathematics Meets the Physical Trades

In the summer of 1858, sailors aboard two steamships met in the middle of the North Atlantic, spliced two coils of copper cable together, and lowered the joint into two miles of dark ocean water. As HMS *Agamemnon* steamed eastward toward Ireland and USS *Niagara* steamed westward toward Newfoundland, they paid out the first telegraph line ever to connect two continents.

Below deck on the *Agamemnon* sat a thirty-four-year-old Scottish physicist named William Thomson, who would later become Lord Kelvin. All across Europe and America, newspapers celebrated the end of distance. In university classrooms, electricity had been taught as a weightless, instantaneous fluid: when you closed a switch in London, a crisp rectangular pulse of current was supposed to appear in New York at the speed of light.

Sitting in the ship's cabin watching a tiny mirror suspended on a silk thread, Thomson saw something very different. Two thousand miles of copper wire wrapped in tree resin and submerged in icy salt water did not behave like a blackboard diagram. The copper resisted the flow of electrons, while the surrounding ocean acted like a giant capacitor that soaked up electrical charge. By the time a sharp Morse code dot traveled from Ireland to Newfoundland, it arrived as a slow,gentle swell that took seconds to rise and fade.

Worse still, the chief electrician in London refused to believe Thomson's equations. Convinced that a sluggish signal simply needed a harder push, he connected giant high-voltage induction coils to the wire. Within weeks, the electrical pressure burned through the insulation on the ocean floor, and the Atlantic cable went silent.

Twenty-five years later, speaking to the Institution of Civil Engineers in London, Lord Kelvin summed up the lesson of that summer at sea: until you can measure what you are speaking about and express it in numbers, your knowledge is meager and unsatisfactory. That is the subject of this book.

Where Classroom Equations Meet the Real World

In a math classroom, the world is wonderfully tidy. Angles always start on the flat horizontal axis on the right side of the page and sweep counterclockwise. Wires carry electricity without getting warm. Whole numbers can grow forever without running out of room. Sound and radio waves travel in straight lines without bouncing off walls, and recipes can be doubled just by pouring two cups instead of one.

The moment you step outside the classroom—into a forest canyon, a ship's engine room, a bakery at four in the morning, or a hospital intensive-care ward—nature introduces rules of its own.

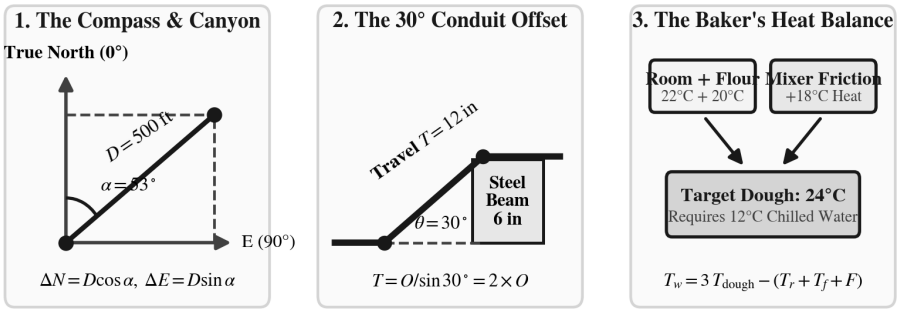


Figure 0.1: *Three everyday physical puzzles where classroom geometry and physics meet the real world.*

Look at the three everyday puzzles in **Figure 0.1**:

1. **The Surveyor's Compass and the Canyon:** Suppose you stand on the rim of a river canyon holding a magnetic compass. Unlike a classroom protractor that puts 0° on the right, every compass needle on Earth points to **True North (0°)** at the top of the dial, and its angles increase *clockwise* toward East (90°), South (180°), and West (270°). When a surveyor sights a target 500 feet away at a clockwise compass angle of 53°, that simple change of direction swaps the familiar roles of sine and cosine.

2. **The Electrician's Bent Steel Pipe:** Imagine running a rigid steel pipe along a factory wall when you meet a 6-inch-thick steel beam blocking your path. To climb over the beam cleanly, the electrician bends the pipe upward at 30° and then bends it back flat at the top. Because the sine of 30° is exactly 1/2, climbing 6 inches vertically requires placing the two bends exactly 2 × 6 = 12 inches apart along the pipe!

3. **The Baker's Spinning Mixer:** When a bakery kneads two hundred loaves of bread dough in a heavy steel spiral mixer, the friction of the spinning steel hook heats the dough by 18°C during mixing. If the dough comes out too warm, the yeast races out of control and the bread collapses. To hit an exact target dough temperature of 24°C, the baker works backward from the room, flour, and friction temperatures to calculate how cold the tap water must be—and how many kilograms of crushed ice must melt inside the bowl.

Every chapter in this book is built around real stories like these. You do not need to be an engineer to enjoy them. If you have ever wondered how an airline pilot lands crabwise in a crosswind, how a movie camera blurs the background behind an actor's face, how a computer stores a negative number using only zeros and ones, or how a pharmacist tailors medicine to a patient's kidneys, the mathematics is waiting right beneath the surface.

Concepts First, Worked Examples Second, Keystrokes Third

In every skilled trade, people who use the same physical law every day eventually give it a short nickname. A land surveyor does not say, *"Let us apply the Pythagorean theorem and four-quadrant arc-tangent to our Northing and Easting differences."* Instead, the surveyor calls that check an **Inverse** (INV). A pipefitter calls the geometry of bending pipe over a beam an **Offset** (OFS). A cinematographer calls the zone of sharp focus **Depth of Field** (DOF), and a baker calls scaling a recipe to flour weight **Baker's Percentage** (BK%).

To make every field welcoming to newcomers and rewarding for experts, each chapter in this book follows the same three-step rhythm as Volume II (*Thinking in the Stack*):

1. **The Physical Story and the Mathematical Idea:** We begin each chapter with a human story and a physical puzzle, building the geometry or physics from first principles without jargon.
2. **Worked Examples by Hand:** Next, we put real numbers into the idea and trace how the quantities fit together step by step, so you can see *why* the answer makes physical sense.
3. **Bringing the Idea to the Calculator:** Finally, we show how that exact concept lives on the StackCalc Domain Series (**SC-35** through **SC-95**), where those three-letter trade nicknames are engraved right on the keys.

Forward Chains and Backward Puzzles: One Story, Two Ways of Thinking

Why do practitioners in the field use a stack calculator with dedicated domain keys instead of a basic four-function calculator or a spreadsheet? Because field problems almost always come in pairs: first you **chain measurements forward**, and then a real-world limit forces you to **solve the same problem backward**.

Let us trace a single concrete example to see how both modes work.

Suppose an electrician is wiring a 240 V motor at the end of a 150-foot pier using medium-gauge copper wire (#10 AWG). Copper is a wonderful conductor, but it is not perfect: every 1,000 feet of #10 AWG copper wire has an electrical resistance of 1.24 ohms (Ω). Because electricity must travel 150 feet out to the motor and 150 feet back along the return wire, the current travels through 300 feet of copper in total.

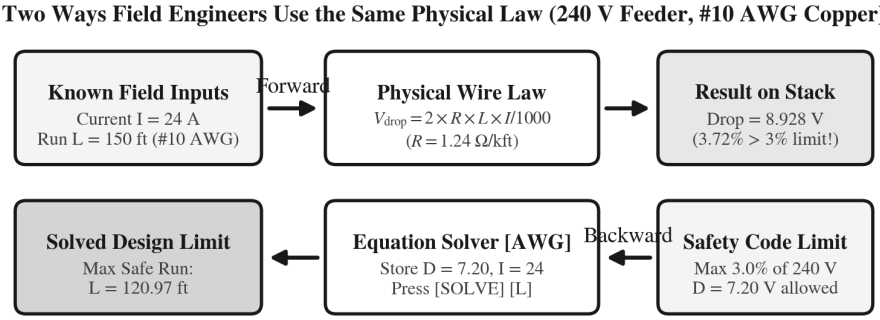


Figure 0.2: *Forward pipeline versus backward design on a 150-foot copper feeder.*

1. Thinking Forward on the Four-Register Stack

When the motor draws 24 amperes of current, how much voltage is lost as heat along the 300-foot round trip? By Ohm's law, voltage drop is simply resistance times current.

On the four-register RPN stack (**Figure 0.3**), four visible lines—Registers T, Z, Y, and X from top to bottom—hold your numbers as you build the calculation from the inside out:

Register T (Top)	Holds earlier subtotals or 4th-level physical arguments
Register Z	Stages 3rd argument (e.g., Load Current I = 24 A)
Register Y	Stages 2nd argument (e.g., One-Way Run L = 150 ft)
Register X (Active)	Live entry & immediate result (e.g., Gauge 10 -> 8.928 V drop)

Figure 0.3: *The four-register RPN stack (T, Z, Y, X) staging intermediate physical quantities.*

Watch how each physical step happens right in Register *X* at the bottom of the screen:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
10 [AWG]	0	0	0	1.24	Look up copper resistance for #**10 AWG** (1.24 Ω per 1,000 ft)
2 [×] **150** [×]	0	0	0	372	Multiply by 2 × 150 ft round-trip distance (372 Ω·ft)
24 [×] **1000** [÷]	0	0	0	8.928	Multiply by 24 A current and divide by 1,000 ft (8.928 V lost)
240 [÷] **100** [×]	0	0	0	**3.72**	Compare 8.928 V against the 240 V supply (3.72% drop)

If all three numbers—current (24 A), one-way length (150 ft), and wire gauge (**10**)—are already sitting on the stack in Registers *Z*, *Y*, and *X*, pressing [AWG] performs that entire round-trip calculation in a single keystroke and puts **8.928** volts directly into Register *X*.

(And if a calculation ever needs more than four shelves, opening [↗] [FLAGS] *lets you switch stack depth from classic four-level* [4-LV] *to eight-level* [8-LV] *or unlimited deep stack* [INF]*.)*

2. Thinking Backward in the Equation Solver Workspace

Now comes the twist that happens every day on a real job site.

The National Electrical Code recommends losing no more than **3.0% of your supply voltage** on a branch circuit so motors do not overheat. On a 240 V line, 3.0% is 0.03 × 240 = **7.20 V**. Our 150-foot wire lost 8.928 V (3.72%), which is too high!

So the electrician asks the backward question: **If I am only allowed to lose 7.20 V, what is the longest one-way distance *L* I can run #10 AWG wire at 24 A?**

Instead of rearranging algebra on a scrap of cardboard, pressing [AWG] on an empty stack opens its **Equation Solver Workspace**. The top line of the screen shows the wire-drop formula (**D = 2×R×L×I / 1000**), and the row of softkeys right below the screen becomes the formula's named variables:

- [D] for Voltage Drop (7.20 V)
- [R] for Wire Resistance (1.24 Ω/kft, loaded automatically when you key **10** [AWG])
- [L] for One-Way Length (our unknown!)
- [I] for Current (24 A)

You simply type the numbers you know into their softkeys—**7.20** [D] and **24** [I]—and then press [→] [SOLVE] [L]. The calculator solves the equation backward and answers immediately: L = **120.97 feet**.

Every instrument in this book works the same way. You can chain values forward on the RPN stack, open a bidirectional Equation Solver Workspace to solve backward for any variable, or press the Reference Table key ([→] [▦]) to inspect live field tables like survey coordinates, wire gauges, or the periodic table of elements.

Nine Worlds of Applied Mathematics

This volume is organized into **nine Units**—one for each domain instrument (**SC-35** through **SC-95**)—spanning **twenty-seven narrative Chapters**. Each Unit introduces a field of human endeavor, and each Chapter follows three connected themes along a single real-world story:

Unit & Chapters	Domain & Instrument	Narrative Trajectory & Core Mathematical Themes
Unit I (Chapters 1–3)	**Geomatics, Land Surveying & Flight Navigation** (**SC-35**)	**Ch 1:** Retracing a lost 18th-century forest boundary using clockwise compass azimuths, slope reduction, and Gauss's shoelace area formula • **Ch 2:** Designing a mountain highway switchback curve, parabolic crest, and canyon spirit-leveling loop • **Ch 3:** Landing a bush plane at night in a high-desert crosswind while balancing density altitude, center of gravity, and fuel endurance
Unit II (Chapters 4–6)	**Electrical Power, Conduit Geometry & Industrial Trades** (**SC-14**)	**Ch 4:** Commissioning a harbor gantry crane feeder across Ohm's law voltage drop, Nikola Tesla's three-phase AC power triangle, and parallel resistance • **Ch 5:** Bending parallel steel conduits over boiler-room beams with trigonometry, miter angles, and lathe cutting speeds • **Ch 6:** Sizing chemical-plant process pipes, centrifugal pump horsepower, control valve flow, and 4–20 mA sensor loops
Unit III (Chapters 7–9)	**Digital Systems, Bitwise Logic & Embedded Timing** (**SC-16**)	**Ch 7:** Debugging a stuck spacecraft motor interrupt across binary/hexadecimal nibbles, two's complement rings, bit masks, and fixed-point numbers • **Ch 8:** Catching a corrupted telemetry packet via byte-order endian swaps, IEEE-754 floating-point bits, and CRC checksum division • **Ch 9:** Designing a 12-bit sensor interface with RC filter time constants, PWM timer waves, and IPv4 network subnets
Unit IV (Chapters 10–12)	**Cinema Optics, Broadcast Timecode & Live Acoustics** (**SC-24**)	**Ch 10:** Keeping a live television broadcast in sync across NTSC 29.97 drop-frame timecode arithmetic and frame rates • **Ch 11:** Pulling focus on an anamorphic night shoot using thin-lens hyperfocal geometry, depth of field, and rotary shutter angles • **Ch 12:** Aligning outdoor stadium delay speaker towers with decibel power addition, sound speed in warm air, and room modes
Unit V (Chapters 13–15)	**Radio-Frequency Engineering, Transmission Lines & Link Budgets** (**SC-50**)	**Ch 13:** Diagnosing a storm-damaged mountain radio tower through decibel-milliwatts, wave reflections, and standing wave ratio • **Ch 14:** Matching a detuned antenna with quarter-wave transmission-line stubs, Smith chart impedance, and LC resonance • **Ch 15:** Closing a deep-space and microwave satellite link across Friis free-space path loss, receiver noise floors, and Shannon capacity
Unit VI (Chapters 16–18)	**Wet-Lab Chemistry, Stoichiometry & Optical Spectroscopy** (**SC-60**)	**Ch 16:** Preparing a pure protein reagent across Avogadro's molar bridge, 118-element formula weights, and serial dilutions • **Ch 17:** Taming acid-base swings with Sørensen's logarithmic pH scale, weak-acid dissociation, and Henderson–Hasselbalch buffers • **Ch 18:** Measuring molecules with light and heat via Beer–Lambert absorbance, ideal gas pressure, Nernst redox cells, and centrifuges

Unit & Chapters	Domain & Instrument	Narrative Trajectory & Core Mathematical Themes
Unit VII (Chapters 19–21)	**Commercial Baking, Dough Thermodynamics & Brewing Science** (**SC-78**)	**Ch 19:** Scaling an artisan sourdough bakery from a single test loaf to a 200-loaf production batch using flour-anchored baker's percentages and levain splits • **Ch 20:** Controlling fermentation clockwork through mixer friction calorimetry, latent heat of fusion in crushed ice, and prep-yield economics • **Ch 21:** Brewing an all-grain craft ale across mash strike-water thermodynamics, Plato sugar gravity, and Tinseth hop bitterness
Unit VIII (Chapters 22–24)	**Clinical Dosing, Renal Clearance & Pharmacokinetics** (**SC-80**)	**Ch 22:** Individualizing hospital chemotherapy doses across geometric body surface area, Devine ideal weight, and Cockcroft–Gault kidney clearance • **Ch 23:** Titrating intensive-care infusions across weight-based microgram pumps, vial reconstitution, and pediatric 4-2-1 fluid rules • **Ch 24:** Tracking drug half-life decay, steady-state vancomycin troughs, and blood-gas electroneutrality in the ICU
Unit IX (Chapters 25–27)	**Quantitative Finance, Fixed-Income Valuation & Six Sigma Quality** (**SC-95**)	**Ch 25:** Evaluating a solar microgrid investment across Fibonacci's present-value discounting, mortgage amortization, and internal rate of return • **Ch 26:** Pricing Treasury bonds, Macaulay duration, compound growth, and break-even manufacturing volume • **Ch 27:** Guiding an aerospace factory to Six Sigma quality with Gaussian bell-curve quantiles, confidence intervals, and capability indices
Appendices A–B	**Reference Compendium**	**App. A:** Cross-Domain Formula & Governing Equation Compendium • **App. B:** Complete Keycap, Softkey & Operator Directory across **SC-35** through **SC-95**

We begin on the land and in the sky in **Unit I: Geomatics, Land Surveying & Flight Navigation**.

Unit I: Geomatics, Land Surveying & Flight Navigation

In the autumn of 1763, the astronomer Charles Mason and the surveyor Jeremiah Dixon landed at Philadelphia. They carried a six-foot brass zenith sector crafted by John Bird and a transit fitted with a micrometer screw. For eighty years, the Penn and Calvert families had disputed the border across the timbered ridges and marshes of the Chesapeake peninsula. Mason and Dixon could not simply stretch an iron chain along an idealized paper parallel. In a classroom, points sit on a frictionless plane where $+x$ points East, $+y$ points North, and angles sweep counterclockwise. In the wild, dense forests, steep ravines, and atmospheric refraction force a crew to anchor every sightline to the stars. Within a decade, the London instrument maker Jesse Ramsden perfected his geared circular dividing engine (1775). By engraving brass circles to fractions of an arcminute, compact optical theodolites could carry celestial accuracy across continents.

Because navigation on land and sea grew around the magnetic compass and Polaris, field geometry reverses both classroom conventions. When a surveyor levels a theodolite over a monument or an aviator reads a directional gyro, zero (0°) points to **True North**. From that meridian, the graduated azimuth circle increases **clockwise** through 90° at East, 180° at South, and 270° at West. Starting at the vertical axis and rotating clockwise reflects the plane across the 45° Northeast diagonal, **swapping sine and cosine**. North-South displacement (**Latitude**, ΔN) projects along the cosine of the azimuth, while East-West displacement (**Departure**, ΔE) projects along the sine. Whenever a coordinate pair must be transposed between (N, E) and (E, N) conventions, [↶] [N↔E] (or [x⇄y]) swaps Registers X and Y in one keystroke.

The long lever arm of an outdoor sightline also explains why field crews preserve **sexagesimal notation** (Degrees, Minutes, and Seconds **DDD.MMSS**, where $1^\circ = 60'$ and $1' = 60''$). One arcsecond ($1'' = 1/3600^\circ$) is the tiny angle subtended by a quarter coin viewed from more than three miles (5 km) away. Over a routine boundary sightline of 2,000 ft, a one-arcminute error ($1'$) shifts the far stake laterally by $2000 \sin(1') \approx 0.582$ ft—nearly seven inches. A careless rounding error of just 0.05° ($3'$) displaces a property corner by almost two feet.

The **SC-35 Geomatics, Land Surveying & Flight Navigation** calculator is built for this outdoor world. Each chapter in this Unit follows a single field expedition and weaves three related mathematical themes into one narrative arc. You will first explore the physical geometry and worked mathematical examples, and then see how the SC-35 executes them on the stack and in its bidirectional Equation Solver workspaces:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 1	Retracing a 1794 colonial boundary across a steep 6° 15′ Pine Barrens ravine to recover a buried brass cornerstone and certify parcel acreage	Sexagesimal conversions, slope reduction, North-clockwise forward and inverse COGO, Bowditch closure, three-wire stadia, and Gauss's shoelace formula	1. Sexagesimal Sightlines & Ravine Slope Reduction • 2. Forward & Inverse North-Clockwise Coordinates • 3. Polygon Traverse Closure & Shoelace Acreage
Chapter 2	Threading a mountain highway curve ($R = 1{,}200$ ft, $\Delta = 28.5°$) and parabolic crest around a granite canyon bluff while leveling from the river floor	Circular horizontal curve layout, parabolic vertical curve elevations, earthwork prism volume, and three-wire differential spirit leveling	1. Circular Highway Curves & Deflection Chords • 2. Parabolic Vertical Crests & Earthwork Prisms • 3. Three-Wire Spirit Leveling & Roadway Grades
Chapter 3	Guiding a twin-turboprop across a high-desert pass against a 28 kt wind at 40° off course into a 6,200 ft mountain strip at 28°C	Aeronautical velocity vector triangles, crab angle, time-speed-distance endurance, aircraft center-of-gravity moment, and barometric density altitude	1. Wind Vector Triangles & Crosswind Crab Angles • 2. Aircraft Center-of-Gravity Balance Moments • 3. Barometric Lapse Rates & Density Altitude

Chapter 1: The Lost Cornerstone in the Pine Barrens

On a crisp November morning in southern New Jersey's Pine Barrens, a two-person boundary crew stands beside a charred pitch-pine stump with a 1794 metes-and-bounds deed unrolled across the hood of their truck. A timber company and the state park service are adjudicating the southeastern corner of a historic bog-iron furnace tract. The original colonial surveyor recorded the boundary in Edmund Gunter's 66-foot chains, sexagesimal compass bearings (**DDD.MMSS**), and stone-wall offsets measured in feet and fractional inches. The legal description ends at a brass-capped granite cornerstone set beside a cedar creek. Over two centuries of wildfire and alluvial wash have buried that stone beneath three feet of sandy loam on the far side of a cedar-choked ravine.

The party chief cannot stretch a flat steel tape across the gulf. Instead, the crew sets their total station over Witness Station A on the north rim and sights along azimuth 47° 22′ 30″ across an inclined slope distance of 412.68 ft at a vertical angle of 6° 15′ 00″. To recover the buried monument and certify the parcel's legal area, the crew must work through three linked mathematical problems:

1. Reduce the inclined ravine shot to flat horizontal ground distance $D_H = 410.10$ ft and vertical rise $\Delta V = 44.92$ ft, then project North and East coordinates to the witness oak on the opposite bluff.
2. Solve the inverse coordinate problem between the recovered witness oak and the historic deed coordinates to find the exact compass bearing and tape distance to the buried brass cap.
3. Verify benchmark rod readings with three-wire stadia leveling, close the four-sided boundary polygon around the tract, and compute the enclosed legal acreage.

1.1 Sexagesimal Sightlines, Slope Reduction, and North-Clockwise Projections

Every boundary retracement begins by translating raw optical circle readings, architectural stone-wall ties, and inclined laser rangefinder shots into consistent decimal degrees and horizontal surveyor's feet on a local tangent plane. Once reduced to the horizontal plane, each sightline projects along the North-clockwise azimuth circle into orthogonal Northing and Easting displacements.

1.1.1 Sexagesimal Angles, Architectural Lengths, and Ravine Slope Reduction

A sexagesimal angle read from a theodolite glass circle or historic deed plat consists of integer degrees D°, integer arcminutes $M' \in [0,\ 59]$, and arcseconds $S'' \in [0,\ 60)$. Because one degree contains 60 arcminutes and $60 \times 60 = 3{,}600$ arcseconds, converting between packed sexagesimal notation **DDD.MMSS** (where the first two digits after the decimal point store arcminutes MM and the next two store arcseconds SS, converted directly via [↰] [▸DEG] / [↰] [▸HR] and [↱] [▸HMS]) and decimal degrees ϑ_{dec} evaluates the base-60 polynomial:

$$\theta_{\text{dec}} = D + \frac{M}{60} + \frac{S}{3600} \qquad \textbf{(Eq. 1.1)}$$

In this expression, D is the whole-degree component, M is the arcminute count divided by 60 min/deg, and S is the arcsecond count divided by 3,600 sec/deg. For a historic deed azimuth of 47° 22′ 30″ (written in packed field notation as **47.2230**), dividing the 22 arcminutes by 60 yields 0.366667°, dividing the 30 arcseconds by 3,600 yields 0.008333°, and summing with 47° via [↖] [▸DEG] produces $\vartheta_{\text{dec}} = \mathbf{47.375°}$. Reversing the conversion via [↗] [▸HMS] on 47.375° extracts the integer part $D = 47°$, multiplies the fractional remainder 0.375 × 60 = 22.5′ to isolate $M = 22′$, and multiplies the remaining 0.5′ × 60 = 30″ to recover **47.2230** (47° 22′ 30″).

Linear tie measurements between colonial stone walls, iron witness pipes, and Gunter's chain calls (1 chain = 4 rods = 66.0 ft, converted via [↖] [→ch] and [↗] [→rod]) require a companion bridge between trades. Geodetic coordinates express length in **surveyor's decimal feet** (L_{ft}, graduated in tenths and hundredths of a foot), whereas structural foundations and masonry ties specify **whole feet *F*, whole inches *I*, and binary fractional inches *N*/*D*** with denominator $D \in (2, 4, 8, 16, 32, 64)$ (operational shorthand [↖] [f-i]):

$$L_{\text{ft}} = F + \frac{I + \frac{N}{D}}{12} \qquad \textbf{(Eq. 1.2)}$$

Here F represents whole feet, I represents whole inches, and N/D represents the binary fraction of an inch divided by 12 in/ft. When the 1794 furnace plat calls for a witness offset of 24 ft 9-3/8 in from the millrace foundation wall, combining the fractional inch 3/8 = 0.375 in with 9 in gives 9.375 in, dividing by 12 in/ft via [↖] [f-i] yields 0.78125 ft, and adding 24 ft produces the surveyor's coordinate length $L_{\text{ft}} = \mathbf{24.78125\ ft}$.

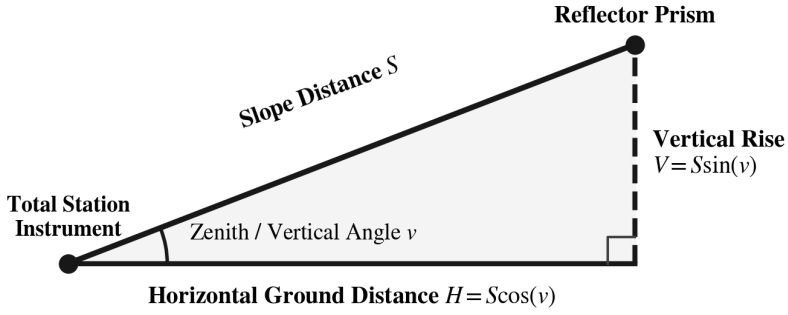


Figure 1.1: *Right-triangle trigonometric reduction of inclined slope distance S to horizontal ground distance H and vertical elevation difference V.*

When the total station on the north rim of the Pine Barrens ravine sights a prism across an inclined **slope distance** S at a vertical slope angle ν (measured above + or below - the horizontal horizon), legal property boundaries and map projections depend on the flat **horizontal ground distance** D_H and the **vertical elevation difference** ΔV (operational shorthand [SLP]):

$$D_H = S\cos(\nu), \qquad \Delta V = S\sin(\nu) \qquad \textbf{(Eq. 1.3)}$$

In this right-triangle reduction, S is the slant range in Register Y, ν is the vertical inclination angle in Register X, D_H is the reduced horizontal distance placed into Register X, and ΔV is the vertical elevation difference placed into Register Y. Sighting across the

cedar ravine over a slope distance of S = 412.68 ft at a vertical angle of ν = 6° 15′ 00″ (**6.15** in **DDD.MMSS**, or 6.25° via [↰] [▸DEG]) projects a horizontal ground distance of D_H = **410.10 ft** in Register X and a vertical elevation difference of ΔV = **44.92 ft** in Register Y.

1.1.2 The North-Clockwise Meridian Frame and Forward Coordinate Projection

In plane surveying, every boundary station is indexed by an ordered pair of orthogonal coordinates (N, E), where **Northing** (N) measures distance north of a reference origin and **Easting** (E) measures distance east of the origin, while directions are measured as a full-circle **Azimuth** α ∈ [0°, 360°) clockwise from True North (0°). Projecting a known horizontal ground distance D_H along a known clockwise azimuth α from a starting station $A(N_A,\ E_A)$ to determine the North change (ΔN, **Latitude**), the East change (ΔE, **Departure**), and the target coordinates $B(N_B,\ E_B)$ constitutes the **Forward Coordinate Geometry Problem** ([TRV]):

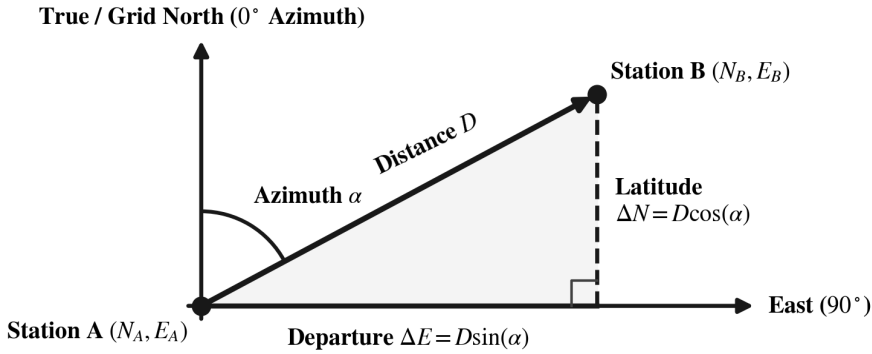


Figure 1.2: *North-clockwise geodetic azimuth frame and orthogonal decomposition of sightline distance D into latitude change and departure.*

$$\Delta N = D_H\cos(\alpha),\qquad \Delta E = D_H\sin(\alpha),\qquad N_B = N_A + \Delta N,\qquad E_B = E_A + \Delta E \qquad \textbf{(Eq. 1.4)}$$

Because cos α is positive in the northern quadrants (270° < α < 90°) and negative in the southern quadrants (90° < α < 270°), while sin α is positive in the eastern quadrants (0° < α < 180°) and negative in the western quadrants (180° < α < 360°), these North-clockwise projection identities assign the correct algebraic signs in all four compass quadrants without manual lookup tables. This orthogonal decomposition of a polar vector (α, D_H) into coupled cosine and sine components is mathematically identical to decomposing apparent alternating-current power S into real working power $P = S\cos\varphi$ and reactive magnetizing power $Q = S\sin\varphi$ in Unit II (**SC-14**) or resolving complex impedance reflections in Unit V (**SC-50**).

On the **SC-35**, [TRV] expects the clockwise **Azimuth α in Register *Y*** and the **Horizontal Distance D_H in Register *X***, pushes the North latitude change $\Delta N = D_H$ cos α into Register Y and the East departure $\Delta E = D_H$ sin α into Register X, and automatically accumulates the leg into the running traverse registers ($\Sigma_{\Delta N}$ and $\Sigma_{\Delta E}$). When our Pine Barrens crew stages the converted azimuth α = 47.375° in Register Y and the reduced horizontal distance D_H = 410.10 ft in Register X and presses [TRV], the instrument deposits ΔN = 410.10 cos(47.375°) = **277.75 ft** into Register Y and ΔE = 410.10

$\sin(47.375^\circ)$ = **301.76 ft** into Register X.

1.2 Inverse Monument Recovery, Traverse Closure, and Shoelace Acreage

Once the crew carries coordinates across the ravine to the witness oak, they must solve the inverse problem—turning known coordinate pairs back into a compass bearing and tape distance to pinpoint the buried brass cornerstone—check their benchmark rod transfer with three-wire stadia leveling, and close the four-sided perimeter loop to certify its legal area.

1.2.1 Inverse Coordinate Geometry and Legal Quadrant Bearings

Running the Forward projection in reverse—starting from two known monument coordinates (N_A, E_A) and (N_B, E_B) and recovering the straight-line horizontal distance D and clockwise North azimuth $\alpha \in [0^\circ, 360^\circ)$ between them—is the **Inverse Coordinate Geometry Problem** (INV). From the coordinate displacements $\Delta N = N_B - N_A$ in Register Y and $\Delta E = E_B - E_A$ in Register X, the Euclidean norm gives D in Register X, while the two-argument arctangent resolves the azimuth in Register Y:

$$\Delta N = N_B - N_A, \qquad \Delta E = E_B - E_A, \qquad D = \sqrt{(\Delta N)^2 + (\Delta E)^2}, \qquad \alpha = \operatorname{atan2}(\Delta E, \Delta N) \pmod{360^\circ} \qquad \textbf{(Eq. 1.5)}$$

Whereas geodetic computations use 0°–360° clockwise azimuths, colonial and modern metes-and-bounds property deeds record directions as **Quadrant Bearings** (BRG). A quadrant bearing begins on either the **North (N)** or **South (S)** meridian and turns an acute angle $\vartheta \in [0^\circ, 90^\circ]$ toward either the **East (E)** or **West (W)**:

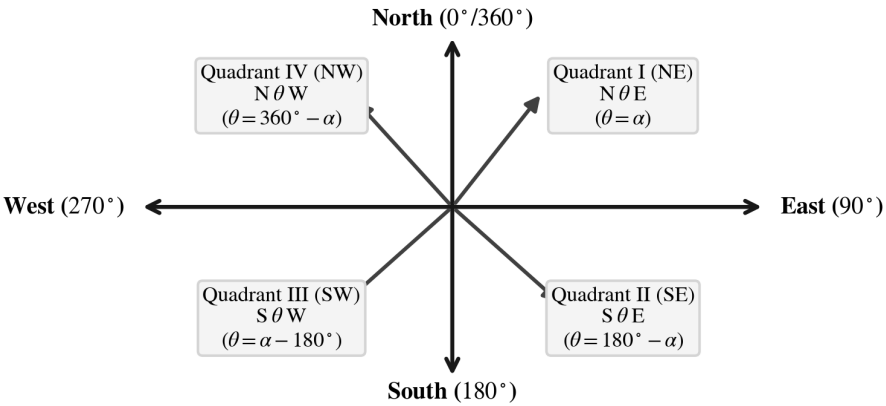


Figure 1.3: *Four-quadrant surveyor bearing angles (NE, SE, SW, NW) measured acute from the North-South meridian.*

In Quadrant I (NE, $0^\circ \le \alpha \le 90^\circ$), the acute meridian angle is $\vartheta = \alpha$ (**N** ϑ **E**). In Quadrant II (SE, $90^\circ < \alpha \le 180^\circ$), $\vartheta = 180^\circ - \alpha$ (**S** ϑ **E**). In Quadrant III (SW, $180^\circ < \alpha \le 270^\circ$), $\vartheta = \alpha - 180^\circ$ (**S** ϑ **W**). And in Quadrant IV (NW, $270^\circ < \alpha < 360^\circ$), $\vartheta = 360^\circ - \alpha$ (**N** ϑ **W**). Between the recovered Witness Oak at (N_1 = 1000.00 ft, E_1 = 1000.00 ft) and the historic deed coordinates of the buried Pine Barrens cornerstone at (N_2 = 640.00 ft, E_2 = 520.00 ft), the coordinate differences ΔN = -360.00 ft and ΔE = -480.00 ft yield an inverse horizontal distance of $D = \sqrt{(-360.00)^2 + (-480.00)^2}$ = **600.00 ft** (9.0909 chains via ↖ →ch). Because both displacements are negative, the sightline lies in Quadrant III (SW) with acute meridian angle $\vartheta = \arctan(480/360) \approx 53.1301^\circ = 53^\circ\ 07'\ 48''$, corresponding to

a full-circle azimuth of $\alpha = 180^\circ + 53.1301^\circ =$ **233.1301°** and a legal deed bearing of **S 53° 07′ 48″ W**.

1.2.2 Closed Polygon Traverses, Three-Wire Stadia Leveling, and Shoelace Acreage

With the lost cornerstone uncovered, the survey crew checks the vertical datum from the creek benchmark using three-wire stadia leveling ([↰] [LVL]) and walks the closed perimeter loop around the parcel, accumulating each boundary leg into the **@COGO** ledger via [TRV] and [↰] [PT+] and closing back onto the Point of Beginning (POB).

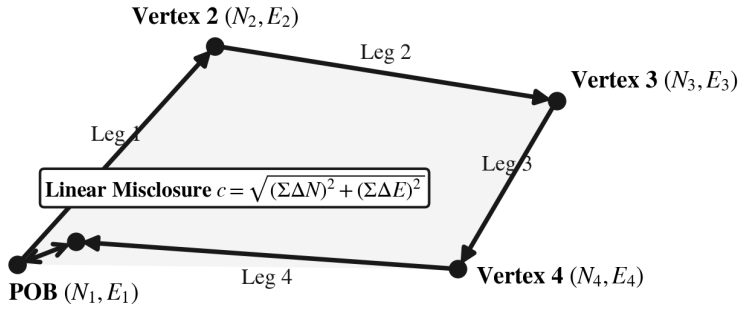


Figure 1.4: *Closed three-leg polygon traverse illustrating vector summation and the linear misclosure gap at the Point of Beginning.*

In an exact closed polygon of n sides, the vector sum of all North latitudes $\sum_{\Delta N} = \sum_{i=1}^{n} \Delta N_i$ and the vector sum of all East departures $\sum_{\Delta E} = \sum_{i=1}^{n} \Delta E_i$ both equal zero. Because field measurements contain small random optical and atmospheric errors, the accumulated traverse returns to a coordinate slightly displaced from the POB. The hypotenuse of that residual gap is the **Linear Error of Closure** E_c ([↻] [CLS]), and dividing total perimeter $P = \sum_{i=1}^{n} D_i$ by E_c defines the **Relative Precision Ratio**. Once closure meets statutory accuracy standards, Nathaniel Bowditch's 1807 **Compass Rule** distributes the misclosure across each leg i in proportion to its length fraction $D_i \,/\, P$:

$$E_c = \sqrt{(\Sigma_{\Delta N})^2 + (\Sigma_{\Delta E})^2}, \qquad \text{Precision Ratio} = 1 : \frac{P}{E_c}, \qquad \delta N_i = -\Sigma_{\Delta N}\left(\frac{D_i}{P}\right), \qquad \delta E_i = -\Sigma_{\Delta E}\left(\frac{D_i}{P}\right) \tag{Eq. 1.6}$$

Here E_c is the linear misclosure magnitude in feet, P is the total polygon perimeter, and (δN_i, δE_i) are the additive latitude and departure corrections applied to leg i. From the sequential vertex coordinates (N_1, E_1), (N_2, E_2), ..., (N_n, E_n) of a closed simple polygon (with (N_{n+1}, E_{n+1}) = (N_1, E_1)), **Gauss's Shoelace Formula** evaluates the enclosed planar area A (displayed live as [↻] [△] alongside [↻] [CLS] inside the **@COGO** reference table [↻] [▦], while [↻] [△] evaluates right-triangle and sub-parcel areas), which converts to statutory acreage via [↰] [→ac] (43,560 ft²/acre) and metric hectares via [↻] [→ha] (10,000 m²/ha):

$$A = \frac{1}{2}\left|\sum_{i=1}^{n}(E_i N_{i+1} - E_{i+1} N_i)\right|, \qquad \text{Acres} = \frac{A_{\text{ft}^2}}{43{,}560} \tag{Eq. 1.7}$$

For a four-leg rectangular boundary loop of total perimeter P = 3,200.00 ft (spanning 600.00 ft North-South by 1,000.00 ft East-West) whose field measurements return with net

latitude misclosure $\sum_{\Delta N}$ = -0.12 ft and net departure misclosure $\sum_{\Delta E}$ = +0.16 ft, the linear error of closure is $E_c = \sqrt{((-0.12)^2 + 0.16^2)}$ = **0.20 ft** ([R/S] [CLS]), yielding a relative precision ratio of 1 : (3200 / 0.20) = **1 : 16,000**, and Gauss's shoelace sum across the adjusted corners in [R/S] [▦] gives an enclosed area of A = **600,000 ft²**, or 600,000 / 43,560 **≈ 13.7741 acres** (5.5742 ha).

1.3 Retracing the Pine Barrens Boundary on the Stack and Solver Workspace

On the **SC-35**, two-component survey operators ([SLP], [TRV], [INV]) and three-wire stadia leveling ([↖] [LVL]) execute immediately on the RPN stack while [↖] [N↔E] and [x⇄y] swap (N, E) ↔ (E, N) axes whenever needed. At the same time, the **@COGO** reference table ([R/S] [▦]) provides live polygon [R/S] [△] and [R/S] [CLS] readouts, and pressing [↖] [LVL] or [R/S] [△] when $X = 0$, $Y = 0$ (or after [R/S] [SOLVE]) opens the bidirectional Equation Solver workspaces (**LVL: E = H + B - F** with variable softkeys [E], [H], [B], [F], and **TRI: A = B × H / 2** with variable softkeys [A], [B], [H]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
[↖] [▶DEG] / [R/S] [▶HMS] / [↖] [▶HR]	Packed Sexagesimal Conversion	—	—	**DDD.MMSS** or ϑ_{dec}	$\vartheta_{\text{dec}} = D + M/60 + S/3600$
[↖] [f-i]	Architectural Foot-Inch Parser	—	Feet F	Inches I (or packed)	$L_{\text{ft}} = F + (I + N/D)/12$
[SLP]	Slope Distance Reduction	—	Slope S	Inclination ν°	$Y = \Delta V = S \sin \nu$, $X = D_H = S \cos \nu$
[TRV]	Forward North-Clockwise COGO	—	Azimuth α°	Distance D_H	$Y = \Delta N = D_H \cos \alpha$, $X = \Delta E = D_H \sin \alpha$ (accumulates leg)
[INV]	Inverse Coordinate Geometry	—	Latitude ΔN	Departure ΔE	$Y = \alpha$, $X = D = \sqrt{((\Delta N)^2 + (\Delta E)^2)}$
[BRG]	Acute Quadrant Bearing	—	—	Azimuth α°	Maps [0°, 360°) → ϑ ∈ [0°, 90°]
[↖] [LVL]	3-Wire Stadia & Leveling Solver	Top wire T	Mid wire M	Bot wire B_w	$Y = (T+B_w)/2$, $X = 100(T-B_w)$; [↖] [LVL] workspace ([E], [H], [B], [F])
[↖] [PT+]	Polygon Traverse Accumulator	—	Latitude ΔN	Departure ΔE	Appends vertex to **@COGO** ([R/S] [▦]) & shoelace sum
[R/S] [CLS]	Linear Error of Closure	—	$\sum_{\Delta N}$	$\sum_{\Delta E}$	$E_c = \sqrt{((\sum_{\Delta N})^2 + (\sum_{\Delta E})^2)}$
[R/S] [▦] / [R/S] [△] / [↖] [→ac] / [R/S] [→ha]	Polygon **@COGO** Area & Land Units	—	Base B	Height H / A_{ft^2}	[R/S] [▦] displays shoelace [R/S] [△] & [R/S] [CLS]; [R/S] [△] opens [R/S] [SOLVE] ([A], [B], [H])

1.3.1 Reducing the Ravine Slope Shot, Projecting Coordinates, and Checking Three-Wire Stadia

Standing at Witness Station A, the party chief begins with the raw ravine sightline from the field book: an azimuth of 47° 22′ 30″ (**47.2230** [↶] [▸HR] = 47.375°) and an inclined slope distance of S = 412.68 ft at a slope angle of 6° 15′ 00″ (**6.15** [↶] [▸HR] = 6.25°). Keying **412.68** [ENTER] **6.25** [SLP] places the vertical rise ΔV = **44.92 ft** into Register Y and the reduced horizontal distance D_H = **410.10 ft** into Register X. Next, placing the azimuth α = 47.375° in Register Y and D_H = 410.10 ft in Register X and pressing [TRV] [↶] [N↔E] places the northward latitude change ΔN = **277.75 ft** into Register Y and the eastward departure ΔE = **301.76 ft** into Register X. Finally, reading three-wire stadia crosshair intercepts on the creekside benchmark rod—top wire 7.82 ft in Register Z, middle wire 5.42 ft in Register Y, and bottom wire 3.02 ft in Register X—and pressing [↶] [LVL] verifies crosshair balance by placing the expected mid-wire reading **5.42 ft** into Register Y and the horizontal stadia sight distance 100 × (7.82 - 3.02) = **480.00 ft** into Register X:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
412.68 [ENTER] **6.25** [SLP]	0	0	44.92	410.10	Reduce slope shot to vertical ΔV = 44.92 ft (Y) and horizontal D_H = 410.10 ft (X)
47.2230 [↶] [▸HR] [x⇄y] [TRV] [↶] [N↔E]	0	44.92	277.75	301.76	Convert 47° 22′ 30″ → 47.375° into Y, keep D_H = 410.10 in X, and project ΔN = 277.75, ΔE = 301.76
7.82 [ENTER] **5.42** [ENTER] **3.02** [↶] [LVL]	301.85	301.85	5.42	**480.00**	3-wire stadia check: expected mid-wire 5.42 ft in Y and stadia distance 480.00 ft in X

To recover the buried 1794 brass cornerstone at (N_2 = 640.00 ft, E_2 = 520.00 ft) from the Witness Oak at (N_1 = 1000.00 ft, E_1 = 1000.00 ft), the crew keys **640** [ENTER] **1000** [-] **520** [ENTER] **1000** [-] [INV], placing the horizontal stakeout distance D = **600.00 ft** in Register X and the angle in Register Y. Swapping the azimuth to Register X via [x⇄y] and pressing [BRG] [↷] [▸HMS] converts the direction into the acute Southwest deed bearing **53.0748** (**S 53° 07′ 48″ W**).

1.3.2 Auditing Traverse Closure in @COGO ([↷] [▦]) and Solving Sub-Parcel Geometry ([↷] [△], [↶] [LVL])

To close a multi-leg boundary loop and evaluate its enclosed acreage, each [TRV] or [↶] [PT+] entry accumulates coordinates into the live **@COGO** ledger. For a three-leg right-triangular sub-parcel starting at (0, 0) with Leg 1 running 300.00 ft at Azimuth 0° (Due North), Leg 2 running 400.00 ft at Azimuth 90° (Due East), and Leg 3 running 500.00 ft at Azimuth 233.1301° back to the origin, staging Azimuth in Y and Distance in X for each [TRV] call populates the live **@COGO** ledger in [↷] [▦] (**CLOS** = **0**, **AREA** = **60000**) and converts the area to acres via [↶] [→ac]:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↖] [CLEAR] [CLΣ]	0	0	0	0	Clear **@COGO** traverse statistical and shoelace vertex registers
0 [ENTER] **300** [TRV] [↖] [PT+]	0	0	0	1	Project Leg 1 ($\alpha_1 = 0°$, $D_1 = 300 \rightarrow \Delta N_1 = 300$, $\Delta E_1 = 0$) and store Vertex 1
90 [ENTER] **400** [TRV] [↖] [PT+]	0	0	400	2	Project Leg 2 ($\alpha_2 = 90°$, $D_2 = 400 \rightarrow \Delta N_2 = 0$, $\Delta E_2 = 400$) and store Vertex 2
233.1301 [ENTER] **500** [TRV] [↖] [PT+]	0	400	-400.00	3	Project Leg 3 (ΔN_3 = -300, ΔE_3 = -400); [→] [▦] reports **CLOS = 0**, **AREA = 60000**
60000 [↖] [→ac]	0	400	-400.00	**1.3774**	Convert enclosed shoelace area 60,000 ft² from [→] [▦] to 1.3774 acres

When the full four-leg Pine Barrens traverse of total perimeter P = 3,200.00 ft returns to the Point of Beginning with net latitude misclosure $\sum_{\Delta N}$ = -0.12 ft and net departure misclosure $\sum_{\Delta E}$ = +0.16 ft, pressing [→] [CLS] (or **0.12** [+/−] [ENTER] **0.16** [→] [CLS]) resolves the linear error of closure E_c = **0.20 ft** in Register X, and keying **3200** [x⇄y] [÷] yields the relative precision denominator **16,000** (1 : 16,000). Furthermore:

- **Triangle Sub-Parcel Solver ([→] [△]):** If the county register requires carving off a triangular creekside easement of exact area A = 43,560 ft² (1.0000 acre) along a B = 300.00 ft baseline, pressing [→] [SOLVE] [→] [△] opens the [→] [SOLVE] **Equation Solver Workspace** displaying **A = B × H / 2** with variable softkeys [A] ([LFU0]), [B] ([LFU1]), and [H] ([LFU2]). Storing **43560** [A] and **300** [B] and pressing [→] [SOLVE] [H] solves the equation backward to return the required perpendicular easement offset H = 2 × 43,560 / 300 = **290.40 ft**.
- **Differential Leveling Solver ([↖] [LVL]):** Pressing [→] [SOLVE] [↖] [LVL] opens the [↖] [LVL] **Equation Solver Workspace** displaying **E = H + B - F** with variable softkeys [E] ([LFU0]), [H] ([LFU1]), [B] ([LFU2]), and [F] ([LFU3]), allowing the crew to solve forward for new station elevation E or backward for a target foresight rod reading F.

1.3.3 Try It on the Stack

Use these three boundary retracement exercises to practice slope reduction, coordinate projection, and traverse closure on the stack:

1. **Slope Reduction and Forward Coordinate Projection:** Reduce slope distance S = 412.68 ft at inclination ν = +6° 15′ 00″ (**6.15** [↖] [▸HR] = 6.25°) via **412.68** [ENTER] **6.25** [SLP] to obtain D_H = 410.10 ft in Register X and ΔV = 44.92 ft in Register Y, then place Azimuth α = 47° 22′ 30″ (**47.2230** [↖] [▸HR] = 47.375°) in Register Y and D_H = 410.10 ft in Register X and press [TRV] [↖] [N↔E]. *(Verified stack results: ΔV = **44.92 ft**, D_H = **410.10 ft**, ΔN = **277.75 ft** in Y, and ΔE = **301.76 ft** in X.)*
2. **Inverse Bearing Between Witness Monuments:** Given two iron boundary pipes at A(1200.00, 1500.00) and B(1585.20, 1912.60), key **1585.20** [ENTER] **1200** [-] **1912.60** [ENTER] **1500** [-] [INV] and [x⇄y] [BRG] [→] [▸HMS] to find the boundary length and legal quadrant bearing. *(Verified Register X results: $D \approx$ **564.4624 ft**, $\alpha \approx$ **46.9669°**, packed DMS bearing **46.5801** or N 46° 58′ 01″ E.)*
3. **Traverse Closure Precision and Acreage Conversion:** A four-leg boundary loop of perimeter P = 2,500.00 ft has net misclosures $\sum_{\Delta N}$ = +0.15 ft and $\sum_{\Delta E}$ = -0.20 ft and an enclosed shoelace area in [→] [▦] of A = 326,700 ft². Key **0.15** [ENTER] **0.20** [+/−] [→] [CLS] for E_c, **2500** [x⇄y] [÷] for the precision ratio, and **326700** [↖] [→ac] for acreage. *(Verified Register X results: E_c = **0.25 ft**, precision **1 : 10,000**, and **7.5000 acres**.)*

Having recovered the Pine Barrens cornerstone and closed the planar boundary traverse, we move from flat timberland into steep mountain terrain in Chapter 2 to fit horizontal and vertical highway curves around a granite bluff, compute earthwork prism volumes, and carry vertical benchmark elevations up a canyon wall.

Chapter 2: Threading the Switchback Grade Through Granite Gorge

High in the Sierra Nevada, a state highway alignment climbing a narrow river canyon encounters an overhanging granite bluff that juts across the projected roadbed. The two straight approach centerlines (tangents) of the mountain highway intersect inside the solid rock face at the Point of Intersection (PI) with an exterior deflection angle of $\Delta = 28^\circ\ 30'\ 00''$ (28.5°). To swing the two-lane roadbed smoothly around the bluff without blasting away half the cliff or exceeding safe lateral acceleration for logging trucks, the resident highway engineer specifies a tangent circular curve of centerline radius $R = 1{,}200.00$ ft.

Before the earthmoving crew can stake the curve or set grade rails for the paving machines, three coupled geometric problems must be solved in the field:

1. Determine how far back from the inaccessible PI to set the Point of Curvature (PC) and Point of Tangency (PT) via the tangent distance T, how long the curved centerline arc L and straight-line long chord LC will be, and how to tighten the curve if a rockfall barrier limits the allowable arc length.

2. Compute the parabolic vertical crest elevations where the $+2.50\%$ approach grade transitions into a -1.75% downgrade over an 800 ft (8-station) vertical curve, along with the cubic yards of rock cut in the trapezoidal prism.

3. Carry a millimeter-grade vertical datum up from a bronze U.S. Geological Survey benchmark on the canyon floor using three-wire differential spirit leveling, and verify the longitudinal road grade and culvert drainage flow.

2.1 Tangential Circular Curves, Vertical Crest Paraboloids, and Earthwork Volumes

When two straight highway or railway centerlines intersect horizontally at a **Point of Intersection (PI)** with an exterior deflection angle Δ, engineers fit a tangent circular arc of radius R between the **Point of Curvature (PC)** and the **Point of Tangency (PT)**. When two longitudinal grades $g_1\%$ and $g_2\%$ meet vertically at a crest or sag, engineers fit a constant-curvature parabola ([↖] [VCV]) and tally excavation yardage between cross-sectional end areas ([↗] [CUT]).

2.1.1 Centerline Arc Length, Tangent Distance, and Long Chord Projections

Because the radial lines drawn from the center of curvature O to the tangent points PC and PT meet the entry and exit tangents at exact right angles (90°), the quadrilateral formed by O, PC, PI, and PT has interior angles 90° at PC and PT and 180° - Δ at PI. Consequently, the central angle subtended at O equals the exterior deflection angle Δ.

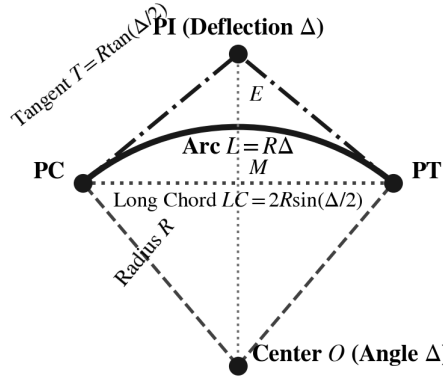


Figure 2.1: *Circular horizontal alignment curve geometry linking the Point of Curvature (PC), Point of Intersection (PI), and Point of Tangency (PT).*

Bisecting the central angle Δ with the radial line from O through the mid-arc apex to PI partitions the layout into two congruent right triangles of half-angle Δ / 2, determining the primary horizontal curve elements ([CRV])—the **Tangent Distance** T from PC (or PT) to PI, the curved centerline **Arc Length** L from PC to PT, and the straight-line **Long Chord** LC connecting PC directly to PT:

$$T = R\tan\left(\frac{\Delta}{2}\right), \qquad L = R \cdot \Delta \cdot \frac{\pi}{180^\circ}, \qquad LC = 2R\sin\left(\frac{\Delta}{2}\right) \tag{Eq. 2.1}$$

In these expressions, R is the centerline radius of curvature in feet, Δ is the exterior deflection angle in decimal degrees, T is the opposite leg of the right triangle of angle Δ / 2 and adjacent side R, L scales the radian measure $\Delta(\pi$ / 180°) by R, and LC doubles the half-chord $R\sin(\Delta$ / 2). For the Granite Gorge highway curve of radius R = 1,200.00 ft in Register Y and deflection angle Δ = 28.5° in Register X (half-angle Δ / 2 = 14.25°), pressing [CRV] yields the tangent setback distance $T = 1200\tan(14.25^\circ) =$ **304.77 ft** in Register Y and the centerline paving arc length $L = 1200 \times 28.5^\circ \times (\pi / 180^\circ) =$ **596.90 ft** in Register X, while the straight-line long chord is $LC = 2400\sin(14.25^\circ) =$ **590.80 ft**.

2.1.2 External Clearance, Parabolic Vertical Curves ([↰] [VCV]), and Prism Earthwork ([↱] [CUT])

To determine how far the curved roadbed swings inward from the granite point at PI and outward from the straight baseline chord connecting PC and PT, the engineer inspects the radial line from O to PI. Subtracting R from the secant hypotenuse R / $\cos(\Delta$ / 2) isolates the **External Distance** E, and subtracting $R\cos(\Delta$ / 2) from R isolates the **Middle Ordinate** M:

$$E = R\left(\frac{1}{\cos(\Delta/2)} - 1\right), \qquad M = R\left(1 - \cos\left(\frac{\Delta}{2}\right)\right) \tag{Eq. 2.2}$$

At the canyon crest, transitioning from an entry grade g_1 (in ft/station, i.e., percent grade) at the Point of Vertical Curvature (PVC) to an exit grade g_2 over a vertical curve of length L stations ([↰] [VCV]) traces the vertical parabola $\text{Elev}(x)$ at distance x stations from PVC, while excavating a rock cut of length L_{cut} (in feet) between cross-sectional end areas A_1 and A_2 (in ft²) yields the average-end-area trapezoidal prism volume V_{cut} ([↱] [CUT]) in bank cubic yards (1 yd³ = 27 ft³):

$$\text{Elev}(x) = \text{PVC} + g_1 x + \left(\frac{g_2 - g_1}{2L}\right)x^2, \qquad V_{\text{cut}} = \frac{A_1 + A_2}{2} \cdot \frac{L_{\text{cut}}}{27} \tag{Eq. 2.3}$$

For a vertical crest curve starting at elevation PVC = 425.00 ft with entry grade g_1 = +2.50%, exit grade g_2 = -1.75%, and curve length L = 8 stations (800 ft), the midpoint elevation at $x = L / 2 = 4$ stations is Elev(4) = 425.00 + 2.5(4) + ((-1.75 - 2.5) / 2(8))(4)² = 425.00 + 10.00 - 4.25 = **430.75 ft** (and at the end of the curve x = 8 stations, PVT = 425.00 + 2.5(8) - 17.00 = **428.00 ft**). Across a L_{cut} = 150.0 ft granite bench cut between end areas A_1 = 320.0 ft² and A_2 = 400.0 ft², [→] [CUT] evaluates V_{cut} = (320 + 400) / 2 × 150/27 = **2,000.00 yd³**.

2.2 Differential Spirit Leveling, Three-Wire Stadia, and Longitudinal Grade

With the horizontal and vertical curves designed around the bluff, the grading crew transfers benchmark elevations up the canyon wall and verifies the vertical climb rate of the pavement.

2.2.1 Backsight, Height of Instrument, and Foresight Benchmark Transfers

To carry a vertical elevation from a Benchmark (BM) of known elevation $H = E_{\text{BM}}$ to an uphill Turning Point (TP) or construction stake, a surveyor sets up a compensated optical level midway between the two stations ([←] [LVL]). Sighting backward along the horizontal optical plane to a graduated leveling rod held plumb on the benchmark yields the **Backsight** reading B, which adds to H to establish the optical **Height of Instrument** (HI = H + B). Turning the telescope forward to a rod held on the new station yields the **Foresight** reading F, which subtracts from HI to determine the new station **Elevation** E:

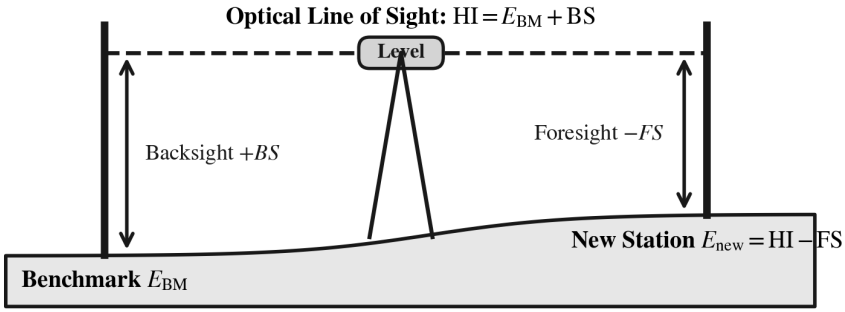


Figure 2.2: *Differential spirit leveling sightline connecting Backsight (BS), Height of Instrument (HI), and Foresight (FS) between benchmarks.*

$$\text{HI} = H + B, \qquad E = \text{HI} - F = H + B - F \tag{Eq. 2.4}$$

In this differential leveling chain, H is the starting benchmark elevation above mean sea level, HI is the elevation of the telescope's horizontal line of sight, and B - F is the net vertical rise (+) or fall (-) between the two rod positions. Setting up above canyon-floor benchmark BM-1 (H = 4,812.40 ft) and reading a backsight of B = 6.85 ft establishes the optical plane at HI = 4812.40 + 6.85 = **4,819.25 ft**. Sighting forward to the curve entry stake at PC (Station **124+50.00** via [→] [STA]) with a center-wire foresight of F = 5.42 ft

determines the PC pavement elevation as $E = 4819.25 - 5.42 =$ **4,813.83 ft**.

2.2.2 Three-Wire Stadia Interval Verification and Longitudinal Grade Percentage

To guard against rod reading blunders and balance backsight and foresight distances, precision optical levels engrave two short **stadia crosshairs**—a **Top wire** T and a **Bottom wire** B_w—spaced symmetrically above and below the **Middle leveling wire** M at an optical stadia interval factor of $K = 100$. Pushing all three crosshair readings onto the stack (T in Z, M in Y, B_w in X) and pressing [↶] [LVL] verifies that the expected mean $(T + B_w)/2$ in Register Y matches M within 0.005 ft and places the horizontal stadia sight distance $D = 100(T - B_w)$ in Register X, while comparing the elevation change ΔE_{vert} across a horizontal alignment run L defines the longitudinal **Grade Percentage** ([→] [GR%]):

$$\left| M - \frac{T + B_w}{2} \right| \le 0.005 \text{ ft}, \qquad D = 100 \times (T - B_w), \qquad \text{GR\%} = \left(\frac{\Delta E_{\text{vert}}}{L} \right) \times 100\% \qquad \textbf{(Eq. 2.5)}$$

On the canyon benchmark shot with Top wire $T = 7.82$ ft in Register Z, Middle wire $M = 5.42$ ft in Register Y, and Bottom wire $B_w = 3.02$ ft in Register X, pressing [↶] [LVL] places the expected mid-wire mean $(7.82 + 3.02)/2 =$ **5.42 ft** in Register Y and the stadia distance $100(7.82 - 3.02) =$ **480.00 ft** in Register X. Carrying the level loop up to PT across a +35.814 ft rise over the $L = 596.90$ ft curve arc and pressing [→] [GR%] confirms a longitudinal climb of GR% = (35.814 / 596.90) × 100% = **+6.00%**.

2.3 Staking the Granite Gorge Alignment on the Stack and Solver Workspace

On the **SC-35**, route geometry, earthwork, and leveling operators work in both immediate RPN stack mode and interactive Equation Solver Workspace mode ([CRV] on [CRV], [↶] [LVL] on [↶] [LVL], and [FLW] on [FLW]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
[CRV]	Circular Curve Tangent, Arc & Solver	—	Radius R	Deflection Δ°	$Y = T = R\tan(\Delta/2)$, $X = L = R\Delta\pi/180°$; [CRV] ([L], [R], [D])
[↶] [VCV]	Parabolic Vertical Curve Elevation	g_1% (PVC in T)	g_2%	Length L (stns)	$\text{Elev}(x) = \text{PVC} + g_1 x + ((g_2 - g_1) / 2L)x^2$
[→] [CUT]	Trapezoidal Prism Earthwork Volume	Area A_1 (ft²)	Area A_2 (ft²)	Length L (ft)	$V = (A_1 + A_2) / 2 \cdot (L/27)$ (yd³)
[↶] [LVL]	3-Wire Stadia & Leveling Solver	Top wire T	Mid wire M	Bot wire B_w	$Y = (T+B_w)/2$, $X = 100(T-B_w)$; [↶] [LVL] ([E], [H], [B], [F])
[→] [GR%]	Longitudinal Percentage Grade	—	Rise ΔE_{vert}	Run L	$\text{GR\%} = (\Delta E_{\text{vert}} / L) \times 100\%$

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
FLW	Culvert & Channel Continuity Solver	—	Area A (ft²)	Velocity V (ft/s)	$Q = A \times V$ (ft³/s); FLW (Q, A, V)
→ STA	Highway Stationing Formatter	—	—	Distance L_{ft}	Formats 12,450.00 ft as **124+50.00** (124.50 stns)

2.3.1 Immediate Stack Execution and Inverse Curve Solving in CRV (CRV) and ↖ VCV

To lay out the R = 1,200.00 ft horizontal curve with deflection angle $\Delta = 28.5°$, evaluate the parabolic vertical crest curve (PVC = 425.00 ft, $g_1 = +2.5\%$, $g_2 = -1.75\%$, $L = 8$ stations), and compute the 150 ft rock cut volume between end areas A_1 = 320 ft² and A_2 = 400 ft², the engineer executes:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1200 ENTER **28.5** CRV	0	0	304.77	596.90	Compute tangent $T = 1200 \tan(14.25°) = 304.77$ ft (Y) and arc $L = 596.90$ ft (X)
425.00 ENTER **2.5** ENTER **1.75** +/− ENTER **8** ↖ VCV	425	425	425	430.75	Compute parabolic crest elevation Elev(4) = 430.75 ft (PVT = 428.00 ft at $x = 8$)
320 ENTER **400** ENTER **150** → CUT	430.75	430.75	430.75	**2000.00**	Compute trapezoidal prism volume $V = (320+400) / 2 \cdot (150/27) =$ 2,000.00 yd³

Now suppose a canyon rockfall catchment wall restricts the maximum allowable curved roadbed arc length to L = 450.00 ft. Pressing → SOLVE CRV opens the **Bidirectional Equation Solver Workspace**, displaying **HCRV: HORIZ CURV** and **L = R × Δ × π / 180** on Line 1 with variable softkeys L (LFU0), R (LFU1), and D (LFU2):

- **Inverse Solve for Deflection Angle Δ (D):** Keying **450** L (LFU0) and **1200** R (LFU1) and pressing → SOLVE D (LFU2) solves backward for the maximum deflection angle $\Delta = 450 \times 180° / 1200\ \pi \approx$ **21.4859°** (**21.2909** via → ▸HMS, or 21° 29′ 09″).
- **Inverse Solve for Curve Radius R (R):** If the 28.5° deflection angle is fixed by the canyon walls (**28.5** D) while L = 450.00 ft (**450** L), pressing → SOLVE R (LFU1) solves backward for the tightened curve radius $R = 450 \times 180° / (28.5° \times \pi) \approx$ **904.6746 ft**.

2.3.2 Running the Three-Wire Canyon Leveling Loop and ↖ LVL Solver (↖ LVL)

At the base of Granite Gorge, the rodman holds the stadia rod on PC while the instrument operator reads Top wire T = 7.82 ft, Middle wire M = 5.42 ft, and Bottom wire B_w = 3.02 ft. Loading those three readings in descending order (**7.82** ENTER **5.42** ENTER **3.02** ↖ LVL) verifies rod symmetry, places the expected mid-wire foresight $F =$ **5.42 ft** in Register Y and the stadia distance $D =$ **480.00 ft** in Register X, and chains into the benchmark transfer from H = 4,812.40 ft (B = 6.85 ft) and the +35.814 ft climb over 596.90 ft to PT:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
7.82 ENTER **5.42** ENTER **3.02** ↶ LVL	0	0	5.42	480.00	Verify 3-wire symmetry; expected mid-wire $F = 5.42$ ft in Y, distance $D = 480.00$ ft in X
x⇄y **4812.40** ENTER **6.85** +	480	480.00	5.42	4819.25	Compute Height of Instrument HI = 4812.40 + 6.85 = 4,819.25 ft
x⇄y -	480	480	480.00	4813.83	Subtract foresight $F = 5.42$ ft to find $E_{PC} = 4{,}813.83$ ft
35.814 ENTER **596.90** → GR%	480	480.00	4813.83	**6.0000**	Evaluate longitudinal road grade GR% = (35.814 / 596.90) × 100% = +6.00%

In the ↶ LVL Equation Solver Workspace (→ SOLVE ↶ LVL, **E = H + B - F** with variable softkeys E, H, B, F), the crew can also solve backward for the target foresight rod reading F needed to set a blue-top grading stake at a specified design elevation E: storing benchmark elevation **4812.40** H, backsight **6.85** B, and design grade elevation **4813.83** E and pressing → SOLVE F returns the exact target rod reading $F =$ **5.4200 ft**.

2.3.3 Try It on the Stack

Practice circular route curve staking, parabolic vertical curves, and three-wire differential leveling with these three field problems:

1. **Highway Curve Layout from Radius and Deflection Angle:** For a circular alignment with radius $R = 1{,}200.00$ ft and deflection angle $\Delta = 28.5°$, evaluate **1200** ENTER **28.5** CRV to find tangent distance T in Register Y and arc length L in Register X, then compute long chord $LC = 2R \sin(14.25°)$. *(Verified stack results: $T =$ **304.77 ft** in Y, $L =$ **596.90 ft** in X, and $LC \approx$ **590.80 ft**.)*

2. **Parabolic Vertical Curve and Earthwork Cut Volume:** Evaluate **425.00** ENTER **2.5** ENTER **1.75** +/- ENTER **8** ↶ VCV for the midpoint parabolic elevation at $x = 4$ stations, and compute the cubic-yard volume of a 180 ft prism cut between end areas $A_1 = 250$ ft² and $A_2 = 380$ ft² via **250** ENTER **380** ENTER **180** → CUT. *(Verified Register X results: Elev(4) =* **430.75 ft** *and $V =$* **2,100.00 yd³**.*)*

3. **Three-Wire Benchmark Transfer and Grade Check:** Verify foresight stadia wires $T = 7.82$ ft, $M = 5.42$ ft, $B_w = 3.02$ ft via **7.82** ENTER **5.42** ENTER **3.02** ↶ LVL, transfer benchmark elevation $H = 842.15$ ft with backsight $B = 5.32$ ft in the ↶ LVL workspace (**842.15** H **5.32** B **5.42** F → SOLVE E), and find the grade percentage (→ GR%) of a -6.00 ft fall over that 480.00 ft stadia distance (**6** +/- ENTER **480** → GR%). *(Verified results: expected mid-wire* **5.42 ft** *in Y, stadia distance $D =$* **480.00 ft** *in X, $E =$* **842.05 ft**, *and GR% =* **-1.2500%**.*)*

From terrestrial sightlines and canyon switchbacks, we lift off into the third dimension of moving air masses, aircraft weight-and-balance moments, and barometric lapse rates in Chapter 3.

Chapter 3: Night Approach Under a Mountain Crosswind

At dusk over the high desert of eastern Oregon, a twin-turboprop medical transport aircraft cruises at 14,200 ft Mean Sea Level (MSL) along a desired true course of 045° (Northeast) at a True Airspeed of $V_{\mathrm{TAS}} = 145$ knots over a $D = 128$ NM mountain leg. Ahead lies a high-elevation valley airstrip at a Pressure Altitude of $P = 6{,}200$ ft, flanked by basalt rimrock. As a dry cold front spills over the Cascades, winds aloft blow at $V_w = 28$ knots at an angle of $\vartheta = 40°$ off the aircraft's nose. Down on the valley floor, the automated weather station reports a late-afternoon desert Outside Air Temperature of $T = 28$°C.

Before descending through the mountain pass, the captain must resolve three interconnected flight-deck calculations that have governed air navigation since Lieutenant Philip Dalton patented the circular E-6B flight computer in 1933:

1. Decompose the 28 kt wind at $\vartheta = 40°$ into its longitudinal headwind component (21.45 kt) and lateral crosswind component (18.00 kt), determine the Wind Correction Angle (WCA) and Ground Speed (GS), and verify that the aircraft can cover the 128 NM leg within its fuel endurance window.

2. Verify the aircraft's longitudinal center-of-gravity moment ($M = W \times A$) after loading a patient litter in the cabin.

3. Evaluate the runway's Density Altitude at Pressure Altitude $P = 6{,}200$ ft and Outside Air Temperature $T = 28$°C to determine the aircraft's true landing roll and single-engine climb margin.

3.1 Aeronautical Wind Triangles, Crab Angles, and Weight-and-Balance Moments

An aircraft in flight moves within an air mass that is itself moving across the ground. Navigating along a fixed geographic track and maintaining longitudinal pitch stability require decomposing wind vectors into orthogonal headwind and crosswind components and balancing mass moments about the aircraft datum.

3.1.1 Wind Vector Decomposition, Crosswind Limits, and Wind Correction Angles

When an aircraft aligns with a runway or flies a cross-country course of heading ψ_{course}, a wind of speed V_w (in knots, converted via [→] [→kt]) blowing from direction ψ_{wind} forms a relative wind angle $\vartheta = \psi_{\mathrm{wind}} - \psi_{\mathrm{course}}$ with the flight path. Staging wind speed V_w in Register Y and relative wind angle ϑ in Register X and pressing [WIND] simultaneously decomposes the wind vector into its longitudinal **Headwind Component** V_{hw} in Register Y and its lateral **Crosswind Component** V_{xw} in Register X (while [→] [XWC] isolates the crosswind component directly):

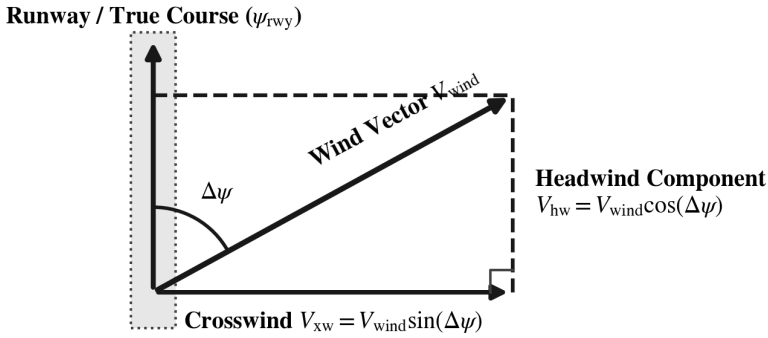


Figure 3.1: *Aeronautical wind vector decomposition into longitudinal headwind and lateral runway crosswind components.*

$$V_{hw} = V_w\cos(\theta), \quad V_{xw} = V_w\sin(\theta), \quad \text{WCA} = \arcsin\left(\frac{V_{xw}}{V_{TAS}}\right), \quad \text{GS} = V_{TAS}\cos(\text{WCA}) - V_{hw} \qquad \textbf{(Eq. 3.1)}$$

In these velocity-triangle equations, V_{TAS} is the aircraft's True Airspeed through the air mass, WCA is the **Wind Correction Angle** (crab angle) derived from the Law of Sines so the aircraft's lateral airspeed component $V_{TAS}\sin(\text{WCA})$ cancels the crosswind drift V_{xw}, and GS is the resulting Ground Speed along the desired track. With a $V_w = 28$ kt wind blowing at $\vartheta = 40°$ off the course (**28** [ENTER] **40** [WND]), the instrument places the headwind component $V_{hw} = 28\cos(40°) = \mathbf{21.45\ kt}$ into Register Y and the crosswind component $V_{xw} = 28\sin(40°) = \mathbf{18.00\ kt}$ into Register X. Crabbing into the wind at $V_{TAS} = 145$ kt by WCA $= \arcsin(18.00 / 145) \approx \mathbf{7.13°}$ ([↶] [ASIN]) holds the desired track over the ridge at a net ground speed of GS $= 145\cos(7.13°) - 21.45 \approx 143.88 - 21.45 = \mathbf{122.43\ kt}$.

3.1.2 Endurance Time-Speed-Distance ([TSD]) and Aircraft Weight & Balance ([↶] [CG])

Across any navigation leg of distance D (in nautical miles, [↶] [→NM], where 1 NM = 6,076.12 ft = 1,852 m) flown at ground speed S = GS (in knots, NM/hr) over elapsed time T (in hours), the **Time–Speed–Distance** relation ([TSD], workspace variables [D], [S], [T]) is $D = S \times T$. Simultaneously, loading passengers, medical litters, and fuel aboard the aircraft shifts the longitudinal **Center of Gravity** ([↶] [CG]) relative to the manufacturer's reference datum: each item of weight W (in pounds) located at station arm A (in inches aft of datum) generates a pitching **Moment** $M = W \times A$ ([↶] [CG], workspace variables [M], [W], [A]), and dividing the total aircraft moment $\sum M$ by total gross weight $\sum W$ locates the center-of-gravity station A_{CG}:

$$D = S \times T, \qquad M = W \times A, \qquad A_{CG} = \frac{\sum M_i}{\sum W_i} \qquad \textbf{(Eq. 3.2)}$$

Over the $D = 128.0$ NM mountain leg at ground speed $S = 122.43$ kt, solving [TSD] (**D** = **S** × **T**) for time T yields $T = 128.0 / 122.43 \approx \mathbf{1.0455\ hr}$ (62.7 min). Meanwhile, a $W = 220$ lb medical litter locked into cabin station arm $A = 145.0$ in adds a pitching moment of $M = 220 \times 145.0 = \mathbf{31{,}900\ lb{\cdot}in}$ via [↶] [CG].

3.2 International Standard Atmosphere, Pressure Altitude, and Density Altitude

Aerodynamic lift, propeller thrust, and turbine mass flow depend on the local mass density ρ of the air column rather than the geometric height above sea level read from a map.

3.2.1 Barometric Lapse Rates and Standard Temperature at Altitude

Under the **International Standard Atmosphere (ISA)**, standard sea-level barometric pressure is $P_0 = 29.92$ inHg (1013.25 hPa) and standard sea-level temperature is $T_0 = 15°C$ (59°F), with temperature decreasing through the troposphere at a standard lapse rate of 2°C per 1,000 ft of pressure altitude P:

$$P = E_{\text{field}} + 1000 \times (29.92 - \text{AltSet}), \qquad T_{\text{ISA}}(P) = 15^\circ\text{C} - 2^\circ\text{C} \times \left(\frac{P}{1000}\right) \qquad \textbf{(Eq. 3.3)}$$

At our high-desert destination airstrip where the Pressure Altitude is P = 6,200 ft, the standard atmospheric temperature for that altitude is T_{ISA} = 15 - 2 × 6.2 = 15 - 12.4 = **2.6°C**.

3.2.2 Thermal Expansion and High-Elevation Density Altitude ([↰] [ALT])

By the ideal gas law ($\rho = P / (R_{\text{specific}}\ T)$, explored at the molecular scale in Unit VI on the **SC-60**), heating an air parcel at constant pressure causes it to expand and become less dense. Each 1°C of warming above the standard altitude temperature $T_{\text{ISA}}(P)$ = 15 - 0.002 P thins the air by the equivalent of +120 ft of altitude gain, defining the **Density Altitude** ([↰] [ALT]) from Pressure Altitude P (in feet) in Register Y and Outside Air Temperature T (in °C) in Register X:

$$\text{DA} = P + 120 \times (T - T_{\text{ISA}}(P)) = P + 120 \times \left(T - \left(15 - 2 \times \frac{P}{1000}\right)\right) \qquad \textbf{(Eq. 3.4)}$$

At the high-desert strip where Pressure Altitude is P = 6,200 ft (T_{ISA} = 2.6°C) and the afternoon Outside Air Temperature is T = 28°C, the +25.4°C thermal deviation above standard temperature adds 120 × 25.4 = +3,048 ft, raising the true Density Altitude via **6200** [ENTER] **28** [↰] [ALT] to DA = 6200 + 3048 = **9,248 ft** and warning the crew to use high-altitude landing flare speeds and full reverse thrust.

3.3 Flying the High-Desert Ridge on the Stack and Solver Workspace

On the **SC-35**, flight-navigation keys operate both as immediate two-register stack shortcuts and as bidirectional Equation Solver workspaces ([WND] on [WND] with variables [X], [V], [A]; [↰] [ALT] on [↰] [ALT] with variables [D], [P], [T]; [TSD] on [TSD] with variables [D], [S], [T]. And [↰] [CG] on [↰] [CG] with variables [M], [W], [A]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
WND	Headwind, Crosswind & Solver	—	Wind V_w (kt)	Angle ϑ°	$Y = V_w \cos\vartheta$, $X = V_w \sin\vartheta$; WND (X, V, A)
→ XWC	Crosswind Component	—	Wind V_w (kt)	Angle ϑ°	$V_{\text{xw}} = V_w \sin(\vartheta)$
TSD	Time–Speed–Distance Solver	—	Speed S (kt)	Time T (hr)	$D = S \times T$; TSD workspace (D, S, T)
↰ CG	Weight & Balance Moment Solver	—	Weight W (lb)	Arm A (in)	$M = W \times A$; ↰ CG workspace (M, W, A)
↰ ALT	Density Altitude & Solver	—	Press Alt P (ft)	OAT T (°C)	DA $= P + 120(T - (15 - 0.002\,P))$; ↰ ALT (D, P, T)

3.3.1 Solving the Wind Triangle, Density Altitude, and Weight & Balance on the Stack

Cruising at $V_{\text{TAS}} = 145$ kt with a $V_w = 28$ kt wind at angle $\vartheta = 40°$ off course, keying **28** ENTER **40** WND places the headwind component $V_{\text{hw}} =$ **21.45 kt** in Register Y and the crosswind component $V_{\text{xw}} =$ **18.00 kt** in Register X. Dividing V_{xw} by 145 and pressing ↰ ASIN gives the crab angle WCA = **7.13°**. Next, keying **6200** ENTER **28** ↰ ALT evaluates the destination Density Altitude DA = **9,248 ft** at Pressure Altitude $P = 6{,}200$ ft and $T = 28$°C, and keying **220** ENTER **145** ↰ CG computes the 31,900 lb·in cabin litter moment:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
28 ENTER **40** WND	0	0	21.45	18.00	Resolve headwind $V_{\text{hw}} = 21.45$ kt in Y and crosswind $V_{\text{xw}} = 18.00$ kt in X
145 ÷ ↰ ASIN	0	0	21.45	7.13	Compute Wind Correction Angle WCA = $\arcsin(18.00/145) \approx 7.13°$
6200 ENTER **28** ↰ ALT	21.4492	21.45	7.13	9248	Compute Density Altitude DA = 6200 + 120(28 - 2.6) = 9,248 ft
220 ENTER **145** ↰ CG	7.1302	7.13	9248	**31900**	Compute cabin litter moment $M = 220$ lb × 145 in = 31,900 lb·in

3.3.2 Inverse Solves in the TSD, ↰ CG, WND, and ↰ ALT Workspaces

To verify fuel endurance, baggage station limits, crosswind limits, and thermal departure ceilings using the **Bidirectional Equation Solver Workspaces**:

- **Inverse Groundspeed Solve in TSD (TSD, D = S × T):** When remaining non-reserve fuel fixes maximum leg endurance time to $T = 0.90$ hr (54.0 min) across the $D = 128.0$ NM mountain pass leg, opening → SOLVE TSD displays **D = S × T** with variable softkeys D (LFU0), S (LFU1), and T (LFU2). Keying **128** D and **0.90** T and pressing → SOLVE S (LFU1) solves backward for the minimum required groundspeed $S = 128.0 / 0.90 \approx$ **142.2222 kt**.

- **Inverse Cargo Weight Limit in ↰ CG (↰ CG, M = W × A):** If the aft baggage bay at arm $A = 180.0$ in (**180** A) is restricted to a maximum moment contribution of $M = 27{,}000$ lb·in (**27000** M), pressing → SOLVE W solves backward for the maximum allowable aft cargo weight $W = 27{,}000 / 180.0 =$ **150.00 lb**.

- **Inverse Temperature Ceiling in ↰ ALT (↰ ALT):** Opening → SOLVE ↰ ALT binds softkeys D (LFU0), P (LFU1), and T (LFU2), allowing the crew to store the maximum chart density altitude D and pressure altitude P and press → SOLVE T for the maximum outside air temperature.

3.3.3 Try It on the Stack

Consolidate your mastery of the **SC-35** flight-navigation keys and solver workspaces with these three cockpit problems:

1. **Wind Vector Decomposition and Crab Angle:** Flying at true airspeed V_{TAS} = 145 kt with a V_w = 28 kt wind at relative angle ϑ = 40°, key **28** [ENTER] **40** [WND] to find headwind V_{hw} in Register Y and crosswind V_{xw} in Register X, then key **145** [÷] [↶] [ASIN] for WCA. *(Verified stack results: V_{hw} = **21.45 kt** in Y, V_{xw} = **18.00 kt** in X, and WCA ≈ **7.13°**.)*

2. **Inverse Wind Angle Limit in [WND] and Density Altitude ([↶] [ALT]):** On an aircraft with a 15.0 kt maximum demonstrated crosswind limit (**15** [X]) landing in a V = 30.0 kt wind (**30** [V]), open the [WND] workspace ([↷] [SOLVE] [WND]) and press [↷] [SOLVE] [A] to find the maximum allowable wind angle ϑ_{max}. Then evaluate **6200** [ENTER] **28** [↶] [ALT] for density altitude at P = 6,200 ft and T = 28°C. *(Verified Register X results: ϑ_{max} = **30.00°** and DA = **9,248 ft**.)*

3. **Endurance Leg Check ([TSD]) and Weight & Balance ([↶] [CG]):** Use [TSD] to find the required groundspeed S to cover D = 165 NM (**165** [D]) in T = 1.25 hr (**1.25** [T] [↷] [SOLVE] [S]), and use [↶] [CG] (**350** [ENTER] **120** [↶] [CG]) to compute the moment of 350 lb of fuel at station arm 120 in. *(Verified Register X results: S = **132.00 kt** and M = **42,000 lb·in**.)*

With terrestrial surveying, circular route alignments, and flight-deck vector navigation complete on the **SC-35**, we turn from the open landscape to industrial power distribution, conduit geometry, and machine-shop instrumentation in **Unit II: Electrical Power, Conduit Geometry & Industrial Trades**.

Unit II: Electrical Power, Conduit Geometry & Industrial Trades

In 1826, in a physics laboratory at the Jesuit Gymnasium of Cologne, Georg Simon Ohm set aside the unstable zinc-copper acid cells of Alessandro Volta. In their place, he wired a bismuth-copper thermocouple—one junction immersed in boiling water, the other packed in crushed ice—across test wires of varying lengths and diameters. In textbook diagrams, wires are drawn as frictionless black lines of zero resistance. Alternating current behaves like direct current, steel conduits turn sharp corners without bending radii, and industrial sensors report directly in pounds per square inch. Through his torsion galvanometer, Ohm proved in *Die galvanische Kette, mathematisch bearbeitet* (1827) that every metallic conductor opposes electron flow in direct proportion to its length L and in inverse proportion to its cross-sectional area A. Whenever an industrial feeder carries current I over hundreds of feet of copper or aluminum wire, the conductor's series resistance R drops a portion of the supply voltage ($V_{\text{drop}} = IR$). It also converts that electrical energy into Joule heat ($P_{\text{heat}} = I^2\,R$) trapped inside the raceway.

Sixty years after Ohm's experiments, Nikola Tesla's 1888 polyphase rotating magnetic field, Mikhail Dolivo-Dobrovolsky's 1891 three-phase transmission line, and Charles Proteus Steinmetz's 1893 phasor calculus established **Three-Phase Alternating Current** as the backbone of industrial power. Single-phase power pulsates to zero twice every cycle (120 times per second on a 60 Hz grid). Three sinusoidal phase voltages staggered by 120° never cross zero simultaneously, delivering smooth torque to heavy induction motors. Vector addition across 120° phase angles introduces the geometric factor $\boldsymbol{\sqrt{3} \approx 1.73205}$. Meanwhile, inductive magnetic fields shift current behind voltage by phase angle φ, scaling real working power by the **Power Factor** $\text{PF} = \cos\varphi$.

Around those electrical feeders, tradespeople solve coupled geometric, mechanical, and fluid problems across the plant floor. Electricians bend rigid steel and thin-wall EMT conduit around structural steel beams using right-triangle cosecant multipliers and NEC Chapter 9 cross-sectional area limits. Machinists follow Frederick Winslow Taylor's 1906 metal-cutting laws (*On the Art of Cutting Metals*) to spin carbide end-mills at the exact rotational speed, tap drill diameter, and bolt-circle chord spacing required to cut steel flanges without tool chatter. Simultaneously, instrumentation and fluid-power technicians size control valves, centrifugal pumps, and noise-immune 4–20 mA current loops.

The **SC-14 Industrial Trades, Electrical & Mechanical** calculator brings these shop-floor and job-site laws into one pocket instrument. In each chapter of this Unit, we follow a single industrial troubleshooting story through three connected mathematical themes—first building the physical intuition and worked examples, and then executing them on the SC-14 stack and in its bidirectional Equation Solver workspaces:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 4	Diagnosing why a 480 V three-phase gantry crane trips under load at the end of a 450 ft #**2 AWG** copper pier feeder and correcting its power factor	Circular mils, NEC Table 8 wire resistance, $1\varphi/3\varphi$ line voltage drop, $\sqrt{3}$ Wye phasors, AC power triangles, and parallel conductors	1. Circular Mils & Feeder Voltage Drop • 2. AC Power Triangles & Capacitor Compensation • 3. Three-Phase $\sqrt{3}$ Phasors & Parallel Conductors
Chapter 5	Routing a parallel bank of four 1-in. EMT conduits over a 6 in. boiler-room I-beam and 4 in. steam riser, then verifying NEC 40% raceway fill	Cosecant offset travel, half-angle tangent conduit shrinkage, 3D rolling offsets, 3-point saddle geometry, and NEC cross-sectional fill	1. Cosecant Offset Travel & Axial Shrinkage • 2. Compound 3D Rolling Offsets & Three-Point Saddles • 3. NEC 40% Raceway Fill & Bracket Fabrication
Chapter 6	Silencing chatter on a *varnothing* 0.500 in. carbide end-mill cutting 4140 steel, verifying hydraulic clamping force, and calibrating a cooling valve and 4–20 mA loop	Tangential surface speed, spindle RPM, thread tap drills, bolt circles, pipe continuity, valve C_v, pump horsepower, and 4–20 mA scaling	1. Rotary Cutting Speeds, Tap Drills & Bolt Circles • 2. Hydraulic Force, Pipe Continuity & Valve C_v • 3. Centrifugal Pump Power & 4–20 mA Process Loops

Chapter 4: The Voltage Drop on Pier 14's Crane Feeder

At a container terminal on Pier 14, a newly commissioned 480 V three-phase ship-to-shore gantry crane rated at $S = 75$ kVA ($I_L \approx 90.21$ A at lagging power factor PF = 0.80) trips its under-voltage relay whenever the hoist motor lifts a loaded forty-foot container at full speed. The marine electrical contractor fed the pier-end crane disconnect through a one-way conduit run of $L = 450$ ft from the substation switchgear using #2 AWG copper conductors. Under NEC Table 310.16, #2 AWG copper is rated for 115 A at 75°C—seemingly plenty of thermal capacity for a 90.21 A load.

What the contractor overlooked is the physical resistance of 900 feet of round-trip copper path along the pier. Sizing a long feeder solely by ampacity without checking **Voltage Drop** over a 450 ft run invites two compounding failures. First, series $I\,R$ voltage drop starves the crane's 480 V terminals during starting surges. Second, because an AC induction hoist motor delivers constant mechanical shaft power ($P = \sqrt{3}\ V_{LL}\ I_L \cos\varphi$), a drop in terminal voltage V_{LL} forces the motor to draw **higher line current** I_L, increasing $I^2\,R$ heating inside the conduit until the thermal overload trips. To fix Pier 14's crane feeder before the next cargo ship docks, the port electrical engineer must work through three linked electrical analyses:

1. Evaluate the circular-mil cross-section and NEC Table 8 resistance of #2 AWG copper to find the exact single-phase and three-phase line voltage drop over the 450 ft run.
2. Decompose the crane's 75 kVA apparent load into real working power (60 kW) and reactive magnetizing power (45 kVAR), and size a parallel capacitor bank to raise the power factor.
3. Solve backward for the maximum allowable conduit length and minimum wire gauge, and compare upsizing the feeder against running two #1/0 AWG conductors in parallel per phase.

4.1 Conductor Resistivity, Circular Mils, and Feeder Voltage Drop

Sizing a long industrial feeder requires linking the geometric cross-section of a round metallic conductor to its ohmic resistance per thousand feet and evaluating the series voltage drop across single-phase and three-phase paths.

4.1.1 Circular Mil Cross-Sections and American Wire Gauge Resistivity

In Euclidean geometry, a cylindrical wire of diameter d has cross-sectional area $A = (\pi/4)d^2$. To eliminate the transcendental factor $\pi/4$ from field cable tables, North American electrical engineering defines the **Circular Mil** (CM) as the area of a circle whose diameter is exactly one mil (1 mil = 0.001 in), so area in circular mils equals the square of the diameter in mils:

$$\mathrm{CM} = d_{\text{mils}}^2 = (1000 \cdot d_{\text{inches}})^2, \qquad R_{\text{wire}} = \frac{K \cdot L}{\mathrm{CM}} \tag{Eq. 4.1}$$

In these expressions, d_{mils} is the conductor diameter in thousandths of an inch, L is the conductor length in feet, and K is the material's **specific resistance constant** (in Ω·cmil/ft at 75°C operating temperature), equal to $K_{\text{Cu}} = 12.9$ Ω·cmil/ft for uncoated copper and $K_{\text{Al}} = 21.2$ Ω·cmil/ft for aluminum (both listed in the NEC reference table). Solid and stranded building conductors are indexed on the logarithmic **American Wire Gauge** (AWG) scale, in which smaller gauge numbers designate thicker conductors that double in cross-sectional area every three gauge steps down. NEC Chapter 9, Table 8 tabulates the direct-current resistance R_{1000} in **ohms per thousand feet** (Ω/kft at 75°C) alongside NEC Table 310.16 75°C copper ampacities:

Wire Size (AWG)	Circular Mils (CM)	Copper Resistance R_{1000} (Ω/kft at 75°C)	75°C Copper Ampacity (A)
#14 AWG (14)	4,110	3.07 Ω/kft (3.14 coated)	20 A (15 A OCPD rule)
#12 AWG (12)	6,530	1.93 Ω/kft (1.98 coated)	25 A (20 A OCPD rule)
#10 AWG (10)	10,380	1.24 Ω/kft (1.29 coated)	35 A (30 A OCPD rule)
#8 AWG (8)	16,510	0.778 Ω/kft	50 A
#6 AWG (6)	26,240	0.491 Ω/kft	65 A
#4 AWG (4)	41,740	0.308 Ω/kft	85 A
#2 AWG (2)	66,360	0.194 Ω/kft	115 A
#1 AWG (1)	83,690	0.154 Ω/kft	130 A
#1/0 AWG (0)	105,600	0.122 Ω/kft	150 A

4.1.2 Single-Phase and Three-Phase Feeder Voltage Drop and Parallel Conductors

In a single-phase circuit located a **one-way** physical conduit distance L (in ft) from the panelboard, load current I (in A) travels L feet out along the ungrounded hot conductor and L feet back along the return conductor, traversing a total series path of $2L$ feet.

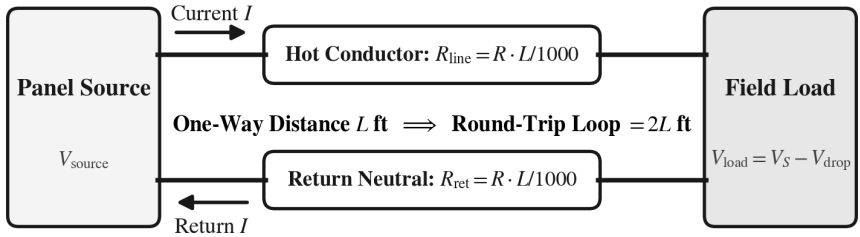


Figure 4.1: *Single-phase two-wire feeder loop resistance model showing outgoing hot conductor and return neutral path.*

Whether expressed via NEC Table 8 resistance R_{1000} (Ω/kft) or specific resistivity K and circular mils CM, the single-phase line **Voltage Drop** $V_{\text{drop},1\varphi}$ (AWG) and its balanced three-phase counterpart $V_{\text{drop},3\varphi}$ (where return currents share phases separated by 120°, replacing the round-trip factor 2 with $\sqrt{3} \approx 1.73205$), together with the combined resistance $R_{\parallel}$ of two parallel conductors or loads R_1 and R_2 ($x \parallel y$), obey:

$$V_{\text{drop},\,1\phi} = \frac{2 \cdot L \cdot I \cdot R_{1000}}{1000} = \frac{2 \cdot K \cdot L \cdot I}{\text{CM}}, \qquad V_{\text{drop},\,3\phi} = \frac{\sqrt{3} \cdot L \cdot I \cdot R_{1000}}{1000}, \qquad R_1 \| R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} \qquad \textbf{(Eq. 4.2)}$$

Here R_{1000} is the NEC Table 8 conductor resistance in Ω/kft, L is the one-way run length in feet, I is the line current in amperes, and $R_1 \parallel R_2 = (R_1^{-1} + R_2^{-1})^{-1}$ is the harmonic-mean parallel operator—the exact same reciprocal sum that governs thin-lens focal length relations ($1/f = 1/d_o + 1/d_i$) in Unit IV (**SC-24**) and shunt reactance networks in Unit V (**SC-50**). On Pier 14's three-phase 480 V crane feeder carrying $I = 90.211$ A over $L = 450$ ft of #2 AWG copper ($R_{1000} = 0.194$ Ω/kft), the single-phase equivalent drop is $V_{\text{drop},1\varphi} = 2 \times 450 \times 90.211 \times 0.194 \,/\, 1000 \approx 15.75$ V and the balanced three-phase line voltage drop is $V_{\text{drop},3\varphi} = \sqrt{3} \times 450 \times 90.211 \times 0.194 \,/\, 1000 \approx \mathbf{13.64\ V}$ (2.84% at rated steady current), which spikes to **20.46 V** (4.26%, starving the crane terminals down to 459.5 V) during a 150% hoist starting surge. Paralleling two #1/0 AWG conductors per phase ($R_1 = R_2 = 0.122$ Ω/kft, so $0.122 \parallel 0.122 = \mathbf{0.061\ \Omega/kft}$ via [x ∥ y]) cuts the steady three-phase voltage drop to $\sqrt{3} \times 450 \times 90.211 \times 0.061 \,/\, 1000 \approx \mathbf{4.29\ V}$ (0.89%).

4.2 Three-Phase Rotating Phasors, Power Triangles, and Reactive Compensation

Understanding why the crane draws 90.21 A to perform 60 kW of mechanical lifting work requires examining the phase angle between alternating voltage and current and the $\sqrt{3}$ geometry of three-phase Wye phasors.

4.2.1 Alternating Current Phase Shift, Power Factor, and Capacitor Sizing

In a resistive load such as an immersion heater, alternating current rises and falls in phase with alternating voltage. In an inductive load such as the Pier 14 gantry crane hoist motor, the alternating magnetic field stores and returns reactive energy every quarter-cycle, causing the RMS line current I to lag behind the RMS supply voltage V by a phase angle φ. Because only the current component in phase with voltage ($I \cos \varphi$) performs real mechanical or thermal work, AC power forms a right triangle governed by the dimensionless **Power Factor** $\text{PF} = \cos \varphi \in [0, 1]$ ([PF]), linking **Real Working Power** P (in watts, W, or kilowatts, kW, converted via [↖] [→kW] and [↗] [→HP]), **Reactive Magnetizing Power** Q (in volt-amperes reactive, VAR, or kVAR), and **Apparent Power** S (in volt-amperes, VA, or kVA):

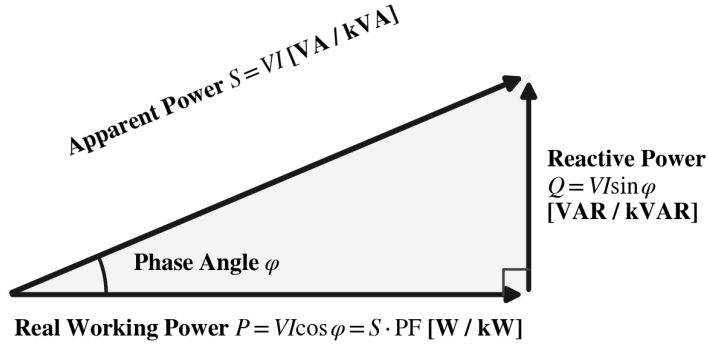


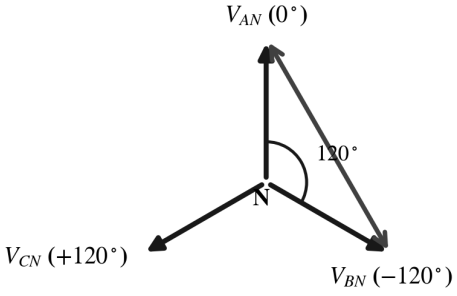
Figure 4.2: *Alternating-current power triangle relating real working power P, reactive magnetizing power Q, and apparent volt-amperes S.*

$$P = V \cdot I \cdot \text{PF} = S\cos(\phi), \qquad Q = S\sin(\phi) = \sqrt{S^2 - P^2}, \qquad \text{PF} = \cos(\phi) = \frac{P}{S} \qquad \textbf{(Eq. 4.3)}$$

In this power triangle, S is the hypotenuse that sizes transformers, switchgear, and feeder conductors, P is the adjacent horizontal leg converted into shaft work and heat, and Q is the opposite vertical leg oscillating between the source and the motor's magnetic field—forming the exact electrical counterpart to the North-clockwise surveying projection ($\Delta N = D\cos\alpha$, $\Delta E = D\sin\alpha$) and aeronautical wind triangle in Unit I (**SC-35**). On Pier 14's crane rated at apparent power $S = 75.0$ kVA and lagging power factor PF = 0.80 ($\varphi = \arccos(0.80) \approx 36.87°$, $\sin\varphi = 0.60$), the real working power is $P = 75.0 \times 0.80 =$ **60.0 kW** (80.46 HP via [→] [→HP]) and the reactive magnetizing demand is $Q = 75.0 \times 0.60 =$ **45.0 kVAR**. Installing a parallel 25.0 kVAR capacitor bank at the crane disconnect supplies 25.0 kVAR locally, reducing net reactive draw from the feeder to $Q_{\text{new}} = 45.0 - 25.0 = 20.0$ kVAR, shrinking apparent feeder demand to $S_{\text{new}} = \sqrt{(60.0^2 + 20.0^2)} \approx$ **63.25 kVA**, and raising the corrected power factor to $\text{PF}_{\text{new}} = 60.0 / 63.2456 \approx$ **0.9487**.

4.2.2 Three-Phase Wye Phasor Geometry and Motor Full-Load Current

In a balanced three-phase Wye (Y) distribution system, three transformer windings share a common grounded neutral center point N and radiate outward at phase angles 0°, 120°, and 240°. If each winding develops a phase-to-neutral magnitude V_{LN}, the line-to-line voltage V_{LL} measured between any two outer phase conductors follows from the Law of Cosines across the 120° isosceles triangle formed by Neutral N, Phase A, and Phase B:



Line-to-Line Phasor Law

$V_{AB} = 2V_{LN}\sin(60°) = \sqrt{3}\,V_{LN}$

$120\text{ V} \times \sqrt{3} = 207.8\text{ V} \approx 208\text{ V}$

$277\text{ V} \times \sqrt{3} = 479.8\text{ V} \approx 480\text{ V}$

Figure 4.3: *Symmetric 120-degree three-phase Wye phasor diagram showing the sqrt(3) vector relationship between line-to-neutral and line-to-line voltage.*

$$V_{LL} = \sqrt{V_{LN}^2 + V_{LN}^2 - 2V_{LN}^2\cos(120°)} = V_{LN}\sqrt{2 - 2(-0.5)} = \sqrt{3} \cdot V_{LN} \qquad \textbf{(Eq. 4.4)}$$

This geometric ratio $\sqrt{3} \approx 1.73205$ produces the two standard commercial voltage systems in North America: multiplying $V_{LN} = 120$ V by $\sqrt{3}$ gives 207.85 V (208Y/120 V), and multiplying $V_{LN} = 277$ V by $\sqrt{3}$ gives 479.78 V (480Y/277 V). Because each of the three phases delivers apparent power $V_{LN}\,I_L = (V_{LL}/\sqrt{3})I_L$, total three-phase apparent power across all three phases is $3(V_{LL}/\sqrt{3})I_L = \sqrt{3}\;V_{LL}I_L$. Expressing three-phase apparent power S_{kVA} in kilovolt-amperes ([→] [kVA]) to solve for per-phase line current I_L, or converting rated mechanical output horsepower HP (1 HP = 746 W) and fractional efficiency $\eta \in (0, 1]$ into three-phase motor **Full-Load Amps** I_{FLA} ([→] [FLA]), gives:

$$S_{\text{kVA}} = \frac{\sqrt{3} \cdot V_{LL} \cdot I_L}{1000}, \qquad I_L = \frac{S_{\text{kVA}} \times 1000}{\sqrt{3} \cdot V_{LL}}, \qquad I_{\text{FLA}} = \frac{\text{HP} \times 746}{\sqrt{3} \cdot V_{LL} \cdot \eta} \qquad \textbf{(Eq. 4.5)}$$

Here V_{LL} is the line-to-line RMS voltage, I_L is the per-phase line current in amperes, and 746 W/HP converts mechanical horsepower into electrical watts. On Pier 14's 480 V three-phase feeder supplying $S_{\text{kVA}} = 75$ kVA, dividing 75,000 VA by $\sqrt{3} \times 480 \approx 831.384$ V via [→] [kVA] yields the uncorrected line current $I_L \approx$ **90.21 A** (90.211 A). After capacitor correction to $S_{\text{new}} = 63.2456$ kVA, the feeder line current drops to $I_{L,\text{new}} \approx$ **76.07 A**.

4.3 Diagnosing the Pier 14 Feeder on the Stack and [AWG] Solver Workspace

On the **SC-14**, electrical power keys operate both as immediate RPN stack shortcuts—computing 1-phase copper voltage drop $V_{\text{drop}} = 2 \cdot L \cdot I \cdot R_{1000} / 1000$ when current I is in Register Z, length L (in ft) is in Register Y, and wire gauge (**14**..**0**) is in Register X via [AWG], scaling three-phase apparent power by $\sqrt{3}\ V_{LL}$ via [→] [kVA], evaluating AC real power via [PF], and combining parallel impedances via [x‖y]—and as bidirectional Equation Solver workspaces ([AWG], [→] [FLA], [XEQ], [PF], [XEQ], and [←] [MTR]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Workspace Variables
[AWG]	NEC Copper Voltage Drop & Solver	Current I (A)	Length L (ft)	Gauge G	$V_{\text{drop}} = 2 \cdot L \cdot I \cdot R_{1000} / 1000$; [AWG] ([D], [R], [L], [I], [G])
[PF]	AC Working Power & Power Factor	Voltage V	Current I	Power Factor PF	3-arg $P = V \cdot I \cdot \text{PF}$ (2-arg P/S); [PF] ([P], [V], [I], [F])
[→] [kVA]	Three-Phase Transformer Current	—	Power S (kVA)	Voltage V_{LL}	$I_L = (S \times 1000) / (\sqrt{3} \cdot V_{LL})$; [XEQ] ([S], [V], [I])
[→] [FLA]	Three-Phase Motor Full-Load Amps	—	Power P (HP)	Voltage V_{LL}	$I = P \cdot 746 / (\sqrt{3} \cdot V \cdot E)$; [→] [FLA] ([I], [P], [V], [E])
[←] [Ω]	Electrical Power & Ohm's Law	—	Voltage V	Current I	$P = V \times I$; [XEQ] workspace ([P], [V], [I])
[←] [MTR]	Rotary Motor Shaft Horsepower	—	Torque T (lb·ft)	Speed N (RPM)	$H = T \times N / 5252$; [←] [MTR] workspace ([H], [T], [N])
[x‖y]	Parallel Resistance / Impedance	—	Value R_1	Value R_2	$R_1 \parallel R_2 = (R_1 \cdot R_2) / (R_1 + R_2)$

4.3.1 Evaluating Feeder Current, NEC Table 8 Resistance, and Line Voltage Drop on the Stack

To diagnose Pier 14's 75 kVA, 480 V crane feeder across $L = 450$ ft of #2 AWG copper, the engineer first computes the three-phase line current $I_L \approx$ **90.2110 A** via **75** [ENTER] **480** [→] [kVA], evaluates the 1-phase copper voltage drop across $L = 450$ ft of #2 AWG ($R_{1000} = 0.194$ Ω/kft) via **450** [ENTER] **2** [AWG] (**15.7508 V**), and scales by $\sqrt{3}/2$ to obtain the balanced three-phase voltage drop $V_{\text{drop},3\varphi} \approx$ **13.6406 V**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
75 [ENTER] **480** [↷] [kVA]	0	0	0	90.2110	Compute 3-phase crane current I_L = 75,000 / ($\sqrt{3}$ × 480) ≈ 90.211 A
450 [ENTER] **2** [AWG]	0	0	0	15.7508	Compute 1-phase #**2 AWG** copper drop 2 × 450 × 90.211 × 0.194 / 1000 ≈ 15.75 V
3 [√x] [×] **2** [÷]	0	0	0	**13.6406**	Scale by $\sqrt{3}/2$ to obtain 3-phase drop $V_{\text{drop},3\varphi}$ ≈ 13.64 V (2.84% of 480 V)

Next, the engineer evaluates the real power of a single-phase 240 V, 18 A auxiliary air compressor at PF = 0.85 using the three-argument [PF] operator (**240** [ENTER] **18** [ENTER] **0.85** [PF] → **3,672 W**), looks up #1/0 AWG copper (**0** [AWG] → 0.122 Ω/kft), combines two parallel #1/0 AWG conductors per phase via [ENTER] [x‖y] (**0.061 Ω/kft**), and verifies that the upgraded parallel feeder drops only **4.2889 V** (0.89%):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
240 [ENTER] **18** [ENTER] **0.85** [PF]	0	0	0	3672	Evaluate real power P = 240 × 18 × 0.85 = 3,672 W
0 [AWG] [R/ON]	0	0	3672	0.122	Bind #**1/0 AWG** copper resistance (0.122 Ω/kft) and return to stack
[ENTER] [x‖y]	0	0	3672	0.061	Combine two parallel #**1/0 AWG** wires per phase (0.122 ‖ 0.122 = 0.061 Ω/kft)
90.211 [×] **450** [×] **3** [√x] [×] **1000** [÷]	0	0	3672	**4.2889**	Compute parallel 3-phase feeder drop $V_{\text{drop},3\varphi}$ ≈ 4.29 V (0.89%)

4.3.2 Inverse Maximum Run Length and Gauge Selection in the Bidirectional [AWG] Workspace

In field design, an electrician often knows the wire gauge on the truck and the load current, and needs to solve **backward** for the **maximum allowable one-way conduit run length *L*** that keeps voltage drop within the NEC 3.0% limit (D = 0.03 × 240 V = 7.20 V on a 240 V circuit)—or knows the run length L and current I and needs the **minimum NEC copper wire gauge G**. Pressing [AWG] with **0** in X (or typing a wire gauge and pressing [AWG]) opens the **Bidirectional [AWG] Equation Solver Workspace**, displaying **AWG: VOLT DROP** and the governing law **D = 2×R×L×I / 1000** on Line 1 with five variable softkeys [D] ([LFU0]), [R] ([LFU1]), [L] ([LFU2]), [I] ([LFU3]), and [G] ([LFU4]):

- **Automatic NEC Table 8 Resistance Binding via [G] & Inverse Run Length ([L]):** Entering wire gauge **12** (#12 AWG) and pressing [G] ([LFU4]) or [AWG] automatically looks up the NEC Table 8 uncoated copper resistance (**1.93** Ω/kft) and binds both [G] and [R] (or you can store **1.98** [R] for coated copper at 75°C). For a 240 V pier winch control circuit carrying I = 28 A with an NEC 3% maximum voltage drop of D = 7.20 V, keying **12** [G], **7.20** [D] ([LFU0]), and **28** [I] ([LFU3]) and pressing [XEQ] (or [XEQ] [L]) solves the equation backward for the maximum one-way conduit length:

$$L_{\max} = \frac{1000 \times D}{2 \cdot R \cdot I} = \frac{1000 \times 7.20}{2 \times 1.93 \times 28} \approx \mathbf{66.6173}\ \text{ft} \quad (\text{or } \mathbf{64.9351}\ \text{ft at coated } R = 1.98\ \Omega/\text{kft}) \qquad \textbf{(Eq. 4.6)}$$

- **Inverse Maximum Load Current ([I]) & Automatic Wire Gauge Selection ([G]):** If the conduit run is fixed at L = 100.0 ft (**100** [L]) on #12 AWG (R = 1.93 Ω/kft) with D = 7.20 V (**7.20** [D]), pressing [XEQ] [I] ([LFU3]) solves backward for the maximum permissible continuous load current $I_{\max}$ = 7200 / (2 × 1.93 × 100) ≈ **18.6528 A**. Conversely, with **7.20** [D], **125** [L], and **24** [I] bound, solving for [G] selects the most economical standard NEC copper gauge (#8 AWG, R = 0.778 Ω/kft ≤ 1.20 Ω/kft) whose drop stays within 7.20 V.

4.3.3 Try It on the Stack

Test your mastery of feeder resistance, inverse [AWG] run-length solving, and three-phase currents with these three electrical design problems:

1. **Three-Phase Feeder Current and Copper Voltage Drop:** On a 480 V three-phase feeder supplying 45 kVA over $L = 220$ ft of #6 AWG copper wire, use **45** [ENTER] **480** [↗] [kVA] to find I_L, evaluate **220** [ENTER] **6** [AWG] ($R_{1000} = 0.491$ Ω/kft), and scale by $\sqrt{3}/2$ to obtain $V_{\text{drop},3\varphi} = \sqrt{3}\ R_{1000}\ L\ I_L\ /\ 1000$. *(Verified Register X results: I_L* ***≈ 54.1266 A****, R_{1000} =* ***0.491 Ω/kft****, $V_{drop,1\varphi}$* ***≈ 11.6935 V****, and $V_{drop,3\varphi}$* ***≈ 10.1269 V****, or 2.11%.)*

2. **Inverse Maximum Conduit Length in the [AWG] Solver:** A 120 V single-phase lighting circuit carries $I = 16$ A with a 3.0% ($D = 3.60$ V) drop limit on #10 AWG copper (**10** [AWG] → $R = 1.24$ Ω/kft). Store **3.60** [D] and **16** [I] in the [AWG] workspace and press [XEQ] to find the maximum allowable one-way conduit length $L_{\max}$. Then check the drop at $L = 125$ ft on #8 AWG (**8** [G], **125** [L], [XEQ] [D]). *(Verified Register X results: #10 AWG L_{max}* ***≈ 90.7258 ft****, and #8 AWG at 125 ft drops* ***3.1120 V*** *or 2.59%* ***≤*** *3.0%.)*

3. **Motor Full-Load Current in [↗] [FLA] and NEC Continuous Duty Sizing:** A 480 V three-phase induction motor rated at $P = 30$ HP operates at efficiency $\eta = 0.90$ and power factor PF = 0.86. Open the [↗] [FLA] workspace (**0** [ENTER] **0** [↗] [FLA]: **I** = **P×746/(√3×V×E)**), store **30** [P], **480** [V], and **0.774** [E] ($\eta \times \text{PF} = 0.90 \times 0.86$), press [XEQ] for I_{FLA}, and multiply by **1.25** [×] for NEC continuous-duty conductor sizing. *(Verified Register X results: I_{FLA}* ***≈ 34.7790 A*** *and 125% minimum ampacity* ***43.4738 A****.)*

With the feeder conductors sized for acceptable voltage drop, we step into the plant boiler room in Chapter 5 to bend the steel conduit raceways that carry them around structural I-beams and steam headers.

Chapter 5: Bending Four Conduits Around the Boiler-Room I-Beam

Inside the central boiler plant of a regional hospital, a journeyman electrician is routing a parallel rack of four 1-in. Electrical Metallic Tubing (EMT) raceways along a concrete wall at elevation 9 ft 6 in. above the floor. Forty-eight inches (4 ft 0 in.) ahead of the main splitter trough, a structural steel wide-flange I-beam projects $O = 6.0$ in. out from the wall face. Twenty feet further down the run, a cylindrical high-pressure steam riser of height $O_{\text{pipe}} = 4.0$ in. crosses perpendicular to the wall before the rack rolls 6.0 in. upward and 8.0 in. laterally into an elevated motor control center (MCC).

If the electrician marks the first bend on a stick of EMT at 48 in. from the coupling without accounting for **Conduit Shrinkage**, the diagonal climb up the 30° offset pulls the far end of the pipe backward by 1-1/2 inches. The second bend will crash directly into the flange of the steel I-beam. Before bending a single stick of tubing or setting up the tugger winch, the crew must solve three coupled shop and field layout problems:

1. Calculate the center-to-center **Travel Distance** and axial shrinkage pullback around the 6 in. I-beam, and solve backward for the required bend angle when a tight ceiling bay limits the diagonal travel length.

2. Compute the compound 3D hypotenuse of the **Rolling Offset** into the MCC and the outer leg travel of the **Three-Point Saddle** over the 4 in. steam pipe.

3. Verify that twelve #10 AWG THHN conductors stay within the NEC Chapter 9 40% **Conduit Fill** ceiling inside 1-in. EMT, and size the trapeze bracket's spindle speed, thread tap drill, and bolt circle chord spacing.

5.1 Cosecant Offset Multipliers, Axial Shrinkage, and Rolling Offsets

Every conduit offset bend forms a right triangle in space whose opposite leg is the obstruction depth O, whose hypotenuse along the diagonal pipe is the travel distance T, and whose difference between hypotenuse and adjacent horizontal projection is the axial shrinkage S_{shrink}.

5.1.1 Right-Triangle Offset Travel and Half-Angle Tangent Shrinkage

When a conduit running along a wall encounters a structural beam or duct of perpendicular depth O (in inches), the electrician bends the pipe outward at shoe angle ϑ and bends it back parallel to the wall at an equal return angle ϑ.

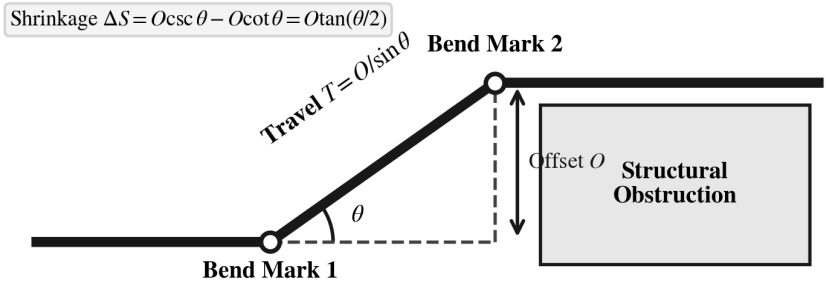


Figure 5.1: *Conduit offset bend geometry showing obstacle depth O, bend angle theta, travel length T, and linear conduit shrinkage.*

In the right triangle formed by the perpendicular offset depth O (opposite side) and the bend angle ϑ (denoted A in the OFS solver workspace **T = O / SIN(A)** opened via OFS), the center-to-center **Travel Distance** T along the diagonal pipe between the two bend marks (OFS) is the cosecant hypotenuse O csc ϑ. Because the conduit previously spanned the full length T along the wall but now projects horizontally across only the adjacent side O cot ϑ, bending the offset pulls the far end of the pipe backward toward the starting coupling by the axial **Conduit Shrinkage** S_{shrink}, which simplifies by the trigonometric half-angle identity csc ϑ - cot ϑ = (1 - cos ϑ) / sin ϑ = tan(ϑ / 2):

$$T = \frac{O}{\sin(\theta)} = O\csc(\theta), \qquad S_{\text{shrink}} = T - O\cot(\theta) = O\tan\left(\frac{\theta}{2}\right) \qquad \textbf{(Eq. 5.1)}$$

In these expressions, O is the perpendicular offset depth in inches, ϑ is the bender shoe angle in degrees, T is the distance measured along the tubing between the two bend marks, and S_{shrink} is the length added to the obstruction setback when placing the Obstruction-Approach mark first (while a 3-argument OFS call with offset O in Register Z, angle ϑ in Register Y, and bender deduct D in Register X returns net travel T - D). Standard conduit bender shoes engrave five primary bend angles, each with its cosecant travel multiplier and half-angle tangent shrinkage per inch of offset depth:

Bend Angle ϑ	Travel Multiplier (csc ϑ = 1/sin ϑ)	Exact Shrinkage Factor (tan(ϑ/2))	Field Rule Shrinkage per Inch of Offset
10°	5.759 ≈ 6.0	0.0875 in/in	1/16 in per inch
22.5°	2.613 ≈ 2.6	0.1989 in/in	3/16 in per inch
30°	**2.000**	0.2679 in/in	**1/4 in** (0.25 in) per inch
45°	**1.414** (√2)	0.4142 in/in	**3/8 in** (0.375 in) per inch
60°	1.155	0.5774 in/in	9/16 in per inch

To clear the boiler-room I-beam of depth O = 6.0 in. located D_{obs} = 48.0 in. (4 ft 0 in.) from the splitter box using ϑ = 30° bends, keying **6** ENTER **30** OFS returns the travel distance between marks T = 6.0 / sin(30°) = **12.00 in.** and the shrinkage (at 1/4 in/in field rule, or 6 tan 15° = **1.6077 in.** exact) is S_{shrink} = 6.0 × 0.25 = **1.50 in.** (1-1/2 in.). Adding S_{shrink} to the 48.0 in. setback places the second bend mark (at the beam edge) at 48.0 + 1.50 = **49-1/2 in.** and the first bend mark at 49.50 - 12.00 = **37-1/2 in.** from the end of the conduit.

5.1.2 Compound Rolling Offsets Across Horizontal and Vertical Planes

When a conduit run must simultaneously shift vertically by a **Rise** R_{vert} and horizontally by a lateral **Roll** R_{horiz} to enter a misaligned cabinet, the two perpendicular displacements form a right triangle in the cross-sectional plane perpendicular to the conduit axis. By the Pythagorean theorem, rotating the conduit bender in that plane aligns the shoe with the **True Offset Depth** O_{true} ([↷] [FIT] or [↶] [MIT]), which then scales by csc ϑ via [OFS] to yield the rolling travel distance T_{roll}:

$$O_{\text{true}} = \sqrt{R_{\text{vert}}^2 + R_{\text{horiz}}^2}, \qquad T_{\text{roll}} = \frac{O_{\text{true}}}{\sin(\theta)} = \csc(\theta)\sqrt{R_{\text{vert}}^2 + R_{\text{horiz}}^2} \qquad \textbf{(Eq. 5.2)}$$

Here R_{vert} and R_{horiz} are the orthogonal vertical and horizontal shifts in inches, O_{true} is the diagonal offset depth in the rotated bending plane evaluated via [↷] [FIT] or [↶] [MIT], and T_{roll} is the center-to-center distance between bend marks along the conduit evaluated via [OFS]. Where the boiler-room rack rises $R_{\text{vert}} = 6.0$ in. and rolls laterally $R_{\text{horiz}} = 8.0$ in. into the MCC knockouts, keying **6** [ENTER] **8** [↷] [FIT] returns the true diagonal offset $O_{\text{true}} = \sqrt{(6.0^2 + 8.0^2)} = \mathbf{10.00\ in.}$, and keying **30** [OFS] bends that compound offset at $\vartheta = 30°$ with a travel distance of $T_{\text{roll}} = 10.00 \,/\, \sin(30°) = \mathbf{20.00\ in.}$ between marks.

5.2 Three-Point Saddle Geometry, NEC Raceway Fill, and Shop Bracket Machining

After clearing the I-beam and planning the rolling entry, the electrician must hop the conduit bank over the cylindrical 4.0 in. steam pipe, verify that twelve #10 AWG conductors satisfy NEC cross-sectional fill inside 1-in. EMT, and drill and tap the steel trapeze mounting plates.

5.2.1 Symmetric Three-Point Saddle Bends Over Cylindrical Obstructions

When a conduit strapped flat against a wall must step over a single cylindrical pipe of height O and return to the same mounting surface, electricians form a **Three-Point Saddle** consisting of a center crown bend of angle $\vartheta_{\text{center}} = 45°$ placed directly over the obstruction centerline, flanked by two symmetric outer return bends of $\vartheta_{\text{outer}} = \vartheta_{\text{center}} \,/\, 2 = 22.5°$:

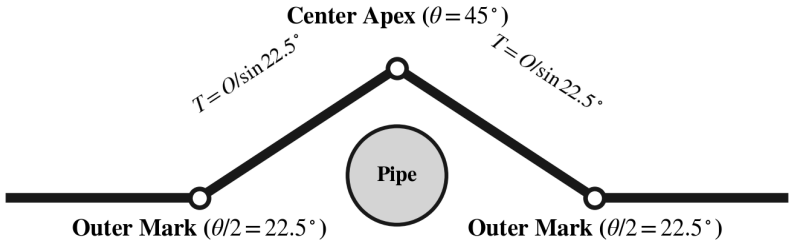


Figure 5.2: *Three-point conduit saddle bend geometry with a 45-degree center bend and symmetric 22.5-degree outer return bends.*

$$T_{\text{saddle}} = \frac{O}{\sin(22.5^\circ)} = O\csc(22.5^\circ) \approx 2.6131 \cdot O, \qquad S_{\text{saddle}} = 2 \cdot O\tan(11.25^\circ) \approx \tfrac{3}{16}\text{ in.} \times O \qquad \textbf{(Eq. 5.3)}$$

In this symmetric geometry, each outer sloped leg rises through angle $\vartheta_{\text{outer}} = 22.5^\circ$ over height O, so keying **4** [ENTER] **22.5** [OFS] places the two outer 22.5° marks at distance $T_{\text{saddle}} \approx 2.613\ O$ (or 2.5 in/in field rule) on either side of the center 45° mark, while total axial shrinkage across both sloped legs adds 3/16 in. (0.1875 in.) per inch of obstruction height to the center mark distance. Over the boiler room's $O = 4.0$ in. steam pipe centered 120.0 in. (10 ft 0 in.) down the wall, total saddle shrinkage is 4 × (3/16 in.) = **0.75 in.** (3/4 in.), placing the center 45° mark at 120.75 in. and the two outer 22.5° return marks at $T_{\text{saddle}} = 4.0 / \sin(22.5^\circ) \approx$ **10.45 in.** (10-7/16 in. via [↶] [f-i], or 10.0 in. by the 2.5× field rule) on either side of the center mark.

5.2.2 Cross-Sectional Raceway Fill Limits, Pipe Wall Schedule, and Bracket Layout

Once a conduit run is installed, pulling insulated conductors around multiple bends generates capstan friction proportional to $e^{\mu\,\Sigma\vartheta}$ and traps $I^2\,R$ heat inside the tube. To prevent insulation abrasion during pulling and ensure adequate air space for heat dissipation, **NEC Chapter 9, Table 1** (accessible directly via [↷] [▦] on the **SC-14**: **1W FILL = 53% MAX**, **2W FILL = 31% MAX**, **3+W FIL = 40% MAX**) limits the cross-sectional **Conduit Fill Ratio** to 53% for 1 conductor, 31% for 2 conductors, and **40%** for 3 or more conductors:

$$\text{Fill}\% = \left(\frac{\sum_{k=1}^{N} A_{\text{wire},k}}{A_{\text{conduit}}} \right) \times 100\% \le 40\%, \qquad N_{\max} = \left\lfloor \frac{0.40 \cdot A_{\text{conduit}}}{A_{\text{wire}}} \right\rfloor \qquad \textbf{(Eq. 5.4)}$$

Here A_{conduit} is the total internal cross-sectional area of the raceway from NEC Chapter 9, Table 4 (while [↶] [SCH] and [↷] [WAL] evaluate pipe inner diameter and wall thickness $OD - ID$), A_{wire} is the insulated conductor area from NEC Chapter 9, Table 5, and $N_{\max}$ is the maximum whole number of identical conductors permitted at the 40% ceiling:

Trade Size (EMT)	Total Internal Area (100%)	1 Wire Fill (53%)	2 Wires Fill (31%)	3+ Wires Fill (40%)
1/2 in EMT	0.304 in²	0.161 in²	0.094 in²	0.122 in²
3/4 in EMT	0.533 in²	0.283 in²	0.165 in²	0.213 in²
1 in EMT	0.864 in²	0.458 in²	0.268 in²	0.346 in²
1-1/4 in EMT	1.496 in²	0.793 in²	0.464 in²	0.598 in²
1-1/2 in EMT	2.036 in²	1.079 in²	0.631 in²	0.814 in²
2 in EMT	3.356 in²	1.778 in²	1.040 in²	1.342 in²

In our boiler-room run of 1-in. EMT ($A_{\text{conduit}} = 0.864$ in², 40% allowance = 0.3456 in²), each #10 AWG THHN conductor has an insulated cross-sectional area of $A_{\text{wire}} = 0.0211$ in². Pulling twelve #10 AWG THHN conductors occupies 12 × 0.0211 = **0.2532 in²**, giving a fill ratio of Fill% = (0.2532 / 0.864) × 100% ≈ **29.31%** (well below the 40.0% limit, which permits up to ⌊ 0.3456 / 0.0211 ⌋ = **16** conductors). Meanwhile, to fabricate the trapeze header's steel adapter ring—milling with a $D = 0.500$ in. cutter at SFM = 400 ft/min ([↶] [SFM]), tapping 1/2-13 UNC anchor holes ([↶] [TAP]), and drilling an $N = 8$-hole pattern on a $D = 6.00$ in. bolt circle ([↷] [BC])—the shop keys **400** [ENTER] **0.5** [↶] [SFM] (**3,055.77 RPM**), **0.5** [ENTER] **13** [↶] [TAP] (**0.4231 in.** tap drill), and **6** [ENTER] **8** [↷] [BC] ($C = 6\sin(22.5^\circ) =$ **2.2961 in.** chord spacing).

5.3 Laying Out the Boiler-Room Conduit Bank on the Stack and [OFS] Solver

On the **SC-14**, [OFS] operates both as an immediate stack shortcut—consuming offset depth O in Register Y and bend angle $\vartheta°$ in Register X (**6** [ENTER] **30** [OFS] $\rightarrow$ 12.00 in.) or 3-argument offset O in Z, angle $\vartheta°$ in Y, and deduct D in X—and as a bidirectional Equation Solver Workspace (**BEND: CONDUIT**, displaying **T = O / SIN(A)** on Line 1 with softkeys [T], [O], and [A], pre-initialized to the standard $A = 30°$ bender shoe angle):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Workspace Variables
[OFS]	Conduit Offset Travel & Solver	(Opt. Offset O)	Offset O (in)	Angle $\vartheta°$ (or Deduct D)	$T = O$ / $\sin(A)$ (3-arg T - D); [OFS] ([T], [O], [A])
[←] [MIT] / [→] [FIT]	True 3D Rolling Offset Hypotenuse	—	Rise R_{vert}	Roll R_{horiz}	$O_{\text{true}} = \sqrt{(R_{\text{vert}}{}^2 + R_{\text{horiz}}{}^2)}$
[←] [SCH] / [→] [WAL]	Pipe Schedule & Wall Thickness	—	Outer Dia OD	Inner Dia / Wall	ID or wall thickness difference Y - X
[←] [SFM]	Spindle Speed & Surface Velocity	—	Speed SFM	Diameter D (in)	$N = (3.8197 \cdot \text{SFM})$ / D; [←] [SFM] ([N], [S], [D])
[←] [TAP]	75% Thread Tap Drill Diameter	—	Major D (in)	Pitch p or TPI	$D_{\text{tap}} = D$ - 1/TPI (if $X > 3$) or D - p
[→] [BC]	Bolt Circle Chord Spacing & Solver	—	Circle Dia D	Hole Count N	$C = D \sin(180° / N)$; [→] [BC] ([C CM], [D], [N])
[←] [f-i] / [→] [▦]	Architectural **FF.IISS** & NEC Table	—	(Opt. Feet)	Decimal / Inches	Converts decimal feet ↔ **FF' II-S/16"**; opens NEC fill table

5.3.1 Computing Offset Travel, Shrinkage Marks, and Rolling Offsets on the Stack

To lay out the $O = 6.0$ in. I-beam offset at $\vartheta = 30°$ (48.0 in. from the splitter trough) and the subsequent 6.0 in. × 8.0 in. rolling offset into the MCC at 30°, the electrician evaluates **6** [ENTER] **30** [OFS] (T = **12.00 in.**), adds the 1/4 in/in shrinkage ($6 \times 0.25 = 1.50$ in.) to 48.0 in. to place the two bend marks at **49.50 in.** and **37.50 in.**, and uses [→] [FIT] followed by [OFS] to compute the rolling offset travel T_{roll} = **20.00 in.**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
6 [ENTER] **30** [OFS] [C CM]	0	0	0	12	Compute 30° I-beam offset travel $T = 6$ / $\sin(30°) = 12.00$ in.
48 [ENTER] **6** [ENTER] **0.25** [×] [+]	12	12	12	49.5	Add shrinkage $S = 1.50$ in. to 48.0 in. for Mark 2 at 49.50 in.
[x⇄y] [-]	12	12	12	37.5	Subtract travel $T = 12.00$ in. to locate Mark 1 at 37.50 in.
6 [ENTER] **8** [→] [FIT] **30** [OFS] [C CM]	12	12	37.5	**20**	Compute true rolling offset $O_{\text{true}} = 10$ in. ([→] [FIT]) and 30° travel $T_{\text{roll}} = 20.00$ in.

5.3.2 Inverse Bend Angle Solving in [OFS] and Verifying NEC Conduit Fill

Suppose a fire-sprinkler main hung just ahead of an overhead structural beam restricts the available diagonal travel length of a $O = 6.0$ in. offset to at most $T = 10.0$ in. (so a standard 30° bend with $T = 12.00$ in. will not fit in the ceiling bay). Pressing **0** [ENTER] **0** [OFS] opens the **Bidirectional** [OFS] **Equation Solver Workspace**, displaying **BEND: CONDUIT** and **T = O / SIN(A)** on Line 1 with variable softkeys [T] ([LFU0]), [O] ([LFU1]), and [A] ([LFU2]). Storing the constrained travel length **10** [T] ([LFU0]) and offset height **6** [O] ([LFU1]) and pressing [XEQ] [A] ([LFU2]) solves the cosecant equation backward for the minimum required bend angle:

$$A = \arcsin\left(\frac{O}{T}\right) = \arcsin\left(\frac{6.0}{10.0}\right) \approx \mathbf{36.8699^\circ} \qquad \textbf{(Eq. 5.5)}$$

This tells the electrician immediately to step up from the 30° bender groove to a custom 37° digital protractor bend (or the 45° shoe mark, where **6** [O] **45** [A] [XEQ] [T] gives $T \approx$ **8.4853 in.**).

Next, to step over the $O = 4.0$ in. steam pipe using a 45°–22.5° three-point saddle (**4** [ENTER] **22.5** [OFS]) and verify the NEC conduit fill of twelve #10 AWG THHN conductors (0.0211 in² each) inside 1-in. EMT ($A_{\text{conduit}} = 0.864$ in²), the crew chains the saddle travel and area ratio on the stack:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
4 [ENTER] **22.5** [OFS] [C ON]	0	0	0	10.4525	Compute outer 22.5° saddle leg travel $T_{\text{saddle}} = 4/\sin(22.5°) \approx 10.45$ in.
12 [ENTER] **0.0211** [×]	0	0	10.4525	0.2532	Compute total cross-sectional area of 12 **#10 AWG** THHN wires (0.2532 in²)
0.864 [÷] **100** [×]	0	0	10.4525	29.3056	Divide by 1-in. EMT area (0.864 in²) to confirm Fill% ≈ 29.31% ≤ 40%
0.3456 [ENTER] **0.0211** [÷]	10.4525	10.4525	29.3056	**16.3791**	Divide 40% allowance (0.3456 in²) by 0.0211 in² (⌊ 16.38 ⌋ = 16 max wires)

5.3.3 Try It on the Stack

Practice conduit offset geometry, inverse [OFS] workspace solves, and shop bracket machining with these three field layout exercises:

1. **Comparing Thirty-Degree and Forty-Five-Degree Offsets:** For an obstruction depth of $O = 6.5$ in. (6-1/2 in.), compare the travel distance ([OFS]) and shrinkage at $\vartheta = 30°$ (**6.5** [ENTER] **30** [OFS]) against $\vartheta = 45°$ (**6.5** [ENTER] **45** [OFS]). *(Verified Register X results: T_{30} =* ***13.00 in.*** *with 1.625 in. field shrinkage; T_{45}* ***≈ 9.1924 in.*** *with 2.4375 in. field shrinkage.)*

2. **Inverse Saddle Bend Angle in a Constrained Ceiling Bay:** To clear a rectangular HVAC duct of depth $O = 9.0$ in. where ceiling hangers restrict maximum diagonal travel between bends to $T = 18.0$ in., open [OFS] (**0** [ENTER] **0** [OFS]), store **18** [T] and **9** [O], and press [XEQ] [A] for the required bend angle. Then compare with the travel at $A = 22.5°$ (**22.5** [A] [XEQ] [T]). *(Verified Register X results: constrained angle $A =$* ***30.00°****, and 22.5° travel $T \approx$* ***23.5182 in.*** *with exact shrinkage $S_{shrink} \approx$* ***1.7902 in.****.)*

3. **Trapeze Bracket Spindle Speed, Tap Drill, and Bolt Circle:** For the steel trapeze adapter ring, evaluate spindle speed for SFM = 400 ft/min on a $D = 0.500$ in. end-mill (**400** [ENTER] **0.5** [↰] [SFM]), tap drill diameter for 1/2-13 UNC (**0.5** [ENTER] **13** [↰] [TAP]), and chord spacing for $N = 8$ holes on a $D = 6.00$ in. bolt circle (**6** [ENTER] **8**

[↷] [BC]). *(Verified Register X results: $N \approx$ **3,055.77 RPM**, $D_{tap} \approx$ **0.4231 in.**, and $C \approx$ **2.2961 in.**.)*

With the boiler-room conduit bank bent and pulled, we move to the machine shop and instrumentation bench in Chapter 6 to silence a chattering carbide milling cutter, size a coolant control valve and pump, and calibrate a drifting hydraulic pressure loop.

Chapter 6: The Chattering Milling Cutter and the Drifting Pressure Loop

In the heavy-machinery maintenance shop adjacent to the boiler plant, a toolmaker is milling an 8-bolt high-pressure steam flange out of annealed 4140 alloy steel on a vertical CNC machining center using a *varnothing* 0.500 in. (1/2 in.) 4-flute solid carbide end-mill. Halfway through the roughing pass, the cutter emits a high-pitched harmonic shriek—chatter—leaving washboard ripples along the flange wall while the control panel flags an intermittent low-pressure alarm on the hydraulic workholding fixture.

Two distinct shop-floor problems have converged at the milling table. First, the operator accidentally left the spindle speed set to 3,056 RPM from an earlier aluminum job (SFM = 400 ft/min)—four times the 100 Surface Feet per Minute cutting velocity rating of 4140 alloy steel—while starving the 4-flute cutter of its proper f_t = 0.003 in/tooth chip load. Second, on the hydraulic fixture clamping the flange and the heat-exchanger cooling loop, the 4–20 mA pressure transmitter (calibrated for 0–3,000 psi) is reading 62.5% of span (1,875 psi) across a *varnothing* 2.00 in. hydraulic clamping cylinder. To restore smooth cutting and verify the hydraulic and coolant systems, the shop team works through three related themes:

1. Calculate the true spindle RPM and linear feed rate in inches per minute for 4140 alloy steel, then determine the tap drill size and center-to-center chord spacing for N = 8 threaded mounting holes (1/2-13 UNC) around the pitch circle.
2. Verify the hydraulic clamping force in pounds across the *varnothing* 2.00 in. cylinder, check pipe flow continuity, and size the heat-exchanger cooling control valve (C_v).
3. Calculate centrifugal coolant pump brake horsepower and net positive suction head, and audit the 4–20 mA pressure transmitter loop and Pt100 coolant RTD temperature sensor.

6.1 Rotary Cutting Kinematics, Thread Tapping, and Bolt Circle Trigonometry

Eliminating chatter and laying out precision flange patterns require coupling rotary tangential velocity with per-tooth chip thickness, thread pitch geometry, and circular chord trigonometry.

6.1.1 Surface Feet per Minute, Spindle Speed, and Chip-Load Feed Rate

When a lathe turns a cylindrical bar of diameter D (in inches) or a milling machine spins an end-mill of diameter D at spindle rotational speed N (in Revolutions Per Minute, RPM), each revolution carries a cutting tooth around the outer circumference $\pi\ D$ inches = $\pi\ D$ / 12 feet. Multiplying by N rev/min defines the tangential cutting velocity in **Surface Feet per Minute** ([↰] [SFM]), and inverting that relation determines the required **Spindle Speed** N ([↰] [SFM] / [↱] [RPM]), the linear table **Feed Rate** F_m (in in/min) for a cutter with Z flutes and chip load f_t (in in/tooth), and the 75%-thread **Tap Drill Diameter** D_{tap} ([↰] [TAP]) for a screw of major diameter D_{major} and pitch P_{tpi} (threads per inch, or metric/decimal pitch p when $X \leq 3$):

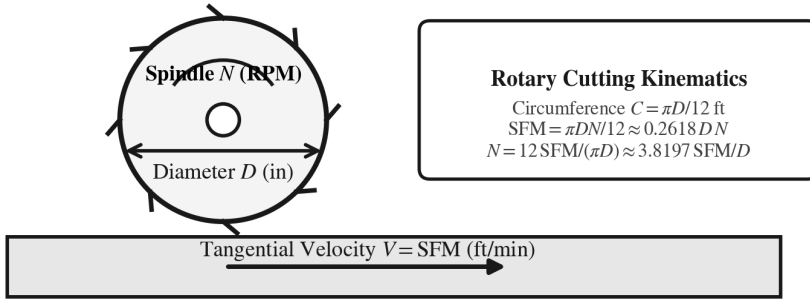


Figure 6.1: *Rotary milling cutter kinematics relating cutter diameter D, spindle speed RPM, and tangential surface feet per minute (SFM).*

$$\text{SFM} = \frac{\pi \cdot D \cdot N}{12} \Rightarrow N = \frac{12 \cdot \text{SFM}}{\pi \cdot D} \approx \frac{3.8197 \cdot \text{SFM}}{D}, \qquad F_m = N \cdot f_t \cdot Z, \qquad D_{\text{tap}} = D_{\text{major}} - \frac{1}{P_{\text{tpi}}} \qquad \textbf{(Eq. 6.1)}$$

In these machining equations, D is the cutter or workpiece diameter in inches, 12 / π ≈ 3.8197186 (3.82 in the ↰ SFM workspace **N = 3.82 × S / D** with softkeys N, S, D) converts feet per minute into inches per revolution, $f_t \cdot Z$ is the linear advance per spindle revolution, and subtracting one pitch length 1 / P_{tpi} from the major diameter D_{major} yields the standard 75% unified national thread tap drill diameter. While an aluminum roughing pass at SFM = 400 ft/min on a D = 0.500 in. cutter runs at **400 ENTER 0.5** ↰ SFM → **3,055.77 RPM**, milling 4140 steel rated at SFM = 100 ft/min with a D = 0.500 in. 4-flute end-mill and a chip load of f_t = 0.003 in/tooth requires reducing the spindle speed via **100 ENTER 0.5** ↰ SFM to N = 12 × 100 / (π × 0.500) **≈ 763.94 RPM** and setting the table feed rate to F_m = 763.9437 × 0.003 × 4 **≈ 9.17 in/min**. Tapping the 1/2-13 UNC flange holes (D_{major} = 0.500 in., P_{tpi} = 13 TPI) via **0.5 ENTER 13** ↰ TAP requires a tap drill diameter of D_{tap} = 0.500 - 1/13 **≈ 0.4231 in.** (27/64 in. = 0.4219 in.).

6.1.2 Bolt Circle Chord Indexing and Right-Triangle Fixture Layout

When drilling N equally spaced bolt holes around a pitch circle of diameter D_{bolt} (radius R = D_{bolt}/2), the central angle between adjacent holes is $\Delta\vartheta$ = 360° / N. Bisecting the isosceles triangle formed by the circle center and two adjacent holes yields two right triangles with half-angle $\Delta\vartheta$ / 2 = 180°/N and hypotenuse R = D_{bolt}/2:

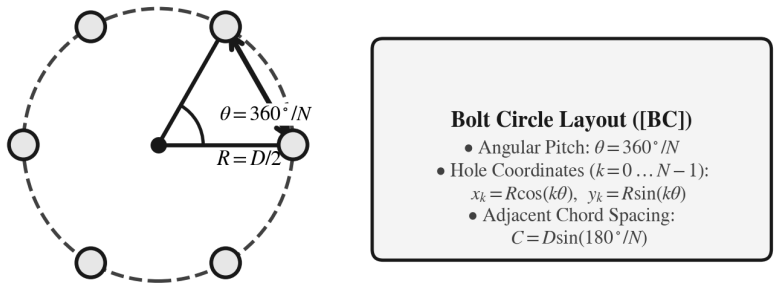


Figure 6.2: *Bolt circle hole layout showing pitch radius R, angular pitch 360/N, and center-to-center chord distance C.*

$$C = 2R\sin\left(\frac{180^\circ}{N}\right) = D_{\text{bolt}}\sin\left(\frac{180^\circ}{N}\right), \qquad X_k = \frac{D_{\text{bolt}}}{2}\cos\left(\theta_0 + k\cdot\frac{360^\circ}{N}\right), \qquad Y_k = \frac{D_{\text{bolt}}}{2}\sin\left(\theta_0 + k\cdot\frac{360^\circ}{N}\right) \qquad \textbf{(Eq. 6.2)}$$

Here C is the straight-line center-to-center **Chord Distance** ([↗] [BC], which also opens the [↗] [BC] workspace **C = D × SIN(180 / N)** with softkeys [C ON], [D], [N]) between adjacent holes used for caliper verification, and (X_k, Y_k) are the Cartesian CNC table coordinates of hole $k \in (0, 1, ..., N\text{-}1)$ starting from reference angle ϑ_0. On an N = 8-bolt hub with a D_{bolt} = 6.00 in. pitch circle (180°/8 = 22.5°), keying **6** [ENTER] **8** [↗] [BC] returns C = 6.00 sin(22.5°) **≈ 2.2961 in.**, while on the outer D_{bolt} = 10.00 in. steam flange (**10** [ENTER] **8** [↗] [BC]), the chord spacing is C = 10.00 sin(22.5°) **≈ 3.8268 in.**.

6.2 Pipe Flow Continuity, Valve Coefficient, Pump Power, and Process Loops

Holding a heavy steel forging stationary against cutting forces and monitoring its coolant and hydraulic circuits unite fluid continuity, Bernoulli's orifice flow law, Net Positive Suction Head, and Kirchhoff's current law.

6.2.1 Pipe Continuity, Valve Flow Coefficient, Pump Power, and Suction Head

In the cooling header supplying the machining center, incompressible fluid of cross-sectional pipe area A moving at velocity V obeys the **Pipe Flow Continuity Law** $Q = A \times V$ ([PIP], opening the [PIP] workspace with softkeys [Q], [A], [V]). Downstream, a control valve of **Flow Coefficient** C_v ([↗] [Cv]) passes Q GPM ([↖] [→GPM]) of fluid of specific gravity $G = \rho / \rho_w$ ([↗] [PW]) across pressure drop ΔP (in psi, [↖] [→PSI]), lifting Q GPM against H feet of total dynamic head requires hydraulic **Pump Power** P_{BHP} ([PMP], where 33,000 ft·lb/min ÷ 8.333 lb/gal = 3,960), and preventing cavitation at the pump impeller eye requires maintaining **Available Net Positive Suction Head** NPSH_a ([↗] [NPH]) above the pump's required threshold:

$$Q = A\cdot V, \qquad Q = C_v\sqrt{\frac{\Delta P}{G}}, \qquad P_{\text{BHP}} = \frac{Q\cdot H}{3960}, \qquad \text{NPSH}_a = P_{\text{atm}} - P_{\text{vap}} \pm z_s - h_f \qquad \textbf{(Eq. 6.3)}$$

In these fluid-power relations, Q scales with pipe area and velocity in [PIP] and with the square root of valve pressure drop ΔP in [↗] [Cv] ([↗] [Cv] workspace softkeys [Q], [C ON], [P], [G]), P_{BHP} measures the water horsepower delivered by [PMP] ([PMP] workspace softkeys

[P], [Q], [H]), and [↷] [NPH] evaluates available suction head margin so vapor bubbles do not implode against the impeller vanes. Meanwhile, with the fixture's *varnothing* 2.00 in. hydraulic clamping cylinder ($A_{\text{piston}} = (\pi/4)(2.00)^2 = \pi \approx 3.1416\ \text{in}^2$) pressurized to $P_{\text{psi}} =$ 1,875 psi, the clamping force holding the 4140 flange is $F_{\text{clamp}} = 1875 \times \pi \approx$ **5,890.49 lbf** ([↷] [→lbf]). A coolant control valve with $C_v = 25$ at $\Delta P = 16$ psi ($G = 1.0$) delivers $Q = 25\sqrt{16} =$ **100.0 GPM** via **25** [ENTER] **16** [↷] [C↓], and pumping $Q = 198$ GPM against $H =$ 100 ft of head requires $P_{\text{BHP}} = (198 \times 100)/3960 =$ **5.00 HP** via **198** [ENTER] **100** [PMP].

6.2.2 Noise-Immune Four-to-Twenty Milliampere Loops and Platinum RTDs

Industrial plants transmit pressure, temperature, and flow readings from field sensors to programmable logic controllers (PLCs) as a **4–20 mA direct-current signal** ([mA]) rather than a 0–10 V voltage signal for two physical reasons: first, by Kirchhoff's Current Law, series loop current is identical at every point along a 2,000 ft cable regardless of wire voltage drop. Second, elevating 0% of calibrated span to a **live zero of 4.0 mA** allows the controller to distinguish a 0 psi reading (4.0 mA) from a severed wire (0.0 mA).

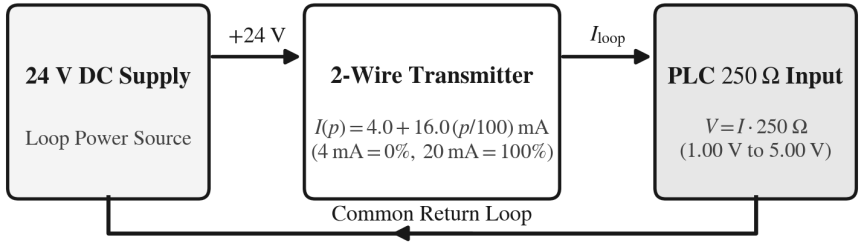


Figure 6.3: *Industrial 4-to-20 mA current loop circuit with 24 V supply, two-wire field transmitter, and 250-ohm PLC sense resistor.*

Across the 16.0 mA active span between 4.0 mA (0%) and 20.0 mA (100%), loop current I_{mA} scales linearly with process percentage Pct $\in [0, 100]$ via [mA], while an industrial **Pt100 Resistance Temperature Detector** ([↶] [RTD])—standardized to ice-point resistance $R_0 = 100.0\ \Omega$ at 0°C with IEC 60751 platinum temperature coefficient $\alpha = 0.00385\ \Omega/(\Omega\cdot°\text{C})$—follows the linear Callendar–Van Dusen relation at temperature T (in °C):

$$I_{\text{mA}} = 4.0 + 16.0 \times \left(\frac{\text{Pct}}{100}\right), \qquad \text{Pct} = \left(\frac{I_{\text{mA}} - 4.0}{16.0}\right) \times 100\%, \qquad R(T) = R_0(1 + \alpha \cdot T) = 100 + 0.385 \cdot T \qquad \textbf{(Eq. 6.4)}$$

Here Pct is the percentage of calibrated transmitter span, I_{mA} is the loop current in milliamperes evaluated via [mA], and $R(T)$ is the Pt100 sensor resistance in ohms evaluated via [↶] [RTD]. When the fixture's 0–3,000 psi hydraulic transmitter is pressurized to 1,875 psi (Pct = (1875 / 3000) × 100% = 62.5%), keying **62.5** [mA] gives the calibrated loop current $I_{\text{mA}} = 4.0 + 16.0(0.625) =$ **14.00 mA** (developing 14.0 mA × 250 Ω = 3.50 V across the PLC's 250 Ω sense resistor), confirming that the transmitter is accurate and the intermittent alarm was caused by a loose terminal screw rather than hydraulic pressure loss.

6.3 Dialing In the Milling Fixture and Fluid Loop on the Stack and Solver Workspaces

On the **SC-14**, machining, fluid-power, and instrumentation keys operate both as immediate stack evaluators and as bidirectional Equation Solver workspaces ([↖] [SFM] with **N = 3.82 × S / D**, [↗] [BC] with **C = D × SIN(180 / N)**, [PIP] with **Q = A × V**, [↗] [C$_v$] with **Q = C × √(P / G)**, and [PMP] with **P = Q × H / 3960**):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Workspace Variables
[↖] [SFM] / [↗] [RPM]	Spindle Speed & Surface Velocity	—	Speed SFM	Diameter D (in)	$N = (3.8197 \cdot \text{SFM}) / D$; [↖] [SFM] ([N], [S], [D])
[↖] [TAP]	75% Thread Tap Drill Diameter	—	Major D_{major}	Pitch p or TPI	$D_{\text{tap}} = D_{\text{major}}$ - 1 / TPI (if $X > 3$) or D_{major} - p
[↗] [BC]	Bolt Circle Chord Spacing & Solver	—	Circle Dia D_{bolt}	Hole Count N	$C = D_{\text{bolt}} \sin(180° / N)$; [↗] [BC] ([C$_{ON}$], [D], [N])
[PIP]	Pipe Continuity Flow & Solver	—	Pipe Area A	Velocity V	$Q = A \times V$; [PIP] workspace ([Q], [A], [V])
[↗] [C$_v$]	Control Valve Flow & Solver	—	Coeff C_v	Drop ΔP (psi)	$Q = C\sqrt{(P / G)}$; [↗] [C$_v$] workspace ([Q], [C$_{ON}$], [P], [G])
[PMP] [NPH] / [↗]	Hydraulic Pump Power & Suction Head	—	Flow Q (GPM)	Head H (ft)	$P = Q \cdot H / 3960$; [PMP] ([P], [Q], [H]) & NPSH$_a$
[mA] [RTD] / [↖]	Process Current & Pt100 Resistance	—	—	Pct (%) or T (°C)	$I_{\text{mA}} = 4 + 0.16$ Pct, $R = 100 + 0.385\ T$

6.3.1 Solving Spindle Speed, Tap Drill, and Bolt Circle Chords on the Stack

To compare the aluminum spindle setting (**400** [ENTER] **0.5** [↖] [SFM] → **3,055.77 RPM**) against 4140 alloy steel at SFM = 100 ft/min (**100** [ENTER] **0.5** [↖] [SFM] → **763.94 RPM**), compute the 1/2-13 UNC tap drill size (**0.5** [ENTER] **13** [↖] [TAP] → **0.4231 in.**), and evaluate the 8-hole bolt circle chord spacing on a 6.00 in. hub (**6** [ENTER] **8** [↗] [BC] → **2.2961 in.**), the machinist executes:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
400 [ENTER] **0.5** [↖] [SFM] [C$_{ON}$]	0	0	0	3055.7749	Compute spindle speed at 400 SFM: $N = (3.8197 \times 400)/0.5 \approx$ 3,055.77 RPM
100 [ENTER] **0.5** [↖] [SFM] [C$_{ON}$]	0	0	3055.7749	763.9437	Reduce spindle speed to 100 SFM for 4140 steel (763.94 RPM)
0.5 [ENTER] **13** [↖] [TAP]	3055.77	3055.7749	763.9437	0.4231	Compute **1/2-13 UNC** tap drill diameter $D_{\text{tap}} = 0.500$ - 1/13 ≈ 0.4231 in.
6 [ENTER] **8** [↗] [BC] [C$_{ON}$]	763.94	763.9437	0.4231	**2.2961**	Compute 8-bolt chord spacing on 6 in. circle ($C = 6 \sin 22.5° \approx$ 2.2961 in.)

6.3.2 Calibrating the Loop on the Stack and Solving Backward in [→] [Cv] and [PMP]

Turning to the hydraulic workholding fixture (P_{psi} = 1,875 psi on a 0–3,000 psi span) and the shop coolant loop (C_v = 30 valve at ΔP = 9 psi pumping against H = 88 ft of head), the technician converts 1875 / 3000 = 62.5% span into **14.00 mA** via [mA], checks the Pt100 sensor at 150°C (**157.75 Ω** via [↖] [RTD]), evaluates valve flow Q = **90.00 GPM** via [→] [Cv], and chains directly into **88** [PMP] (→ **2.00 HP**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1875 [ENTER] **3000** [÷] **100** [×] [mA]	0	0	0	14	Convert 62.5% pressure span to loop current I_{mA} = 14.00 mA
150 [↖] [RTD]	0	0	14	157.75	Compute Pt100 RTD resistance at 150°C (R = 157.75 Ω)
30 [ENTER] **9** [→] [Cv] [G/ON]	14	14	157.75	90	Compute coolant valve flow $Q = 30\sqrt{9}$ = 90.00 GPM
88 [PMP] [G/ON]	14	14	157.75	**2**	Compute pump hydraulic horsepower P = (90 × 88)/3960 = 2.00 HP

When plant engineering upgrades the heat-exchanger manifold, the technician uses the **Bidirectional Equation Solver Workspaces** for [PIP] (**PIP: PIPE FLOW**, **Q = A × V**), [→] [Cv] (**CV: VALVE FLOW**, **Q = C × √(P / G)**), and [PMP] (**PMP: PUMP BHP**, **P = Q × H / 3960**) to solve backward for required component ratings:

- **Inverse Valve Pressure Drop & Minimum C_v in [→] [Cv] (Q = C × √(P / G)):** Suppose a quench manifold requires Q = 100.0 GPM (**100** [Q] on [LFU0]) of synthetic heat-transfer brine of specific gravity G = 1.08 (**1.08** [G] on [LFU3]) through a valve with C_v = 35.0 (**35** [G/ON] on [LFU1]). Pressing [XEQ] [P] ([LFU2]) solves backward for the required pressure drop across the valve:

$$\Delta P = G \cdot \left(\frac{Q}{C_v}\right)^2 = 1.08 \times \left(\frac{100.0}{35.0}\right)^2 \approx \mathbf{8.8163}\ \text{psi} \qquad \textbf{(Eq. 6.5)}$$

Conversely, if header pressure drop is limited to ΔP = 16.0 psi (**16** [P]) for Q = 150.0 GPM (**150** [Q]) of water (**1** [G]), pressing [XEQ] [G/ON] ([LFU1]) solves backward for the minimum required valve coefficient $C_v = 150.0 / \sqrt{(16.0)} =$ **37.50**.

- **Inverse Discharge Head in [PMP] (P = Q × H / 3960):** When an existing P = 5.00 HP (**5** [P] on [LFU0]) coolant pump must deliver Q = 180.0 GPM (**180** [Q] on [LFU1]), pressing [XEQ] [H] ([LFU2]) solves backward for the maximum vertical discharge head the pump can sustain: $H_{\text{max}} = 3960 \times 5.00 / 180.0 =$ **110.00 ft**.

6.3.3 Try It on the Stack

Complete your mastery of the **SC-14** industrial trades calculator with these three machine-shop and process-control exercises:

1. **Milling Spindle Speed and Table Feed Rate:** Compute spindle speed N ([↖] [SFM]) and feed rate $F_{\text{m}} = N \cdot f_t \cdot Z$ for a D = 0.625 in. (5/8 in.) 4-flute carbide end-mill cutting alloy steel at SFM = 120 ft/min (**120** [ENTER] **0.625** [↖] [SFM]) with chip load f_t = 0.0025 in/tooth. *(Verified Register X results: $N \approx$ **733.3860 RPM** on the stack or **733.44 RPM** in the [↖] [SFM] workspace, and $F_{\text{m}} \approx$ **7.3339 in/min**.)*

2. **Twelve-Hole Flange Bolt Circle and Inverse Cutter Diameter:** Find the adjacent hole chord spacing ([→] [BC]) for a 12-hole pattern on a D_{bolt} = 9.50 in. pitch

circle (**9.5** [ENTER] **12** [↗] [BC]), and use the [↖] [SFM] workspace ($\mathbf{N} = \mathbf{3.82} \times \mathbf{S} \,/\, \mathbf{D}$) to solve backward for the maximum cutter diameter [D] that can run at spindle speed N = 764 RPM (**764** [N]) without exceeding surface speed S = 150 ft/min (**150** [S], [XEQ] [D]). *(Verified Register X results: $C \approx$* ***2.4588 in.*** *and D_{max} =* ***0.7500 in.****.)*

3. **Process Loop Calibration and Inverse Valve Sizing:** For a Pt100 RTD at T = 125°C on a 0–200°C transmitter, find sensor resistance (**125** [↖] [RTD]) and loop current (**62.5** [mA]), and use the [↗] [C▾] and [PMP] workspaces to find flow Q (**35** [C ON] **25** [P] **1** [G] [XEQ] [Q]) and water brake horsepower (**175** [Q] **132** [H] [XEQ] [P]). *(Verified Register X results: $R =$* ***148.125 Ω****, $I_{mA} =$* ***14.00 mA****, $Q =$* ***175.00 GPM****, and BHP $\approx$* ***5.8333 HP****.)*

From macroscopic copper feeders, bent steel raceways, and 4–20 mA analog current loops, we cross into the digital registers, bitwise masks, and clock dividers that govern embedded silicon in **Unit III: Digital Systems, Bitwise Logic & Embedded Timing**.

Unit III: Digital Systems, Bitwise Logic & Embedded Timing

In 1703, Gottfried Wilhelm Leibniz published *Explication de l'Arithmétique Binaire* in the *Mémoires de l'Académie Royale des Sciences*. He demonstrated that every integer can be represented using only the two figures **0** and **1** through successive powers of two. More than two centuries later, in his 1937 master's thesis at the Massachusetts Institute of Technology (*A Symbolic Analysis of Relay and Switching Circuits*), twenty-one-year-old Claude Shannon proved that George Boole's 1854 algebra of logical propositions maps one-to-one onto physical electrical switches wired in series, in parallel, and through inverting relays. When John von Neumann, Arthur Burks, and Herman Goldstine drafted their 1945–1946 reports on the EDVAC and the Institute for Advanced Study computer, they united Leibniz's base-2 positional notation with Shannon's switching gates. They also standardized **two's complement** binary arithmetic across fixed-width hardware registers.

In pure mathematics, the set of integers Z extends without bound in both directions, and the continuum of real numbers R is infinitely dense between any two points. An algebraist never asks how many bits wide an equation is, whether shifting a number right preserves its sign, or whether a variable's memory address is divisible by sixteen. Inside a physical microprocessor, an operating-system kernel, or a network router, every number lives inside a hardware register built from a fixed number W of bistable transistor flip-flops. Each flip-flop holds a low voltage (**0**) or a high voltage (**1**) across standard **Word Sizes** of $W \in (8,\ 16,\ 32,\ 64)$ bits. Because a W-bit register has 2^W states, hardware integer arithmetic operates over the finite **modular ring** $\boldsymbol{Z\ /\ 2^W\ Z}$. Just as a five-digit mechanical odometer wraps from **99999** to **00000**, an 8-bit unsigned register holding **1111 1111** (255) wraps to **0000 0000** (0) when incremented by 1. Similarly, a 16-bit signed register incremented past +32,767 (**0x7FFF**) wraps into -32,768 (**0x8000**), triggering hardware **Signed Arithmetic Overflow**.

In the decades following von Neumann's architecture, Richard Hamming (1950) formalized bit-level Hamming weight, William Wesley Peterson (1961) introduced Galois-field polynomial **Cyclic Redundancy Checks**, and William Kahan (1985) led the standardization of **IEEE 754 floating-point** representation.

The **SC-16 Systems Programming, Embedded & Cyber Operations** calculator brings these register-level and mixed-signal laws together in a single instrument. Across the three chapters of this Unit, we follow a robotics firmware team from bringing up a 32-bit motor controller to debugging a serial telemetry link and calibrating a networked vibration sensor array:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 7	Debugging a 16/32-bit motor controller timer register where an unmasked bitfield and a logical-vs-arithmetic shift bug corrupt signed velocity commands	Positional radix compression, $Z/2^{W}Z$ two's complement rings, Boolean masking, logical vs. arithmetic shifts, and $Q_{m.n}$ fixed-point	1. Hexadecimal Nibbles & Two's Complement Rings • 2. Bitwise Boolean Masking & Peripheral Control Registers • 3. Logical vs. Arithmetic Shifts & Fixed-Point Scaling
Chapter 8	Unpacking a big-endian 4-byte telemetry packet into an IEEE 754 float, verifying its CRC checksum, partitioning CIDR subnets, and checking UART baud drift	Little/Big-Endian byte swaps, IEEE 754 **binary32** bitfields, GF(2) polynomial division, CIDR netmask synthesis, and 16× UART divisors	1. Network Byte Order & IEEE 754 Floating-Point Decode • 2. Galois-Field GF(2) Polynomials & CRC Checksums • 3. CIDR Subnet Masks & 16× Oversampled UART Clocks
Chapter 9	Synthesizing timer PWM carriers, sizing an RC anti-aliasing filter and 12-bit ADC front end for a 3.3 V vibration sensor, and aligning DMA ring buffers	Timer prescalers, RC settling time constants ($\tau = RC$), uniform 2^{N} ADC quantization, Hamming weight, leading zeros, and memory alignment	1. Hardware Timer Prescalers & PWM Carrier Synthesis • 2. Uniform 2^{N} ADC Quantization & RC Filter Settling • 3. Bit Counting, CIDR Subnets & Power-of-Two DMA Alignment

Chapter 7: The Stuck Interrupt Flag in the Motor Controller

On a robotics test bench, an embedded firmware engineer is bringing up a 32-bit ARM Cortex-M microcontroller that drives a precision linear actuator. Two baffling bugs appear during closed-loop testing. First, when the initialization routine writes a 4-bit clock-divider setting of **10** (**0xA**) into bits **[11:8]** of the 32-bit Timer Control Register (initially holding **0x00240505**), the interrupt-enable flag at bit 0 stays stuck and the old prescaler bits in **[11:8]** corrupt the new setting. The developer used a raw bitwise OR without first masking and clearing the target bitfield.

Second, whenever the optical encoder commands a slow reverse jog of V = -16 counts/tick inside a 16-bit signed velocity loop (where -16 is stored in two's complement as **0xFFF0**), the proportional damping filter divides the velocity by 4 using a 2-bit **Logical Shift Right** instead of a sign-preserving **Arithmetic Shift Right**. Zero-filling the top two bits turns **0xFFF0** (-16) into **0x3FFC** (+16,380), slamming the actuator forward at full saturation. To squash both bugs at the register transfer level, the engineer works through three foundational themes:

1. Translate between binary nibbles and hexadecimal words, and verify 16-bit two's complement negation and signed overflow boundaries.

2. Construct a surgical 4-bit clear-and-merge bitmask for bits **[11:8]** that updates the clock divider while preserving every neighboring bit.

3. Compare logical zero-fill shifts against sign-preserving arithmetic shifts across signed and unsigned words, and verify $Q_{m.n}$ fixed-point filter gains.

7.1 Positional Radix Compression and Fixed-Width Two's Complement Rings

At the register transfer level, every value stored in memory or across an internal CPU bus is a sequence of W bits that can be viewed in any positional radix and interpreted as an unsigned magnitude or a signed two's complement integer.

7.1.1 Binary Nibbles, Octal Triads, and Hexadecimal Register Notation

In any positional number system of integer radix $b \geq 2$—where $b = 16$ for Hexadecimal ([HEX]), $b = 10$ for Decimal ([↖] [DEC]), $b = 8$ for Octal ([↖] [OCT]), and $b = 2$ for Binary ([↗] [BIN])—a k-digit word $(d_{k-1}\ d_{k-2}\ \ldots\ d_1\ d_0)_b$ with digits $d_i \in (0,\ 1,\ \ldots,\ b\text{-}1)$ represents the scalar integer V obtained by summing each digit weighted by its positional power b^i. When [HEX] mode is active on the **SC-16**, the six top-row softkeys [LFU0]–[LFU5] automatically bind to the hexadecimal digits [A], [B], [C ON], [D], [E], and [F] for rapid single-keystroke entry of 16-, 32-, and 64-bit register words. Because $16 = 2^4$, partitioning a binary string into blocks of **four bits (nibbles)** from the least significant bit ($i = 0$) upward maps each 4-bit block $(b_{4m+3}\ b_{4m+2}\ b_{4m+1}\ b_{4m})_2$ with internal weights (8, 4, 2, 1) onto a single hexadecimal digit $H_m \in [0,\ 15]$ (**0**–**9**, [A]–[F]), while $8 = 2^3$ maps each 3-bit triad onto a single octal digit (**0**–**7**):

$$V = \sum_{i=0}^{k-1} d_i \cdot b^i, \qquad \sum_{j=0}^{3} b_{4m+j} \cdot 2^{4m+j} = 2^{4m}(8b_{4m+3} + 4b_{4m+2} + 2b_{4m+1} + 1b_{4m}) = H_m \cdot 16^m \qquad \textbf{(Eq. 7.1)}$$

In this radix decomposition, each hexadecimal digit H_m summarizes four adjacent transistor states without cross-boundary carry arithmetic:

Hex (HEX)	Binary Nibble (→ BIN)	Decimal (← DEC)	Octal (← OCT)	Hex (HEX)	Binary Nibble (→ BIN)	Decimal (← DEC)	Octal (← OCT)
0	**0000**	0	**00**	**8**	**1000**	8	**10**
1	**0001**	1	**01**	**9**	**1001**	9	**11**
2	**0010**	2	**02**	**A** (LFU0)	**1010**	10	**12**
3	**0011**	3	**03**	**B** (LFU1)	**1011**	11	**13**
4	**0100**	4	**04**	**&** (LFU2)	**1100**	12	**14**
5	**0101**	5	**05**	**D** (LFU3)	**1101**	13	**15**
6	**0110**	6	**06**	**E** (LFU4)	**1110**	14	**16**
7	**0111**	7	**07**	**F** (LFU5)	**1111**	15	**17**

16-Bit Register (WSZ = 16)**: Hex** 0x4A3F = $19{,}007_{10}$

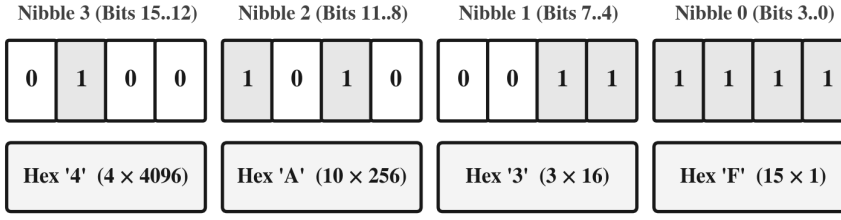


Figure 7.1: *Partitioning a 16-bit hardware register (0x4A3F) into four 4-bit hexadecimal nibbles.*

To translate a 16-bit hexadecimal register snapshot **0x4A3F** into binary (→ BIN) and decimal (← DEC), each hex digit expands into its 4-bit nibble under the (8, 4, 2, 1) weights—**4** = 0100_2, **A** (10 = 8+2) = 1010_2, **3** (2+1) = 0011_2, and **F** (8+4+2+1) = 1111_2—giving **0100 1010 0011 1111**, while summing the powers of 16 gives $V = 4(16^3) + 10(16^2) + 3(16^1) + 15(16^0) = 16{,}384 + 2{,}560 + 48 + 15 = \mathbf{19{,}007}$.

7.1.2 Modular Integer Rings, Two's Complement Negation, and Signed Overflow

In a fixed W-bit register (→ WSZ), an **Unsigned Integer** (→ UNS) assigns positive weight $+2^i$ to every bit b_i from $i = 0$ through W - 1. To encode negative numbers without building separate subtraction circuitry, digital processors use **Two's Complement** (← NEG) representation, in which the most significant bit b_{W-1} carries **negative weight** $\mathbf{-2^{W-1}}$ while the lower W - 1 bits retain positive weights $+2^i$:

$$V_{\text{unsigned}} = \sum_{i=0}^{W-1} b_i \cdot 2^i, \qquad V_{\text{signed}} = -b_{W-1} \cdot 2^{W-1} + \sum_{i=0}^{W-2} b_i \cdot 2^i, \qquad -X \equiv 2^W - X \equiv (\sim X) + 1 \pmod{2^W} \quad \textbf{(Eq. 7.2)}$$

Because adding any W-bit word X to its bitwise one's complement $\sim X$ ([SHIFT] [~], where every **0** flips to **1** and every **1** flips to **0**) produces a word of W ones equal to 2^W - 1, adding 1 yields X + (($\sim X$) + 1) = $2^W \equiv 0 \pmod{2^W}$. Consequently, the additive inverse -X in W-bit two's complement ([SHIFT] [NEG]) is formed by inverting all W bits and adding 1:

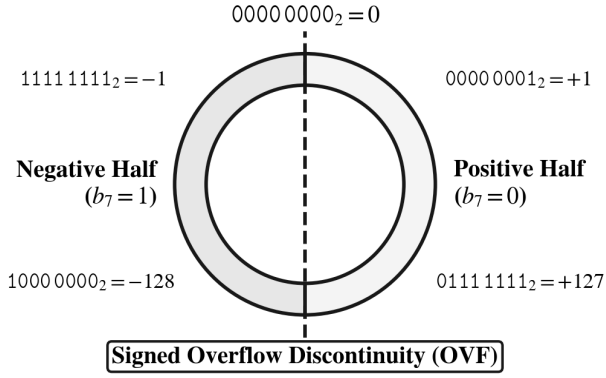


Figure 7.2: *Eight-bit modular two's complement integer ring showing sign symmetry around zero and the signed overflow boundary.*

Because zero (**0x00**) consumes one of the 2^{W-1} non-negative bit patterns, every signed W-bit two's complement range is asymmetric by one count, extending from -2^{W-1} (**0x80…0**) to +2^{W-1} - 1 (**0x7F…F**):

Word Size ([SHIFT] [WSZ])	Bits W	Unsigned Range ([SHIFT] [UNS]: 0 to 2^W - 1)	Signed Two's Complement Range (-2^{W-1} to +2^{W-1}-1)
Byte	8	0 to 255 (**0xFF**)	-128 (**0x80**) to +127 (**0x7F**)
Halfword / **Word**	16	0 to 65,535 (**0xFFFF**)	-32,768 (**0x8000**) to +32,767 (**0x7FFF**)
Word / **DWord**	32	0 to 4,294,967,295 (**0xFFFFFFFF**)	-2,147,483,648 (**0x80000000**) to +2,147,483,647 (**0x7FFFFFFF**)
Quadword	64	0 to 1.84467×10^{19}	-9.22337×10^{18} to $+9.22337 \times 10^{18}$

In our motor controller's 16-bit velocity register ([SHIFT] [WSZ] = 16), +16 is **0x0010** (**0000 0000 0001 0000**). Inverting all 16 bits gives **0xFFEF** and adding 1 via [SHIFT] [NEG] yields -16 ≡ **0xFFF0** (65,520 unsigned). Similarly, in 8-bit mode ([SHIFT] [WSZ] = 8), -42 is formed from +42 = 0x2A (**0010 1010**) via ~ 0x2A + 1 = 0xD5 + 1 = **0xD6** (214 unsigned). When two W-bit words A and B sum to $S = (A + B) \bmod 2^W$, **Signed Arithmetic Overflow** ([SHIFT] [OVF]) occurs if and only if two operands of the same sign bit ($A_{W-1} = B_{W-1}$) produce a sum of the opposite sign bit ($S_{W-1} \neq A_{W-1}$):

$$\text{OVF} = (A_{W-1} \text{ AND } B_{W-1} \text{ AND } (\sim S_{W-1})) \mid ((\sim A_{W-1}) \text{ AND } (\sim B_{W-1}) \text{ AND } S_{W-1}) \quad \textbf{(Eq. 7.3)}$$

Here A_{W-1}, B_{W-1}, and S_{W-1} are the sign bits of the two addends and their modular sum. In 8-bit signed arithmetic ([-128, +127]), adding +100 (**0x64**) and +50 (**0x32**) produces **0x96**. Because two positive operands ($A_7 = B_7 = 0$) yielded $S_7 = 1$, the processor asserts [SHIFT] [OVF] (**1**) as +150 wraps across +127 / -128 to 150 - 256 = **-106**.

7.2 Bitwise Boolean Masking, Shifts, and Fixed/Floating-Point Encoding

Manipulating packed peripheral control registers without disturbing adjacent interrupt flags requires combining Boolean truth tables with contiguous bitfield mask generators, distinguishing logical zero-fill shifts from arithmetic sign-extending shifts, and converting between IEEE-754 floating-point bit patterns ([FLT]) and $Q_{m.n}$ fixed-point integers ([↶] [Q.n], [↷] [FXP]).

7.2.1 Boolean Logic Gates and Peripheral Control Register Masking

Bitwise operators apply Boolean logic gates in parallel across each bit index $i \in (0, \ldots, W\text{-}1)$ of two W-bit words Y and X:

Bit Y_i	Bit X_i	[↯] (Y_i & X_i)	[|] (Y_i | X_i)	[⊛] ($Y_i \oplus X_i$)	[↷] [~] (~ X_i)	[↶] [~↯] ~ (Y_i & X_i)	[↶] [~|] ~ (Y_i | X_i)
0	**0**	**0**	**0**	**0**	**1**	**1**	**1**
0	**1**	**0**	**1**	**1**	**0**	**1**	**0**
1	**0**	**0**	**1**	**1**	—	**1**	**0**
1	**1**	**1**	**1**	**0**	—	**0**	**0**

To isolate or modify a multi-bit hardware field of width w bits starting at bit offset s, the **Bitfield Mask Generator** ([↶] [MSK]) shifts 2^w - 1 (w contiguous **1** bits) left by s positions:

$$\mathrm{MSK}(s,w) = ((1 \ll w) - 1) \ll s, \qquad R_{\text{updated}} = (R \text{ AND } {\sim}\mathrm{MSK}(s,w)) \mid ((V_{\text{field}} \ll s) \text{ AND } \mathrm{MSK}(s,w)) \quad \textbf{(Eq. 7.4)}$$

In this read-modify-write identity, R is the initial control register value, MSK(s, w) is the positive mask covering bits $[s{+}w\text{-}1 : s]$, **~** MSK(s, w) is the inverted clear mask with **0**s across the target field and **1**s everywhere else, and $V_{\text{field}} \ll s$ aligns the new setting into position. For our motor controller's 4-bit clock-divider field ($w = 4$) at bit offset $s = 8$ (bits **[11:8]**), the mask is MSK(8, 4) = ((1 ≪ 4) - 1) ≪ 8 = 15 × 256 = **0x0F00** (3,840). Clearing bits **[11:8]** of **0x00240505** with **~** 0x0F00 leaves **0x00240005**, and merging **0xA** shifted left by 8 (**0x0A00**) via [|] yields **0x00240A05** (or **0xA05** on the low 16-bit halfword).

7.2.2 Logical Shifts, Sign-Preserving Arithmetic Shifts, Rotations, and Fixed-Point Scaling

Shifting a binary word Y left or right by k bit positions scales the integer by 2^k, where the fill rule for the vacated bits distinguishes unsigned magnitudes from signed two's complement velocities:

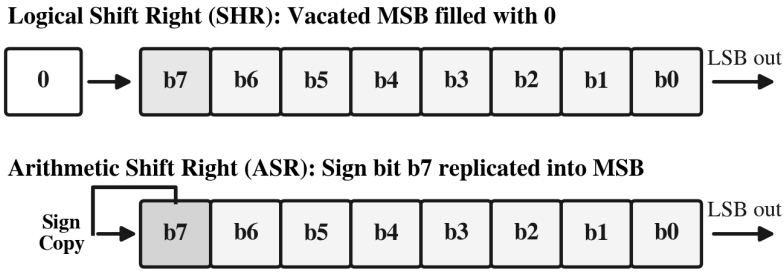


Figure 7.3: *Comparison of Logical Shift Right (SHR, zero-fill) and Arithmetic Shift Right (ASR, sign-bit replication).*

$$Y \ll k = (Y \cdot 2^k) \bmod 2^W \quad ([\ll]), \qquad Y \gg k = \left\lfloor \frac{Y_{\text{unsigned}}}{2^k} \right\rfloor \quad ([\gg]), \qquad Y \gg k = \left\lfloor \frac{Y_{\text{signed}}}{2^k} \right\rfloor \quad ([\text{f}]\ [\text{ASR}]) \qquad \textbf{(Eq. 7.5)}$$

Here **Logical Shift Left** ([≪]) shifts bits toward the most significant end and fills vacated low bits with **0**; **Logical Shift Right** ([≫]) shifts toward the least significant end and fills vacated high bits with **0** (dividing an unsigned word by 2^k); **Arithmetic Shift Right** ([↰] [ASR]) shifts right while **replicating the original sign bit b_{W-1}** into all vacated high bits to preserve two's complement sign; **Circular Rotate** ([↱] [ROR]) rotates bits across the W-bit word. And single-bit manipulation keys **Bit Set** ([↰] [SB], $Y \mid (1 \ll X)$), **Bit Clear** ([↰] [CB], $Y \,\&\, \sim (1 \ll X)$), and **Bit Test** ([↱] [B?], $(Y \gg X) \,\&\, 1$) operate directly on individual flag bits. Furthermore, on the **SC-16**:

- **Bidirectional IEEE-754 Single-Precision (binary32) Conversion ([FLT]):** When Register X holds a 32-bit integer bit pattern ($>=$ **0x00800000**, such as **0x41480000** = 1,095,237,632), pressing [FLT] decodes it into its floating-point value (**12.5**). When Register X holds a floating-point value such as **12.5**, pressing [FLT] encodes it into its IEEE-754 32-bit integer pattern (**1095237632** = **0x41480000**).
- **Fixed-Point $Q_{m.n}$ Encode ([↰] [Q.n]) & Decode ([↱] [FXP]):** With a floating-point number in Register Y and fractional bit count n in Register X, pressing [↰] [Q.n] converts to the rounded Q-format integer round($Y \cdot 2^X$) (e.g., **0.75** [ENTER] **15** [↰] [Q.n] → **24,576** = 0x6000), while pressing [↱] [FXP] converts a Q-format integer in Register Y with fractional bits n in Register X back to its real floating-point value $Y \,/\, 2^X$.

7.3 Debugging the Motor Controller Register and Velocity Shift on the Stack

The **SC-16** brings multi-radix display switching ([HEX], [↰] [DEC], [↰] [OCT], [↱] [BIN]), configurable hardware word widths ([↱] [WSZ]: 8, 16, 32, 64-bit), two's complement negation ([↰] [NEG]), Boolean logic gates ([&], [|], [⊕], [↱] [~], [↰] [~&], [↰] [~|]), single-bit and contiguous bitfield manipulators ([↰] [SB], [↰] [CB], [↱] [B?], [↰] [MSK]), logical/arithmetic shifts ([≪], [≫], [↰] [ASR], [↱] [ROR]), and IEEE-754 / $Q_{m.n}$ converters ([FLT], [↰] [Q.n], [↱] [FXP]) together on the RPN stack:

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
HEX ↶ DEC ↶ / OCT ↷ BIN	Active Radix Selection	—	—	Value X	Displays stack in base $b \in$ (16, 10, 8, 2) (LFU0–LFU5 = A–F in HEX)
↷ WSZ / ↷ UNS / ↷ OVF	Word Width, Unsigned & Overflow	—	—	Width W / Word X	Masks modulo 2^W ($W \in$ (8,16,32,64)), unsigned cast, signed overflow check
↶ NEG	Two's Complement Negation	—	—	Word X	$-X \equiv (\sim X + 1) \bmod 2^W$
& | ⊕ ↶ ~& ↶ ~|	Binary Bitwise Logic	—	Word Y	Word X	$Y \,\&\, X$, $Y \mid X$, $Y \oplus X$, $\sim(Y \,\&\, X)$, $\sim(Y \mid X)$
↷ ~	One's Complement Inversion	—	—	Word X	$(\sim X) \bmod 2^W$
↶ SB ↶ CB / ↷ B?	Single-Bit Set / Clear / Test	—	Word Y	Bit index k	$Y \mid 2^k$, $Y \,\&\, \sim 2^k$, $(Y \gg k) \,\&\, 1$
↶ MSK	Contiguous Bitfield Mask	—	Offset s	Width w	$\text{MSK}(s, w) = ((1 \ll w) - 1) \ll s$
≪ ≫ ↶ ASR ↷ ROR	Bitwise Shifts & Rotation	—	Word Y	Count k	$Y \ll k$, $Y \gg k$ (zero-fill), $Y \gg k$ (sign-fill), circular rotate
FLT / ↶ Q.n / ↷ FXP	IEEE-754 & $Q_{m.n}$ Fixed-Point	—	Value Y	Word X or Bits n	**0x41480000** ↔ 12.5 (FLT); $\text{round}(Y \cdot 2^X)$ (↶ Q.n); $Y/2^X$ (↷ FXP)

7.3.1 Masking and Merging the Timer Control Field in Hexadecimal

To configure the 4-bit clock divider at bits **[11:8]** ($s = 8$, $w = 4$) with **0xA** (10_{10}, entered via softkey A on LFU0) on a control register holding **0x0005**, the engineer switches to HEX, generates **8** ENTER **4** ↶ MSK (**0xF00**), shifts A left by 8 bits via A ENTER **8** ≪ (**0xA00**), and merges with **5** via | to obtain **0xA05**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
HEX **8** ENTER **4** ↶ MSK	0	0	0	0xF00	Generate 4-bit mask at bit offset 8 (MSK(8,4) = 0x0F00)
5 ENTER	0	0xF00	0x5	0x5	Stage initial control register value **0x0005**
A ENTER **8** ≪	0xF00	0xF00	0x5	0xA00	Shift hex digit A (LFU0) left by 8 bits to **0x0A00**
|	0xF00	0xF00	0xF00	**0xA05**	Merge shifted field into register (0x0005 *lor* 0x0A00 = 0x0A05)

When extracting bits **[11:8]** in the inverse direction from a returned status word **0x002405A0** at peripheral address **0x40013800** + **0x1C** = **0x4001381C**, keying HEX **2405** A **0** ENTER **8** ENTER **4** ↶ MSK & **8** ≫ masks **0x0500** and shifts it down to **0x5**.

7.3.2 Fixing the Signed Velocity Right Shift in Sixteen-Bit Mode

Next, setting the word width to 16 bits (HEX **10** ↷ WSZ, since **0x10** = 16_{10}) and negating **+16** (**0x10**) with ↶ NEG produces **0xFFF0** (-16). Executing **2** ↶ ASR preserves the sign bit to give **0xFFFC** (-4), whereas **2** ≫ reproduces the runaway positive value **0x3FFC** (+16,380):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
HEX **10** g WSZ	0	0	0	0x10	Set active word width to 16 bits (**0x10** = 16_{10})
10 ↶ NEG	0	0	0x10	0xFFF0	Negate **0x0010** (+16) to 16-bit two's complement **0xFFF0** (-16)
ENTER **2** ↶ ASR	0	0	0x10	0xFFFC	Arithmetic right shift by 2 bits preserves sign (**0xFFFC** = -4)
f f f **0** ENTER **2** ≫	0x10	0x10	0xFFFC	**0x3FFC**	Logical right shift by 2 bits zero-fills high bits (**0x3FFC** = +16,380)

7.3.3 Try It on the Stack

Practice register masking, two's complement negation, and IEEE-754 / $Q_{m.n}$ conversion with these three firmware debugging exercises:

1. **Peripheral Control Register Field Modification:** A 32-bit SPI control register holds **0x00000301**. Using ↶ MSK, g ~, &, ≪, and | in HEX mode, clear the 3-bit baud-rate field at bits **[5:3]** (shift $s = 3$, width $w = 3$) and insert the new setting **0x5** (5_{10}) into bits **[5:3]**. *(Verified Register X results: mask MSK(3,3) = **0x38**, cleared register **0x301**, shifted setting 5 ≪ 3 = **0x28**, and final merged register **0x329**.)*

2. **Sixteen-Bit Signed Sensor Delta Division and Bit Set/Clear:** A 16-bit encoder delta equals Δ = -36 (+**36** = **0x0024**). Key ↶ DEC **16** g WSZ HEX **24** ↶ NEG to find its 16-bit two's complement hex word and **2** ↶ ASR to divide by 4 with sign extension. Then in 8-bit mode (↶ DEC **8** g WSZ HEX), set bit 7 of **0x5A** (**5** A ENTER **7** ↶ SB), clear bit 1 (**1** ↶ CB), and test bit 3 (**3** g B?). *(Verified Register X results: -36 ≡ **0xFFDC**, -36 ≫ 2 = -9 ≡ **0xFFF7**, ↶ SB = **0xDA**, ↶ CB = **0xD8**, and g B? = **0x1**.)*

3. **Bidirectional IEEE-754 Float and Fixed-Point Q_{15} Round-Trip:** In ↶ DEC mode, encode **12.5** into its 32-bit IEEE-754 integer pattern (**12.5** FLT → **1,095,237,632** = 0x41480000) and press FLT again to decode back to **12.5**. Then encode **0.75** into Q_{15} (**0.75** ENTER **15** ↶ Q.n → **24,576**) and decode back via **15** g FXP (→ **0.75**).

With the internal control register and velocity loop operating as designed, we trace the actuator's serial telemetry link and industrial network subnet across the system bus in Chapter 8.

Chapter 8: The Corrupt Sensor Packet Across the Serial Bus

Once the linear actuator is running, the microcontroller streams 4-byte temperature and current telemetry packets over an RS-485 serial bus at 115,200 bits/s to a Little-Endian ARM monitoring gateway connected to an isolated **/24** (**255.255.255.0**) industrial Ethernet VLAN. On the monitoring console, however, a normal heatsink temperature reading of +12.25°C (and +12.50°C on Channel 2) suddenly displays as 6.02×10^{-39} °C, and one out of every thousand frames fails its **Cyclic Redundancy Check**.

Inspecting the raw serial capture on a logic analyzer reveals three layers of embedded systems and network physics at work:

1. The sensor node transmitted the 32-bit IEEE 754 floating-point word **0x41440000** in Big-Endian **Network Byte Order** (**41 44 00 00**), but the Little-Endian host CPU read the four bytes backward as **0x00004441** without executing an **Endian Byte Swap** before decoding the **IEEE 754 Single-Precision Float**.
2. Verifying packet integrity requires evaluating the polynomial division remainder over GF(2) (**CRC8/CCITT** with polynomial **0x07** and IEEE 802.3 32-bit frame verification).
3. Auditing the gateway's CIDR subnet mask and host capacity alongside the 16× oversampled UART baud divisor reveals whether integer rounding in the baud clock (115,384.6 bps, +0.16% error) or an un-terminated RS-485 line caused the frame errors.

8.1 Network Byte Order, Floating-Point Bitfields, and Galois-Field Checksums

Transmitting fractional measurements reliably across heterogeneous processors requires reconciling byte endianness, unpacking sign-exponent-mantissa bitfields, and computing polynomial division remainders over GF(2).

8.1.1 Endian Byte Swapping and Single-Precision Floating-Point Encoding

When multi-byte words cross between x86 or ARM processors—which store the least significant byte at the lowest memory address in **Little-Endian** order—and network or industrial fieldbus protocols in **Big-Endian** (Network Byte Order, most significant byte first), **Byte-Order Endian Swap** (

) reverses the four octets of a 32-bit word (or the two bytes of a 16-bit halfword):



Figure 8.1: *Little-Endian host byte order versus Big-Endian network byte order across a 32-bit word (SWP).*

Once the 32-bit word is in native host order, the **IEEE 754 Single-Precision (binary32)** standard ([FLT], alongside half-precision [↰] [F16] and double-precision [↱] [F64]) partitions bits **[31:0]** into a 1-bit sign $S \in (0, 1)$ at bit 31, an 8-bit unsigned biased exponent $E \in [1, 254]$ at bits **[30:23]** (with bias 127, so the true binary exponent is $e = E - 127$), and a 23-bit fractional mantissa $M \in [0, 2^{23}\text{-}1]$ at bits **[22:0]** with an implicit leading 1 before the binary point. On microcontrollers without floating-point hardware, $\boldsymbol{Q_{m.n}}$ **Fixed-Point Scaling** ([↰] [Q.n] and [↱] [FXP]) instead converts between a real value V_{real} and an integer Q_{int} scaled by 2^{n}:

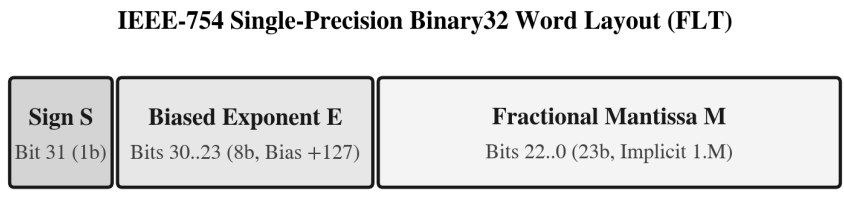


Figure 8.2: *IEEE-754 single-precision 32-bit floating-point bitfield allocation (sign bit, 8-bit biased exponent, and 23-bit mantissa).*

$$V_{\text{float}} = (-1)^S \times \left(1 + \tfrac{M}{2^{23}}\right) \times 2^{E-127}, \quad Q_{\text{int}} = \text{round}(V_{\text{real}} \cdot 2^n) \ \text{([f] [Q.n])}, \quad V_{\text{real}} = \tfrac{Q_{\text{int}}}{2^n} \ \text{([g] [FXP])} \qquad \textbf{(Eq. 8.1)}$$

In the byte-swapped 32-bit telemetry word **0x41440000** (**0100 0001 0100 0100 0000 0000 0000 0000**), sign bit $S = 0$ indicates a positive value, biased exponent $E = 1000\ 0010_2 = 130$ gives true exponent $e = 130 - 127 = +3$, and the fractional mantissa bits **.1000100... _2** evaluate to $2^{-1} + 2^{-5} = 0.5 + 0.03125 = 0.53125$, yielding $V_{\text{float}} = +1.53125 \times 2^3 = +\mathbf{12.25°C}$ (while **0x41480000** = 1,095,237,632 decodes via [FLT] to $+1.5625 \times 2^3 = +\mathbf{12.50°C}$, and pressing [FLT] on **12.5** encodes back to **1095237632** = 0x41480000).

8.1.2 Polynomial Division over Galois Field Two and Cyclic Redundancy Checks

To detect burst noise on serial sensor buses and Ethernet links, a **Cyclic Redundancy Check** ([↰] [CRC]) treats a binary message frame as a polynomial over the two-element Galois field GF(2), where addition and subtraction are both carry-free [⊕] (⊕), and computes the remainder modulo a generator polynomial:

- **8-Bit Sensor Bus CRC8/CCITT ($G(x) = x^8 + x^2 + x + 1$, Polynomial 0x07):** For compact 1-wire and RS-485 sensor headers, each bit shift tests the MSB (b_7) of the 8-bit accumulator and XORs with **0x07** when $b_7 = 1$.
- **32-Bit Frame CRC-32 (G_{ref} = 0xEDB88320):** For IEEE 802.3 Ethernet and telemetry frames, starting from C_0 = 0xFFFFFFFF ⊕ X, the reflected Linear Feedback Shift Register (LFSR) shifts right and XORs with **0xEDB88320** whenever the LSB is **1**, followed by final bitwise inversion ~ C:

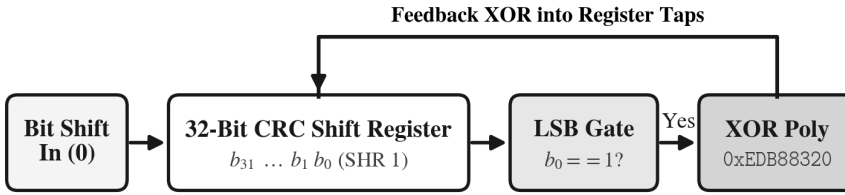


Figure 8.3: *Reflected 32-bit Linear Feedback Shift Register (LFSR) datapath for IEEE 802.3 CRC-32 polynomial division.*

$$C_{i+1} = (C_i \gg 1) \oplus G_{\text{poly}} \text{if } (C_i \text{ AND } 1) = 1,\ C_i \gg 1 \text{if } (C_i \text{ AND } 1) = 0 \qquad \textbf{(Eq. 8.2)}$$

8.2 CIDR Subnet Mask Synthesis and Asynchronous Serial Baud Divisors

Routing telemetry frames from the RS-485 gateway onto the plant Ethernet backbone requires synthesizing CIDR netmasks and host capacities ([NET]) and verifying that the 16× oversampled UART baud clock ([⟳] [BPS]) remains within framing tolerance.

8.2.1 CIDR Prefix Netmasks and Usable Host Counts

In **Classless Inter-Domain Routing** ([NET]), an IPv4 subnet prefix [NET] ($P \in [0, 32]$) assigns the upper P bits of a 32-bit IPv4 word to the network mask and leaves 32 - P bits for host addresses. Pressing [NET] with prefix P in Register X evaluates both the **32-bit Netmask** in Register Y and the **Usable Host Count** N_{hosts} in Register X:

$$\text{Netmask}(P) = (2^{32} - 2^{32-P}) = \text{0xFFFFFFFF} \ll (32 - P), \qquad N_{\text{hosts}} = 2^{32-P} - 2 \qquad \textbf{(Eq. 8.3)}$$

For a standard /**24** industrial VLAN (**24** [NET]), the 32-bit netmask pushed to Register Y is **4294967040** (**0xFFFFFF00**, or **255.255.255.0**) and the usable host count in Register X is 2^{32-24} - 2 = **254 hosts** (excluding the **.0** network address and **.255** broadcast address).

8.2.2 Sixteen-Times Oversampled Asynchronous Serial Divisors and Clock Drift

Because a **Universal Asynchronous Receiver/Transmitter** ([→] [BPS]) transmits data without a shared clock wire, the receiver oversamples the incoming line at 16 times the target symbol rate B (in bits/s) so it can align its sample point with the center (8th tick) of each bit cell. Dividing the peripheral clock f_{clk} (in Register Y) by $16 \cdot B$ (B in Register X) and rounding to the nearest integer yields the hardware baud divisor D:

$$D = \text{round}\left(\frac{f_{\text{clk}}}{16 \cdot B}\right), \qquad B_{\text{actual}} = \frac{f_{\text{clk}}}{16 \cdot D}, \qquad \text{Err}\% = \left(\frac{B_{\text{actual}} - B}{B}\right) \times 100\% \qquad \textbf{(Eq. 8.4)}$$

Here D is the integer value written to the baud-rate register via [→] [BPS], B_{actual} is the synthesized bit rate, and Err% is the fractional clock skew (which must remain within $\pm$ 2.0% over a 10-bit start-data-stop frame to prevent stop-bit framing errors). On our $f_{\text{clk}} =$ 48 MHz bus targeting $B =$ 115,200 bps, keying **48000000** [ENTER] **115200** [→] [BPS] evaluates 48,000,000 / (16 × 115,200) = 26.0417 and rounds to $D =$ **26**. The actual baud rate is 48,000,000 / (16 × 26) **≈ 115,384.6 bps**, a clock drift of +0.16%—proving that the UART timing is well within tolerance and the CRC errors were caused by an un-terminated RS-485 bus end rather than baud-rate skew.

8.3 Unpacking the Telemetry Frame and Auditing Subnets and Clocks on the Stack

On the **SC-16**, [→] [SWP], [FLT], [←] [CRC], [NET], and [→] [BPS] evaluate 32-bit telemetry words, CIDR prefixes, and serial clock dividers directly on the RPN stack:

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression
[→] [SWP]	Endian Byte-Order Swap	—	—	Word X	Reverses byte order (**B3..B0 ↔ B0..B3**) across W bits
[FLT] / [←] [F16] / [→] [F64]	IEEE-754 Float Decode & Encode	—	—	Word or Float X	Decodes **0x41480000** → 12.5; encodes **12.5** → 1,095,237,632 (**0x41480000**)
[←] [CRC]	Cyclic Redundancy Check	—	—	Byte / Word X	Evaluates polynomial CRC remainder (**CRC8**/**CCITT** **0x07** / CRC-32)
[NET]	CIDR Subnet Mask & Usable Hosts	—	—	Prefix P	Pushes netmask 2^{32}-2^{32-P} to Y, and usable hosts 2^{32-P}-2 to X
[→] [BPS]	16× UART Baud Divisor	—	Clock f_{clk}	Target Baud B	$D = \text{round}(f_{\text{clk}} / (16 \cdot B))$
[←] [÷R] / [→] [MOD] / [→] [GCD]	Integer Quotient, Modulo & GCD	—	Dividend Y	Divisor X	Integer quotient & remainder ([←] [÷R]), $Y \bmod X$ ([→] [MOD]), $\gcd(Y, X)$ ([→] [GCD])
[←] [→MHz] [→] [→GHz] [←] [→cyc] [→] [→ns] [←] [→ms] [→] [→µs]	Clock & Timing Conversions	—	—	Frequency / Time X	Converts between Hz, MHz/GHz, ms/µs/ns, and CPU cycles

8.3.1 Byte-Swapping the Telemetry Word, Decoding IEEE-754, and Checking CIDR and CRC

In 32-bit [HEX] mode ([↶] [DEC] **32** [↷] [WSZ] [HEX]), loading the mis-ordered Little-Endian capture **0x00004441** and pressing [↷] [SWP] restores the Big-Endian word **0x41440000**. Pressing [FLT] decodes the true heatsink temperature $+$**12.25°C**, keying **12.5** [FLT] encodes $+12.5$°C to **1095237632** (**0x41480000**), and keying **24** [NET] pushes netmask **4294967040** (**0xFFFFFF00**) to Register Y and **254** usable hosts to Register X:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↶] [DEC] **32** [↷] [WSZ] [HEX] **4441** [↷] [SWP]	0	0	0x20	0x41440000	Byte-swap Little-Endian **0x00004441** into Big-Endian **0x41440000**
[FLT]	0	0	32	12.25	Decode IEEE 754 **binary32** pattern **0x41440000** to $+12.25$
[↶] [DEC] **12.5** [FLT]	0	32	12.25	1095237632	Encode float **12.5** into IEEE 754 bit pattern **1095237632** (**0x41480000**)
24 [NET]	12.25	1095237632	4294967040	**254**	Compute /**24** netmask in Y (**0xFFFFFF00**) and 254 usable hosts in X

8.3.2 Auditing the Sixteen-Times Oversampled UART Baud Divisor on the Stack

Switching to [↶] [DEC] mode, the engineer verifies the 115,200 bps UART divisor on the 48 MHz peripheral bus via **48000000** [ENTER] **115200** [↷] [BPS] ($\rightarrow D =$ **26**) and computes the resulting $+0.1603\%$ baud-rate drift:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↶] [DEC] **48000000** [ENTER] **115200** [↷] [BPS]	0	0	0	26	Compute 16× oversampled UART integer divisor $D = 26$
16 [×] **48000000** [x⇄y] [÷]	0	0	0	115384.6154	Compute actual synthesized baud rate $B_{\text{actual}} = 48{,}000{,}000 \,/\, (16 \times 26)$
115200 [−] **115200** [÷] **100** [×]	0	0	0	**0.1603**	Confirm $+0.1603\%$ UART clock drift (well within $\pm$ 2.0% limit)

8.3.3 Try It on the Stack

Reinforce your command of endianness, floating-point bitfields, CIDR prefixes, and UART clock dividers with these three exercises:

1. **Network Byte Order Swap on an IPv4 Address:** A 32-bit Little-Endian socket buffer holds **0xC0A80164** (**192.168.1.100**). Key [↶] [DEC] **32** [↷] [WSZ] [HEX] [6 OFL] **0** [A] **80164** [↷] [SWP] to reverse the four octets into Big-Endian Network Byte Order, and swap a 16-bit port number **0x1234** in 16-bit mode ([↶] [DEC] **16** [↷] [WSZ] [HEX] **1234** [↷] [SWP]). *(Verified Register X results: 32-bit swap **0x6401A8C0** and 16-bit swap **0x3412**.)*

2. **Decoding IEEE-754 Floats and Fixed-Point Scaling:** Decode the 32-bit IEEE 754 hex words **0x42280000** and **0xC0B00000** via [HEX] <**WORD**> [FLT], and encode the real filter coefficient -0.375 into 16-bit Q_{15} fixed-point format via [↶] [DEC] **0.375**

[+/-] [ENTER] **15** [↶] [Q·n]. *(Verified Register X results:* **0x42280000** → ***+42.00***, **0xC0B00000** → ***-5.50***, *and* Q_{15} = ***-12,288*** *or* **0xD000**.*)*

3. **CIDR Subnet Evaluation and High-Speed UART Divisor:** Evaluate **24** [NET] and **26** [NET] to compare /**24** and /**26** usable host capacities, and on a microcontroller running at f_{clk} = 72,000,000 Hz (72 MHz), find the 16× UART divisor for 115,200 bps via **72000000** [ENTER] **115200** [↷] [BPS]. *(Verified results:* /**24** *yields* ***Y = 4,294,967,040*** *(***0xFFFFFF00***),* ***X = 254 hosts****;* /**26** *yields* ***X = 62 hosts****. And* D_{UART} = ***39***.*)*

Having restored the serial telemetry link and verified the gateway subnet, we complete Unit III in Chapter 9 by synthesizing timer PWM carriers, calibrating the analog-to-digital vibration sensor front end, and aligning DMA memory buffers.

Chapter 9: Partitioning the Factory Floor Subnet and ADC Front End

To expand predictive maintenance across the assembly line, the systems engineering team is deploying a cluster of fifty-four networked vibration-monitoring nodes along the factory floor while driving the actuator's H-bridge power stage with a 1.0 kHz **Pulse-Width Modulation** carrier derived from a 48 MHz crystal oscillator (and a 20.0 kHz ultrasonic servo loop on a 168 MHz high-speed timer bus). Each sensor node samples a piezoelectric accelerometer through a first-order RC anti-aliasing filter feeding a 12-bit Successive-Approximation **Analog-to-Digital Converter** referenced to $V_{\text{ref}} = 3.30$ V. It then packs the quantized samples into 53-byte telemetry descriptors aligned to 16-byte Direct Memory Access (DMA) cache boundaries, and transmits them over an isolated industrial Ethernet VLAN carved from the **192.168.10.0/24** address space around gateway address **192.168.10.200/26**.

Before flashing the node firmware or programming the managed Ethernet switch, the lead systems architect must verify three boundary conditions:

1. Synthesize the 1.0 kHz PWM carrier on the 48 MHz peripheral bus, and solve backward for the 16-bit auto-reload period A needed to run an ultrasonic 20.0 kHz loop on a 168 MHz timer bus.

2. Compute the step size and digital code of the 12-bit ADC across 3.30 V, invert a live DMA sample register code ($D = 2{,}894$) back to its physical sensor voltage, and verify the RC filter time constant ($\tau = RC$) and -3 dB cutoff frequency.

3. Inspect bit weights and power-of-two buffer capacities using Hamming weight and leading-zero count, confirm /**26** usable host capacity, and align 53-byte descriptors and 300-byte ring buffers in memory.

9.1 Hardware Timer Prescalers, Mixed-Signal ADC Quantization, and RC Settling

Bridging crystal oscillator clocks and continuous analog sensor voltages to discrete digital registers relies on integer timer prescaling, uniform 2^N quantization, exponential RC charge settling, and combinatorial bit-counting primitives.

9.1.1 Hardware Timer Prescalers and Pulse-Width Modulation Carrier Synthesis

A hardware timer divides the peripheral bus clock f_{clk} (GO ON, in Hz) through an integer **Prescaler** P (P) to produce a counter tick clock $f_{\text{tick}} = f_{\text{clk}} \,/\, P$, which then counts up to an integer **Auto-Reload Period** ARR (A) to synthesize a **Pulse-Width Modulation** (PWM) carrier of frequency f_{PWM} (F):

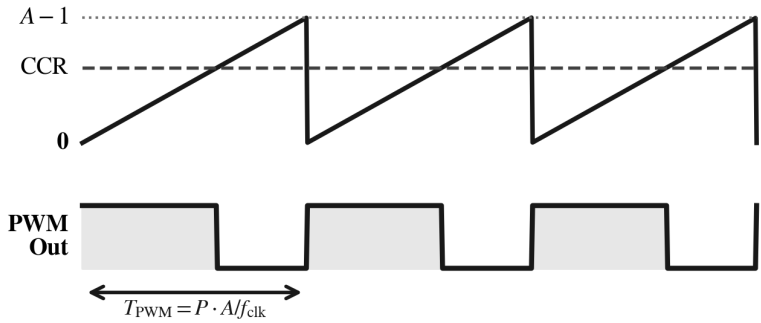


Figure 9.1: *Microcontroller timer up-counter sawtooth waveform, auto-reload period A, compare threshold CCR, and PWM output pulse train.*

$$f_{\text{tick}} = \frac{f_{\text{clk}}}{P}, \qquad f_{\text{PWM}} = \frac{f_{\text{clk}}}{P \cdot \text{ARR}} \Leftrightarrow \text{ARR} = \frac{f_{\text{clk}}}{P \cdot f_{\text{PWM}}}, \qquad \text{Duty}\% = \left(\frac{\text{CCR}}{\text{ARR}}\right) \times 100\% \qquad \textbf{(Eq. 9.1)}$$

On the immediate RPN stack, pressing [PWM] with f_{clk} in Register Z, prescaler P in Register Y, and auto-reload period ARR in Register X computes $f_{\text{PWM}} = f_{\text{clk}} \; / \; (P \times \text{ARR})$ and opens the [PWM] **Equation Solver Workspace** (**F = C / (P × A)**) with variable softkeys [F] ([LFU0]), [C ON] ([LFU1]), [P] ([LFU2]), and [A] ([LFU3]). Driven by a f_{clk} = 48 MHz (48,000,000 Hz) bus clock with prescaler P = 48 and auto-reload period ARR = 1,000, keying **48000000 [ENTER] 48 [ENTER] 1000 [PWM]** generates f_{PWM} = **1,000.0 Hz** (1.0 kHz). Conversely, when upgrading the servo stage to an ultrasonic F = 20,000 Hz (20 kHz) switching frequency on a C = 168,000,000 Hz (168 MHz) timer bus with prescaler P = 4, solving **F = C / (P × A)** backward for [A] via [↷] [SOLVE] [A] yields ARR = 168,000,000 / (4 × 20,000) = **2,100 counts**.

9.1.2 Successive-Approximation Quantization, ENOB, and Half-LSB RC Settling

An N-bit **Analog-to-Digital Converter** ([↶] [ADC]) partitions its full-scale reference voltage V_{ref} into 2^N uniform quantization steps of width V_{LSB} (the Least Significant Bit voltage), mapping a continuous analog input voltage $V_{\text{in}} \in [0,\ V_{\text{ref}})$ to the integer digital code $D \in [0,\ 2^N - 1]$ (while [↷] [DAC] reconstructs V_{out} from a 12-bit code):

$$V_{\text{LSB}} = \frac{V_{\text{ref}}}{2^N}, \qquad D = \left\lfloor \frac{V_{\text{in}}}{V_{\text{ref}}} \cdot 2^N \right\rfloor = \left\lfloor \frac{V_{\text{in}}}{V_{\text{LSB}}} \right\rfloor \qquad \textbf{(Eq. 9.2)}$$

In these quantization equations, N is the converter resolution in bits, 2^N is the total number of discrete code bins (4,096 for a 12-bit converter or 65,536 for a 16-bit converter), and V_{LSB} is the smallest voltage change that increments D by 1. Because ideal uniform quantization introduces a sawtooth error bounded by $\pm(1/2)\ V_{\text{LSB}}$ with root-mean-square noise $V_{q,\text{rms}} = V_{\text{LSB}} \; / \; \sqrt{12}$, a full-scale sine wave of RMS amplitude $V_{\text{ref}} \; / \; (2\sqrt{2})$ achieves a theoretical Signal-to-Noise-and-Distortion ratio $\text{SINAD}_{\text{dB}} = 6.0206\ N + 1.7609$ dB, defining the converter's **Effective Number of Bits**:

$$\text{ENOB} = \frac{\text{SINAD}_{\text{dB}} - 1.7609}{6.0206}, \qquad \tau = R \cdot C, \qquad f_c = \frac{1}{2\pi RC}, \qquad t_{\text{acq}} \geq (N+1)\ln(2)\,\tau \qquad \textbf{(Eq. 9.3)}$$

To prevent front-end charge-injection droop from degrading ENOB, the external RC anti-aliasing and sample-and-hold network ([↶] [τ]) of time constant $\tau = R\ C$ and -3 dB

cutoff frequency $f_c = 1 \,/\, 2\pi\ R\ C$ must settle following a full-scale input step $V_{\text{step}} = V_{\text{ref}}$ until the residual tracking error $V_{\text{err}}(t) = V_{\text{ref}}\ e^{-t/\tau}$ falls below (1/2) LSB $= V_{\text{ref}} \,/\, 2^{N+1}$—requiring at least $(N + 1)\ \ln(2) \approx 9.0109$ time constants for a 12-bit converter. This exponential relaxation law $e^{-t/\tau}$ is mathematically identical to first-order pharmacokinetic drug elimination $C(t) = C0\ e^{-ke\ \ t}$ in **Unit VIII: Clinical Nephrology, Pharmacokinetics & Infusion Critical Care**, where the elimination rate constant k_e plays the exact reciprocal role of the RC time constant $1/\tau$, while the 6.02 dB/bit scaling of ENOB mirrors the logarithmic spectral efficiency $\log_2(1 + \text{SNR})$ of the Shannon–Hartley channel capacity theorem in **Unit V: Radio-Frequency Engineering, Transmission Lines & Link Budgets**. On our vibration node's 12-bit [↰] [ADC] with reference $V_{\text{ref}} = 3.30$ V, the step size is $V_{\text{LSB}} = 3.30 \,/\, 4096 \approx$ **0.80566 mV/LSB** (805.66 µV), a DC bias input of $V_{\text{in}} = 1.65$ V maps to mid-scale code $D = \lfloor\ (1.65 \,/\, 3.30) \times 4096\ \rfloor =$ **2,048** (**0x800**), and a live DMA register sample $D = 2{,}894$ (**0xB4E**) inverts in the [↰] [ADC] workspace (**D = V / R × 2^N**, softkeys [D], [V], [R], [N]) to $V_{\text{in}} = 2894 \times 3.30 \,/\, 4096 =$ **2.3316 V**.

9.1.3 Population Count, Hamming Distance, and Count Leading Zeros

Two unary bit-inspection operators implemented in single-cycle silicon hardware—**Population Count** ([↰] [*B]) and **Count Leading Zeros** ([↱] [CLZ])—analyze the bit distribution of a W-bit word X:

$$\text{POP}(X) = \sum_{i=0}^{W-1} b_i, \qquad d_H(Y,X) = \text{POP}(Y \oplus X), \qquad \lfloor \log_2(X) \rfloor = W - 1 - \text{CLZ}(X) \quad (X > 0) \qquad \textbf{(Eq. 9.4)}$$

Here [↰] [*B] (the **Hamming weight**) sums all **1** bits in X, $\text{POP}(Y \oplus X)$ measures the **Hamming distance** (number of differing bit positions) between words Y and X, and [↱] [CLZ] counts the contiguous **0** bits preceding the highest **1** bit in a W-bit register, directly determining the base-2 logarithm ([1/x]) and the smallest power of two $2^{W\ -\ \text{CLZ}(X\ -\ 1)}$ ([↱] [2^x]) capable of holding a DMA ring buffer of size X. For the 32-bit word **0x41440000** (**0100 0001 0100 0100 0000 0000 0000 0000**), keying [↰] [*B] returns **4** (bits 30, 24, 22, and 18) and keying [↱] [CLZ] returns **1** (bit 31 is **0**).

9.2 Classless Inter-Domain Routing Subnets and Memory Bus Alignment

Partitioning an IPv4 address block into collision-free factory subnets ([NET]) and aligning DMA packet descriptors in RAM ([↱] [ALN]) both apply power-of-two bitmask boundaries.

9.2.1 Prefix Bitmasks, Usable Host Capacity, and Broadcast Boundaries

In **Classless Inter-Domain Routing** ([NET]), a 32-bit IPv4 address written with prefix length [NET] ($P \in [0, 32]$) allocates the upper P bits to the fixed **Network Prefix** (where the subnet mask pushed to Register Y has [↰] [*B] $= P$ leading **1** bits) and leaves $H = 32 - P$ bits for **Host Addresses**:

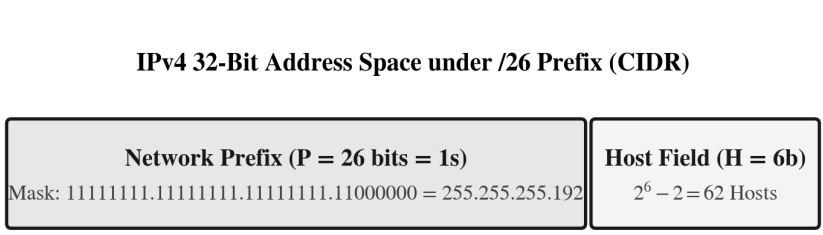


Figure 9.2: *IPv4 32-bit address partitioning under a /26 CIDR prefix into a 26-bit network mask and 6-bit host field.*

$$\text{Block Size} = 2^{32-P}, \qquad N_{\text{hosts}} = 2^{32-P} - 2 \quad (0 \le P \le 30) \tag{Eq. 9.5}$$

Because the all-zeros host pattern (0) is reserved for the **Subnet Network ID** and the all-ones host pattern (2^{32-P} - 1) is reserved for the **Subnet Broadcast Address**, pressing [NET] on a [NET] prefix pushes the 32-bit netmask to Register Y and $N_{\text{hosts}} = 2^{32-P}$ - 2 usable **Host Addresses** to Register X. For our factory floor /**26** subnet (**26** [NET], leaving $H = 6$ host bits and fourth-octet mask 256 - 2^6 = 192, or **255.255.255.192** = 0xFFFFFFC0), the total block size is $2^6 = 64$ addresses and the usable capacity is N_{hosts} = 64 - 2 = **62 usable hosts**—accommodating all 54 vibration nodes and the **.200** gateway with 7 addresses to spare. Because ⌊ 200 / 64 ⌋ × 64 = 3 × 64 = 192, the **192.168.10.200/26** block starts at Network ID **192.168.10.192**, spans usable hosts **192.168.10.193** through **192.168.10.254**, and terminates at Broadcast Address **192.168.10.255**.

9.2.2 Power-of-Two DMA Ring Buffers and Natural Structure Alignment

Inside system memory, CPU load/store units and DMA engines require packet descriptors to begin at byte addresses A divisible by a power-of-two **Alignment Boundary** $B \in$ (2, 4, 8, 16, 32, 64, 4096) ([SHIFT] [ALN]), or round a minimum ring-buffer size S up to the next power of two S_{pow2}, with binary storage keys ([SHIFT] [→B], [SHIFT] [→bit], [SHIFT] [→KiB], [SHIFT] [→MiB], [SHIFT] [→GiB], [SHIFT] [→TiB]) converting between raw bit counts and IEC power-of-two byte capacities (1 KiB = 2^{10} = 1,024 B):

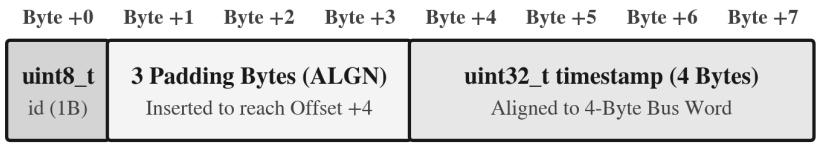


Figure 9.3: *Natural 4-byte memory bus alignment of a C structure showing compiler-inserted padding bytes (ALN).*

$$A_{\text{aligned}} = \left\lceil \frac{A}{B} \right\rceil \times B = (A + B - 1) \text{ AND } \sim(B - 1), \qquad S_{\text{pow2}} = 2^{\lceil \log_2 S \rceil} \quad \textbf{(Eq. 9.6)}$$

Here $(A + B - 1)$ & $\sim (B - 1)$ clears the lower $\log_2 B$ address bits to round A up to the next multiple of B. Aligning our vibration node's raw 53-byte packet descriptor ($A = 53$) up to a $B = 16$-byte DMA burst boundary via **53** [ENTER] **16** [↷] [ALN] yields $A_{\text{aligned}} = \lceil 53 / 16 \rceil \times 16 = 4 \times 16 = \mathbf{64\ bytes}$, while rounding a 300-byte circular sample queue up to the next power of two via **0** [ENTER] **300** [↷] [ALN] yields $S_{\text{pow2}} = 2^9 = \mathbf{512\ bytes}$ (0.5 KiB via [↶] [→KiB]).

9.3 Configuring Timer PWM, the Vibration Node, and Subnet Block on the Stack

The **SC-16** instrument combines immediate RPN stack evaluation for [PWM] carrier synthesis, [↶] [ADC] quantization, [↶] [τ] RC filter timing, bit population counting ([↶] [#B]), leading-zero inspection ([↷] [CLZ]), CIDR usable-host and netmask evaluation ([NET]), DMA memory alignment ([↷] [ALN]), and binary storage scaling ([↶] [→B], [↷] [→bit], [↶] [→KiB], [↷] [→MiB], [↶] [→GiB], [↷] [→TiB]) with the **Bidirectional Equation Solver Workspace** ([↷] [SOLVE] / [XEQ]) on [PWM] (**F = C / (P × A)**), [↶] [ADC] (**D = V / R × 2^N**), and [↶] [τ] (**T = R × C**):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Solver Schema
[PWM]	Timer PWM Frequency & Solver	Clock f_{clk}	Prescaler P	Period ARR	Stack: $f_{\text{PWM}} = f_{\text{clk}} / (P \cdot \text{ARR})$; [PWM] ([F], [C ON], [P], [A])
[↶] [ADC]	ADC Quantization & Code Solver	(Opt. V_{in})	V_{in} or V_{ref}	V_{ref} or Bits N	Stack: code / V_{LSB}; [↶] [ADC]: **D = V / R × 2^N** ([D], [V], [R], [N])
[↶] [τ]	RC Filter Time Constant & Cutoff	—	Resistance R	Capacitance C	Stack: $\tau = RC$ in Y, $f_c = 1/(2\pi RC)$ in X; [↶] [τ]: **T = R × C** ([T], [R], [C ON])
[↶] [#B]	Population Count (Hamming Weight)	—	—	Word X	$\text{POP}(X) = \sum_{i=0}^{W-1} b_i$ (**1** bits)
[↷] [CLZ]	Count Leading Zeros	—	—	Word X	Contiguous **0** bits from MSB b_{W-1} downward
[↷] [2^x] / [lg]	Binary Power & Base-2 Logarithm	—	—	Value X	2^X and $\log_2(X)$
[NET]	CIDR Subnet Mask & Usable Hosts	—	—	Prefix P	Netmask $2^{32}-2^{32-P}$ in Y, usable hosts $2^{32-P} - 2$ in X
[↷] [ALN]	Memory & Power-of-Two Alignment	—	Address A	Boundary B	$\lceil A / B \rceil \times B$ (or $2^{\lceil \log_2 X \rceil}$ if $Y = 0$)
[↶] [→KiB] / [↶] [→B] / [↷] [→bit]	Binary Storage Unit Scaling	—	—	Count X	Convert between bits, bytes, and IEC KiB/MiB/GiB/TiB

Pressing [↷] [▦] on the **SC-16** also opens the non-destructive systems reference table of live multi-radix decompositions ([HEX], [↶] [DEC], [↶] [OCT], [↷] [BIN], IEEE 754 32-bit,

SIGN, **EXP**, **MANT**, **Q16.16**, **ALIGN 4B**, **ALIGN 8B**), ASCII boundaries, and integer ceilings:

Constant Name	Hexadecimal ([HEX])	Decimal ([↶] [DEC])	Systems Engineering Meaning
0x20 / **0x30** / **0x41** / **0x61**	**0x20** / **0x30** / **0x41** / **0x61**	32 / 48 / 65 / 97	ASCII **' '**, **'0'**, **'A'**, and **'a'** byte codes
INT8 / **UINT8**	**0x80..0x7F** / **0xFF**	-128..127 / 255	8-bit signed and unsigned boundaries
INT16 / **UINT16**	**0x8000** / **0xFFFF**	-32,768 / 65,535	16-bit signed and unsigned boundaries
INT32	**0x7FFFFFFF**	2,147,483,647	32-bit signed two's complement maximum

9.3.1 Synthesizing PWM Carrier Frequencies and Quantizing the Vibration Sensor on the Stack

To verify the 1,000.0 Hz PWM carrier (**48000000** [ENTER] **48** [ENTER] **1000** [PWM]), compute the 12-bit vibration sensor [↶] [ADC] step size at $V_{\text{ref}} = 3.30$ V (**3.3** [ENTER] **12** [↶] [ADC] **1000** [×] $\rightarrow$ **0.8057 mV/LSB**), and inspect the set bits ([↶] [*B]) and leading zeros ([↷] [CLZ]) of the 32-bit word **0x41440000**, the architect executes:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↶] [DEC] **48000000** [ENTER] **48** [ENTER] **1000** [PWM] [C ON]	0	0	0	1000	3-arg stack shortcut: $f_{\text{PWM}} = 48{,}000{,}000 / (48 \times 1000) = 1{,}000$ Hz
3.3 [ENTER] **12** [↶] [ADC] [C ON] **1000** [×]	0	0	1000	0.8057	Compute 12-bit step size $V_{\text{LSB}} = (3.3 / 4096) \times 1000 \approx 0.8057$ mV
1650 [x⇄y] [÷]	0	0	1000	2048	Divide 1,650 mV input by V_{LSB} to obtain mid-scale code $D = 2{,}048$
32 [↷] [WSZ] [HEX] **41440000** [ENTER] [↶] [*B]	2048	32	0x41440000	0x4	Count set **1** bits in **0x41440000** ([↶] [*B] = 4)
[x⇄y] [↷] [CLZ]	2048	32	0x4	**0x1**	Count leading zeros of **0x41440000** in 32-bit register ([↷] [CLZ] = 1)

9.3.2 Solving Backward in the [PWM], [↶] [ADC], and [↶] [τ] Workspaces

When configuring the 20 kHz ultrasonic servo timer on a 168 MHz bus ([PWM] workspace **F = C / (P × A)**), inverting a raw 12-bit DMA accelerometer code $D = 2{,}894$ ([↶] [ADC] workspace **D = V / R × 2^N**, pre-populated with $R = 3.3$ V and $N = 12$), or sizing the anti-aliasing series resistor R driving a $C = 1.0$ nF (10^{-9} F) capacitor for $\tau = 1.0$ µs ([↶] [τ] workspace **T = R × C**), the architect solves directly for the unknown parameter:

Keystroke	Solver Display / Register X	What Happens in the [PWM], [↶] [ADC], & [↶] [τ] Workspaces
[↶] [DEC] **0** [ENTER] **0** [PWM]	**F = C / (P × A)**	Open the [PWM] Solver Workspace (**PWM: TIMER FREQ**)
168000000 [C ON] **4** [P] **20000** [F]	20,000.0000	Bind $C = 168$ MHz ([LFU1]), $P = 4$ ([LFU2]), and $F = 20$ kHz ([LFU0])
[↷] [SOLVE] [A]	**2,100.0000**	Solve backward for 16-bit auto-reload period $A = 168{,}000{,}000 / (4 \times 20{,}000) =$ **2,100**
0 [ENTER] **0** [↶] [ADC]	**D = V / R × 2^N**	Open the [↶] [ADC] Solver Workspace (pre-populated with $R = 3.3$ V, $N = 12$)

Keystroke	Solver Display / Register X	What Happens in the PWM, ↶ ADC, & ↶ τ Workspaces
2894 D	2,894.0000	Store the captured DMA sample register code $D = 2{,}894$ into softkey D (LFU0)
↷ SOLVE V	**2.3316**	Solve backward for analog input voltage $V = 2894 \times 3.3 \,/\, 2^{12} =$ **2.3316 V**
0 ENTER **0** ↶ τ	**T = R × C**	Open the RC Time Constant (↶ τ) Solver Workspace with softkeys T, R, C ON
1 E **6** +/− T **1** E **9** +/− C ON	1.0000×10^{-9}	Store target time constant $T = 1.0$ µs into T and $C = 1.0$ nF into C ON
↷ SOLVE R	**1,000.0000**	Solve backward for anti-aliasing series resistance $R = T \,/\, C =$ **1,000 Ω** (1.0 kΩ)

9.3.3 Partitioning the Factory Subnet and Aligning DMA Packet Descriptors

In ↶ DEC mode, keying **26** NET (or **32** ENTER **26** - ↷ 2^x **2** -) confirms that the /**26** factory floor subnet provides **62** usable IPv4 host addresses (with netmask **4294967232** = 0xFFFFFFC0 in Register *Y*), keying **53** ENTER **16** ↷ ALN aligns the 53-byte telemetry header up to a **64-byte** DMA boundary, and keying **0** ENTER **300** ↷ ALN rounds the 300-byte sample queue up to the next power of two at **512 bytes** (**0.5 KiB** via ↶ →KiB):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
↶ DEC **26** NET	0	0	4294967232	62	Evaluate /**26** netmask in *Y* (**0xFFFFFFC0**) and 62 usable IPv4 hosts in *X*
22 NET	4294967232	62	4294966272	1022	Evaluate plant-wide /**22** supernet netmask in *Y* and 1,022 usable hosts in *X*
53 ENTER **16** ↷ ALN	4294967232	4294966272	1022	64	Align 53-byte packet descriptor up to 16-byte boundary (64 bytes)
0 ENTER **300** ↷ ALN	1022	64	0	**512**	Round 300-byte DMA ring request up to next power of two (512 bytes)

9.3.4 Try It on the Stack

Complete Unit III with these three mixed-signal, networking, and memory-alignment exercises:

1. **Precision Sixteen-Bit ADC Step Resolution and Backward Voltage Inversion:** A precision strain-gauge bridge uses a 16-bit ↶ ADC ($2^{16} = 65{,}536$) with a $V_{\text{ref}} = 2.50$ V reference. Key ↶ DEC **2.5** ENTER **16** ↶ ADC C ON **1000000** × on the stack to find the step size V_{LSB} in microvolts (µV), then open ↶ ADC (**0** ENTER **0** ↶ ADC), store **2.5** R, **16** N, and **49152** D (**0xC000**), and press ↷ SOLVE V to invert the code back to volts. *(Verified Register X results: $V_{LSB} \approx$ **38.1470 µV** and $V =$ **1.8750 V**.)*

2. **Subnet Capacity and Broadcast Boundary Check:** Compute the usable host count and netmask (NET) for a /**25** subnet (**25** NET) and a /**28** equipment management subnet (**28** NET), and determine the block size (2^{32-28} via **4** ↷ 2^x) of the /**28** subnet. *(Verified Register X results: /**25** yields **126 hosts**; /**28** yields **14 hosts** with a block size of **16 addresses**.)*

3. **Cache-Line Descriptor Alignment and Power-of-Two Sizing:** Align a 108-byte DMA descriptor to a 64-byte cache line (**108** [ENTER] **64** [g] [ALN]), align a 210-byte buffer to a 32-byte boundary (**210** [ENTER] **32** [g] [ALN]), and round a 1,200-byte FIFO request up to the next power of two (**0** [ENTER] **1200** [g] [ALN]) and convert it to KiB ([↖] [→KiB]). *(Verified Register X results:* ***128 bytes****,* ***224 bytes****, and* ***2,048 bytes*** *=* ***2.0000 KiB****.)*

Having traced how silicon registers, bitfields, and crystal oscillators process digital words on the **SC-16**, we move in **Unit IV: Cinema Optics, Broadcast Timecode & Live Acoustics** to optical wavefronts, SMPTE drop-frame timecode, and acoustic decibels.

Unit IV: Cinema Optics, Broadcast Timecode & Live Acoustics

On June 15, 1878, at Leland Stanford's Palo Alto Stock Farm in California, the English photographer Eadweard Muybridge lined a whitewashed wooden shed with twenty-four wet-plate stereoscopic cameras spaced twenty-one inches apart. Across the dirt track, fine linen threads stretched from each brass lens shutter to an electrical contact switch. When the trotting mare Sallie Gardner dashed through the threads at thirty-six feet per second, she tripped the electromagnetic shutters in rapid succession. Each camera froze a discrete slice of motion at an exposure shorter than one two-thousandth of a second. For millennia, painters had depicted galloping horses with both forelegs extended forward and both hind legs stretched backward like a rocking horse. Muybridge's silver-gelatin silhouettes proved that when all four hooves leave the ground simultaneously, they gather beneath the horse's belly. Projected onto a screen through his spinning glass-disk zoopraxiscope, those discrete static photographs fused in the viewer's visual cortex into continuous, living motion. Half a century later, optical sound-on-film tracks required a constant linear transport velocity past an exciter lamp. Motion-picture studios responded by standardizing theatrical projection at twenty-four frames per second, establishing the foundational temporal quantum of cinema.

While cinema remained anchored to integer mechanical sprockets, the birth of color television in 1953 forced broadcast engineers on the National Television System Committee (NTSC) into a subtle frequency compromise. Since the 1940s, North American monochrome television had scanned thirty frames per second, locked to the 60 Hz alternating-current utility grid so power-supply hum bars would remain stationary on cathode-ray tubes. Squeezing a 3.579545 MHz chrominance subcarrier beside the 4.5 MHz frequency-modulated audio carrier threatened to create beat interference on millions of existing black-and-white sets. To prevent that interference, engineers reduced the horizontal line rate and frame rate by one part in a thousand (1000/1001). Slowing the video frame rate to $30/1.001 \approx 29.97003$ fps caused a thirty-frame-per-second counter to fall 3.6 seconds (108 frames) behind atomic wall-clock time every hour. That drift motivated the Society of Motion Picture and Television Engineers (SMPTE) in 1967 to invent mixed-radix timecode and Drop-Frame label reconciliation.

Capturing and projecting a story across space demands an equally tight coordination between refractive optics and human sensory physiology. While the visual cortex forgives discrete temporal sampling at twenty-four frames per second, its spatial acuity and brightness adaptation operate over delicate geometric and logarithmic scales. Light rays refracted through a rectilinear cinema prime project a three-dimensional scene onto a flat Super 35 or Full-Frame silicon sensor across an angular field of view. Meanwhile, object points outside the plane of focus spread into blur disks whose permissible diameter—the Circle of Confusion (C)—sets the hyperfocal distance and conjugate depth-of-field limits. Simultaneously, camera assistants and lighting gaffers balance iris f-stops, rotary shutter angles, neutral-density glass, and blackbody color temperatures along base-two Exposure Value stops and micro-reciprocal degrees. A striking mathematical isomorphism unites photometry, acoustics, and spectroscopy. A neutral-density filter of optical density $D = 0.3010 \times \text{stops} = -\log_{10}(T_{\text{trans}})$ obeys the same base-ten exponential attenuation law as Beer–Lambert UV-Vis absorbance $A = \varepsilon\, b\, c = -\log_{10}(I/I_0)$ in wet-lab chemistry (Unit VI) and free-space path loss in radio-frequency link budgets (Unit V).

Human hearing reverses the visual perceptual hierarchy, trading coarse angular localization for millisecond temporal discrimination. Between 1895 and 1898, Harvard physicist Wallace Clement Sabine worked through quiet midnight hours hauling hundreds of Sanders Theatre seat cushions into the reverberant Fogg Lecture Hall. By timing the decay of a 512 Hz organ pipe with a chronograph, he formulated the architectural reverberation law. Fifty years later, Helmut Haas demonstrated the precedence effect in 1949. When two identical sound waves arrive at a listener's ears within a window of 1 ms to 30 ms, the auditory cortex fuses them into a single sonic image anchored to the first-arriving wavefront.

The **SC-24 Cinema, Broadcast, Optics & Live Acoustics** calculator unites these temporal, optical, and acoustic laws for post-production editors, camera assistants, gaffers, and front-of-house sound engineers. Each chapter in this Unit follows a single production story through three related mathematical themes—building the physical concepts and worked examples first, and then executing them on the SC-24 stack and in its Equation Solver workspaces:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 10	Reconciling a live multi-reel documentary broadcast with a +3.6 s/hr NTSC drift and a 120 fps slow-motion wildlife insert	Mixed-radix **HH:MM:SS:FF** arithmetic, 1000/1001 NTSC pulldown, SMPTE drop-frame rules, and overcrank speed ramps	1. Mixed-Radix Timecode & Frame Accounting • 2. High-Speed Overcrank & 3:2 Telecine Cadence • 3. NTSC 1000/1001 Drift & Drop-Frame Reconciliation
Chapter 11	Holding focus on a Super 35 Steadicam close-up at dusk while balancing shutter angle, ND filtration, and Mired color gels	Rectilinear field of view, Circle of Confusion, hyperfocal conjugates, base-2 Exposure Value, Mired shifts, and inverse-square lux	1. Rectilinear Framing & Angular Field of View • 2. Hyperfocal Distance & Depth-of-Field Conjugates • 3. Base-2 Exposure Reciprocity, ND Stops & Mired Gels
Chapter 12	Time-aligning a 45 m stadium delay tower at 25°C and mapping SPL decay, boom-mic reach, and incoherent PA summation	Thermal speed of sound, Haas precedence delay, acoustic wavelength, inverse-square decibel loss, incoherent power summation, and Sabine RT_{60}	1. Thermal Speed of Sound & Haas Delay Alignment • 2. Inverse-Square SPL Decay & Boom-Mic Reach • 3. Incoherent Decibel Summation & Sabine Reverberation

Chapter 10: The Missing Three-Point-Six Seconds at the Broadcast Desk

Twenty minutes before a live coast-to-coast documentary premiere goes to air in New York, the master-control transmission producer notices a discrepancy between the edit bay sheet and the network's rubidium-locked studio clock. Two documentary reels cut on a non-linear timeline—Reel A running **01:14:28:12** and Reel B running **00:52:45:20** at a nominal 30 fps—must be joined end-to-end with a high-speed wildlife B-roll sequence shot at 120 fps over 180 captured frames and conformed to theatrical 24 fps playback before the live satellite commercial break triggers at the top of the hour. Yet the archival field camera that recorded the primary interview reel had its timecode generator set to 29.97 fps Non-Drop-Frame (NDF) rather than broadcast Drop-Frame.

On a live network feed where satellite transponders and automated local-affiliate commercial inserters switch on the millisecond, a mismatch between nominal frame labels and physical wall-clock seconds is fatal. When a 29.97 fps video stream is indexed by a counter that requires thirty full frames to advance its seconds digit, every nominal hour of **01:00:00:00** on the timecode display actually consumes 3603.6 seconds of real earth rotation. By the end of a two-hour documentary, an uncompensated Non-Drop-Frame timeline runs more than seven seconds past the top-of-the-hour network break, clipping the closing credits in mid-sentence across two hundred affiliate stations.

To rescue the broadcast rundown before the countdown reaches zero, the post-production engineer works through three coupled temporal calculations:

1. Sum the two main reel addresses across mixed sexagesimal and frame radices, and convert between packed timecode (**HH.MMSSFF**), total integer frames, and decimal seconds.
2. Scale the 120 fps overcranked wildlife insert to its true 24 fps screen duration and exposure compensation.
3. Scale the nominal program runtime through the 1000/1001 NTSC pulldown ratio, reconcile SMPTE's minute-boundary Drop-Frame counting rule, and solve backward for the exact frame budget allowed by the network's commercial window.

10.1 Mixed-Radix Frame Indexing and Overcranked Time Scaling

Every motion-picture camera and broadcast deck records time as a discrete sequence of still images. Translating between the human language of hours, minutes, and seconds and the machine language of integer frame pulses requires a mixed-radix coordinate system coupled with linear rate scaling.

10.1.1 Sexagesimal Timecode Rollovers and Integer Frame Conversion

To give every individual frame on a reel a unique, human-readable address across cameras, dual-system field audio recorders, and editing timelines, the Society of Motion Picture and Television Engineers standardized an eight-digit mixed-radix index known as **Timecode**:

$$\mathbf{HH:MM:SS:FF} \quad \text{(Hours : Minutes : Seconds : Frames)} \tag{Eq. 10.1}$$

In this four-field representation, the first three fields obey sexagesimal Babylonian clock arithmetic—sixty seconds per minute (SS ∈ [00, 59]) and sixty minutes per hour (MM ∈ [00, 59])—whereas the rightmost frame field (FF) counts modulo the nominal integer frame rate R, cycling from 00 up to R - 1 (such as 00–23 for cinema 24 fps, 00–24 for European PAL/SECAM 25 fps, or 00–29 for North American 30 fps video). On the **SC-24**, a timecode address is entered as a packed decimal literal **HH.MMSSFF** (so **01:14:28:12** is keyed as **1.142812**). Any timecode address converts to a cumulative integer frame count F ([↶] [→FRM], using the frame rate staged in Register Y or the cinema default R = 24 fps) and back to packed timecode ([↷] [→TC]) or decimal elapsed seconds S ([TC]) by expanding the sexagesimal fields into total seconds and multiplying or dividing by the framebase R:

$$S = 3600 \cdot \mathrm{HH} + 60 \cdot \mathrm{MM} + \mathrm{SS} + \frac{\mathrm{FF}}{R}, \qquad F = S \times R = (3600 \cdot \mathrm{HH} + 60 \cdot \mathrm{MM} + \mathrm{SS})R + \mathrm{FF} \qquad \textbf{(Eq. 10.2)}$$

Here S is the elapsed duration in seconds (s), R is the frame rate in frames per second (fps), and F is the exact count of frames since **00:00:00:00**. When adding ([↷] [+TC]) or subtracting ([↷] [-TC]) two packed timecode durations A and B, arithmetic proceeds from right to left across the four radices, carrying +1 second whenever the frame sum reaches or exceeds R, carrying +1 minute whenever seconds reach 60, and carrying +1 hour whenever minutes reach 60.

Consider the two documentary reels sitting on the broadcast desk at a nominal framebase of R = 30 fps: Reel A spans **01:14:28:12** (**1.142812**) and Reel B spans **00:52:45:20** (**0.524520**). Summing the rightmost frame fields gives 12 + 20 = 32 frames. Dividing by R = 30 leaves a remainder of **:02** frames and carries +1 s into the seconds column. Summing the seconds column gives 28 + 45 + 1 = 74 s, which leaves **:14** seconds and carries +1 min. Summing the minutes column gives 14 + 52 + 1 = 67 min, which leaves **:07** minutes and carries +1 hr. Summing the hours column gives 01 + 00 + 1 = 02 hr, yielding a combined master reel address of **02:07:14:02** (**2.071402**). Expanded into total seconds and frames at R = 30 fps, Reel A contains 3600(1) + 60(14) + 28 + 12/30 = 4468.40 s (134,052 frames) and Reel B contains 60(52) + 45 + 20/30 = 3165.667 s (94,970 frames), whose integer sum is 229,022 frames, or 7634.067 nominal seconds.

10.1.2 Variable Frame Rate Overcranking and Speed Ramp Ratios

In early hand-cranked silent cameras, an operator who turned the crank faster than normal captured more frames per second than the theater projector later replayed, stretching brief physical events into slow motion on screen—a craft term preserved today as **overcranking**. Conversely, cranking slower than the projection rate compressed long events into fast-motion time-lapse, known as **undercranking**. When a camera captures a scene at acquisition frame rate R_{shoot} over a recording duration t_{shoot} (collecting $F_{\text{total}} = R_{\text{shoot}} \cdot t_{\text{shoot}}$ total frames) and the editing timeline plays those frames back one-for-one at project base rate R_{play}, conservation of frame count F_{total} governs the screen playback duration t_{play}, the time-stretch magnification factor M_{ramp}, and the base-two exposure stop loss $\Delta\text{EV}_{\text{fps}}$ ([FPS]):

$$F_{\text{total}} = R_{\text{shoot}} \cdot t_{\text{shoot}} = R_{\text{play}} \cdot t_{\text{play}}, \qquad M_{\text{ramp}} = \frac{R_{\text{shoot}}}{R_{\text{play}}}, \qquad \Delta\text{EV}_{\text{fps}} = \log_2\left(\frac{R_{\text{shoot}}}{R_{\text{play}}}\right) \qquad \textbf{(Eq. 10.3)}$$

In these relations, R_{play} is the base delivery timeline frame rate staged in Register Y (such as 24 fps), R_{shoot} is the camera sensor acquisition rate staged in Register X (such as 60 fps or 120 fps), M_{ramp} is the dimensionless slow-motion playback factor deposited in Register X by [FPS], and $\Delta\text{EV}_{\text{fps}}$ is the exposure stop loss deposited in Register Y by [FPS] (because raising the frame rate at a fixed rotary shutter angle shortens each frame's integration time proportionally).

For the documentary's high-speed wildlife B-roll shot—where a peregrine falcon strike was captured at $R_{\text{shoot}} = 120$ fps for $t_{\text{shoot}} = 1.50$ s, yielding $F_{\text{total}} = 120 \times 1.50 = 180$ frames—conforming those 180 frames onto a theatrical $R_{\text{play}} = 24$ fps timeline (**24** [ENTER] **120** [FPS]) stretches the time axis by $M_{\text{ramp}} = 120 / 24 = \mathbf{5.0}\times$ in Register X (with an overcrank exposure loss of $\log_2(5) \approx \mathbf{2.3219}$ **EV** in Register Y, or $60 / 24 = \mathbf{2.5}\times$ and $\log_2(2.5) \approx \mathbf{1.3219}$ **EV** for a 60 fps shot), producing a screen playback duration of $t_{\text{play}} = 180 / 24 = \mathbf{7.50}$ **s** (**00:00:07:12**, or **0.000712** at 24 fps). If that same 180-frame burst is dropped directly onto the network's 30 fps broadcast master timeline, the time magnification becomes $M_{\text{ramp}} = 120 / 30 = \mathbf{4.0}\times$ and the insert runs for $t_{\text{play}} = 180 / 30 = \mathbf{6.00}$ **s** (**00:00:06:00**).

10.2 NTSC Subcarrier Pulldown and Drop-Frame Reconciliation

When a broadcast facility operates on North American fractional frame rates (23.976 fps, 29.97 fps, or 59.94 fps), the rate at which frames pass through the switcher no longer equals a whole integer per second. Reconciling integer frame labels with continuous atomic time requires tracking the 1000/1001 frequency offset ([↷] [DF]) and applying SMPTE's deterministic label-skipping algorithm.

10.2.1 The 1953 Color Subcarrier Offset and Wall-Clock Drift

When the NTSC added color to black-and-white television in 1953, engineers needed to interleave a high-frequency color burst subcarrier (f_{sc}) inside the existing 6 MHz channel without causing audible buzz in the 4.5 MHz sound carrier or visible dot crawl on monochrome screens. Choosing the sound carrier to equal 286 times the horizontal line rate f_H required pulling the horizontal scan rate down by a factor of 1000/1001 (from 15,750 Hz to 15,734.2657 Hz), which pulled the vertical field rate down from 60 Hz to $60/1.001 \approx 59.94$ Hz and the frame rate down from 30 fps to R_{NTSC}:

$$R_{\text{NTSC}} = \frac{30}{1.001} = \frac{30{,}000}{1001} \approx 29.97002997 \text{ fps}, \qquad t_{\text{real}} = t_{\text{nominal}} \times 1.001, \qquad \Delta t_{\text{drift}} = t_{\text{nominal}} \times 0.001 \qquad \textbf{(Eq. 10.4)}$$

In this pulldown relationship, $t_{\text{nominal}} = F / 30$ is the apparent duration indicated by a Non-Drop-Frame (NDF) counter that increments one second every 30 frames, $t_{\text{real}} = F / R_{\text{NTSC}}$ is the true wall-clock elapsed time in seconds ([↷] [DF]), and $\Delta t_{\text{drift}} = t_{\text{real}} - t_{\text{nominal}}$ is the cumulative clock error. Similarly, when theatrical 24 fps cinema masters are transferred to NTSC video via 2:3 telecine pulldown ([↷] [3:2]), the cinema frame rate is pulled down by 1000/1001 to $24 / 1.001 \approx 23.976024$ fps.

Because each physical NTSC video frame lasts $1.001 / 30 \approx 33.3667$ ms instead of 33.3333 ms, every 1000 seconds of nominal Non-Drop-Frame timecode actually takes 1001 seconds of wall-clock time to play out. Over a nominal one-hour program ($t_{\text{nominal}} = 3600$ s, or $F = 3600 \times 30 = 108{,}000$ frames), the true elapsed wall-clock duration is $t_{\text{real}} = 3600 \times 1.001 = \mathbf{3603.6}$ **s**. When the Non-Drop-Frame studio monitor reads **01:00:00:00**, the actual broadcast has run $\Delta t_{\text{drift}} = +3.6$ seconds ($3.6 \text{ s} \times 30 \text{ fps} = \mathbf{108}$ **frames**) past the top of the hour. Across our documentary's two-reel nominal runtime of $t_{\text{nominal}} = 7634.067$ s (**02:07:14:02**), playing those 229,022 frames at 29.97003 fps consumes $t_{\text{real}} = 7634.067 \times 1.001 = \mathbf{7641.701}$ **s** of real time—an accumulated drift of +7.634 seconds (229 frames).

10.2.2 The SMPTE Drop-Frame Counting Rule Across Minute Boundaries

To force the eight-digit timecode display to match wall-clock time at 29.97 fps without cutting or deleting a single recorded image from the video stream, SMPTE invented **Drop-Frame** (DF) timecode. Because an NDF counter accumulates +108 extra frames of lag every hour (60 minutes), the counter must skip ahead by 108 numerical frame labels per hour, or 108 / 60 = 1.8 frame labels per minute. Since a timecode display can only skip whole integer labels, skipping 2 frame labels at the start of every minute would drop 2 × 60 = 120 labels per hour—overcorrecting by 12 labels. Dropping 2 labels on nine out of every ten minutes removes 2 × 9 = 18 labels every ten minutes, or 18 × 6 = 108 labels per hour:

$$N_{\text{dropped}} = 2 \times \left(M_{\text{total}} - \left\lfloor \frac{M_{\text{total}}}{10} \right\rfloor \right), \qquad F_{\text{DF}} = 107{,}892 \cdot \text{HH} + 1798 \cdot \text{MM} + 2\left\lfloor \frac{\text{MM}}{10} \right\rfloor + 30 \cdot \text{SS} + \text{FF} \qquad \textbf{(Eq. 10.5)}$$

Here $M_{\text{total}} = 60 \cdot \text{HH} + \text{MM}$ is the total number of elapsed minute boundaries, N_{dropped} is the cumulative count of skipped frame numbers (labels **:00** and **:01** omitted at the start of every minute except minutes **00**, **10**, **20**, **30**, **40**, and **50**), and F_{DF} is the exact physical frame index corresponding to a Drop-Frame address **HH:MM:SS;FF** (written by convention with a semicolon before the frame field).

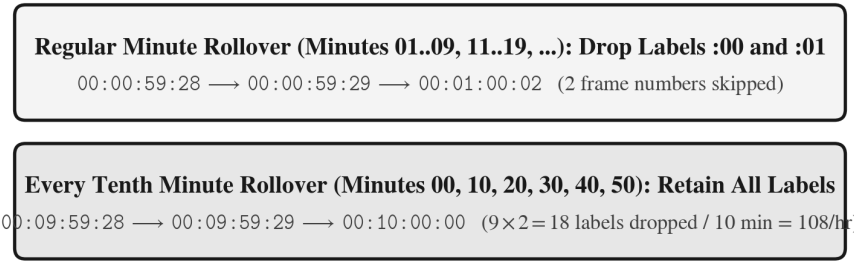


Figure 10.1: *SMPTE 29.97 drop-frame timecode minute-boundary label reconciliation across regular minutes and every tenth minute.*

At the end of the first minute, a Drop-Frame counter advances directly from **00:00:59;29** to **00:01:00;02**, skipping the numbers **:00** and **:01** while every recorded video frame remains intact. At the tenth minute boundary, it counts normally from **00:09:59;29** to **00:10:00;00**. Over one full hour (**01:00:00;00**), $M_{\text{total}} = 60$ and $N_{\text{dropped}} = 2(60 - 6) = 108$ dropped labels, so **01:00:00;00** corresponds to physical frame $F_{\text{DF}} = 108{,}000 - 108 =$ **107,892 frames**. At the true NTSC rate of 30,000 / 1001 fps, those 107,892 frames occupy 107,892 × 1.001 / 30 = **3599.9964 s**, locking the studio timecode display to atomic time within 3.6 milliseconds (0.1 frame) per hour.

10.3 Synchronizing Broadcast Timelines on the Stack and in the Solver Workspace

On the **SC-24**, packed **HH.MMSSFF** timecode arithmetic ([↗] [+TC], [↗] [-TC]), immediate frame-to-timecode conversions ([↖] [→FRM], [↗] [→TC]), overcrank playback and exposure scaling ([FPS]), Drop-Frame drift scaling ([↗] [DF]), and the bidirectional [TC] Equation Solver Workspace ($\mathbf{F} = \mathbf{S} \times \mathbf{R}$, equivalently $\mathbf{S} = \mathbf{F} \,/\, \mathbf{R}$) work together across any configured stack depth ([4-LV], [8-LV], or [INF] via [↗] [FLAGS]) while logging cue points to the **@CUES** edit decision list in [↗] [▦].

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU2])	Governing Expression
[FPS]	Immediate Stack	Register Y: Base rate R_{play}, Register X: Capture rate R_{shoot}	$X = R_{\text{shoot}} \,/\, R_{\text{play}}$ (playback factor), $Y = \log_2(R_{\text{shoot}} \,/\, R_{\text{play}})$ (EV stop loss)
[↗] [+TC]	Immediate Stack	Register Y: **HH.MMSSFF** A, Register X: **HH.MMSSFF** B	Mixed-radix sum $A + B$ (logs cue to **@CUES** in [↗] [▦])
[↗] [-TC]	Immediate Stack	Register Y: **HH.MMSSFF** A, Register X: **HH.MMSSFF** B	Mixed-radix difference A - B with sexagesimal/frame borrow
[↖] [→FRM]	Immediate Stack	Register Y: Rate R (or 24 fps default), Register X: **HH.MMSSFF**	Expands **HH.MMSSFF** into integer frame count F
[↗] [→TC]	Immediate Stack	Register Y: Rate R (or 24 fps default), Register X: Frame count F	Packs integer frame count F into **HH.MMSSFF**
[↗] [DF]	Immediate Stack	Register X: Nominal NDF duration t_{nom}	Scales by 1.001 NTSC pulldown: $t_{\text{real}} = 1.001 \cdot X$
[TC]	Stack / Solver	Softkeys: [F] Frames, [S] Seconds, [R] Rate	Bidirectional solver for $F = S \times R$ ($S = F \,/\, R$; default $R = 24$)

10.3.1 Immediate Stack Trace: Packed Timecode Rollovers, Overcrank Scaling, and NTSC Drift

At the broadcast desk, the producer keys Reel A (**1.142812** for **01:14:28:12**) and Reel B (**0.524520** for **00:52:45:20**), sums them across mixed radices with [↗] [+TC], expands the combined **HH.MMSSFF** address into integer frames at 30 fps with [↖] [→FRM], evaluates the 1.001 NTSC wall-clock overrun with [↗] [DF], and checks a 60 fps overcranked B-roll clip against 24 fps playback with [FPS]:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1.142812 [ENTER] **0.524520** [↗] [+TC]	0	0	0	2.071402	Sum Reel A and Reel B across mixed radices to **02:07:14:02** (**2.071402** at 30 fps)
30 [x⇄y] [↖] [→FRM]	0	0	0	229022	Expand **02:07:14:02** at $R = 30$ fps into $F = 229{,}022$ total frames
30 [÷]	0	0	0	7634.0667	Convert frame total into nominal NDF seconds $t_{\text{nom}} = 7634.067$ s
[↗] [DF]	0	0	0	7641.7007	Scale by 1.001 NTSC pulldown: true wall-clock runtime $t_{\text{real}} = 7641.701$ s
24 [ENTER] **60** [FPS]	0	7641.7007	1.3219	**2.5000**	Compute $60/24 = 2.5\times$ slow-motion factor in X and 1.3219 EV stop loss in Y

Subtracting $t_{\text{nom}} = 7634.067$ s from $t_{\text{real}} = 7641.701$ s reveals the +7.634 s (229-frame) Non-Drop-Frame overrun, which the editor converts directly into a packed 30 fps trim duration via **30** [ENTER] **229** [↗] [→TC] → **0.000719** (**00:00:07:19**).

10.3.2 Bidirectional Solver Workspace ([TC]): Solving Backward for Frame Budgets

Suppose the network's automated satellite switcher allocates a strict broadcast act window of $S = 1650.0$ seconds (27 minutes 30 seconds of real atomic time), and the editor needs to know the exact total frame budget F available when delivering at fractional NTSC $R = 29.97003$ fps versus cinema $R = 23.976024$ fps, and then solve forward for the playback duration S of the $F = 180$-frame falcon slow-motion insert at $R = 24$ fps. Pressing [TC] opens the **TC: TIMECODE** Equation Workspace on Line 1 (**F = S × R**, equivalently **S = F / R**) and binds softkeys [LFU0]–[LFU2] to [F], [S], and [R]:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU2])	Register X Result & Physical Meaning
[TC] (open workspace)	**TC: F = S × R**	[F] [S] [R]=24	Opens [TC] solver with cinema default R = 24.0 fps pre-bound
1650 → [S]	**TC: F = S × R**	[F] [S]=1650 [R]=24	Stores broadcast act duration S = 1650.0 s
29.97003 → [R]	**TC: F = S × R**	[F] [S]=1650 [R]=29.97	Stores fractional NTSC broadcast frame rate R = 29.97003 fps
[XEQ] → [F]	**F = 49450.55**	[F]=49451 [S]=1650 [R]=29.97	Solves for exact NTSC act capacity F = **49,450.5 frames** (**00:27:30;00** DF)
23.976024 → [R], [XEQ] → [F]	**F = 39560.44**	[F]=39560 [S]=1650 [R]=23.98	Re-solves at 23.976 fps for cinema pulldown budget F = **39,560.4 frames**
180 → [F], **24** → [R], [XEQ] → [S]	**S = 7.5000**	[F]=180 [S]=7.5 [R]=24	Solves backward for 24 fps playback duration S = 180 / 24 = **7.50 s**

Because [S], [R], and [F] remain bound in the workspace, changing a single delivery specification—such as switching [R] from **29.97003** to **23.976024** or **25** for European PAL delivery—recomputes the frame budget in two keystrokes without re-entering the segment duration.

10.3.3 Try It on the Stack

Practice mixed-radix timecode rollovers, NTSC drop-frame reconciliation, and bidirectional [TC] solver workflows with these three broadcast desk exercises:

1. **Two-Hour Feature NTSC Drift in Seconds and Packed Timecode:** A feature film transfer runs for a nominal 2 hours (**2.000000** = 7200 s) on a 30 fps Non-Drop-Frame timeline. Compute the integer frame count (**30** [ENTER] **2.000000** [↶] [→FRM]), the true wall-clock duration at 29.97 fps (**30** [÷] [↷] [DF]), and the accumulated drift in frames (**7200** [-] **30** [×]). *(Verified Register X results: F = **216,000 frames**, t_{real} = **7207.2 s**, and drift ΔF = **216 frames** = **0.000706** via* **30** [ENTER] **216** [↷] [→TC]*.)*

2. **Commercial Act Trim via Packed Timecode Subtraction:** An act must fit inside a **00:22:30:00** (**0.223000**) broadcast window at 30 fps, but the current cut runs **00:24:12:18** (**0.241218**). Find the required trim as packed timecode and total frames (**0.241218** [ENTER] **0.223000** [↷] [-TC] **30** [x⇄y] [↶] [→FRM]). *(Verified Register X results: packed trim* **0.014218** *(**00:01:42:18**) and ΔF = **3078 frames**.)*

3. **High-Speed Phantom Overcrank at 240 fps in the [TC] Solver:** A tabletop liquid-pour commercial is shot at $R_{\text{shoot}} = 240$ fps for $S = 0.85$ s and played back at $R_{\text{play}} = 23.976$ fps. In the [TC] Solver Workspace, store **0.85** in [S] and **240** in [R], solve via [XEQ] → [F], then store **23.976** in [R] and solve via [XEQ] → [S]. *(Verified results: F*

$= \boldsymbol{204\ frames}$ *and screen playback duration* $S \approx \boldsymbol{8.5085\ s}$*.)*

With the broadcast master clock and frame counters locked to atomic time, we move from the edit suite onto the exterior location set in Chapter 11 to hold optical focus and balance twilight exposure on a moving Steadicam close-up.

Chapter 11: Holding Focus on the Steadicam Close-Up at Dusk

Out on a coastal bluff twenty minutes after sunset, a Steadicam operator and first assistant camera (1st AC) are preparing the final dramatic close-up of the shooting day. The camera body holds an ANSI/SMPTE Super 35 digital cinema sensor of active horizontal width $W = 24.89$ mm (permissible Circle of Confusion $C = 0.025$ mm for critical 4K Super 35 acquisition, or $C = 0.029$ mm–0.030 mm on Full-Frame 36 mm reference gates) fitted with an $F = 50$ mm prime lens set wide enough for twilight at aperture $f/2.8$ ($N = 2.8$). As the actor walks along a cliff path at a nominal focus distance of $s = 3.00$ m (3000 mm) while the camera glides backward at 24 fps with a 180° rotary shutter ($T = 1/48$ s), the 1st AC must know the horizontal field of view, the hyperfocal distance, and the exact near and far focus boundaries down to the centimeter so the actor's eyes never soften into blur. Immediately afterward, the operator switches to an $F = 35$ mm wide-angle prime for a running tracking shot and asks what exact f-stop N will lock the hyperfocal distance at $H = 4.50$ m (4500 mm) so everything from 2.25 m to the horizon stays sharp without pulling focus.

At the same time, the director of photography and gaffer are racing the dropping horizon light. Ten minutes earlier, while shooting against the bright western sky, the camera carried an ND 0.9 neutral-density filter in the matte box to cut three stops of daylight so the lens could stay at $f/2.8$. Now, as twilight deepens, the gaffer brings up a 5600 K daylight HMI Fresnel fixture on a condor lift and a 3200 K tungsten practical lantern inside the cabin window, requiring a micro-reciprocal degree gel calculation to match the two blackbody sources and a rotary shutter check against the 60 Hz generator feeding the set.

Every decision on that bluff couples geometric optics to logarithmic photometry:

1. Determine the horizontal angular field of view across the Super 35 sensor and its Full-Frame equivalent focal length.
2. Calculate the hyperfocal distance and near/far sharpness boundaries around the actor at 3.00 m, then solve backward for the f-stop that locks hyperfocal focus at 4.50 m on the 35 mm tracking shot.
3. Balance rotary shutter integration time, base-two Exposure Value stops, neutral-density optical attenuation, and Mired color-correction gels.

11.1 Rectilinear Framing, Circle of Confusion, and Depth of Field

A motion-picture prime lens acts as a geometric transformer between three-dimensional set space and the flat two-dimensional plane of the film gate or silicon sensor. Two right-triangle relationships govern what the audience sees: the chief-ray field-of-view triangle across the sensor width W, and the marginal-ray blur cone across the iris aperture diameter $D_{\text{iris}} = F \,/\, N$.

11.1.1 Sensor Format Geometry, Crop Factors, and Angular Field of View

When chief rays from a distant scene pass through the rear nodal point of a rectilinear lens focused at infinity, they form an inverted planar image at axial distance F (the lens focal length, in millimeters, mm) behind the nodal point.

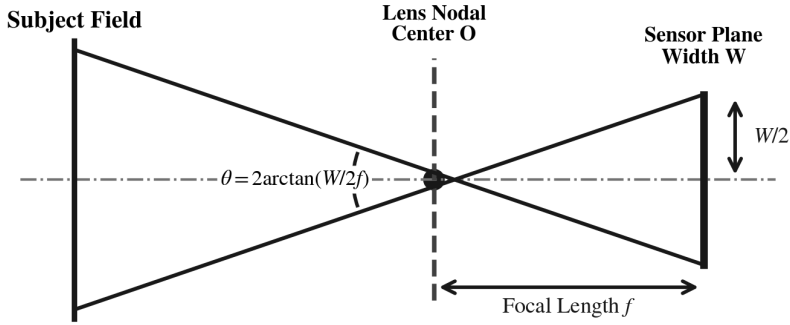


Figure 11.1: *Rectilinear pinhole lens projection mapping sensor width W and focal length f to angular field of view theta.*

The optical axis bisects the horizontal width W (in mm) of the active sensor or film aperture into two congruent right triangles, each having an opposite leg of half-width $W / 2$ and an adjacent leg of focal length F. Doubling the half-angle inverse tangent defines the horizontal angular **Field of View** ([FOV]), denoted A in degrees (°), while the ratio of the 36 mm× 24 mm Full-Frame diagonal (43.2666 mm) or horizontal width to a smaller format's dimension defines the format **Crop Factor** ([↰] [→35], [↱] [→16], and aspect ratio [↱] [W:H]):

$$A = 2\arctan\left(\frac{W}{2F}\right), \qquad W_{\text{scene}} = 2s\tan\left(\frac{A}{2}\right) = s\left(\frac{W}{F}\right), \qquad F_{\text{equiv}} = F \times K_{\text{crop}} \qquad \textbf{(Eq. 11.1)}$$

In these expressions, W is the active horizontal sensor width in mm, F is the lens focal length in mm, A is the horizontal coverage angle in degrees, s is the subject distance along the optical axis, and K_{crop} scales focal length to its 35 mm Full-Frame ([↰] [→35], $K_{\text{crop}} \approx 1.5\times$ from Super 35) or Super 16 ([↱] [→16], $K_{\text{crop}} \approx 2.88\times$) equivalent angle of view.

On our Steadicam's ANSI/SMPTE Super 35 sensor ($W = 24.89$ mm) fitted with an $F = 50$ mm prime lens, the half-width tangent ratio is $W / (2F) = 24.89 / 100 = 0.2489$. Taking the inverse tangent gives half-angle $\arctan(0.2489) \approx 13.977°$, which doubles to a horizontal angular field of view of $A \approx$ **27.954°**. At the actor's distance of $s = 3.00$ m, the frame spans a horizontal scene width of $W_{\text{scene}} = 3.00 \times (24.89 / 50) =$ **1.493 m**—a classic medium close-up with breathing room for the cliff horizon. On a Full-Frame VistaVision sensor ($W = 36.0$ mm), that same 50 mm lens opens up to $A = 2\arctan(36 / 100) \approx$ **39.598°**, whereas matching the Super 35 framing on Full-Frame requires scaling focal length by $36.0 / 24.89 \approx 1.4464$ to an $F \approx$ **72.3 mm** lens.

11.1.2 Hyperfocal Distance and Conjugate Sharpness Boundaries

While the nodal chief rays set the framing angle A, the outer marginal rays passing through the finite iris diameter $D_{\text{iris}} = F \,/\, N$ (where N is the dimensionless aperture f-number or T-stop) govern sharpness along the depth axis. In paraxial Gaussian optics, a lens focused at distance s brings only object points lying on that exact plane to a mathematical point on the sensor. Points closer or farther than s converge slightly behind or in front of the sensor plane, intersecting the silicon surface as a circular blur disk known as the **Circle of Confusion** (C).

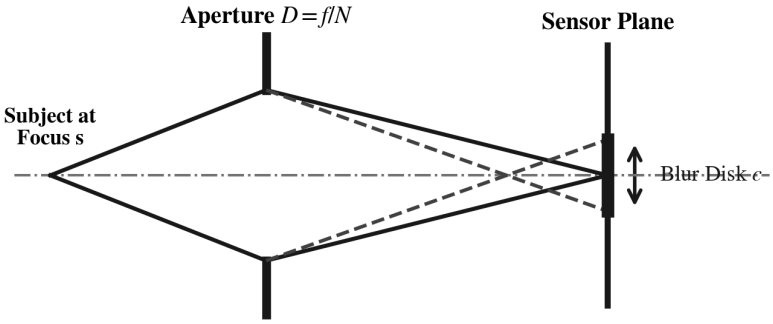


Figure 11.2: *Optical ray-cone convergence through aperture diameter $D = f/N$ forming a blur circle of confusion c on the sensor plane.*

Because human visual acuity is finite when viewing a theater screen from a standard seating distance, any blur disk smaller than a critical diameter C (standardized at $C = 0.029$ mm–0.030 mm for 35 mm Full-Frame reference gates and $C = 0.025$ mm for Super 35 cinema work) appears indistinguishable from a sharp point. The axial distance H at which focusing the lens places the far boundary of acceptable sharpness at optical infinity (∞) is the **Hyperfocal Distance** ([DOF], [↰] [HFD]), which governs the **near focus limit** D_{near}, **far focus limit** D_{far}, and total **Depth of Field** $\Delta D = D_{\text{far}}$ - D_{near} for any subject distance $s < H$:

$$H = \frac{F^2}{N \cdot C}, \qquad D_{\text{near}} = \frac{H \cdot s}{H + s}, \qquad D_{\text{far}} = \frac{H \cdot s}{H - s}, \qquad \Delta D = D_{\text{far}} - D_{\text{near}} = \frac{2Hs^2}{H^2 - s^2} \qquad \textbf{(Eq. 11.2)}$$

Here F is the focal length in mm, N is the f-stop number, C is the Circle of Confusion diameter in mm, H is the hyperfocal distance (in mm in the [DOF] solver workspace, or divided by 1000 in meters on the immediate stack), and s is the focus distance in the same length units as H. When the focus ring is set directly to the hyperfocal mark ($s = H$), the denominator of D_{far} vanishes (H - $H = 0 \Rightarrow D_{\text{far}} \to \infty$) while D_{near} reduces to $H^2 \,/\, (2H) = H \,/\, 2$, rendering everything from half the hyperfocal distance to the horizon acceptably sharp.

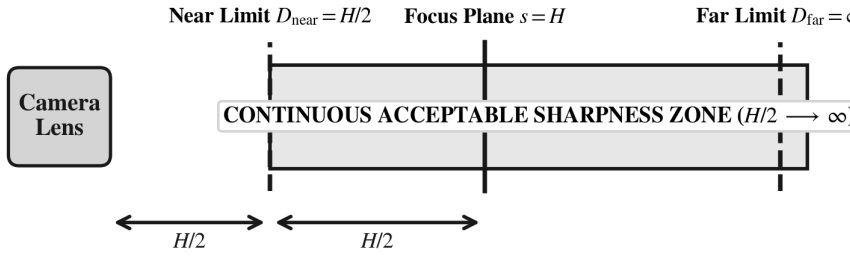


Figure 11.3: *Hyperfocal focus geometry ($s = H$) extending continuous acceptable sharpness from $H/2$ to optical infinity.*

On our Steadicam close-up with $F = 50$ mm and $N = 2.8$ using the $C = 0.030$ mm reference criterion (**50** ENTER **2.8** ENTER **0.03** DOF), the hyperfocal distance is $H = 50^2 / (2.8 \times 0.030) = 2500 / 0.084 \approx$ **29,761.9 mm ≈ 29.762 m** (or $H \approx$ **30.788 m** with the 2-argument default $C = 0.029$ mm). With the lens focused on the actor at $s = 3.00$ m, the conjugate sharpness boundaries lie at:

$$D_{near} = \frac{29.762 \times 3.00}{29.762 + 3.00} = \frac{89.286}{32.762} \approx \mathbf{2.725}\ \text{m}, \qquad D_{far} = \frac{29.762 \times 3.00}{29.762 - 3.00} = \frac{89.286}{26.762} \approx \mathbf{3.336}\ \text{m} \qquad \textbf{(Eq. 11.3)}$$

The total depth-of-field window around the actor at 3.00 m spans $\Delta D = 3.336 - 2.725 =$ **0.611 m** (61.1 cm)—extending 27.5 cm in front of the actor's eyes and 33.6 cm behind them. Conversely, when the operator mounts an $F = 35$ mm prime on the Super 35 body ($C = 0.025$ mm) and wants to lock the hyperfocal distance at $H = 4.50$ m $= 4500$ mm, solving $H = F^2 / (N \cdot C)$ backward for N yields $N = F^2 / (H \cdot C) = 35^2 / (4500 \times 0.025) = 1225 / 112.5 =$ **10.889** (just under $f/11$).

11.2 Logarithmic Exposure Reciprocity, Rotary Shutters, and Color Mireds

Once the focal length F and aperture N are chosen to establish framing and depth of field, the camera and lighting crews must regulate photon flux and spectral color temperature so the sensor records balanced exposure and uniform white point as dusk settles.

11.2.1 Base-Two Exposure Value, Shutter Angles, Neutral Density, and Flicker-Free Windows

Because human brightness perception and silicon photo-site well capacity scale geometrically, cinematography measures exposure changes in base-two **stops**, where each +1 stop step doubles photon exposure and each -1 stop step halves it. Combining relative aperture irradiance ($\propto 1/N^2$) and frame integration time T (in seconds) onto a unified base-two logarithmic scale defines **Exposure Value** (EV, ↰ →EV at ISO 100), while a spinning motion-picture rotary shutter of open sector angle $\vartheta_{shutter}$ (↰ SHT, where 180° is standard) sets integration time T as a fraction of the frame period 1 / FPS, and gray **Neutral Density** (↱ →ND) filters attenuate excess light according to base-ten optical density D:

$$N^2 = T \cdot 2^E \Leftrightarrow E = \log_2\left(\frac{N^2}{T}\right), \qquad T = \frac{\vartheta_{shutter}}{360^\circ \cdot \text{FPS}}, \qquad D = -\log_{10}(T_{trans}) = \log_{10}(2) \times \text{stops} \approx 0.30103 \times \text{stops} \qquad \textbf{(Eq. 11.4)}$$

In these photometric equations, N is the lens f-number, T is the exposure integration time in seconds, E is the Exposure Value (EV_{100}), $\vartheta_{shutter} \in (0°, 360°]$ is the rotary shutter angle, D is the dimensionless neutral-density filter rating ([↷] [→ND] converts stops s in Register X into optical density $D = 0.3 \times s$ in Register X and filter factor 2^s in Register Y), and $T_{trans} = 2^{-stops} = 10^{-D}$ is the fractional light transmittance. Notice the exact mathematical bridge between cinematography and analytical chemistry: just as an ND 0.9 filter ($D = 0.9$) reduces transmitted irradiance by 3 stops to $T_{trans} = 2^{-3} = 0.125$, a biochemical sample with UV-Vis absorbance $A = 0.903$ in Unit VI (**SC-60**) transmits $10^{-0.903} = 12.5\%$ of incident monochromatic light.

When filming under alternating-current discharge lamps or LED pulse-width-modulated drivers operating at line frequency f_{line} (60 Hz in North America or 50 Hz in Europe), the lamp intensity pulses at 2 f_{line} (120 Hz or 100 Hz). Avoiding rolling exposure bands across the frame ([↶] [FLK]) requires choosing a shutter integration time T that integrates an integer number $k \in (1, 2, 3, ...)$ of half-cycles:

$$T_{\text{flicker-free}} = \frac{k}{2 f_{line}}, \qquad \theta_{\text{flicker-free}} = 360° \cdot \text{FPS} \cdot \left(\frac{k}{2 f_{line}}\right) \qquad \textbf{(Eq. 11.5)}$$

On our Steadicam setup rolling at FPS = 24 fps with a standard $\vartheta_{shutter} = 180°$ rotary shutter (**24** [ENTER] **180** [↶] [SHT]), the frame integration time is $T = 180° / (360° \times 24) = 1/48$ s **≈ 0.020833 s** in Register X with denominator 48 in Register Y. Under an unballasted 60 Hz source (120 Hz ripple), choosing $k = 2$ half-cycles sets $T = 1/60$ s, corresponding to a flicker-free shutter angle of $\vartheta_{shutter} = 360° \times 24 / 60 =$ **144°**. At $f/2.8$ ($N = 2.8$) and $T = 1/48$ s, the camera's native Exposure Value at ISO 100 is $E = \log_2(2.8^2 \times 48) = \log_2(376.32) \approx$ **8.556 EV**. When the matte box held an ND 0.9 filter (**3** [↷] [→ND] $\rightarrow D = 0.9$ in X and filter factor 8× in Y, cutting **3.0 stops**), the lens accommodated a bright dusk sky at 8.556 + 3.0 = **11.556 EV** without stopping down the iris.

11.2.2 Inverse-Square Set Illuminance, Footcandles, and Mired Gel Shifts

To supplement the fading sky once the ND 0.9 filter is pulled from the matte box, the gaffer positions a spotlight of axial luminous intensity I (in candelas, cd) at throw distance D (in meters) from the actor, delivering incident metric **illuminance** E_{lx} ([LUX], [↷] [→lux], in lux, lm/m²) or imperial **footcandles** E_{fc} ([↶] [→fc], in lm/ft², where 1 fc = 10.76391 lx) across a conical beam diameter W_{beam} ([↶] [BM°]):

$$E_{lx} = \frac{I}{D^2}, \qquad E_{fc} = \frac{E_{lx}}{10.76391}, \qquad W_{beam} = 2D \tan\left(\frac{\theta_{beam}}{2}\right) \qquad \textbf{(Eq. 11.6)}$$

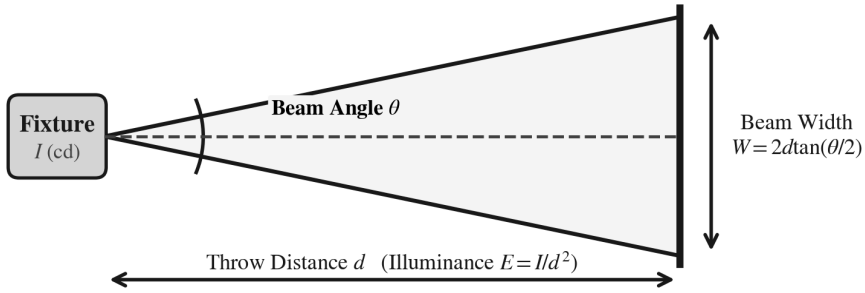


Figure 11.4: *Inverse-square illuminance falloff and conical beam diameter W = 2d tan(theta/2) over throw distance d.*

Simultaneously, the gaffer must reconcile the spectral color temperature of the 3200 K tungsten cabin lantern with the 5600 K daylight twilight and HMI key light. Because Wien's radiation displacement law ties peak blackbody wavelength to the reciprocal of absolute temperature T_K, scaling $1 \,/\, T_K$ by one million defines **Micro-Reciprocal Degrees** ([↖] [→MRD], $M = 10^6 \,/\, T_K$) and the linear **Mired filter shift** ΔM ([↗] [GEL]) required to convert source temperature T_{source} to target temperature T_{target}:

$$M = \frac{10^6}{T_K}, \qquad \Delta M = \frac{10^6}{T_{\text{target}}} - \frac{10^6}{T_{\text{source}}} \tag{Eq. 11.7}$$

Here a positive shift ($\Delta M > 0$) calls for an amber **Color Temperature Orange** (CTO) gel to warm a daylight source toward tungsten, whereas a negative shift ($\Delta M < 0$) calls for a blue **Color Temperature Blue** (CTB) gel to cool a $T_{\text{source}} = 3200$ K tungsten lamp ($M_{\text{source}} = 10^6 \,/\, 3200 = 312.50$ Mireds) up to $T_{\text{target}} = 5600$ K daylight ($M_{\text{target}} = 10^6 \,/\, 5600 = 178.57$ Mireds), yielding $\Delta M = 178.57 - 312.50 =$ **-133.93 Mireds** (a Full CTB gel).

11.3 Pulling Focus and Balancing Dusk Exposure on the Stack and in the Solver Workspace

On the **SC-24**, optical framing ([FOV], [↗] [W:H], [↖] [→35], [↗] [→16]), hyperfocal depth of field ([DOF], [↖] [HFD]), exposure reciprocity ([EV], [↖] [SHT], [↖] [→EV], [↗] [→ND]), and lighting photometry ([LUX], [↗] [→lux], [↖] [→fc], [↗] [GEL], [↖] [→MRD]) operate both as immediate RPN stack shortcuts and as bidirectional Equation Solver Workspaces ([FOV], [DOF], [EV], [LUX]).

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU3])	Governing Expression
[FOV]	Stack / Solver	Stack: $Y = W$, $X = F$ • Solver: [A] [W]=36 [F]	$A = 2\arctan(W \,/\, (2F))$ (Full-Frame $W = 36$ mm default)
[DOF]	Stack / Solver	Stack: $Z = F$, $Y = N$, $X = C$ (or $Y = F$, $X = N$) • Solver: [H] [F] [N] [C ON]	$H = F^2 \,/\, (N \cdot C) \,/\, 1000$ m on stack ($H = F^2 \,/\, (N \cdot C)$ mm in solver)
[EV]	Stack / Solver	Stack: $Y = N$, $X = T$ • Solver: [N] [E] [T]	$N^2 = 2^E \cdot T \Leftrightarrow E = \log_2(N^2 \,/\, T)$
[↖] [SHT]	Immediate Stack	Register Y: Frame rate fps, Register X: Shutter angle $\vartheta°$	$X' = (\vartheta \,/\, 360°) \,/\, \text{fps (s)}$, $Y' = 1 \,/\, X'$ (shutter denominator)

Key / Solver	Mode	Inputs / Softkeys (LFU0–LFU3)	Governing Expression
f →ND / ← →EV	Immediate Stack	Register X: Attenuation in stops (f →ND) or lux (← →EV)	$X = 0.3 \times$ stops, $Y = 2^{\text{stops}}$ (f →ND); $\text{EV}_{100} = \log_2(\text{lx}/2.5)$ (← →EV)
LUX / f →lux / ← →fc	Stack / Solver	Stack: $Y = I$, $X = D$ (LUX) • Solver: E I D	$E = I \,/\, D^2$ (LUX); unit conversion 1 fc = 10.7639 lx (f →lux, ← →fc)
f GEL / ← →MRD	Immediate Stack	Register Y: Source T_{src} (K), Register X: Target T_{dst} (K)	Mired shift $\Delta M = 10^6/X - 10^6/Y$ (f GEL); $M = 10^6/X$ (← →MRD)

11.3.1 Immediate Stack Trace: Super 35 Framing, Conjugate Limits, and Dusk Photometry

Before calling camera roll on the 50 mm Steadicam close-up, the 1st AC and gaffer evaluate the Super 35 horizontal field of view ($W = 24.89$ mm, $F = 50$ mm), the hyperfocal distance in meters at $f/2.8$ with $C = 0.030$ mm (**50** ENTER **2.8** ENTER **0.03** DOF $\rightarrow$ 29.762 m), the shutter integration time (**24** ENTER **180** ← SHT $\rightarrow$ 0.020833 s) and native Exposure Value (EV), the ND 0.9 filter density (**3** f →ND), and the 3200 K $\rightarrow$ 5600 K Mired shift (f GEL):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
24.89 ENTER **50** FOV	0	0	0	27.9542	Super 35 horizontal field of view A = 27.954° at $F = 50$ mm
50 ENTER **2.8** ENTER **0.03** DOF	27.9538	27.9538	27.9542	29.7619	Hyperfocal distance $H = 50^2$ / (2.8 × 0.03 × 1000) = 29.762 m at $f/2.8$
24 ENTER **180** ← SHT	27.9538	29.7619	48	0.020833	180° shutter at 24 fps gives t = 0.020833 s in X and denominator 48 in Y
2.8 x⇄y EV	29.7619	29.7619	48	8.5558	Native Exposure Value $E = \log_2(2.8^2 / 0.020833) = 8.556$ EV at 1/48 s
3 f →ND	29.7619	48	8	0.9000	Convert 3 stops into neutral density $D = 0.9$ in X and filter factor 8× in Y
3200 ENTER **5600** f GEL	8	8	0.9000	**-133.9286**	Mired gel shift ΔM = -133.93 Mireds (Full CTB) for 3200 K $\rightarrow$ 5600 K

11.3.2 Bidirectional Solver Workspaces (DOF, FOV, and EV): Solving Backward for Aperture and Focal Length

Now the Steadicam operator switches to an $F = 35$ mm prime for the running cliffside tracking shot on the Super 35 sensor ($C = 0.025$ mm) and asks for the exact T-stop / f-number N that locks the hyperfocal distance at $H = 4.50$ m (4500 mm), followed by the exact focal length F required on Full-Frame ($W = 36.0$ mm) to match a 27.954° horizontal field of view. Opening the DOF and FOV Equation Workspaces solves both inverse optical problems directly without manual algebra:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar (LFU0–LFU3)	Register X Result & Physical Meaning
DOF (open workspace)	**DOF: H = F² / (N × C)**	H F N S/ON	Opens hyperfocal depth-of-field solver workspace
4500 → H, **35** → F	**DOF: H = F² / (N × C)**	H=4500 F=35 N S/ON	Stores target $H = 4500$ mm (4.50 m) and lens $F = 35$ mm

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU3])	Register X Result & Physical Meaning
0.025 → [C ON]	**DOF: H = F² / (N × C)**	[H]=4500 [F]=35 [N] [C ON]=.025	Stores Super 35 critical Circle of Confusion $C = 0.025$ mm
[XEQ] → [N]	**N = 10.8889**	[H]=4500 [F]=35 [N]=10.89 [C ON]=.025	Solves backward for required f-stop $N = 35^2 / (4500 \times 0.025) =$ **10.89** ($\approx f/11$)
[FOV] (open workspace)	**FOV: A = 2×ATAN(W/(2×F))**	[A] [W]=36 [F]	Opens lens angle workspace with Full-Frame $W = 36.0$ mm pre-bound
27.9542 → [A], [XEQ] → [F]	**F = 72.3182**	[A]=27.95 [W]=36 [F]=72.32	Solves backward for Full-Frame focal length $F =$ **72.32 mm** matching Super 35 50 mm

Similarly, opening the [EV] workspace (**N² = 2^E × T**) allows the gaffer to store target exposure $E = 11.5558$ EV in [E] and shutter time $T = 0.020833$ s in [T], then execute [XEQ] → [N] to find the unattenuated iris stop $N =$ **7.924** ($f/8$, confirming that 3 stops of ND 0.9 filtration return the lens to $f/2.8$).

11.3.3 Try It on the Stack

Test your mastery of cinema lens geometry, hyperfocal depth of field, base-two exposure stops, and Mired color balancing with these three on-set problems:

1. **Super 35 Portrait Prime vs. Inverse Circle of Confusion Audit:** On a Super 35 sensor ($W = 24.89$ mm), compute the horizontal field of view for an $F = 85$ mm lens (**24.89** [ENTER] **85** [FOV]), and then use the [DOF] Solver Workspace (**H = F² / (N × C)**) with $F = 85$ mm in [F], $N = 4.0$ in [N], and $C = 0.025$ mm in [C ON] to solve via [XEQ] → [H] in millimeters (and on the stack in meters via **85** [ENTER] **4** [ENTER] **0.025** [DOF]). *(Verified results: A ≈ **16.659°** and H = **72,250 mm** = **72.25 m**.)*

2. **European 25 fps Exterior Exposure and 5-Stop ND Filter:** On a 25 fps European shoot with a 180° shutter (**25** [ENTER] **180** [↶] [SHT] → $T = 1/50$ s $= 0.02$ s) at $f/11$, evaluate the Exposure Value (**11** [ENTER] **0.02** [EV]) and find the ND filter rating and filter factor required to open the lens by 5 stops to $f/2.0$ (**5** [↷] [→ND]). *(Verified Register X and Y results: E ≈ **12.563 EV**, optical density D = **1.5** in X, and filter factor **32×** in Y for an ND 1.5 filter.)*

3. **Follow-Spot Illuminance in Lux and Footcandles & Daylight-to-Tungsten Gel:** In the [LUX] Solver Workspace (**E = I / D²**), store luminous intensity $I = 48{,}000$ cd in [I] and throw distance $D = 12$ m in [D], solve for illuminance via [XEQ] → [E] in lux, convert to footcandles via [↶] [→fc], and compute the Mired shift to warm a 5600 K lamp to 3200 K (**5600** [ENTER] **3200** [↷] [GEL]). *(Verified results: E_{lx} ≈ **333.333 lx**, E_{fc} ≈ **30.968 fc**, and ΔM ≈ **+133.929 Mireds** Full CTO.)*

Having locked timecode in the broadcast truck and focused the camera on set, we turn in Chapter 12 to the third physical wave domain of live production: aligning acoustic wavefronts, room-mode wavelengths, and decibel levels across an outdoor stadium and broadcast stage.

Chapter 12: Aligning the Stadium Delay Towers Before Downbeat

Ninety minutes before seventy thousand fans fill an open-air football stadium for a summer concert broadcast, the front-of-house (FOH) systems engineer stands beside the mid-field mixing console with a measurement microphone and weather meter. A warm late-afternoon breeze has pushed the ambient air temperature on the turf to $T_C = 25$°C (77°F). Hanging sixty feet above the stage, the main left-and-right line arrays deliver pristine clarity to the front half of the field, but inverse-square spherical spreading attenuates their sound pressure level long before the wave reaches the upper grandstand. To restore vocal intelligibility for the rear audience without blasting the front row at pain-threshold levels, the crew has rigged supplementary delay speaker towers $d = 45.0$ m downfield from the main stage array.

Those downfield towers introduce a classic wave-propagation race condition. Electrical audio signals leave the mixing console over fiber-optic and copper trunks at roughly two hundred thousand kilometers per second, reaching the delay tower amplifiers in less than a microsecond. Meanwhile, the acoustic pressure wave from the main stage array crawls through the 25°C stadium air at roughly 346 meters per second. If the delay tower fires uncompensated, listeners in the back half of the stadium hear the delay tower more than a tenth of a second *before* the main stage wave arrives—shattering the rhythmic pocket into a cavernous slap-back echo and pulling perceived vocal localization away from the performers on stage toward the rigging scaffold overhead.

Before the headliner's downbeat, the systems engineer and broadcast sound mixer must solve three acoustic problems:

1. Compute the thermal speed of sound at 25°C, the acoustic wavelength of low-frequency subwoofer and room modes, and the digital signal processor (DSP) delay time for the 45.0 m tower including a psychoacoustic Haas precedence offset.
2. Trace the inverse-square Sound Pressure Level decay of a stage source both forward to the mix position and **backward** to find the maximum boom-mic reach D_2 before dialogue drops below the broadcast noise floor.
3. Sum the incoherent acoustic power of dual PA arrays in decibels, convert console line levels between decibels and volts, and verify reverberation decay (RT_{60}).

12.1 Acoustic Wave Propagation, Precedence, and Room Decay

Unlike electromagnetic light and radio waves, which travel through vacuum at $c \approx 3 \times 10^8$ m/s, sound is a longitudinal mechanical compression wave whose propagation velocity depends directly on the kinetic translational energy—and therefore the absolute temperature—of air molecules.

12.1.1 Thermal Speed of Sound, Delay Tower Transit Time, and the Haas Window

In dry air treated as an ideal diatomic gas ($\gamma = 1.40$, molar mass $M \approx 0.02897$ kg/mol), Laplace's adiabatic wave law sets the speed of sound v_s proportional to the square root of absolute Kelvin temperature: $v_s = 331.3\sqrt{(1 + T_C / 273.15)}$ m/s. Linearizing that square root across ambient outdoor and indoor temperatures (-20°C to +40°C) yields the standard acoustic velocity law ([→] [v_s]), the propagation transit time Δt_{ms} ([→] [DLY]) across a downfield throw distance d (in meters), and the acoustic wavelength λ ([→] [λ]) at frequency f (in hertz):

$$v_s(T_C) = 331.3 + 0.606 \cdot T_C \quad [\tfrac{m}{s}], \qquad \Delta t_{ms} = \tfrac{d}{v_s} \times 1000, \qquad \lambda = \tfrac{v_s}{f} \qquad \textbf{(Eq. 12.1)}$$

In these expressions, T_C is the ambient air temperature in degrees Celsius (°C), v_s is the acoustic phase velocity in m/s (evaluating to 343.42 m/s ≈ 343 m/s at the standard 20°C room reference recalled by [→] [v_s]), d is the physical distance from the main stage array to the delay tower in meters, Δt_{ms} is the acoustic transit delay in milliseconds (ms), and λ is the physical wavelength in meters.

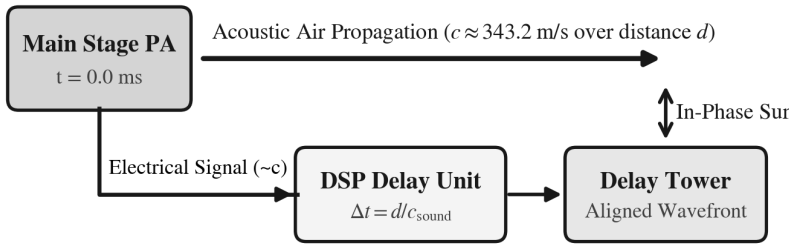


Figure 12.1: *Time-aligned festival sound reinforcement coordinating acoustic air travel from the main stage array with DSP delay towers.*

At our stadium's turf temperature of $T_C = 25$°C, the speed of sound rises to $v_s(25°C) = 331.3 + 0.606(25) = 331.3 + 15.15 = \mathbf{346.45\ m/s}$. Over the $d = 45.0$ m air path to the delay tower (**45** [ENTER] **25** [→] [DLY]), the main stage wavefront requires a transit time of:

$$\Delta t_{25°C} = \tfrac{45.0}{346.45} \times 1000 \approx \mathbf{129.889}\ \text{ms} \qquad \textbf{(Eq. 12.2)}$$

Compare that thermal result with the standard 20°C reference velocity ($v_s = 343.0$ m/s via **45** [→] [DLY]), where the geometric delay is $\Delta t_{20°C} = (45.0 / 343.0) \times 1000 \approx \mathbf{131.195\ ms}$. That 1.31 ms thermal shortening across a 5°C afternoon warming corresponds to 0.45 m of acoustic path length—a full 180° phase cancellation ($\lambda / 2 = 0.45$ m ⇒ $\lambda = 0.90$ m) at $f = 346.45 / 0.90 \approx \mathbf{385\ Hz}$ right in the vocal fundamental region. Furthermore, by **Helmut Haas's 1949 precedence effect**, if the delay tower fires at the exact geometric arrival time (129.89 ms), fans standing near the louder delay tower localize the singer's voice up in the delay rigging rather than on stage. Adding a +12.0 ms **Haas precedence offset** (setting the DSP to 129.89 + 12.0 = **141.89 ms**) ensures the main stage wavefront reaches the listener's ears 12 ms before the louder delay wave. The auditory cortex fuses both arrivals inside the 30 ms Haas integration window and locks spatial localization onto the stage.

12.1.2 Sabine Reverberation Decay and Room Standing-Wave Modes

Inside the enclosed concrete-and-glass VIP broadcast studio beneath the stadium bowl, sound waves reflect repeatedly off parallel walls of spacing L_{room}, exciting axial standing-wave room modes whenever L_{room} equals an integer multiple of half a wavelength ($f_n = n \cdot v_s / (2\ L_{\text{room}})$, or $\lambda_1 = 2\ L_{\text{room}}$). Simultaneously, Wallace Clement Sabine's 1898 architectural law sets the **Reverberation Time** RT_{60} (in seconds)—the time required for reverberant sound energy in a room of volume V (m³) and total absorption area $A = \sum S_i \alpha_i$ (metric sabins, m²) to decay by 60 dB:

$$RT_{60} = \frac{0.161 \cdot V}{A} = \frac{0.161 \cdot V}{\sum_i S_i \alpha_i}, \qquad \lambda = \frac{v_s}{f} \tag{Eq. 12.3}$$

For a broadcast studio of volume $V = 2400$ m³ treated with acoustic baffles providing $A = 480$ metric sabins of absorption, Sabine's formula predicts a tight decay time of $RT_{60} = (0.161 \times 2400) / 480 =$ **0.805 s**, while an $f = 68.6$ Hz bass note at $v_s = 343$ m/s has an acoustic wavelength of $\lambda = 343 / 68.6 =$ **5.00 m** (**68.6** [↻] [λ]).

12.2 Spherical Spreading Attenuation and Incoherent Decibel Summation

To size the power amplifiers for the stadium arrays and position shotgun boom microphones on the broadcast stage, the engineer must calculate how acoustic pressure decays with distance across open air and how multiple speaker arrays combine in decibels.

12.2.1 Inverse-Square Pressure Falloff and Maximum Boom-Mic Distance

Because human hearing responds logarithmically to acoustic pressure across six orders of magnitude—from the quietest audible threshold at reference pressure $P_0 = 20$ μPa (0 dBSPL) to 20 Pa (120 dBSPL)—acoustics defines **Sound Pressure Level** (L, in dBSPL) as $20 \log_{10}(P_{\text{rms}} / P_0)$. As an acoustic wave expands spherically into free space from a compact source, its acoustic intensity ($I \propto P_{\text{rms}}^2$) falls off inversely with the sphere's surface area ($1 / (4\pi D^2)$), so root-mean-square sound pressure falls inversely with distance ($P_{\text{rms}} \propto 1/D$). Expressing that inverse-distance pressure ratio in decibels yields the **Spherical Spreading Attenuation Law** ([↶] [SPL]) in both its 3-operand level form (L_1 in Z, D_1 in Y, D_2 in X) and its inverse target-distance form:

$$L_2 = L_1 - 20\log_{10}\left(\frac{D_2}{D_1}\right) \Leftrightarrow D_2 = D_1 \cdot 10^{(L_1 - L_2)/20} \tag{Eq. 12.4}$$

Here L_1 is the known Sound Pressure Level in dBSPL at reference distance D_1, and L_2 is the Sound Pressure Level at distance D_2 (when called with two arguments D_1 in Y and D_2 in X, [↶] [SPL] returns the relative decibel attenuation $\Delta L = -20 \log_{10}(D_2 / D_1)$). Notice the exact mathematical bridge to radio-frequency engineering in Unit V (**SC-50**): the distance term $-20 \log_{10}(D_2/D_1)$ in acoustic [↶] [SPL] is identical to the $-20 \log_{10}(D)$ inverse-square spreading term in the Friis Free-Space Path Loss equation ([FSL]). Every doubling of distance ($D_2 / D_1 = 2$) reduces sound pressure level by $20 \log_{10}(2) \approx$ **6.0206 dB**.

In our stadium check, a reference cabinet produces $L_1 = 104.0$ dBSPL at $D_1 = 1.0$ m. Projecting that wavefront out to $D_2 = 32.0$ m (2^5, five doublings of distance via **104** [ENTER] **1** [ENTER] **32** [↶] [SPL]) attenuates the level by $20 \log_{10}(32) \approx 30.103$ dB to $L_2 = 104.0 - 30.103 =$ **73.897 dBSPL**. Conversely, on the interview stage, if an actor speaks at $L_1 =$

74.0 dBSPL measured at $D_1 = 0.50$ m and the broadcast sound mixer requires at least L_2 = 63.5 dBSPL at the hypercardioid boom capsule to maintain a 45 dB signal-to-noise margin above the stadium crowd rumble, solving backward for the maximum boom-mic distance D_2 yields $D_2 = 0.50 \times 10^{(74.0 - 63.5)/20} = 0.50 \times 10^{0.525} = 0.50 \times 3.3497 = \mathbf{1.675\ m}$.

12.2.2 Combining Uncorrelated Acoustic Power and Analog Line Levels

When multiple loudspeakers or stage sources radiate into the same listening zone, the combined decibel level depends on whether the waves are **incoherent** (uncorrelated random phases across broadband frequencies) or **coherent** (phase-locked low-frequency waves). Because acoustic power is proportional to pressure squared ($W \propto P_{\text{rms}}^2 \propto 10^{L_i\ /\ 10}$), summing independent acoustic sources of decibel levels L_1 and L_2 requires adding their linear power ratios and taking 10 log₁₀ of the sum via the decibel prefix key [dB] followed by [+] (e.g., **94** [ENTER] **94** [dB] [+] → **97.0103 dB**), while analog broadcast consoles reference signal voltage V_{rms} to $V_0 = \sqrt{(0.600\ \Omega \times 1\ \text{mW})} \approx 0.774597\ \text{V}_{\text{rms}}$ in **dBu** ([↶] [→dBu], [↷] [→V]):

$$L_{\Sigma,\,\text{incoherent}} = 10\log_{10}\left(10^{L_1/10} + 10^{L_2/10}\right), \qquad L_{\text{dBu}} = 20\log_{10}\left(\frac{V_{\text{rms}}}{0.774597}\right), \qquad V_{\text{rms}} = 0.774597 \times 10^{L_{\text{dBu}}/20} \qquad \textbf{(Eq. 12.5)}$$

When two uncorrelated broadband PA hangs each deliver 73.897 dBSPL at the 32.0 m mix position (**73.897** [ENTER] **73.897** [dB] [+]), their acoustic powers add ($W_{\text{total}} = 2\ W_1$), increasing the level by 10 log₁₀(2) ≈ **+3.0103 dB** to L_Σ = **76.9073 dBSPL** (just as two 94.0 dBSPL stage monitors sum via **94** [ENTER] **94** [dB] [+] to **97.0103 dBSPL**). Meanwhile, a professional broadcast console outputting +4.0 dBu drives the fiber-optic stage-box input with $V_{\text{rms}} = 0.774597 \times 10^{4/20} = \mathbf{1.2276\ V_{rms}}$ (**4** [↷] [→V]).

12.3 Time-Aligning Arrays and Evaluating Sound Levels on the Stack and in the Solver Workspace

On the **SC-24**, acoustic delay alignment ([↷] [DLY]), reference sound velocity ([↷] [v_s]), acoustic wavelength ([↷] [λ]), spherical spreading attenuation ([↶] [SPL]), incoherent decibel summation ([dB] [+]), analog studio voltage conversions ([↶] [→dBu], [↷] [→V]), and uncompressed broadcast stream data rates ([BIT]) execute directly on the RPN stack and in the bidirectional Solver Workspace.

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU3])	Governing Expression
[↷] [DLY]	Immediate Stack	Stack: $X = d$ (m) or $Y = d$, $X = T_C$ (°C)	$\Delta t = d$ / 343 × 1000 ms (or d / (331.3 + 0.606 T_C) × 1000 ms)
[↷] [v_s]	Constant / Stack	Pushes reference acoustic speed v_s	v_s = 343.0 m/s (20°C dry air reference)
[↷] [λ]	Immediate Stack	Stack: $X = f$ (Hz) or $Y = v_s$, $X = f$ (Hz)	Acoustic wavelength λ = v_s / f (m, v_s = 343 m/s default)
[↶] [SPL]	Immediate Stack	Stack: $Z = L_1$, $Y = D_1$, $X = D_2$ (or $Y = D_1$, $X = D_2$)	$L_2 = L_1$ - 20 log₁₀(D_2 / D_1) (or ΔL = -20 log₁₀(D_2/D_1))
[dB] [+]	Prefix + Stack	Register Y: Level L_1 (dB), Register X: Level L_2 (dB)	Incoherent power sum $L_\Sigma = 10\ \log_{10}(10^{\,Y/10} + 10^{\,X/10})$
[↶] [→dBu] / [↷] [→V]	Immediate Stack	Register X: Voltage V_{rms} ([↶] [→dBu]) or level L_{dBu} ([↷] [→V])	$L_{\text{dBu}} = 20\ \log_{10}(V$ / 0.7746) and $V = 0.7746 \times 10^{\,L/20}$
[BIT]	Stack / Solver	Stack: $Y = B$, $X = F$ • Solver: [R] [B] [F]	Uncompressed stream rate $R = B \times F$ / 8 MB/s

12.3.1 Immediate Stack Trace: Thermal Delay Alignment, SPL Falloff, and Dual-Array Summation

Standing at the front-of-house console, the systems engineer evaluates the transit delay across $d = 45.0$ m at turf temperature $T_C = 25$°C (**45** [ENTER] **25** [→] [DLY]), adds the +12.0 ms Haas precedence offset, projects the 104.0 dBSPL cabinet from $D_1 = 1.0$ m out to $D_2 = 32.0$ m (**104** [ENTER] **1** [ENTER] **32** [↶] [SPL]), sums two uncorrelated hangs via [dB] [+] (just like **94** [ENTER] **94** [dB] [+] → 97.0103 dB), and checks the 68.6 Hz bass wavelength ([→] [λ]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
45 [ENTER] **25** [→] [DLY]	0	0	0	129.8889	Transit delay across 45 m at 25°C (v_s = 346.45 m/s) = 129.889 ms
12 [+]	0	0	0	141.8889	Add +12.0 ms Haas precedence offset = 141.889 ms DSP delay
104 [ENTER] **1** [ENTER] **32** [↶] [SPL]	141.89	141.89	141.8889	73.8970	Project 104 dBSPL at 1 m out to 32 m: L_2 = 73.897 dBSPL
[ENTER] [dB] [+]	141.89	141.89	141.8889	76.9073	Incoherent power sum of two 73.897 dBSPL hangs = 76.9073 dBSPL
68.6 [→] [λ]	141.89	141.8889	76.9073	**5.0000**	Compute acoustic wavelength λ = 343 / 68.6 = 5.00 m at 20°C reference

12.3.2 Bidirectional Solver Workspaces ([→] [EQN] and [BIT]): Boom-Mic Reach and Broadcast Trunk Bandwidth

When the broadcast audio supervisor asks how far back the boom operator can hold the hypercardioid shotgun microphone (D) before an actor speaking at $S = 74.0$ dBSPL at reference distance $R = 0.50$ m drops below the minimum acceptable dialogue threshold of $L = 63.5$ dBSPL (stored in [→] [EQN] as **L = S - 20×LOG(D/R)** with single-letter softkeys [L], [S], [D], [R]), and what uncompressed data rate [R] ([BIT]) is required to record the concert film's 16 Mb/frame RAW stream at 24 fps, the engineer solves both directly in the Equation Workspace:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU3])	Register X Result & Physical Meaning
[→] [EQN] (**L = S - 20×LOG(D/R)**)	**L = S - 20×LOG(D/R)**	[L] [S] [D] [R]	Selects spherical spreading equation in [→] [EQN]
63.5 → [L], **74** → [S]	**L = S - 20×LOG(D/R)**	[L]=63.5 [S]=74 [D] [R]	Stores target threshold $L = 63.5$ dBSPL and source $S = 74.0$ dBSPL
0.5 → [R], [XEQ] → [D]	**D = 1.6748**	[L]=63.5 [S]=74 [D]=1.67 [R]=0.5	Solves backward for maximum boom-mic reach D = **1.675 m**
[BIT] (open workspace)	**BIT: R = B × F / 8**	[R] [B] [F]	Opens cinema/broadcast stream data-rate workspace
16 → [B], **24** → [F], [XEQ] → [R]	**R = 48.0000**	[R]=48 [B]=16 [F]=24	Solves for uncompressed capture rate $R = 16 \times 24 / 8$ = **48.00 MB/s**

12.3.3 Try It on the Stack

Apply thermal acoustic delay, incoherent decibel summation, inverse spherical spreading, and studio line-level conversions with these three live-sound exercises:

1. **Second-Ring Festival Delay Tower at 30°C:** A rear festival delay tower stands d = 80.0 m from the main stage on a hot afternoon (T_C = 30°C). Find the 30°C transit delay (**80** [ENTER] **30** [↗] [DLY]), and add a +15 ms Haas offset (**15** [+]). *(Verified Register X results: v_S = **349.48 m/s**, Δt_{30} ≈ **228.912 ms**, and DSP ≈ **243.912 ms**.)*

2. **Three-Source Stage Noise Sum and Front-of-House Attenuation:** Combine three uncorrelated stage sources measuring 95 dBSPL, 91 dBSPL, and 86 dBSPL at D_1 = 2.0 m (**95** [ENTER] **91** [dB] [+] **86** [dB] [+]), and project the combined level to the console at D_2 = 24.0 m (**2** [ENTER] **24** [↖] [SPL]). *(Verified Register X results: $L_{\Sigma,2m}$ ≈ **96.830 dBSPL** and L_{FOH} ≈ **75.246 dBSPL**.)*

3. **Broadcast Line-Level Voltage and Maximum PA Throw Distance:** Convert a +18.0 dBu digital console clipping level into RMS volts (**18** [↗] [→V]), and compute the maximum throw distance D_2 at which an array producing L_1 = 132.0 dBSPL at D_1 = 1.0 m still delivers L_2 = 96.0 dBSPL (**132** [ENTER] **96** [−] **20** [÷] **10** [x⇄y] [↖] [y^x]). *(Verified results: V_{rms} ≈ **6.153 V_{rms}** and $D_2 = 10^{36/20}$ ≈ **63.096 m**.)*

Having mastered frame-accurate broadcast clocks, rectilinear cinema optics, and acoustic wave alignment in Unit IV, we ascend in **Unit V: Radio-Frequency Engineering, Transmission Lines & Link Budgets** from mechanical sound waves to high-frequency electromagnetic waves, complex impedance matching, and microwave link budgets.

Unit V: Radio-Frequency Engineering, Transmission Lines & Link Budgets

In the 1880s, the self-taught English physicist and former telegraph operator Oliver Heaviside formulated the *telegrapher's equations* to explain why undersea telegraph cables blurred crisp Morse code pulses into sluggish, overlapping currents. In low-frequency 60 Hz building wiring, where an electromagnetic wave propagating near the speed of light ($c \approx 3 \times 10^8$ m/s) spans a physical wavelength $\lambda = c / f$ of roughly five thousand kilometers, a circuit board ten centimeters wide shares a uniform instantaneous voltage and phase everywhere. Engineers can treat those conductors as simple lumped nodes. Once the operating frequency rises seven orders of magnitude into the radio-frequency (RF) and microwave spectrum—from 146 MHz mountain repeaters ($\lambda \approx 2.05$ m) and 5.8 GHz microwave backhauls ($\lambda \approx 5.17$ cm) to 28 GHz millimeter-wave radar ($\lambda \approx 1.07$ cm)—a coaxial feedline or circuit-board microstrip spans multiple wavelengths. At those frequencies, Heaviside's distributed series inductance L' and shunt capacitance C' govern wave propagation through the line's characteristic impedance $Z_0 = \sqrt{(L' / C')}$.

In May 1929, Bell Telephone Laboratories engineers Lloyd Espenschied and Herman Affel patented the modern concentric coaxial transmission line for wideband long-distance telephony and television. Analyzing an air-insulated copper coaxial cylinder of outer shield inner diameter D and center-conductor diameter d, they discovered a fundamental physical trade-off. High-voltage dielectric breakdown power peaks at a diameter ratio of $D / d \approx 1.65$ ($Z_0 \approx 30\ \Omega$), whereas resistive skin-effect attenuation in the copper conductors reaches its minimum at $D / d \approx 3.59$ ($Z_0 \approx 77\ \Omega$). Two-way radio transmitters and laboratory test instruments standardized on the geometric compromise of $\boldsymbol{Z_0 = 50\ \Omega}$, while receive-only television broadcast and broadband cable networks adopted $\boldsymbol{Z_0 = 75\ \Omega}$ to minimize signal loss over kilometer-long runs. Whenever a forward-traveling wave on a Z_0 line encounters a detuned antenna whose complex load impedance $Z_L = R + jX$ differs from Z_0, boundary continuity of the electric and magnetic fields launches a reflected wave Γ backward toward the transmitter, creating stationary voltage standing waves and return loss. To spare engineers tedious complex-number division when synthesizing quarter-wave transformers and reactive L-networks, RCA engineer Phillip H. Smith invented his conformal transmission-line chart in 1939, mapping every passive impedance ($R \geq 0$) inside the complex unit disk.

Once an electromagnetic wave leaves a matched antenna and radiates across open terrain or space, its power density spreads spherically across millions of square meters. During the 1940s at Bell Labs' Holmdel station in New Jersey, Danish-American radio engineer Harald T. Friis established the two foundational laws of wireless systems engineering. His 1946 **Free-Space Path Loss** transmission equation pairs with Augustin-Jean Fresnel's ellipsoidal diffraction zones to govern line-of-sight microwave links. His 1944 cascaded **Noise Factor** law proves that the noise figure and gain of the very first masthead amplifier above the Johnson–Nyquist thermal noise floor dictate the sensitivity of an entire receiving chain. Two years later, his Bell Labs colleague Claude Elwood Shannon published *A Mathematical Theory of Communication* (1948), establishing the absolute upper bound on error-free data rate that any communication channel of bandwidth B can sustain in the presence of Gaussian thermal noise.

The **SC-50 RF, Microwave, Telecom & Radar** calculator brings these wave-reflection, propagation, and receiver-sensitivity laws into one field instrument. Each chapter in this

Unit follows an end-to-end wireless engineering narrative through three connected mathematical themes:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 13	Diagnosing high reflected power on a 50 Ω mountain repeater coax terminated in $Z_L = 35 - j25\ \Omega$ and designing a reactive match	Decibel-milliwatt scaling, complex reflection Γ, VSWR, Return Loss, Mismatch Loss, Smith Chart L-networks, λ/4 stubs, and LC resonance	1. Complex Impedance & Standing-Wave Reflections • 2. Quarter-Wave Transformers & Smith Chart L-Networks • 3. Lumped LC Tank Resonance & Tuning Solvers
Chapter 14	Closing a 5.8 GHz microwave backhaul across a 12 km forested valley and verifying mid-span pine tree clearance	Friis Free-Space Path Loss, effective isotropic radiated power, logarithmic link budgets, fade margin, and First Fresnel Zone ellipsoid clearance	1. Spherical Wave Spreading & Free-Space Path Loss • 2. Logarithmic Link Budgets & Rain-Fade Margins • 3. Huygens Wavelets & 60% Fresnel Zone Clearance
Chapter 15	Placing a masthead LNA (G_1 = 20 dB, NF_1 = 1.5 dB) ahead of a 6 dB cable loss to rescue a weak satellite telemetry carrier	Johnson–Nyquist thermal noise floor, Friis noise cascade, IIP3/SFDR linearity, Shannon–Hartley capacity, and radar Doppler & R^4 range	1. Thermal Noise Floors & Friis Amplifier Cascades • 2. Third-Order Intermodulation & Dynamic Range • 3. Shannon Channel Capacity, Doppler Shift & R^4 Radar

Chapter 13: The Reflected Wave on the Mountain Repeater Coax

After a February rime-ice storm sweeps across a nine-thousand-foot summit in the Sierra Nevada, the remote telemetry alarm on a 146.0 MHz search-and-rescue VHF repeater trips into high-reflected-power foldback. Driving a 50 W (+47.0 dBm) solid-state power amplifier into a $Z_0 = 50\ \Omega$ coaxial feedline, the repeater normally delivers nearly its full output into a tower-mounted dipole array. When the field engineer snowshoes into the summit shelter and connects a vector network analyzer to the bottom of the feedline, the instrument reveals that ice loading on the antenna elements has detuned the feedpoint to a complex load impedance of $Z_L = 35 - j25\ \Omega$—dropping the radiation resistance to 35 Ω while introducing 25 Ω of capacitive reactance.

On a high-frequency transmission line, an impedance mismatch does far more than reduce radiated signal strength. Because the electric and magnetic fields at the antenna terminals must satisfy Ohm's law for both the 50 Ω cable and the $35 - j25\ \Omega$ load simultaneously, a fraction of the incident voltage wave reflects backward down the coaxial line toward the transmitter. As the backward-traveling reflected wave interferes with the forward-traveling wave, it erects stationary voltage peaks and current troughs along the cable that can punch through coaxial dielectric insulation or destroy the collector junction of the output transistor.

Before climbing the icy tower or soldering an emergency matching network in the shelter, the engineer works through three connected RF analyses:

1. Convert the transmitter power into decibel-milliwatts and evaluate the complex reflection coefficient Γ, Voltage Standing Wave Ratio (VSWR), return loss, and reflected power in watts for $Z_L = 35 - j25\ \Omega$.
2. Determine the physical length of a quarter-wavelength coaxial matching section and the quality factor Q, series inductance, and shunt capacitance of a lumped reactive L-network.
3. Calculate the resonant frequency of an LC tank and solve backward for the exact tuning capacitance needed to cancel parasitic feedpoint inductance.

13.1 Transmission-Line Reflections, Standing Waves, and Decibel Power

When radio-frequency energy travels from a transmitter output stage to an antenna feedpoint, the coaxial cable behaves as a distributed electromagnetic waveguide. Tracking forward and reflected power along that waveguide requires anchoring power measurements to the logarithmic milliwatt scale and solving the boundary equations at the load.

13.1.1 Logarithmic Milliwatt Scaling and Characteristic Impedance

Because radio signals span eighteen orders of magnitude—from kilowatt broadcast transmitters down to femtowatt deep-space telemetry carriers—RF engineering anchors its absolute power scale to **one milliwatt** (1 mW = 10^{-3} W). Any linear power P_{mW} (in milliwatts) or P_{W} (in watts) converts to decibels relative to one milliwatt ([dBm], [↶] [→dBm]) or one watt ([↷] [→dBW]) through the base-ten logarithm and inverts back to linear milliwatts ([↷] [→mW]) via base-ten exponentiation, while across a matched real characteristic impedance Z_0 ([↷] [Z_0], with voltage conversions [↶] [→dBV] and [↷] [→µV]) the root-mean-square RF voltage V_{rms} satisfies Ohm's power law:

$$P_{\mathrm{dBm}} = 10\log_{10}\left(\frac{P_{\mathrm{mW}}}{1\ \mathrm{mW}}\right) = 10\log_{10}(P_{\mathrm{W}}) + 30, \qquad P_{\mathrm{mW}} = 10^{P_{\mathrm{dBm}}/10}, \qquad V_{\mathrm{rms}} = \sqrt{P_{\mathrm{W}} \cdot Z_0} \qquad \textbf{(Eq. 13.1)}$$

In these expressions, P_{dBm} is the absolute power level in dBm, P_{mW} is linear power in milliwatts, $P_{\mathrm{W}} = P_{\mathrm{mW}} / 1000$ is linear power in watts, and Z_0 is the transmission-line characteristic impedance in ohms (Ω, standardized at 50 Ω for transmitters and 75 Ω for video/broadcast coax).

Every +10 dB step multiplies linear power tenfold and every +3.0103 dB step doubles linear power: 0 dBm equals **1.0 mW** ($V_{\mathrm{rms}} \approx 0.2236$ V across 50 Ω), +20 dBm equals **100 mW** (0.10 W), +30 dBm equals 1000 mW = **1.0 W** ($V_{\mathrm{rms}} \approx 7.071$ V across 50 Ω), and our mountain repeater's P_{W} = 50 W (50,000 mW) output corresponds to $P_{\mathrm{dBm}} = 10\ \log_{10}(50{,}000) \approx +\mathbf{46.990\ dBm}$, developing a matched root-mean-square voltage of $V_{\mathrm{rms}} = \sqrt{(50 \times 50)} = \mathbf{50.0\ V}$ across the 50 Ω line.

13.1.2 Complex Reflection Coefficient, Standing Wave Ratio, Return Loss, and Mismatch Loss

When a forward-traveling electromagnetic wave with voltage phasor V^+ and current phasor $I^+ = V^+ / Z_0$ propagates down a lossless transmission line of real characteristic impedance Z_0 and reaches an antenna of complex load impedance $Z_L = R + jX$ ([jX] in [↶] [CMPLX], where R is series resistance and X is inductive $+jX$ or capacitive $-jX$ reactance in ohms), boundary continuity requires total terminal voltage $V_L = V^+ + V^-$ and total terminal current $I_L = (V^+ - V^-) / Z_0$ to satisfy $V_L / I_L = Z_L$. Solving that boundary condition defines the complex **Voltage Reflection Coefficient** Γ ([Γ], inverted back to complex impedance via [Z←Γ]):

$$\Gamma = \frac{V^-}{V^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(R - Z_0) + jX}{(R + Z_0) + jX} = |\Gamma| \angle \theta, \qquad |\Gamma| = \sqrt{\frac{(R - Z_0)^2 + X^2}{(R + Z_0)^2 + X^2}} \qquad \textbf{(Eq. 13.2)}$$

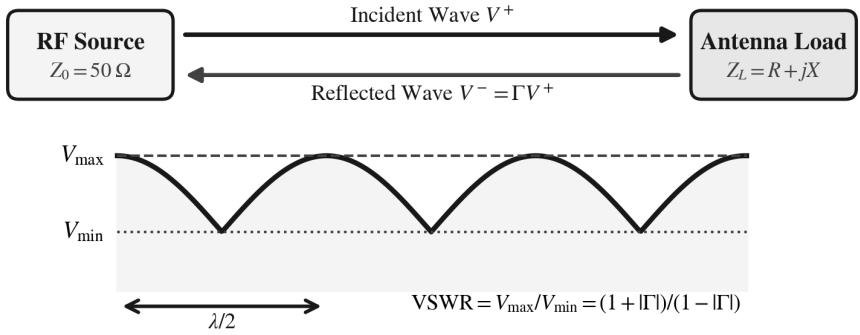


Figure 13.1: *Superposition of forward and reflected traveling waves along a transmission line forming a voltage standing wave envelope (VSWR).*

As the reflected voltage wave V^- travels backward and superposes with the incident wave V^+, constructive interference at half-wavelength (λ / 2) intervals produces stationary voltage maxima $V_{max} = |V^+|(1 + |\Gamma|)$, while destructive interference shifted by a quarter-wavelength (λ / 4) produces voltage minima $V_{min} = |V^+|(1 - |\Gamma|)$. The ratio of peak to trough voltage defines the dimensionless **Voltage Standing Wave Ratio** (SWR, SWR), the decibel attenuation of the reflected power fraction $|\Gamma|^2$ defines the **Return Loss** (RL, in dB), and the decibel reduction in forward power actually delivered into the load ($1 - |\Gamma|^2$) defines the **Mismatch Loss** (ML_{dB}, in dB):

$$VSWR = \frac{V_{max}}{V_{min}} = \frac{1+|\Gamma|}{1-|\Gamma|} \Leftrightarrow |\Gamma| = \frac{VSWR-1}{VSWR+1} \qquad \textbf{(Eq. 13.3)}$$

$$RL_{dB} = -10\log_{10}(|\Gamma|^2) = -20\log_{10}|\Gamma|, \qquad ML_{dB} = -10\log_{10}(1-|\Gamma|^2) \qquad \textbf{(Eq. 13.4)}$$

For our ice-detuned mountain repeater antenna with Z_L = 35 - *j*25 Ω on a Z_o = 50 Ω coaxial feedline, the numerator impedance difference is Z_L - Z_o = -15 - *j*25 Ω, whose magnitude is √((-15)² + (-25)²) = √(225 + 625) = √850 ≈ 29.1548 Ω. The denominator impedance sum is Z_L + Z_o = 85 - *j*25 Ω, whose magnitude is √(85² + (-25)²) = √(7225 + 625) = √7850 ≈ 88.6002 Ω. Dividing numerator by denominator gives the reflection magnitude and phase:

$$|\Gamma| = \frac{\sqrt{850}}{\sqrt{7850}} = \sqrt{\frac{17}{157}} \approx \mathbf{0.3291}, \qquad \theta = \text{atan2}(-25,-15) - \text{atan2}(-25,85) = -120.96^\circ - (-16.39^\circ) = \mathbf{-104.57^\circ} \qquad \textbf{(Eq. 13.5)}$$

Substituting |Γ| ≈ 0.3291 into the standing-wave, return-loss, and mismatch-loss relations yields:

$$VSWR = \frac{1+0.3291}{1-0.3291} = \frac{1.3291}{0.6709} \approx \mathbf{1.981:1}, \quad RL_{dB} = -20\log_{10}(0.3291) \approx \mathbf{9.654}\,\text{dB}, \quad ML_{dB} = -10\log_{10}(1-0.3291^2) \approx \mathbf{0.498}\,\text{dB} \qquad \textbf{(Eq. 13.6)}$$

Because the reflected power fraction is $|\Gamma|^2$ = 17 / 157 ≈ **10.828%**, our 50 W repeater transmitter suffers 50 W × 0.10828 ≈ **5.414 W** (+37.335 dBm) of reflected power slamming back into the amplifier output stage, while 50 W - 5.414 W = **44.586 W** (89.172%) enters the antenna:

SWR	Reflection $\lvert\Gamma\rvert$	Return Loss RL (dB)	Mismatch Loss ML (dB)	Reflected Power $\lvert\Gamma\rvert^2$ (%)	Transmitted Power 1-$\lvert\Gamma\rvert^2$ (%)
1.00 : 1	0.0000	∞ dB	0.000 dB	0.00%	100.00%
1.20 : 1	0.0909	20.83 dB	0.036 dB	0.83%	99.17%
1.50 : 1	0.2000	13.98 dB	0.177 dB	4.00%	96.00%
1.98 : 1	0.3291	9.65 dB	0.498 dB	10.83%	89.17%
3.00 : 1	0.5000	6.02 dB	1.249 dB	25.00%	75.00%
6.00 : 1	0.7143	2.92 dB	3.100 dB	51.02%	48.98%

13.2 Conformal Smith Chart Transformations and Reactive Matching Networks

To eliminate the reflected wave and restore a 1.00 : 1 standing wave ratio on the main 50 Ω feedline, the engineer must insert a lossless impedance-matching network between the coaxial trunk and the antenna terminals.

13.2.1 Normalized Impedance Mapping and Quarter-Wave Transformers

Dividing any complex load impedance $Z_L = R + jX$ by the system reference impedance Z_0 produces the dimensionless normalized impedance $z = Z_L \,/\, Z_0 = (R \,/\, Z_0) + j(X \,/\, Z_0) = r + jx$ (for our 35 - j25 Ω repeater load on a 50 Ω line, $r = 35/50 =$ **0.70** and $x =$ -25/50 = **-0.50**). Under the bilinear Möbius transformation $\Gamma = (z - 1) \,/\, (z + 1)$, Phillip H. Smith showed in 1939 that the entire right half-plane of passive resistances ($r \geq 0$) maps conformally into the unit circle $|\Gamma| \leq 1$ on the **Smith Chart** (Γ, ↻ ▦).

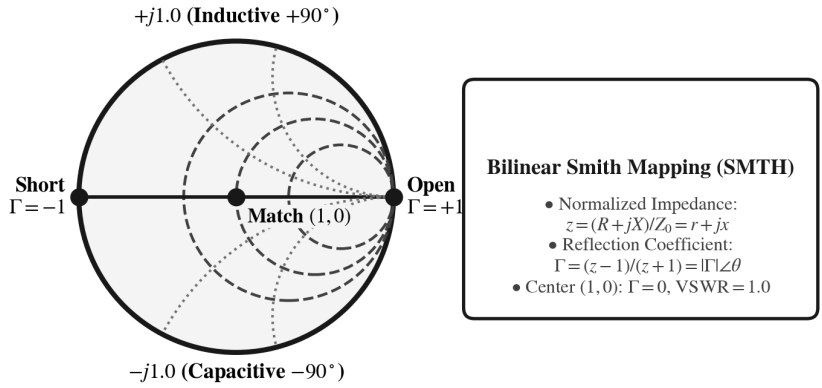


Figure 13.2: *Complex reflection-coefficient plane and normalized impedance loci of the Smith Chart.*

At the center origin of the Smith Chart ($\Gamma = 0 + j0$), the normalized impedance is $z = 1 + j0$ ($Z_L = Z_0$). Moving a physical distance ℓ along a lossless transmission line of wavelength λ (↶ λ/L) rotates the reflection phasor Γ clockwise around the origin at constant radius $|\Gamma|$ through an electrical phase angle of $2\beta\ell = 4\pi\ell \,/\, \lambda$. When the line section has an electrical length of one quarter-wavelength ($\ell = \lambda \,/\, 4$), the rotation angle is $2\beta(\lambda \,/\, 4) = \pi$ radians (180°), which inverts normalized impedance ($z_{\text{in}} = 1 \,/\, z_L$) so that a **Quarter-Wave Transformer** (↻ TLN, and Γ opening the ↻ Z_0 workspace $\mathbf{Z} = \sqrt{(\mathbf{S} \times \mathbf{L})}$) of characteristic impedance Z_{match} matches a real source impedance Z_S to a real load resistance Z_L:

$$\lambda = \frac{c}{f} = \frac{299.792458}{f_{\text{MHz}}}\ \text{[m]}, \qquad \ell_{\lambda/4} = \frac{\lambda}{4} \cdot \text{VF}, \qquad Z_{\text{match}} = \sqrt{Z_S \cdot Z_L} \qquad \textbf{(Eq. 13.7)}$$

Here f_{MHz} is the operating frequency in megahertz, λ is the free-space wavelength in meters, VF = 1 / √(ε_r) is the coaxial cable velocity factor, and Z_{match} is the geometric-mean characteristic impedance in ohms. At the repeater frequency of f = 146.0 MHz, the free-space wavelength is λ = 299.792458 / 146.0 **≈ 2.0534 m** and a free-space quarter-wave whip measures $\ell_{\lambda/4}$ = 2.0534 / 4 **≈ 0.5133 m** (51.33 cm). Cut from foam-dielectric coaxial cable with velocity factor VF = 0.82, the physical quarter-wave stub length is 0.5133 × 0.82 **≈ 0.4209 m** (42.1 cm). To match a Z_S = 50 Ω line to a Z_L = 112.5 Ω antenna resistance, the required quarter-wave section impedance is Z_{match} = √(50 × 112.5) = √5625 = **75.0 Ω**, allowing standard 75 Ω video coax to serve as an exact quarter-wave transformer.

13.2.2 Lumped L-Network Synthesis and Resonant Tank Circuits

When matching a lower resistance R_{low} up to a higher resistance R_{high} at VHF or HF frequencies using compact lumped inductors and capacitors, a two-element reactive **L-network** ([Γ]) places a reactive element X_{series} in series with R_{low} and a complementary opposite-sign reactive element X_{shunt} in parallel across R_{high}. Equivalence of series and parallel resistance-reactance networks defines the nodal **Quality Factor** Q, while inside RF filters and oscillators an inductance L (in henries, H) and capacitance C (in farads, F) resonate at frequency f_0 ([↰] [f_0], with reactance [↰] [XLC] and RC cutoff frequency [↱] [f_C] given by $f_c = 1 / (2\pi\ R\ C)$):

$$Q = \sqrt{\frac{R_{\text{high}}}{R_{\text{low}}} - 1}, \qquad |X_{\text{series}}| = Q \cdot R_{\text{low}}, \qquad |X_{\text{shunt}}| = \frac{R_{\text{high}}}{Q}, \qquad f_0 = \frac{1}{2\pi\sqrt{L \cdot C}} \qquad \textbf{(Eq. 13.8)}$$

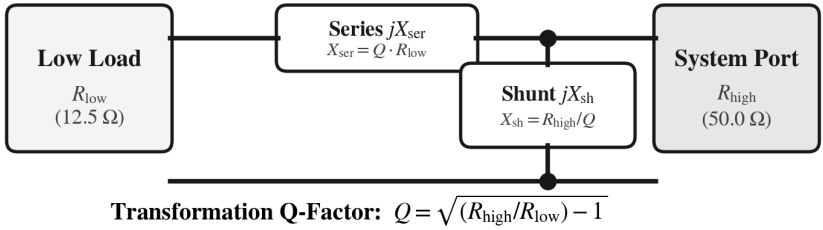


Figure 13.3: *Lossless reactive L-network impedance matching topology transforming a low load resistance to 50 ohms.*

Consider matching our repeater's resistive component R_{low} = 35 Ω up to R_{high} = 50 Ω at f = 146.0 MHz. The required nodal quality factor is Q = √((50 / 35) - 1) = √(0.42857) **≈ 0.6547**, which specifies a net series inductive reactance of $|X_{\text{series}}|$ = 0.6547 × 35 **≈ 22.91 Ω** and a shunt capacitive reactance of $|X_{\text{shunt}}|$ = 50 / 0.6547 **≈ 76.38 Ω**. Because the iced antenna already exhibits -j25 Ω of series capacitive reactance, the series inductor must simultaneously cancel that -25 Ω reactance and supply +22.91 Ω of net inductive reactance, requiring a total inductive reactance of X_L = 25 + 22.91 = **47.91 Ω** (L = 47.91 / (2π × 146 × 10^6) **≈ 52.23 nH**) paired with a shunt capacitor of C = 1 / (2π × 146 × 10^6 × 76.38) **≈ 14.27 pF**. Similarly, when stepping a R_{low} = 10 Ω power-transistor collector up

to R_{high} = 50 Ω, the nodal quality factor is Q = √((50/10) - 1) = **2.00**, giving $|X_{\text{series}}|$ = **20.0 Ω** and $|X_{\text{shunt}}|$ = **25.0 Ω**.

13.3 Diagnosing Mismatches and Synthesizing Matching Networks on the Stack

The **SC-50** combines **Native Complex RF Mode** ([↶] [CMPLX], activating the six top-row softkeys [Γ], [Z←Γ], [jX], [SWR], [RL], [Z₀]) for complex load impedances $Z_L = R + jX$ against toggleable system impedance [→] [Z₀] (50 Ω ↔ 75 Ω), automatic Smith Chart parameter logging ([→] [▦]), decibel-milliwatt scaling ([dBm], [↶] [→dBm], [→] [→mW], [→] [→dBW]), and matching primitives ([SWR], [Γ], [→] [TLN], [↶] [f₀], [→] [fc], [↶] [XLC], [↶] [λ/L]) with the **Bidirectional Equation Solver Workspaces** ([→] [SOLVE] / [XEQ]) on [SWR] (**S = (1 + G) / (1 - G)**, softkeys [S], [G]), [Γ] (**ZMAT: Z = √(S × L)**, softkeys [Z], [S], [L]), and [↶] [f₀] (**F0: F = 1 / (2π√(L×C))**, softkeys [F], [L], [C ON]):

Button / Menu	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Solver Schema
[dBm] / [↶] [→dBm]	Linear Power to dBm	—	—	Power P_{mW} (mW)	$P_{\text{dBm}} = 10\ \log_{10}(P_{\text{mW}})$ (paired with [→] [→dBW], [↶] [→dBV], [→] [→µV])
[→] [→mW]	dBm to Linear Power	—	—	Power P_{dBm} (dBm)	$P_{\text{mW}} = 10^{P_{\text{dBm}}\ /\ 10}$
[↶] [CMPLX]	Native Complex RF Menu	—	Resistance R (Ω)	Reactance X (Ω)	Softkeys: [Γ] [Z←Γ] [jX] [SWR] [RL] [Z₀] ($Z_L = R + jX \leftrightarrow \Gamma$)
[SWR]	Standing Wave Ratio	—	—	$\lvert\Gamma\rvert$ or $Z_L = R + jX$	Stack: VSWR = (1 + |Γ|) / (1 - |Γ|) (fills [→] [▦]); Solver: **S = (1+G)/(1-G)** ([S],[G])
[RL]	Return Loss (dB)	—	—	Γ or Z_L	Softkey in [↶] [CMPLX]: RL_{dB} = -20 $\log_{10}\lvert\Gamma\rvert$
[Γ]	Smith Match & [→] [Z₀] Solver	—	Source Z_S (Ω)	Load Z_L (Ω)	Stack: $Q = \sqrt{((R_{\text{high}}\ /\ R_{\text{low}}) - 1)}$; Opens [→] [Z₀] (**Z = √(S×L)**, [Z],[S],[L])
[→] [TLN]	λ/4 Transformer Impedance	—	Source Z_S (Ω)	Load Z_L (Ω)	Immediate stack: $Z_{\text{match}} = \sqrt{(Z_S \cdot Z_L)}$
[↶] [f₀] / [→] [fc]	LC Resonance & RC Cutoff	—	Inductance L (H)	Capacitance C (F)	Stack: $f_0 = 1\ /\ 2\pi\sqrt{(L \cdot C)}$; Solver: **F = 1 / (2π√(L×C))** ([F],[L],[C ON])
[↶] [λ/L]	Free-Space Wavelength	—	—	Frequency f_{MHz}	λ = 299.792458 / f_{MHz} (m)

13.3.1 Evaluating Complex Repeater Mismatch, Return Loss, and Reflected Watts on the Stack

Inside the mountain repeater shelter, the engineer opens Native Complex Mode ([↶] [CMPLX]), packs the VNA measurement Z_L = 35 - j25 Ω into Register X with softkey [jX], presses [Γ] / [SWR] (or primary [SWR]) to evaluate VSWR and populate the Smith Chart table [→] [▦] (**|GAMMA| = 0.3291**, **GAMMA ANG = -104.57°**, **VSWR = 1.981**, **RET LOSS = 9.654 dB**, **r/z0 = 0.70**, **x/z0 = -0.50**), presses softkey [RL] for Return Loss, and checks the reflected power for the 50,000 mW (50 W) transmitter:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↶] [CMPLX] **35** [ENTER] **25** [+/−] [jX] [+]	0	0	0	35 - *j*25	Pack complex load impedance Z_L = 35 - *j*25 Ω into Register *X*
[SWR]	0	0	0	1.981	Evaluate VSWR = 1.981 : 1 against Z_0 = 50 Ω ([Z_0]) and fill Smith table [↷] [▦]
35 [ENTER] **25** [+/−] [jX] [+] [RL]	0	0	1.981	9.653	Compute Return Loss RL = 9.653 dB from complex load (|Γ| = 0.3291)
50000 [↶] [→dBm] [x⇄y] [-]	0	0	1.981	37.337	Subtract 9.653 dB return loss from +46.990 dBm (50 W) to get reflected dBm
[↷] [→mW] **1000** [÷]	0	0	1.981	**5.416**	Convert +37.337 dBm to milliwatts (5416 mW) and divide by 1000 for 5.416 W

The stack confirms that **5.42 W** of RF power is reflecting back down the line at VSWR = **1.981 : 1**, explaining why the repeater's protection circuit triggered.

13.3.2 Designing Quarter-Wave Stubs, L-Networks, and Resonant Tanks on the Stack

Next, the engineer evaluates the 146.0 MHz wavelength and quarter-wave dimension ([↶] [λ/L]), computes the quarter-wave transformer impedance for a 50 Ω to 112.5 Ω port ([↷] [TLN]), solves an L-network match from 10 Ω to 50 Ω ([Γ]), and checks a 100 nH / 10 pF harmonic trap ([↶] [f_0]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
146 [↶] [λ/L] **4** [÷]	0	0	0	0.5133	Compute free-space λ = 2.0534 m at 146 MHz and divide by 4 for $\ell_{\lambda/4}$ = 0.5133 m
50 [ENTER] **112.5** [↷] [TLN]	0	0	0.5133	75	Quarter-wave transformer impedance $Z_{\text{match}} = \sqrt{(50 \times 112.5)} = 75.0$ Ω
50 [ENTER] **10** [Γ]	0.51334	0.5133	75	2	L-network nodal quality factor $Q = \sqrt{((50/10) - 1)} = 2.00$ (and opens [↷] [Z_0] workspace)
1 [E] **7** [+/−] [ENTER] **1** [E] **11** [+/−] [↶] [f_0]	75	75	2	**1.5915e8**	Resonant frequency f_0 = 159.155 MHz for L = 100 nH and C = 10 pF

Multiplying the L-network quality factor $Q = 2.00$ by $R_{\text{low}} = 10$ Ω gives series reactance $|X_{\text{series}}| =$ **20.0 Ω**, while dividing $R_{\text{high}} = 50$ Ω by $Q = 2.00$ gives shunt reactance $|X_{\text{shunt}}| =$ **25.0 Ω**.

13.3.3 Back-Solving Resonant Tuning Capacitance and Reflection Limits in the Solver Workspace

Suppose an auxiliary 433.92 MHz (4.3392×10^8 Hz) UHF telemetry link exhibits $L = 18.5$ nH (1.85×10^{-8} H) of parasitic feedpoint inductance and the engineer must select the exact series capacitor C that resonates the feedpoint at 433.92 MHz, while also verifying the maximum reflection magnitude |Γ| ([G]) allowed by a VSWR ≤ 1.50 : 1 ([S]) specification. Opening the [↶] [f_0] (**F = 1 / (2π√(L×C))**) and [SWR] (**S = (1 + G) / (1 - G)**) Solver Workspaces solves both inverse problems directly:

Keystroke	Solver Display / Register X	What Happens in the [↖] [f₀] & [SWR] Solver Workspaces
[↖] [f₀]	**F=1/(2π√(L×C))**	Open the LC Resonance ([↖] [f₀]) Solver Workspace with softkeys [F], [L], [C ON]
4.3392 [E] **8** [F]	4.3392×10^{8}	Store target resonant frequency $F = 433.92$ MHz into softkey [F] ([LFU0])
1.85 [E] **8** [+/−] [L]	1.8500×10^{-8}	Store parasitic inductance $L = 18.5$ nH into softkey [L] ([LFU1])
[↗] [SOLVE] [C ON]	$\mathbf{7.2856 \times 10^{-12}}$	Solve backward for matching capacitance $C = 1 / ((2\pi F)^2 L) =$ **7.2856 pF**
[SWR]	**S=(1+G)/(1-G)**	Open the Standing Wave Ratio ([SWR]) Solver Workspace with softkeys [S], [G]
1.5 [S]	1.5000	Store specification limit $S = 1.50$ into softkey [S] ([LFU0])
[↗] [SOLVE] [G]	**0.2000**	Solve backward for reflection magnitude $\lvert\Gamma\rvert = (1.5 - 1)/(1.5 + 1) =$ **0.2000**

13.3.4 Try It on the Stack

Practice complex transmission-line diagnostics, quarter-wave stub sizing, and reactive matching on the stack and in the Solver Workspace with these three RF workbench exercises:

1. **75-Ohm Broadcast Coax Mismatch and Aviation Quarter-Wave Whip:** Open [↖] [CMPLX], toggle the reference impedance to 75 Ω ([Z₀]), pack load impedance $Z_L = 50 + j25$ Ω (**50** [ENTER] **25** [jX]), and evaluate [SWR] (then press [Z₀] again to restore 50 Ω). Next, compute the quarter-wave whip length at the 121.5 MHz aviation emergency frequency (**121.5** [↖] [λ/L] **4** [÷]). *(Verified Register X results: VSWR* ***≈ 1.768 : 1*** *and* $\boldsymbol{\ell_{\lambda/4}}$ ***≈ 0.617 m****.)*

2. **Commercial Specification Limit Return Loss and Reflected Power:** A commercial cellular sector antenna specifies a maximum reflection magnitude of $\lvert\Gamma\rvert = 0.20$. Compute its standing wave ratio (**0.2** [SWR]), return loss (**0.2** [↖] [LOG] **20** [+/−] [×]), and reflected power in watts when driven by a 40 W (40,000 mW) base station (**40000** [↖] [→dBm] [x⇄y] [−] [↗] [→mW] **1000** [÷]). *(Verified Register X results: VSWR =* ***1.50 : 1****, RL* ***≈ 13.979 dB****, and* P_{refl} ***= 1.60 W****.)*

3. **Power Transistor L-Network and Backward Tank Inductance:** Find the L-network quality factor Q and series reactance $\lvert X_{series}\rvert$ required to match a 12.5 Ω transistor output to 50 Ω (**50** [ENTER] **12.5** [Γ] **12.5** [×]), then open [↖] [f₀], store **159.155** [E] **6** [F] and **40** [E] **12** [+/−] [C ON], and press [↗] [SOLVE] [L] to find the required tank inductance L. *(Verified Register X results:* Q ***≈ 1.732***, $\lvert X_{series}\rvert$ ***≈ 21.651 Ω****, and* L ***= 250.0 nH****.)*

With the repeater feedline matched to 50 Ω and full forward power radiating from the tower, we follow the electromagnetic wave into open air in Chapter 14 to close a twelve-kilometer microwave backhaul across a forested mountain valley.

Chapter 14: Clearing the Ridge on a Twelve-Kilometer Microwave Backhaul

To link the mountain repeater shelter back to the county emergency operations center without paying for leased fiber across wilderness terrain, the communications team is commissioning a point-to-point unlicensed $f = 5800$ MHz (5.8 GHz) microwave backhaul across a $d = 12.0$ km forested valley. At each end of the path, a tower crew mounts a parabolic dish antenna delivering +26.0 dBi of isotropic gain ($G = G_{\text{tx}} + G_{\text{rx}} = 52.0$ dB combined net antenna gain after short waveguide jumpers), driven by a microwave transceiver with transmitter output power $P_{\text{tx}} = 250$ mW (+23.979 dBm). To sustain 256-QAM modulation with a 25 dB rain-fade margin above its -78.0 dBm threshold, the receiver at the far end of the valley requires an unfaded signal level of at least $P_{\text{rx}} \geq$ -55.0 dBm.

Standing on the transmit tower platform with binoculars, the lead engineer spots a potential hazard halfway across the valley ($d_{\text{mid}} = 6.0$ km from either tower): a granite ridge crowned with eighty-foot ponderosa pines rises just below the optical sightline between the two dishes. A casual observer might assume that as long as a laser pointer aimed from one dish to the other clears the treetops, the microwave link will suffer no obstruction loss. In wave optics and radio propagation, however, energy does not travel along an infinitely thin geometric pencil ray. Instead, it propagates as an expanding wavefront whose constructive phase coherence fills a cigar-shaped volume of space around the sightline known as the **First Fresnel Zone**.

If the mid-span pine needles intrude into that three-dimensional ellipsoid, waves grazing the treetops travel roughly half a wavelength ($\lambda / 2 \approx 2.58$ cm at 5.8 GHz) farther than the direct optical ray, arriving 180° out of phase at the receiving dish and canceling the direct signal by 6 dB to 15 dB. Before bolting the dishes at their current tower height, the engineer works through three propagation problems:

1. Calculate the Friis Free-Space Path Loss across the 12.0 km valley at 5800 MHz.
2. Evaluate the complete logarithmic link budget and fade margin, and solve backward for maximum hop range and required transmitter power.
3. Compute the First Fresnel Zone radius at mid-span to verify that the sightline clears the pine ridge by the 60% diffraction-free criterion.

14.1 Spherical Wave Spreading and the Friis Transmission Equation

When an electromagnetic wave leaves a transmitting antenna and travels through unobstructed air or vacuum, no energy is dissipated as heat in the vacuum. Instead, the signal weakens with distance because a fixed number of watts is spread over an ever-larger spherical surface while the receiving antenna intercepts only a tiny patch of that sphere.

14.1.1 Isotropic Effective Aperture and Free-Space Path Loss

Imagine a hypothetical **isotropic radiator** emitting total radio-frequency power P_{tx} uniformly in all directions. At distance d (in meters), that radiated power is distributed evenly over a sphere of surface area $4\pi\ d^2$, producing a power flux density of $S = P_{tx}\ /\ (4\pi\ d^2)$ W/m². By electromagnetic reciprocity and thermodynamics, an isotropic receiving antenna immersed in a wave of wavelength $\lambda = c\ /\ f$ (= 299,792,458 m/s) captures an **effective aperture area** of $A_{eff,iso} = \lambda^2\ /\ (4\pi)$. Multiplying the incident flux density S by $A_{eff,iso}$ yields Harald T. Friis's 1946 free-space transmission ratio:

$$\frac{P_{rx,\,iso}}{P_{tx,\,iso}} = \left(\frac{\lambda}{4\pi d}\right)^2 = \left(\frac{c}{4\pi df}\right)^2, \qquad \text{FSPL} = \left(\frac{4\pi df}{c}\right)^2 \qquad \textbf{(Eq. 14.1)}$$

Notice why higher-frequency microwave links suffer higher path loss between isotropic antennas even though vacuum is non-absorptive: as frequency f increases and wavelength $\lambda = c\ /\ f$ shrinks, the electrical capture area $\lambda^2\ /\ (4\pi)$ of an isotropic antenna shrinks with the square of wavelength. Converting the dimensionless inverse ratio FSPL into decibels and expressing path length d_{km} in kilometers ($d = 10^3\ d_{km}$) and carrier frequency f_{MHz} in megahertz ($f = 10^6\ f_{MHz}$), where $20\ \log_{10}(4\pi \times 10^9\ /\ 299{,}792{,}458) \approx 32.4418$ dB, gives the engineering **Free-Space Path Loss** ( ) equation:

$$\text{FSPL}_{dB} = 20\log_{10}(d_{km}) + 20\log_{10}(f_{MHz}) + 32.44 \qquad \textbf{(Eq. 14.2)}$$

In this expression, d_{km} is the line-of-sight hop distance in kilometers, f_{MHz} is the carrier frequency in megahertz, and FSPL_{dB} is the positive isotropic path attenuation in decibels. Doubling either the hop distance d_{km} or the carrier frequency f_{MHz} adds $20\ \log_{10}(2) \approx$ **6.02 dB** of path loss.

For our mountain valley backhaul spanning $d_{km} = 12.0$ km at $f_{MHz} = 5800$ MHz, the distance term contributes $20\ \log_{10}(12.0) \approx 21.584$ dB and the frequency term contributes $20\ \log_{10}(5800) \approx 75.269$ dB. Summing the three terms yields the free-space path loss across the valley:

$$\text{FSPL}_{dB} = 21.584 + 75.269 + 32.44 = \mathbf{129.292}\ \text{dB} \qquad \textbf{(Eq. 14.3)}$$

An attenuation of 129.292 dB means that an isotropic receiving antenna 12 km away would capture only $10^{-12.9292} \approx 1.177 \times 10^{-13}$ (roughly one ten-trillionth) of the transmitted power.

14.1.2 Logarithmic Link Budgets and Fade Margin Accounting

To overcome that thirteen-order-of-magnitude free-space spreading loss, microwave engineers collimate the transmitted wave and enlarge the receiving capture aperture using high-gain parabolic dish antennas. Because logarithms turn multiplication and division of linear power ratios into addition and subtraction of decibels, the **Effective Isotropic Radiated Power** ($\text{EIRP}_{dBm} = P_{tx,dBm} + G_{tx,dBi} - L_{line,dB}$) and end-to-end **Link Budget** ( ) combine transmitter output power P_{tx} (in dBm), total net antenna and amplifier gains $G = G_{tx} + G_{rx}$ (in dBi), and total link losses $L = \text{FSPL}_{dB} + L_{misc}$ (in dB):

$$P_{rx,\,dBm} = P_{tx,\,dBm} + G_{dB} - L_{dB}, \qquad M_{fade} = P_{rx,\,dBm} - S_{min,\,dBm} \qquad \textbf{(Eq. 14.4)}$$

Here $P_{rx,dBm}$ is the unfaded received signal power in dBm, $S_{min,dBm}$ is the receiver's minimum sensitivity threshold for a target modulation scheme and bit-error rate, and M_{fade} is the link's **Fade Margin** in dB against heavy rain, snow, or atmospheric multipath ducting.

On our 12.0 km, 5800 MHz link, the P_{tx} = 250 mW transmitter delivers 10 $\log_{10}(250) \approx$ **+23.979 dBm** (and an EIRP of 23.979 + 26.0 = **+49.979 dBm**). Adding the combined +52.0 dB dish gain and subtracting the 129.292 dB free-space path loss yields the unfaded received signal level:

$$P_{rx,\,dBm} = 23.979\ \text{dBm} + 52.0\ \text{dB} - 129.292\ \text{dB} = \mathbf{-53.313}\ \text{dBm} \qquad \textbf{(Eq. 14.5)}$$

Compared against the receiver's -78.0 dBm demodulation threshold, the link achieves an unfaded fade margin of M_{fade} = -53.313 - (-78.0) = **24.687 dB** (and meets the -55.0 dBm design target with 1.69 dB to spare)—provided the mid-span pine ridge does not introduce diffraction loss.

14.2 Huygens Wavelets and Fresnel Zone Diffraction Clearance

Why does an obstacle that lies *below* the straight-line optical path between two antennas still attenuate a microwave signal? In 1818, the French physicist Augustin-Jean Fresnel answered that question by combining Christiaan Huygens's principle of secondary spherical wavelets with phase interference.

14.2.1 Ellipsoidal Phase Zones and Half-Wavelength Destructive Interference

At any intermediate plane perpendicular to the straight line of sight of length $d = d_1 + d_2$ between a transmitter and a receiver, the advancing wavefront can be decomposed into concentric annular rings—**Fresnel Zones**—defined by the excess path length $\Delta d = (r_{tx} + r_{rx}) - d$ that a secondary wavelet travels from the transmitter through a point at radial offset r_n to the receiver. The boundary of the nth Fresnel Zone is the locus of points where that excess path length equals n half-wavelengths ($\Delta d = n\,\lambda / 2$). At the midpoint of a link of total distance d_{km} (in kilometers, where $d_1 = d_2 = d_{km} / 2$) operating at frequency f_{GHz} (in gigahertz), paraxial geometry ($\Delta d \approx r_1^2 / (2\,d_1) + r_1^2 / (2\,d_2) = 2\,r_1^2 / d = \lambda / 2$) yields the midpoint radius r_1 of the **First Fresnel Zone** (↗ F1, in meters):

$$r_1 = \sqrt{\frac{\lambda \cdot d}{4}} = 17.32\sqrt{\frac{d_{km}}{4 f_{GHz}}} = 8.656\sqrt{\frac{d_{km}}{f_{GHz}}}\ \ [\text{m}] \qquad \textbf{(Eq. 14.6)}$$

In this expression, d_{km} is the total hop length in kilometers, f_{GHz} is the carrier frequency in gigahertz, and r_1 is the midpoint cross-sectional radius in meters of the first phase-coherent ellipsoid.

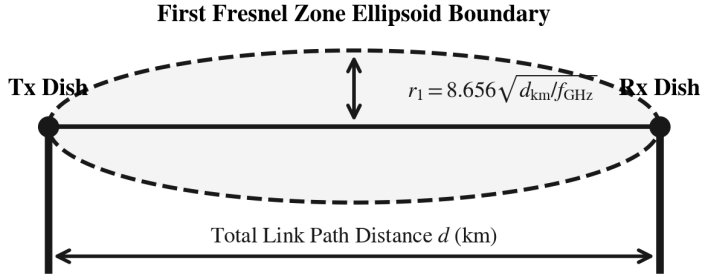


Figure 14.1: *First Fresnel Zone prolate ellipsoid clearance radius r1 between line-of-sight microwave antennas.*

Every Huygens wavelet passing *inside* the First Fresnel Zone ($r < r_1$) travels less than λ / 2 farther than the direct axial ray, so its phase shift is less than 180° and it interferes **constructively** at the receiving dish. If a sharp granite ridge or dense canopy of wet pine needles rises right up to the optical sightline (grazing incidence, 0 m clearance), it blocks the entire bottom half of the First Fresnel Zone and introduces **6.0 dB** of knife-edge diffraction loss, cutting received power to one quarter. Even worse, if a smooth ridge or rooftop sits precisely at the boundary r_1 and reflects the wave with a 180° reflection phase reversal plus the 180° (λ / 2) path-length delay, or if an obstacle blocks the inner 60% of the zone, destructive interference can gouge 10 dB to 20 dB out of the link budget.

14.2.2 Sixty-Percent Clearance Criteria and Earth-Bulge Ridge Audits

In microwave path engineering, free-space propagation (≈ 0 dB diffraction loss) is achieved whenever the optical sightline clears all terrain, foliage, and building obstacles by at least **60% of the First Fresnel Zone radius** ($h_{clear} \geq 0.60\ r_1$) after accounting for atmospheric refraction and Earth curvature:

$$h_{\text{clear, min}} = 0.60 \cdot r_1 = 0.60 \times 17.32\sqrt{\frac{d_{km}}{4 f_{GHz}}}, \qquad h_{\text{earth, m}} = \frac{d_1 \cdot d_2}{2KR_{\oplus}} \approx \frac{d_{km}^2}{68.0} \qquad \textbf{(Eq. 14.7)}$$

Here $h_{clear,min}$ is the required obstacle-free radius in meters for zero diffraction loss, and $h_{earth,m}$ is the midpoint vertical bulge of the spherical Earth in meters under a standard $K = 4/3$ effective Earth-radius atmosphere ($R_{\oplus} \approx 6371$ km).

For our $d_{km} = 12.0$ km valley hop at $f_{GHz} = 5.8$ GHz, the midpoint First Fresnel Zone radius is:

$$r_1 = 17.32\sqrt{\frac{12.0}{4 \times 5.8}} = 17.32\sqrt{\frac{12.0}{23.2}} = 17.32\sqrt{0.51724} \approx \mathbf{12.457}\ \text{m} \qquad \textbf{(Eq. 14.8)}$$

Applying the 60% diffraction-free rule specifies a minimum treetop clearance of $h_{clear,min} = 0.60 \times 12.457 \approx \mathbf{7.474\ m}$ (24.5 ft) below the optical line of sight at mid-span, plus an Earth-curvature bulge of $h_{earth} \approx 12.0^2 / 68.0 \approx 2.12$ m. Because the transit survey shows the eighty-foot mid-span pines sitting 14.0 m below the optical sightline, the link clears the entire First Fresnel Zone (14.0 m > 12.457 m) and the -53.313 dBm free-space link budget holds intact.

14.3 Closing the Valley Link Budget and Clearing the Ridge on the Stack

The **SC-50** chains logarithmic milliwatt conversions ([dBm], [↶] [→dBm], [↷] [→mW]), Friis Free-Space Path Loss ([FSL]), three-register Link Budget summation ([↶] [LNK], taking P_{tx} in Register Z, net antenna gain G in Register Y, and total path loss L in Register X computed in place by [FSL]), and First Fresnel Zone clearance ([↷] [F₁]) directly across the RPN stack, while also providing **Bidirectional Equation Solver Workspaces** ([↷] [SOLVE] / [XEQ]) for [FSL] (**L = 20LOG(D)+20LOG(F)+32.44**, softkeys [L], [D], [F]) and [↶] [LNK] (**R = P + G - L**, softkeys [R], [P], [G], [L]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Solver Schema
[FSL]	Free-Space Path Loss	—	Distance d_{km} (km)	Frequency f_{MHz} (MHz)	Stack: $20\log_{10}(d) + 20\log_{10}(f) + 32.44$; Solver: [L],[D],[F]
[↶] [LNK]	Link Budget Rx Power	Tx Power P_{tx} (dBm)	Net Gain G (dB)	Total Loss L (dB)	Stack: $P_{rx} = P_{tx} + G - L$; Solver: **R = P + G - L** ([R],[P],[G],[L])
[↷] [F₁]	1st Fresnel Zone Radius	—	Distance d_{km} (km)	Frequency f_{GHz} (GHz)	Immediate stack: $r_1 = 17.32\ \sqrt{(d\ /\ (4\ f))} = 8.656\ \sqrt{(d\ /\ f)}$ (m)
[↷] [COX] / [↶] [PAD]	Coax Loss & Resistive Pad	—	Base Loss (dB)	Freq Ratio / Attenuation	$\alpha(f) = \alpha_0\ \sqrt{(f/f_0)}$ ([↷] [COX]); 50 Ω Pi/Tee attenuator ([↶] [PAD])
[↷] [c 0%]	Vacuum Speed of Light	—	—	—	Pushes c = 299,792,458 m/s onto Register X

14.3.1 Computing Free-Space Loss and Received Signal Power Across the Valley on the Stack

To close the 12.0 km, 5800 MHz valley link on the stack, the engineer first converts P_{tx} = 250 mW to +23.979 dBm ([↶] [→dBm]), loads the +52.0 dB combined dish gain, evaluates the 129.292 dB free-space path loss in place with [FSL], and presses [↶] [LNK] to resolve the received signal power P_{rx}:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
250 [↶] [→dBm] **52** [ENTER]	0	23.979	52	52	Convert 250 mW to +23.979 dBm in Z; load dish gain 52 dB into Y
12 [ENTER] **5800** [FSL]	23.9794	23.979	52	129.292	Evaluate FSPL = 129.292 dB at 12 km, 5800 MHz; P_{tx} in Z, G in Y
[↶] [LNK]	23.9794	23.9794	23.9794	**-53.313**	Evaluate link budget P_{rx} = 23.979 + 52 - 129.292 = -53.313 dBm

Notice how [FSL] consumes d_{km} = 12 and f_{MHz} = 5800 from Registers Y and X, dropping P_{tx} = 23.979 dBm into Register Z and G = 52 dB into Register Y so that pressing [↶] [LNK] resolves P_{rx} = **-53.313 dBm** in a single keystroke.

14.3.2 Auditing Mid-Span Pine Tree Clearance and Back-Solving Link Range & Power in the Solver Workspace

With P_{rx} = -53.313 dBm sitting in Register X, the engineer evaluates the midpoint First Fresnel Zone radius r_1 at d_{km} = 12.0 km and f_{GHz} = 5.8 GHz using [↷] [F1] (**12** [ENTER] **5.8** [↷] [F1] **0.6** [×]), and then opens the [FSL] (**L = 20LOG(D)+20LOG(F)+32.44**) and [↶] [LNK] (**R = P + G - L**) Solver Workspaces to answer two inverse planning questions: first, what is the maximum line-of-sight hop distance D ([D]) if rain-fade margin allows a maximum free-space path loss of L = 132.44 dB at F = 5800 MHz. Second, what transmitter power P ([P]) is required on an auxiliary telemetry link (L = 118 dB, G = 34 dBi) to achieve R = -80.0 dBm:

Keystroke	Solver Display / Register X	What Happens on the Stack & in the [FSL] / [↶] [LNK] Solver Workspaces
12 [ENTER] **5.8** [↷] [F1] **0.6** [×]	**7.474**	Evaluate r_1 = 12.457 m via [↷] [F1] and multiply by 0.60 for 7.474 m clearance
[FSL]	**L=20LOG(D×F)+32.44**	Open the Free-Space Path Loss ([FSL]) Solver Workspace with softkeys [L], [D], [F]
132.44 [L]	132.4400	Store maximum allowable path loss L = 132.44 dB into softkey [L] ([LFU0])
5800 [F]	5,800.0000	Store carrier frequency F = 5800 MHz into softkey [F] ([LFU2])
[↷] [SOLVE] [D]	**17.2414**	Solve backward for maximum hop distance $D = 10^{(132.44-32.44)/20}$ / 5800 = **17.2414 km**
[↶] [LNK]	**R=P+G-L**	Open the Link Budget ([↶] [LNK]) Solver Workspace with softkeys [R], [P], [G], [L]
80 [+/−] [R] **34** [G] **118** [L]	118.0000	Store target R = -80 dBm, gain G = 34 dB, and loss L = 118 dB
[↷] [SOLVE] [P] [↷] [→mW]	**2.5119**	Solve backward for P = +4.00 dBm and convert to **2.5119 mW**

14.3.3 Try It on the Stack

Consolidate your mastery of free-space path loss, logarithmic link budgets, and Fresnel diffraction zones with these three field-engineering exercises:

1. **2.4 GHz Ten-Kilometer Campus Link Budget:** A P_{tx} = 100 mW (+20 dBm) bridge operates at f = 2400 MHz (2.4 GHz) across a d = 10.0 km link using two +24 dBi dishes (G = 48 dB). Find the free-space path loss and received power (**100** [↶] [→dBm] **48** [ENTER] **10** [ENTER] **2400** [FSL] [↶] [LNK]). *(Verified Register X results: FSPL* ***≈ 120.044 dB*** *and* P_{rx} ***≈ -52.044 dBm****.)*

2. **Six-Gigahertz Eighteen-Kilometer Fresnel Clearance Audit:** Evaluate the full First Fresnel Zone radius r_1 and the 60% minimum obstacle clearance at mid-span across an 18.0 km hop operating at 6.0 GHz (**18** [ENTER] **6** [↷] [F1] [ENTER] **0.6** [×]), and verify the inverse hop length from r_1 = 15.0 m at 6.0 GHz (**15** [ENTER] **8.656** [÷] [↶] [x²] **6** [×]). *(Verified Register X results:* r_1 ***≈ 15.000 m****, 0.6* r_1 ***≈ 9.000 m****, and* D = ***18.000 km****.)*

3. **Millimeter-Wave 28 GHz Urban Backhaul Loss:** Compute the Free-Space Path Loss over a short d = 2.5 km urban rooftop hop at f = 28,000 MHz (28 GHz) and determine the received power for P_{tx} = +15 dBm and G = 70 dB combined dish gain (**15** [ENTER] **70** [ENTER] **2.5** [ENTER] **28000** [FSL] [↶] [LNK]). *(Verified Register X results: FSPL* ***≈ 129.339 dB*** *and* P_{rx} ***≈ -44.339 dBm****.)*

Having delivered an electromagnetic wavefront across a mountain valley to the receiving antenna, we step inside the receiver front-end in Chapter 15 to rescue a weak satellite telemetry carrier from thermal electron noise and compute the channel's Shannon capacity.

Chapter 15: Pulling a Weak Telemetry Signal Out of Thermal Noise

At a university satellite ground station, an engineering team is attempting to demodulate a weak 437.0 MHz telemetry carrier transmitted by a low-Earth-orbit CubeSat passing overhead at a radial closure velocity of $v = 7500$ m/s over a $B = 1.0$ MHz (10^6 Hz) receiver channel. In the station's initial cabling layout, the roof-mounted Yagi antenna feeds 30 meters of coaxial cable having 6.0 dB of passive insertion loss before reaching the equipment rack indoors. Downstairs on the bench sit a low-noise amplifier ($G_{\text{LNA}} = 20$ dB, noise figure $NF_1 = 1.5$ dB) and a software-defined radio receiver ($NF_{\text{rx}} = 8.0$ dB). With the LNA mounted *downstairs* at the rack after the 6.0 dB cable run, the telemetry waterfall display shows nothing but snow: the downlink signal is buried beneath the receiver system's noise floor.

Why does placing a 20 dB amplifier at the bottom of the coaxial run fail to rescue the signal, whereas climbing onto the roof and bolting that identical amplifier at the antenna masthead *ahead* of the 6.0 dB coaxial cable brings the telemetry packets into crystal-clear lock? Both configurations have the identical net system gain (+20 dB - 6 dB = +14 dB), yet their signal-to-noise ratios differ by more than a factor of three.

As Harald T. Friis proved in 1944, passive coaxial loss placed ahead of the first amplifier attenuates the delicate satellite signal while injecting room-temperature Johnson–Nyquist thermal noise. Moving the LNA to the masthead boosts the signal by 100× (+20 dB) *before* the cable attenuation occurs, dividing the noise contribution of the cable and downstream receiver by one hundred. In this chapter, we follow three connected receiver-design themes:

1. Evaluate the Johnson–Nyquist thermal noise floor across a 1.0 MHz channel and compare the Friis cascaded noise figure with the LNA downstairs versus at the antenna masthead.
2. Quantify third-order intermodulation distortion (IIP3) and Spurious-Free Dynamic Range (SFDR) when a strong nearby paging transmitter threatens to overload the LNA.
3. Calculate the Shannon–Hartley channel capacity limit in megabits per second, and track orbital Doppler shift and monostatic R^4 radar return.

15.1 Thermal Noise Floors, Cascaded Amplifiers, and Receiver Linearity

Every radio receiver is ultimately bounded at the bottom of its dynamic range by the random thermal motion of electrons inside its own conductors, and bounded at the top by non-linear distortion in its active transistors and mixers.

15.1.1 Johnson-Nyquist Thermal Agitation and the Friis Noise Cascade Law

In 1928, Bell Labs physicists John B. Johnson and Harry Nyquist demonstrated through statistical mechanics and fluctuation-dissipation theory that the random thermal agitation of charge carriers inside any matched resistor at absolute temperature T_0 (T_0) delivers an available noise power spectral density of $k_B\,T_0$ watts per hertz of bandwidth, independent of resistance value or frequency across the RF spectrum. At the IEEE standard reference temperature of T_0 = 290 K (16.85°C) with Boltzmann's constant k_B = 1.380649 × 10^{-23} J/K (kB), the thermal noise power per hertz is $k_B\,T_0 \approx$ 4.0039 × 10^{-21} W/Hz = 4.0039 × 10^{-18} mW/Hz, whose base-ten logarithm is 10 $\log_{10}(4.0039 \times 10^{-18}) \approx$ -173.977 dBm/Hz ≈ -174 dBm/Hz. Integrating across an effective receiver noise bandwidth B_{Hz} (in hertz) defines the **Thermal Noise Floor** (Pn, in dBm):

$$N_{\text{floor, dBm}} = 10\log_{10}(k_B T_0 \cdot 1000) + 10\log_{10}(B_{\text{Hz}}) \approx -173.977 + 10\log_{10}(B_{\text{Hz}}) \qquad \textbf{(Eq. 15.1)}$$

When a signal passes through a cascade of amplifier, cable, and mixer stages having linear noise factors F_i = $\text{SNR}_{\text{in},i}$ / $\text{SNR}_{\text{out},i}$ = $10^{NF_i\,/\,10}$ and linear available power gains G_i = $10^{\text{Gain}_i\,/\,10}$, Harald Friis's 1944 cascade formula (NF) determines the overall linear **Total Noise Factor** F_{tot} and decibel **Noise Figure** NF_{tot} = 10 $\log_{10}(F_{\text{tot}})$:

$$F_{\text{tot}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \ldots, \qquad NF_{\text{tot, dB}} = 10\log_{10}(F_{\text{tot}}) \qquad \textbf{(Eq. 15.2)}$$

In this equation, F_1 and G_1 are the linear noise factor and linear power gain of the very first component following the antenna, and each subsequent stage's excess noise (F_k - 1) is divided by the product of all preceding linear gains $G_1\ G_2 \cdots G_{k-1}$. Notice also that a passive lossy component (such as a coaxial cable of insertion loss L_{dB} at 290 K) has linear gain $G_{\text{cable}} = 10^{-L_{\text{dB}}\,/\,10} < 1$ and linear noise factor $F_{\text{cable}} = 1\ /\ G_{\text{cable}} = 10^{+L_{\text{dB}}\,/\,10}$.

Compare the two cabling configurations at our satellite ground station (B_{Hz} = 1.0 MHz = 10^6 Hz, where the thermal noise floor is N_{floor} = -173.977 + 10 $\log_{10}(10^6)$ **≈ -113.98 dBm**):

- **Configuration A (Masthead LNA at the antenna, ahead of the 6 dB cable and 8 dB receiver):** Stage 1 is the LNA (NF_1 = 1.5 dB ⇒ F_1 = 1.4125, G_1 = 20 dB ⇒ 100). The 6.0 dB cable (G_2 = 0.2512, F_2 = 3.9811) and 8.0 dB receiver (F_3 = 6.3096) have a combined downstream noise factor of $F_{23} = F_2 + (F_3 - 1)/G_2$ = 3.9811 + 5.3096 / 0.2512 **≈ 25.119** (14.0 dB). Because the masthead LNA precedes them with G_1 = 100, the total system noise factor is:

$$F_{\text{tot, masthead}} = 1.4125 + \frac{25.119 - 1}{100} = 1.4125 + 0.2412 = \mathbf{1.6537} \quad (NF_{\text{tot}} = 10\log_{10}(1.6537) \approx \mathbf{2.185}\ \text{dB}) \qquad \textbf{(Eq. 15.3)}$$

- **Configuration B (Coaxial cable first, LNA downstairs in the shack):** Stage 1 is the 6.0 dB passive cable (F_1 = 3.9811, G_1 = 0.2512), followed by the LNA (F_2 = 1.4125, G_2 = 100) and receiver (F_3 = 6.3096). Dividing by G_1 = 0.2512 *multiplies* the downstream noise by 3.9811:

$$F_{\text{tot, shack}} = 3.9811 + \frac{1.4125 - 1}{0.2512} + \frac{6.3096 - 1}{0.2512 \times 100} = 3.9811 + 1.6422 + 0.2114 = \mathbf{5.8347} \quad (NF_{\text{tot}} \approx \mathbf{7.660}\ \text{dB}) \qquad \textbf{(Eq. 15.4)}$$

Moving the LNA to the masthead improves the system noise figure from 7.660 dB down to 2.185 dB—a **5.475 dB** (3.53×) improvement in signal-to-noise ratio that pulls the CubeSat telemetry carrier out of the snow!

15.1.2 Third-Order Intermodulation Intercepts and Spurious-Free Dynamic Range

While high LNA gain improves weak-signal sensitivity, excessive gain can overload downstream mixers when strong adjacent broadcast or pager transmitters are present. When two strong interfering tones at frequencies f_1 and f_2 pass through a weakly non-linear amplifier, cubic distortion generates **third-order intermodulation products** at $2f_1 - f_2$ and $2f_2 - f_1$ that fall right back inside the receiver passband and grow 3 dB for every 1 dB increase in input power.

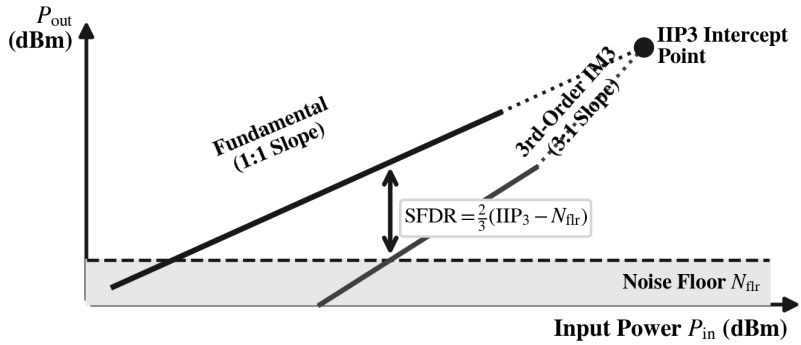


Figure 15.1: *Receiver linearity plot showing fundamental (1:1) and third-order intermodulation (3:1) power slopes, IIP3 intercept, and SFDR.*

Extrapolating the 1:1 fundamental power line and the 3:1 third-order distortion line on a log-log plot defines the **Input Third-Order Intercept Point** (IP3, in dBm), which bounds the receiver's **Spurious-Free Dynamic Range** (FDR, in dB) above the effective noise floor $N_{\text{floor,dBm}}$:

$$\text{SFDR}_{\text{dB}} = \frac{2}{3}(\text{IIP3}_{\text{dBm}} - N_{\text{floor,dBm}}) \qquad \textbf{(Eq. 15.5)}$$

For a receiver front-end with IIP3 = +5.0 dBm and an effective noise floor of N_{floor} = -100.0 dBm, the spurious-free dynamic range spans SFDR = 2/3(5.0 - (-100.0)) = **70.0 dB**.

15.2 Shannon-Hartley Channel Capacity and Radar Doppler Propagation

Once the masthead LNA delivers a clean Signal-to-Noise Power Ratio (SNR) to the demodulator, information theory and relativistic wave kinematics govern how many bits per second the channel can carry and how the satellite's orbital velocity shifts the carrier frequency.

15.2.1 Spectral Efficiency and the Noisy-Channel Coding Limit

In his landmark 1948 theorem, Claude Shannon proved that for a communication channel of bandwidth B (in hertz) perturbed by additive white Gaussian noise where the received signal power is P_{signal} and the total in-band noise power is P_{noise} (giving linear signal-to-noise ratio $\text{SNR} = P_{\text{signal}} \,/\, P_{\text{noise}} = 10^{\text{SNR}_{\text{dB}} \,/\, 10}$), there exist error-correcting codes that can transmit information with arbitrarily small error probability at any rate up to the **Shannon–Hartley Channel Capacity** C (C/B, in bits per second, bps):

$$C = B\log_2(1+\mathrm{SNR}) \quad [\mathrm{bps}], \qquad \eta = \frac{C}{B} = \log_2(1+\mathrm{SNR}) \quad \left[\frac{\mathrm{bits/s}}{\mathrm{Hz}}\right] \qquad \textbf{(Eq. 15.6)}$$

Here B is the channel bandwidth in hertz, SNR is the dimensionless linear signal-to-noise power ratio, C is the maximum theoretical error-free data rate in bps, and $\eta = C \,/\, B$ is the **spectral efficiency** in bits/s/Hz. At high signal-to-noise ratios (SNR $\gg$ 1), spectral efficiency grows as $\eta \approx \log_2(\mathrm{SNR}) = \mathrm{SNR}_{\mathrm{dB}} \,/\, (10 \log_{10} 2) = \mathrm{SNR}_{\mathrm{dB}} \,/\, 3.0103$ bits/s/Hz per power ratio (or $\mathrm{SNR}_{\mathrm{dB}} \,/\, 6.0206$ amplitude bits per sample)—the exact information-theoretic dual of the Effective Number of Bits ENOB = $(\mathrm{SINAD}_{\mathrm{dB}} - 1.7609) \,/\, 6.0206$ of a mixed-signal Analog-to-Digital Converter in **Unit III: Digital Systems, Bitwise Logic & Embedded Timing**.

On our CubeSat ground station's $B = 1.0$ MHz (10^6 Hz) channel, if the masthead LNA upgrade raises the linear signal-to-noise ratio to SNR = 15.0 ($\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10}(15) \approx 11.76$ dB), the spectral efficiency reaches $\eta = \log_2(1 + 15) = \log_2(16) =$ **4.00 bits/s/Hz**, supporting a theoretical error-free channel capacity of $C = 10^6 \times 4.00 =$ **4.00 Mbps** (4,000,000 bps). Even when a deep-space probe operates below the noise floor at SNR = 0.25 (-6.02 dB), $\log_2(1.25) \approx 0.3219$ bits/s/Hz still permits 321.9 kbps across a 1 MHz spread-spectrum channel.

15.2.2 Monostatic Fourth-Power Radar Range and Relativistic Doppler Shift

When a ground station or monostatic radar tracks a moving spacecraft or aircraft of radial velocity v (in m/s) and radar cross-section σ (in m^2) at range R, wave compression shifts the carrier frequency f_0 ([→] [Δf], which takes carrier frequency f_0 in Register Y and closing velocity v in Register X to return the one-way Doppler shift $\Delta f_{1\text{-way}} = f_0\, v \,/\, c$ in Register X and the two-way monostatic radar Doppler shift $\Delta f_{2\text{-way}} = 2\, f_0\, v \,/\, c$ in Register Y), while two-way spherical spreading ($1 \,/\, (4\pi\, R^2)$ outbound and $1 \,/\, (4\pi\, R^2)$ inbound) causes the radar echo power P_{echo} to fall off inversely with the **fourth power of range** ([←] [RCS]):

$$\Delta f_{1-\mathrm{way}} = \frac{f_0 \cdot v}{c}, \qquad \Delta f_{2-\mathrm{way}} = \frac{2 \cdot f_0 \cdot v}{c}, \qquad P_{\mathrm{echo}} = \frac{P_{\mathrm{tx}} G^2 \lambda^2 \sigma}{(4\pi)^3 R^4} \Rightarrow \mathrm{Radar\ Factor} = \frac{\sigma}{(4\pi)^3 R^4} \qquad \textbf{(Eq. 15.7)}$$

For our CubeSat transmitting at $f_0 = 437.0$ MHz (4.37×10^8 Hz) with a radial closure speed of $v = 7500$ m/s (**437** [ENTER] **7500** [→] [Δf]), the one-way telemetry downlink shift in Register X is $\Delta f_{1\text{-way}} = (4.37 \times 10^8 \times 7500) \,/\, 299{,}792{,}458 \approx$ **10,932.6 Hz** (10.93 kHz), and the two-way monostatic radar shift in Register Y is $\Delta f_{2\text{-way}} \approx$ **21,865.1 Hz** (21.87 kHz). Because two-way radar return scales with $\sigma \,/\, ((4\pi)^3\, R^4)$ ([←] [RCS] with range R in Register Y and cross-section σ in Register X), doubling a radar's maximum detection range $R_{\max}$ requires increasing transmitter power sixteen-fold ($2^4 = 16$, or +12.04 dB).

15.3 Rescuing Weak Telemetry Signals and Auditing Channel Capacity on the Stack

The **SC-50** executes two-stage and multi-stage Friis noise cascades ([NF] taking F_1 in Register Z, F_2 in Register Y, and linear gain G_1 in Register X), thermal noise floor evaluation ([→] [Pn], [→] [T₀], [←] [kB]), third-order intermodulation and spurious-free dynamic range ([←] [IP3], [→] [FDR]), Shannon–Hartley channel capacity ([→] [C/B]), and Doppler / monostatic radar cross-section scaling ([→] [Δf], [←] [RCS]) directly on the RPN stack, while providing **Bidirectional Equation Solver Workspaces** ([→] [SOLVE] / [XEQ]) for [NF] (**F = A + (B - 1) / G**, softkeys [F], [A], [B], [G]) and [→] [C/B] (**C = B × LN(1 + S) / LN(2)**, softkeys [S/N], [B], [S]):

Button	Operation	Reg Z	Reg Y	Reg X	Governing Expression & Solver Schema
[NF]	2-Stage Friis Factor	Linear F_1	Linear F_2	Linear Gain G_1	Stack: $F_1 + (F_2 - 1) / G_1$; Solver: $\mathbf{F = A + (B - 1) / G}$ ([F],[A],[B],[G])
[→] [Pₙ]	Thermal Noise Floor	—	—	Bandwidth B_{Hz} (Hz)	Immediate stack: $-173.977 + 10\log_{10}(B_{\text{Hz}})$ (dBm)
[→] [T₀] / [←] [kB]	Thermal Reference & Boltzmann	—	—	—	Pushes $T_0 = 290$ K ([→] [T₀]) or $k_B = 1.380649 \times 10^{-23}$ J/K ([←] [kB])
[→] [FDR] / [←] [IP3]	Spurious-Free Dynamic Range	—	IIP3 (dBm)	N_{floor} (dBm)	$\text{SFDR}_{\text{dB}} = 2/3(\text{IIP3} - N_{\text{floor}})$ ([→] [FDR]); cascaded IIP3 ([←] [IP3])
[→] [C/B]	Shannon–Hartley Capacity	—	Bandwidth B (Hz)	Linear SNR	Stack: $C = B \log_2(1 + \text{SNR})$; Solver: $\mathbf{C = B \times LOG2(1 + S)}$ ([S/N],[B],[S])
[→] [Δf]	One-Way & Two-Way Doppler	—	Carrier f_0 (MHz/Hz)	Velocity v (m/s)	$X = f_0\, v / c$ (1-way Hz), $Y = 2\, f_0\, v / c$ (2-way radar Hz)
[←] [RCS]	Monostatic Radar Factor	—	Range R (m)	Cross-Section σ (m^2)	Two-way geometric factor $\sigma / ((4\pi)^3 R^4)$ (and dB in Y)

15.3.1 Comparing Masthead LNA Placement Against Coaxial Cable Loss on the Stack

To verify the masthead LNA upgrade on the stack, the engineer evaluates the downstream cable-plus-receiver noise factor ($F_2 = 3.9811$, $G_2 = 0.2512$ for the 6 dB cable ahead of $F_3 = 6.3096$ for the 8 dB receiver, giving $F_{23} = 25.119$), and then computes the total linear noise factor F_{tot} with the masthead LNA ($F_1 = 1.4125$, $G_1 = 100$) placed first versus placing a 3 dB cable ($F = 2.0$, $G = 0.5$) before an LNA ($F = 1.25$, $G = 20$):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3.9811 [ENTER] **6.3096** [ENTER] **0.2512** [NF]	0	0	0	25.118	Cable (6 dB) + Receiver (8 dB) combined factor $F_{23} = 25.118$ (14.0 dB)
1.4125 [x⇄y] **100** [NF]	0	0	0	1.654	Masthead LNA ($F_1 = 1.4125$, $G_1 = 100$) ahead of F_{23} yields $F_{\text{tot}} = 1.654$ (2.18 dB)
1.25 [ENTER] **2** [ENTER] **20** [NF]	1.6537	1.6537	1.654	1.3	Reference check: LNA (F=1.25, G=20) before 3 dB coax (F=2) gives $F_{\text{tot}} = 1.30$
2 [ENTER] **1.25** [ENTER] **0.5** [NF]	1.3	1.3	1.3	**2.5**	Reversing order (3 dB coax before LNA) degrades noise factor to $F_{\text{tot}} = 2.50$

15.3.2 Evaluating Thermal Noise, Shannon Capacity, and Orbital Doppler Shift on the Stack

Next, the engineer evaluates the thermal noise floor over the $B = 1.0$ MHz (10^6 Hz) downlink channel ([↗] [P_n]), checks the receiver's Spurious-Free Dynamic Range for IIP3 = +8.0 dBm ([↗] [FDR]), calculates the Shannon–Hartley channel capacity at linear SNR = 15 ([↗] [C/B]), and finds the one-way and two-way Doppler shifts at $f_0 = 437$ MHz and $v = 7500$ m/s ([↗] [Δf]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
8 [ENTER] **1000000** [↗] [P_n]	0	0	8	-113.977	Thermal noise floor N_{floor} = -113.98 dBm at $B = 1$ MHz; IIP3 = 8 in Y
[↗] [FDR]	0	0	0	81.318	Spurious-Free Dynamic Range SFDR = 2/3(8 - (-113.977)) = 81.32 dB
1000000 [ENTER] **15** [↗] [C/B]	0	0	81.318	4000000	Shannon capacity $C = 10^6 \log_2(1 + 15)$ = 4.00 Mbps (4,000,000 bps)
437 [ENTER] **7500** [↗] [Δf]	81.318	4000000	21865.1	**10932.6**	One-way downlink shift $\Delta f = 10{,}932.6$ Hz in X; two-way radar shift 21,865.1 Hz in Y

15.3.3 Back-Solving Required SNR and Maximum Stage Noise Factor in the Solver Workspace

Suppose a high-speed X-band payload downlink must deliver a target error-free data rate of $C = 50$ Mbps (5×10^7 bps) across a channel bandwidth of $B = 20$ MHz (2×10^7 Hz), and the engineer needs the minimum required linear Signal-to-Noise Ratio S ([S]), while also solving backward in the [NF] workspace (**F = A + (B - 1) / G**) for the maximum allowable second-stage noise factor B ([B]) when $F_{tot} = 1.6537$, $A = 1.4125$, and $G = 100$. Opening the [↗] [C/B] and [NF] Solver Workspaces resolves both inverse requirements directly:

Keystroke	Solver Display / Register X	What Happens in the [↗] [C/B] & [NF] Solver Workspaces
[↗] [C/B]	**C=B×LN(1+S)/LN(2)**	Open the Shannon–Hartley ([↗] [C/B]) Solver Workspace with softkeys [C ON], [B], [S]
5 [E] **7** [C ON]	5.0000×10^7	Store required data rate $C = 50$ Mbps (5×10^7 bps) into softkey [C ON] ([LFU0])
2 [E] **7** [B]	2.0000×10^7	Store channel bandwidth $B = 20$ MHz (2×10^7 Hz) into softkey [B] ([LFU1])
[↗] [SOLVE] [S]	**4.6569**	Solve backward for minimum linear SNR = $2^{50/20}$ - 1 = $2^{2.5}$ - 1 = **4.6569** (6.68 dB)
[NF]	**F=A+(B-1)/G**	Open the Friis Noise Factor ([NF]) Solver Workspace with softkeys [F], [A], [B], [G]
1.6537 [F] **1.4125** [A] **100** [G]	100.0000	Store target $F_{tot} = 1.6537$, LNA $F_1 = 1.4125$, and LNA gain $G_1 = 100$
[↗] [SOLVE] [B]	**25.1200**	Solve backward for maximum downstream noise factor F_{23} = **25.12** (14.0 dB)

15.3.4 Try It on the Stack

Test your command of Friis receiver cascades, dynamic range, Shannon capacity, and radar Doppler scaling with these three telemetry and radar exercises:

1. **Three-Stage Receiver Chain and Narrowband SFDR:** Evaluate a three-stage receiver with Stage 1 ($F_1 = 1.25$, $G_1 = 20$), Stage 2 ($F_2 = 2.0$, $G_2 = 0.5$), and Stage 3 ($F_3 = 4.0$) by cascading Stage 2 + Stage 3 first (**2** ENTER **4** ENTER **0.5** NF) and then placing Stage 1 ahead of that result (**1.25** x⇄y **20** NF). Then compute the thermal noise floor and SFDR for IIP3 = +12.0 dBm over a $B = 200$ kHz channel (**12** ENTER **200000** → P_n → FDR). *(Verified Register X results: F_{tot} = **1.60**, N_{floor} ≈ **-120.97 dBm**, and SFDR ≈ **88.65 dB**.)*

2. **20 MHz Wireless Local Loop Shannon Capacity:** A $B = 20$ MHz (2×10^7 Hz) wireless backhaul operates at a linear Signal-to-Noise Ratio of SNR = 31.0 (14.91 dB). Find the theoretical Shannon capacity in megabits per second (**2** E **7** ENTER **31** → C/B **1** E **6** ÷). *(Verified Register X result: C = **100.0 Mbps** at 5.0 bits/s/Hz.)*

3. **X-Band Weather Radar Doppler Shift and Cross-Section Factor:** An X-band radar at $f_0 = 10.0$ GHz (10,000 MHz) tracks an aircraft of cross-section $\sigma = 2.5$ m² at range $R = 5000$ m moving at radial speed $v = 150$ m/s. Compute the one-way and two-way Doppler shifts (**10000** ENTER **150** → Δf) and the two-way geometric radar factor (**5000** ENTER **2.5** ↶ RCS). *(Verified Register X and Y results: $\Delta f_{1\text{-}way}$ ≈ **5,003.5 Hz**, $\Delta f_{2\text{-}way}$ ≈ **10,006.9 Hz**, and geometric factor σ / ($(4\pi)^3$ R^4) ≈ **2.016 × 10^{-18}** or **-176.96 dB**.)*

Having traced electromagnetic waves from coaxial matching networks through free-space Fresnel zones and low-noise receiver cascades in Unit V, we step into the analytical wet lab in **Unit VI: Stoichiometry, Chemical Equilibrium & Wet-Lab Analytics** to count molecules by weighing grams, stabilize enzymatic buffers, and measure protein concentration with ultraviolet light.

Unit VI: Stoichiometry, Chemical Equilibrium & Wet-Lab Analytics

In September 1860, at the First International Congress of Chemists in Karlsruhe, Germany, the Italian chemist Stanislao Cannizzaro distributed a printed pamphlet reviving the 1811 molecular hypothesis of his countryman Amedeo Avogadro. Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules, and elemental gases such as hydrogen, nitrogen, and oxygen travel as diatomic pairs (H_2, N_2, O_2). Every chemical reaction proceeds between discrete integer counts of atoms and molecules. When citric acid neutralizes sodium bicarbonate in aqueous solution ($C_6H_8O_7 + 3\ NaHCO_3 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2\uparrow$), one molecule of citric acid reacts with three molecules of sodium bicarbonate. Yet individual molecules measure roughly a nanometer across and weigh on the order of 10^{-22} grams. By defining the **mole** (mol) as $N_A = 6.02214076 \times 10^{23}$ elementary entities, Avogadro and Cannizzaro built the stoichiometric bridge connecting integer chemical formulas to macroscopic gram masses weighed on an analytical balance and dissolved to known Molarity or diluted across volumetric glassware.

Half a century later in 1909 at the Carlsberg Laboratory in Copenhagen, Danish biochemist Søren Peder Lauritz Sørensen investigated how aqueous hydrogen-ion concentration governs the catalytic speed of malt and yeast enzymes during brewing. Because free proton concentrations $[H^+]$ in water span fourteen orders of magnitude—from 1.0 mol/L in concentrated mineral acid down to 10^{-14} mol/L in caustic alkali—plotting reaction rates against raw negative powers of ten obscured biochemical comparisons. Sørensen introduced the *pondus hydrogenii* scale, compressing exponential proton activity through the negative base-ten logarithm $pH = -\log_{10}[H^+]$. Building on Sørensen's notation and Lawrence J. Henderson's 1908 study of carbonic-acid equilibria in blood plasma, Danish physician Karl Albert Hasselbalch cast the weak-acid dissociation law into logarithmic form in 1916. Together they established the **Henderson–Hasselbalch buffer equation**, which pairs with volumetric titration stoichiometry at the wet-lab bench. Across domains, the Henderson–Hasselbalch logarithmic equilibrium $pH = pK_a + \log_{10}([A^-]/[HA])$ shares its exact mathematical structure with Walther Nernst's electrochemical cell potential ($E = E^\circ - (0.05916/n_e)\log_{10} Q$) and with the decibel gain and power ratios of Unit V.

To count dissolved biomolecules without consuming or destroying the sample, German physicist August Beer proved in 1852 that monochromatic light traversing an absorbing solution attenuates exponentially with both solute concentration and optical path length, uniting Pierre Bouguer and Johann Heinrich Lambert's earlier absorption laws into the **Beer–Lambert spectrophotometric equation**. Inside a quartz cuvette at 280 nm, aromatic tryptophan and tyrosine side chains cast a quantitative optical shadow proportional to protein molarity—obeying the same base-ten logarithmic attenuation law ($A = -\log_{10}(I/I_0) = \varepsilon\ b\ c$) that governs neutral-density camera filters in Unit IV and RF path loss in Unit V. Rounding out the physical-chemistry bench, the ideal gas equation, J. Willard Gibbs's 1876 free-energy criterion for reaction spontaneity, Walther Nernst's 1889 electrochemical potential, Svante Arrhenius's 1889 activation-energy rate law, calorimetry, and Theodor Svedberg's Nobel-winning centrifugal sedimentation equations link microscopic molecular energetics to laboratory instruments.

The **SC-60 Chemistry, Biochemistry & Wet-Lab** calculator brings the periodic table, stoichiometric formula parser, and physical-chemistry solvers directly to the lab bench. Each chapter in this Unit follows a single biochemical preparation and assay workflow

across three connected mathematical themes:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 16	Preparing 250 mL of 0.50 M NaCl and 150 mL of 0.10 M Tris from dry crystals and 2.0 M stock bottles, plus verifying a cofactor dilution by UV absorbance	IUPAC molar mass, elemental mass percentages, hydrate lattices, gravimetric stoichiometry, $C_1\ V_1 = C_2\ V_2$ dilutions, and serial-dilution error propagation	1. IUPAC Formula Weights & Hydrate Crystal Lattices • 2. Gravimetric Molarity & Balance-Pan Weighing • 3. Stock Aliquot Dilutions ($C_1\ V_1 = C_2\ V_2$) & UV Checks
Chapter 17	Formulating a 100 mM conjugate phosphate buffer at pH = 7.40 against a 5.0 mM acid spike and standardizing diprotic H_2SO_4	Water autoionization, Sørensen pH and pK_a, Henderson–Hasselbalch species partitioning, ionic-strength pK_a' shifts, and diprotic titrations	1. Logarithmic Proton Activity (pH & pK_a) • 2. Henderson–Hasselbalch Buffer Partitioning • 3. Acid-Spike Buffer Capacity & Diprotic Titrations
Chapter 18	Measuring purified enzyme concentration at 280 nm ($A = 0.680$), modeling Arrhenius acceleration across a 10 K step, and setting rotor RPM for $12{,}000 \times g$	Beer–Lambert absorbance, molar extinction, ideal gas law, Gibbs ΔG & Nernst E, Arrhenius activation kinetics, and rotor RCF/RPM	1. Exponential Light Attenuation & Beer–Lambert Law • 2. Ideal Gas, Gibbs Free Energy & Nernst Potentials • 3. Arrhenius Thermal Kinetics & Centrifugal Sedimentation

Chapter 16: Weighing the Buffer and Cutting the Stock Aliquot

At eight o'clock on Monday morning in a protein crystallography laboratory, a research biochemist opens the protocol notebook to prepare the day's chromatography elution buffers before loading a purified enzyme column. Two foundational solutions are required at the bench: first, 250 mL ($V = 0.250$ L) of $C = 0.50$ M physiological sodium chloride (NaCl) weighed from anhydrous crystalline reagent on the analytical balance. And second, 150 mL ($V_2 = 0.150$ L) of $C_2 = 0.10$ M Tris buffer—tris(hydroxymethyl)aminomethane, $C_4H_{11}NO_3$—prepared both from dry crystals and by cutting an exact pipette aliquot V_1 from a concentrated $C_1 = 2.0$ M sterile stock bottle without wasting a milliliter of stock reagent. Once the buffers are mixed, the biochemist also checks a diluted nicotinamide cofactor aliquot inside a $b = 1.00$ cm quartz cuvette ($\varepsilon = 6{,}600\ \mathrm{M^{-1}cm^{-1}}$, measured absorbance $A = 0.845$) by solving backward for its exact working concentration C.

Beside the sodium chloride bottle on the reagent shelf sits a jar of dibasic sodium phosphate heptahydrate ($Na_2HPO_4 \cdot 7H_2O$). A novice technician who looks up the formula weight of anhydrous Na_2HPO_4 (141.958 g/mol) and weighs that mass from the heptahydrate jar will shortchange the buffer's actual phosphate molarity by nearly half, because seven water molecules are locked inside every crystal unit cell on the balance pan (268.063 g/mol). In quantitative wet-lab work, every gram weighed on the pan and every microliter aspirated into a pipette tip must account for the exact molecular formula of the reagent in hand.

To complete the morning bench prep, the biochemist works through three interconnected stoichiometric calculations:

1. Sum the IUPAC atomic weights of NaCl, Tris ($C_4H_{11}NO_3$), and hydrated phosphate salts while auditing elemental mass percentages.

2. Compute the exact dry reagent masses W required for the 250 mL NaCl and 150 mL Tris flasks.

3. Determine the exact stock aliquot volume V_1 to pipette from the 2.0 M stock bottle, and verify the cofactor dilution from its UV absorbance.

16.1 Atomic Weights, Formula Stoichiometry, and Hydrated Crystals

Because laboratory balances measure mass in grams while chemical reactions proceed between integer counts of molecules, every wet-lab calculation begins by computing the mass of one Avogadro's number ($N_A = 6.02214076 \times 10^{23}\ \mathrm{mol^{-1}}$, recalled via ↖ Na) of formula units.

16.1.1 IUPAC Standard Atomic Weights and Gravimetric Mole Conversion

Each of the 118 chemical elements in the periodic table is uniquely identified by its **atomic number** $Z \in [1,\ 118]$ ([Z])—the number of protons in its atomic nucleus—and characterized by its **IUPAC Standard Atomic Weight** A_Z (in grams per mole, g/mol, or Daltons, Da), weighted across natural terrestrial isotopic abundances: Hydrogen ($Z = 1$, $A_{\mathrm{H}} = 1.008$), Carbon ($Z = 6$, $A_{\mathrm{C}} = 12.011$), Nitrogen ($Z = 7$, $A_{\mathrm{N}} = 14.007$), Oxygen ($Z = 8$, $A_{\mathrm{O}} = 15.999$), Sodium ($Z = 11$, $A_{\mathrm{Na}} = 22.990$), Phosphorus ($Z = 15$, $A_{\mathrm{P}} = 30.974$), Sulfur ($Z = 16$, $A_{\mathrm{S}} = 32.060$), Chlorine ($Z = 17$, $A_{\mathrm{Cl}} = 35.450$), Potassium ($Z = 19$, $A_{\mathrm{K}} = 39.098$), and Calcium ($Z = 20$, $A_{\mathrm{Ca}} = 40.078$). For any chemical compound whose formula unit ([FML]) contains ν_i atoms of element i with atomic weight A_i, summing the constituent atomic masses defines the compound's **Molar Mass** M (in g/mol), which converts between chemical amount n in moles ([→mol]) and gravimetric mass W in grams ([W]):

$$M = \sum_i \nu_i A_i \quad \left[\frac{\mathrm{g}}{\mathrm{mol}}\right], \qquad W = n \times M \Leftrightarrow n = \frac{W}{M} \tag{Eq. 16.1}$$

In these expressions, ν_i is the integer stoichiometric subscript of element i in the chemical formula, M is the molar mass in g/mol, n is the chemical amount in moles (mol), and W is the physical mass in grams (g) weighed on the balance pan.

For crystalline sodium chloride (NaCl, with $\nu_{\mathrm{Na}} = 1$ and $\nu_{\mathrm{Cl}} = 1$), summing the IUPAC atomic weights gives $M(\mathrm{NaCl}) = 22.990 + 35.450 = \mathbf{58.440\ g/mol}$. For Tris base ($C_4H_{11}NO_3$, containing 4 C, 11 H, 1 N, and 3 O), the molar mass is:

$$M(C_4H_{11}NO_3) = 4(12.011) + 11(1.008) + 1(14.007) + 3(15.999) = 48.044 + 11.088 + 14.007 + 47.997 = \mathbf{121.136}\ \mathrm{g/mol} \tag{Eq. 16.2}$$

16.1.2 Elemental Mass Fractions and Waters of Crystallization in Hydrates

When verifying elemental purity by combustion analysis or preparing trace-metal media, chemists evaluate the **Elemental Mass Percentage** $\mathrm{WT}\%_i$ ([↖] [w/w])—the percentage of one mole's total mass contributed by element i. Furthermore, when an inorganic salt crystallizes from water as a **hydrate** ($\mathrm{Salt} \cdot k\,H_2O$) containing k stoichiometric molecules of water ($M(H_2O) = 2(1.008) + 15.999 = \mathbf{18.015\ g/mol}$) locked inside its solid crystal lattice, those k water molecules contribute directly to the gravimetric molar mass M_{hydrate} on the balance pan before merging into the aqueous solvent upon dissolution:

$$\mathrm{WT}\%_i = \frac{\nu_i A_i}{M} \times 100\%, \qquad M_{\mathrm{hydrate}} = M_{\mathrm{anhydrous}} + k \cdot M(\mathrm{H_2O}) = M_{\mathrm{anhydrous}} + 18.015\,k \tag{Eq. 16.3}$$

In sulfuric acid (H_2SO_4, where $M = 2(1.008) + 32.060 + 4(15.999) = \mathbf{98.072\ g/mol}$), hydrogen accounts for $\mathrm{WT}\%_{\mathrm{H}} = (2.016 / 98.072) \times 100\% = \mathbf{2.056\%}$, sulfur accounts for **32.690%**, and oxygen accounts for **65.254%**. In Tris base ($C_4H_{11}NO_3$, $M = 121.136$ g/mol), carbon accounts for $\mathrm{WT}\%_{\mathrm{C}} = (48.044 / 121.136) \times 100\% = \mathbf{39.661\%}$ and nitrogen accounts for $\mathrm{WT}\%_{\mathrm{N}} = (14.007 / 121.136) \times 100\% = \mathbf{11.563\%}$.

Compare anhydrous dibasic sodium phosphate (Na_2HPO_4), which has molar mass $M_{\mathrm{anhydrous}} = 2(22.990) + 1.008 + 30.974 + 4(15.999) = \mathbf{141.958\ g/mol}$, against its crystalline heptahydrate ($Na_2HPO_4 \cdot 7H_2O$), which adds $7 \times 18.015 = 126.105$ g/mol of lattice water for a total molar mass of $M_{\mathrm{hydrate}} = 141.958 + 126.105 = \mathbf{268.063\ g/mol}$. Because 47.04% of the heptahydrate's weight on the balance pan is water of crystallization, delivering $n = 0.100$ mol of phosphate ions requires weighing $0.100 \times 268.063 = \mathbf{26.806\ g}$ of the heptahydrate crystal rather than 14.196 g.

16.2 Solution Molarity, Stock Dilutions, and Serial Pipetting Error

Once the molar mass M of a dry reagent is established, preparing working solutions at the wet-lab bench relies on the definition of volumetric molarity and the conservation of dissolved solute moles during dilution.

16.2.1 Dissolving Dry Reagents to Target Molarity

When a chemical amount $n = W / M$ (in moles) of solute is dissolved in solvent inside a calibrated volumetric flask and brought to a final total solution volume V (in liters, L, after the solid has dissolved so the solute's partial molar volume is included), the solution **Molarity** C ([↖] [RXN], [↖] [→M], [↗] [→mM], in moles per liter, mol/L or M, or mass concentration [↗] [→g/L]) and the required dry reagent mass W (in grams) satisfy:

$$C = \frac{n}{V} = \frac{W}{M \cdot V} \Leftrightarrow n = C \cdot V \Leftrightarrow W = C \cdot V \cdot M \qquad \textbf{(Eq. 16.4)}$$

Here C is the target molar concentration in mol/L (M), V is the final solution volume in liters (L, where 1 mL = 10^{-3} L), n is the dissolved solute amount in moles, M is the reagent molar mass in g/mol, and W is the dry mass in grams.

For our morning chromatography prep:

- **Physiological Sodium Chloride Flask (V = 250 mL = 0.250 L at C = 0.50 M):** The required chemical amount of NaCl is $n = C \cdot V = 0.50$ mol/L × 0.250 L = **0.125 mol** (computed via [↖] [RXN]). Multiplying by M(NaCl) = 58.440 g/mol gives a balance weight of $W = 0.125 \times 58.440 =$ **7.305 g** (a mass concentration of 7.305 g / 0.250 L = 29.22 g/L via [↗] [→g/L]). If preparing a 2.00 M concentrated NaCl stock in that same 0.250 L flask, $n = 2.00 \times 0.250 = 0.500$ mol and $W = 0.500 \times 58.440 =$ **29.220 g**.
- **Tris Working Buffer from Dry Crystals (V = 150 mL = 0.150 L at C = 0.10 M):** The required chemical amount of Tris ($C_4H_{11}NO_3$) is $n = 0.10$ mol/L × 0.150 L = **0.0150 mol** (15.0 mmol). Multiplying by M(Tris) = 121.136 g/mol specifies a dry crystal weight of $W = 0.0150 \times 121.136 \approx$ **1.817 g**.

16.2.2 Solute Mole Conservation and Serial Dilution Error Propagation

In high-throughput biochemistry laboratories, weighing small milligram quantities of dry buffer salts for every assay introduces balance error and consumes setup time. Instead, researchers prepare concentrated sterile stock bottles (such as C_1 = 2.0 M Tris or 5.0 M NaCl) and pipette an aliquot of volume V_1 into a new volumetric vessel diluted with pure water to final volume V_2 and working concentration C_2.

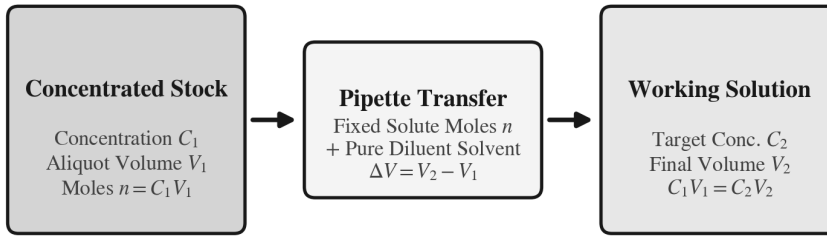


Figure 16.1: *Conservation of dissolved solute moles (n = C1 V1 = C2 V2) during quantitative aliquot dilution.*

Because adding pure water introduces zero additional solute molecules, the total number of solute moles n_{solute} aspirated in the stock aliquot V_1 equals the total number of solute moles present in the final diluted volume V_2, establishing the **Stock Dilution Conservation Law** (C_1V_1):

$$n_{\text{solute}} = C_1 \cdot V_1 = C_2 \cdot V_2 \Leftrightarrow V_1 = \frac{C_2 \cdot V_2}{C_1}, \qquad V_{\text{solvent}} = V_2 - V_1 \tag{Eq. 16.5}$$

In this bilateral conservation equation, C_1 and C_2 may be expressed in any matching concentration unit (M, mM, µM, or mg/mL) and V_1 and V_2 in any matching volume unit (L, mL, or µL), while V_{solvent} is the volume of pure diluent added to bring the aliquot V_1 up to the mark V_2.

Why do analytical chemists prefer a single direct C_1V_1 cut whenever the aliquot volume V_1 falls within a calibrated pipette's range, reserving multi-step serial dilutions for extreme $10^4\times$–$10^6\times$ reductions? When a technician executes k sequential dilutions using a micropipette whose relative volume delivery uncertainty at each transfer is σ_V / V, independent random pipetting errors add in quadrature, causing the cumulative relative concentration uncertainty σ_{C_k} / C_k after k steps to grow as the square root of the step count:

$$\frac{\sigma_{C_k}}{C_k} \approx \sqrt{k}\left(\frac{\sigma_V}{V}\right) \tag{Eq. 16.6}$$

A 6-step serial dilution ($k = 6$) with a ± 1.5% micropipette inflates random concentration error by $\sqrt{6} \times 1.5\% \approx \pm 3.67\%$ (plus 6× any systematic calibration bias), whereas cutting a single direct aliquot ($k = 1$) from a gravimetrically weighed primary stock (MW) holds volumetric error to a single transfer.

To prepare our V_2 = 150 mL (0.150 L) of C_2 = 0.10 M Tris working solution from a C_1 = 2.0 M stock bottle in a single step, the conserved solute amount is n_{solute} = 0.10 M × 150 mL = 15.0 mmol (0.0150 mol), and the required stock aliquot volume is:

$$V_1 = \frac{0.10\ \text{M} \times 150\ \text{mL}}{2.0\ \text{M}} = \mathbf{7.50}\ \text{mL} \quad (0.00750\ \text{L}) \tag{Eq. 16.7}$$

Pipetting 7.50 mL of 2.0 M Tris stock into a calibrated vessel and adding V_{solvent} = 150.0 - 7.50 = **142.50 mL** of deionized water delivers the exact 150 mL of 0.10 M working buffer without touching the analytical balance.

16.3 Weighing Dry Salts and Solving Dilutions on the SC-60

On the **SC-60**, interactive chemical formula parsing ([FML]), 118-element IUPAC atomic weight lookup ([Z]), polyatomic ion recall ([↶] [ION]), elemental mass percentage auditing ([↶] [w/w]), gravimetric mole-mass scaling ([MW], [↷] [→mol]), solution stoichiometry ([↶] [RXN], [↶] [→M], [↷] [→mM], [↷] [→g/L]), and stock aliquot dilutions ([C₁V₁]) operate both as immediate RPN stack shortcuts and as bidirectional Equation Solver workspaces ([↷] [SOLVE] / [XEQ]). When preparing multi-component buffers, switching the stack depth from [4-LV] to [8-LV] keeps every dry salt weight and stock volume on the stack simultaneously.

Key / Softkey	RPN Stack Action	Reg Y	Reg X	Solver Workspace ([↷] [SOLVE] / [XEQ])
[FML] (**0** [Z] or [LFU3] in [MW])	Interactive Formula Tokenizer ([ALP], [NUM], [FML])	—	Formula **FML:**	Computes $M = \sum \nu_i A_i$ (g/mol); binds [M] in [MW]
[Z] / [↶] [ION]	IUPAC Atomic Weight ($Z \in [1,118]$) / Polyatomic Ion Ledger	—	Atomic No. Z	Returns A_Z (g/mol) or opens [↶] [ION] reference
[MW]	Molar Mass Recall & Gravimetric Workspace (**W** = **N** × **M**)	—	Molar Mass M	[MW]: [W], [N], [M], [FML] — solves for W, N, or M
[↶] [w/w] / [↷] [w/v]	Elemental Mass Percentage $\mathrm{WT}\%_i = (\nu_i A_i / M) \times 100\%$	Total M	Component Mass	Evaluates mass fraction **% w/w** or **% w/v**
[↶] [RXN] ([↶] [→M], [↷] [→mM], [↷] [→g/L])	Solution Stoichiometry $n = C \times V$ and unit conversions	Molarity C (M)	Volume V (L)	[↶] [RXN]: [N], [C ON], [V] (**N** = **C** × **V**) — solves for n, C, or V
[C₁V₁]	Stock Dilution $V_1 = (C_2\ V_2) / C_1$ (Z=C_2, Y=V_2, X=C_1)	Volume V_2 or C_2 V_2	Stock C_1	[C₁V₁]: [C ON], [V], [D], [U] (**C** × **V** = **D** × **U**)
[εbc]	Beer–Lambert Absorbance $A = \varepsilon\ b\ c$ (Z=ε, Y=b, X=c)	Path b (cm)	Conc c (M)	[εbc]: [A], [E], [B], [C ON] (**A** = **E** × **B** × **C**)

16.3.1 Parsing Chemical Formulas and Weighing Crystalline Reagents

At the analytical balance, the biochemist first computes the required moles of NaCl (n = 0.50 M × 0.250 L = 0.125 mol via [↶] [RXN]), enters Formula Mode (**FML: _**) to parse NaCl (58.440 g/mol), and multiplies $n \times M$ ([×] or [W] in [MW]) to find the dry gram weight W_{NaCl}. Next, the biochemist computes the moles of Tris (n = 0.10 M × 0.150 L = 0.015 mol), parses C4H11NO3 (**C4H11NO3**), and computes W_{Tris}:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.5 [ENTER] **0.25** [↶] [RXN]	0	0	0	0.125	Compute NaCl moles n = 0.50 M × 0.250 L = 0.125 mol
[MW] [FML] **NaCl** [FML]	0	0	0.125	58.44	Parse NaCl molar mass M = 58.440 g/mol; n = 0.125 mol in Y
[×]	0	0	0	7.305	Evaluate NaCl balance weight W_{NaCl} = 0.125 × 58.440 = 7.305 g
0.1 [ENTER] **0.15** [↶] [RXN]	0	0	7.305	0.015	Compute Tris moles n = 0.10 M × 0.150 L = 0.015 mol

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
MW FML **C4H11NO3** FML	0	7.305	0.015	121.136	Parse Tris ($C_4H_{11}NO_3$) molar mass M = 121.136 g/mol
×	0	0	7.305	**1.817**	Evaluate Tris balance weight W_{Tris} = 1.817 g; W_{NaCl} = 7.305 g in Y

Both analytical balance weights sit side by side on the stack: **7.305 g** of NaCl in Register Y and **1.817 g** of Tris in Register X, while looking up Carbon ($Z = 6$) via **6** Z and pressing ↶ W/W recalls the carbon mass fraction of Tris (**39.661%**) and ↷ ▦ displays the reference ledger.

16.3.2 Cutting Stock Aliquots and Back-Solving Unknown Concentration in the Solver Workspace

Next, the biochemist verifies the molarity of a stock bottle prepared with 0.50 mol (29.220 g) in 0.25 L (÷ or ↶ RXN workspace), and computes the exact aliquot volume V_1 of C_1 = 2.0 M Tris stock required to prepare V_2 = 150 mL at C_2 = 0.10 M using C₁V₁:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.5 ENTER **0.25** ÷	0	0	0	2	Verify stock concentration C_1 = 0.50 mol / 0.25 L = 2.00 M
0.1 ENTER **150** ENTER **2** C₁V₁	2	2	2	7.5	Solve required stock aliquot V_1 = (0.10 × 150) / 2.0 = 7.50 mL
150 x⇄y −	2	2	2	**142.5**	Subtract 7.50 mL aliquot from 150 mL to find V_{solvent} = 142.50 mL water

To verify both the dilution and the chromogenic cofactor aliquot using the **Bidirectional Equation Solver Workspaces** (↷ SOLVE / XEQ):

- Pressing C₁V₁ (**C1V1: C × V = D × U**), binding **2** C ON, **0.1** D, and **150** U, and pressing ↷ SOLVE V directly solves for V_1 = **7.50 mL**.
- Next, to verify the diluted cofactor concentration from its UV-Vis cuvette reading without rearranging the Beer–Lambert law by hand, the biochemist presses εbc, opening the **EBC: A = E × B × C** workspace on Line 1:

Solver Action / Softkey	Workspace Binding	Register X	Physical Meaning
εbc	Opens **EBC: A = E × B × C**	**0.0000**	Softkeys map to A, E, B, C ON (LFU0–LFU3)
6600 → E (LFU1)	ε = 6,600 $M^{-1}cm^{-1}$	**6600**	Bind cofactor molar extinction coefficient at 260 nm
1.00 → B (LFU2)	B = 1.00 cm	**1.0000**	Bind standard 1.00 cm quartz cuvette path length
0.845 → A (LFU0)	A = 0.845 AU	**0.8450**	Bind measured UV-Vis absorbance
↷ SOLVE → C ON (LFU3)	Solves $C = A$ / ($E \cdot B$)	**1.2803e-4**	Unknown cofactor molarity C = **1.280 × 10⁻⁴ M**
↷ →mM	Unit conversion M → mM	**0.1280**	Express verified cofactor concentration as **0.1280 mM** (128.0 μM)

16.3.3 Try It on the Stack

Practice chemical formula parsing, hydrate mass corrections, and stock aliquot dilutions on the stack with these three wet-lab preparation exercises:

1. **Sulfuric Acid Stoichiometry and Hydrogen Mass Percentage:** Parse sulfuric acid (H_2SO_4) in Formula Mode ([MW] [FML]), compute the gram mass required for $n = 0.250$ mol, and check the hydrogen mass percentage ([MW] [FML] **H2SO4** [FML] **0.25** [×] **1** [Z] [↶] [w/w]). *(Verified Register X results: M = **98.072 g/mol**, W = **24.518 g**, and $WT\%_H \approx$ **2.056%**.)*

2. **Glucose and Copper(II) Sulfate Pentahydrate Weighing:** Parse D-glucose ($C_6H_{12}O_6$) and find the mass for 0.150 mol ([MW] [FML] **C6H12O6** [FML] **0.15** [×]). Next, look up Copper ($Z = 29$), add the remaining pentahydrate atoms ($SO_4 \cdot 5H_2O = SO_9H_{10}$), and find the mass for 0.0800 mol (**29** [Z] [MW] [FML] **SO9H10** [FML] [+] **0.08** [×]). *(Verified Register X results: $W_{glucose} \approx$ **27.023 g** and $W_{CuSO_4 \cdot 5H_2O} \approx$ **19.974 g**.)*

3. **Normal Saline Aliquot from 5.0 M Stock:** Determine the volume V_1 (in mL) of $C_1 = 5.0$ M NaCl stock needed to prepare $V_2 = 250$ mL of $C_2 = 0.154$ M isotonic saline (**0.154** [ENTER] **250** [ENTER] **5** [C₁V₁]). *(Verified Register X result: V_1 = **7.70 mL** diluted with 242.30 mL water.)*

With our stock solutions and gravimetric salts weighed, we turn in Chapter 17 to the proton equilibria that keep biological macromolecules folded: formulating a pH 7.40 conjugate buffer against an acid spike and standardizing a diprotic acid titrant.

Chapter 17: Holding pH 7.40 Against an Acid Spike

In the adjacent enzymology bay, a postdoctoral fellow is setting up a kinetic assay for a mammalian kinase whose catalytic histidine residue denatures if the reaction medium drifts more than a tenth of a pH unit away from physiological pH = 7.40. During the thirty-minute phosphorylation reaction, ATP hydrolysis releases n_{spike} = 5.0 mmol (0.0050 mol) of free hydrogen ions (H^+) per liter of assay volume. If that reaction were run in unbuffered physiological saline at pH 7.00 ($[H^+] = 10^{-7.0}$ M), releasing 0.0050 M of strong acid would drive the proton concentration up fifty-thousand-fold to $[H^+]$ = 0.0050 M, crashing the solution acidity to pH = $-\log_{10}(0.0050)$ = **2.301**—as acidic as lemon juice—and killing the enzyme within seconds.

To absorb those released protons like a molecular shock absorber, the researcher formulates V = 1.00 L of a C_{total} = 100 mM (0.100 M) conjugate sodium phosphate buffer ($H_2PO_4^-$ / HPO_4^{2-}, with thermodynamic second dissociation constant pK_{a2} = 7.20 at infinite dilution and apparent pK_a' = 6.86 at 0.15 M physiological ionic strength) poised at target pH = 7.40, while also standardizing a diprotic sulfuric acid (H_2SO_4) solution against primary-standard sodium hydroxide (NaOH) at the burette.

In this chapter, we follow three connected equilibrium themes:

1. Map exponential proton concentrations and weak-acid dissociation constants onto Sørensen's logarithmic pH and pK_a scales.
2. Partition the 100 mM phosphate buffer into exact weak-acid (NaH_2PO_4) and conjugate-base (Na_2HPO_4) molarities and gram weights using the Henderson–Hasselbalch equation.
3. Quantify how little the buffer's pH shifts when the 5.0 mM acid spike enters the vessel, and account for diprotic proton stoichiometry at the titration equivalence point.

17.1 Logarithmic Proton Activity and Conjugate Buffer Equilibria

In aqueous biochemistry, water is far more than an inert background solvent; it is an amphoteric reactant that continuously transfers protons between its own molecules and dissolved weak acids and bases.

17.1.1 Water Autoionization and Sorensen's Logarithmic Acidity Scale

Even in ultra-pure deionized water at 25°C, a tiny fraction of water molecules undergo reversible autoionization ($H_2O \rightleftharpoons H^+ + OH^-$), maintaining a constant thermodynamic ion product $K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$ M^2 (so neutral water has $[H^+] = [OH^-] = 1.0 \times 10^{-7}$ M). Applying S. P. L. Sørensen's 1909 negative base-ten logarithm operator ([pH], inverted via [+/−] [↶] [10ˣ]) maps exponential molar proton concentrations $[H^+]$ and weak-acid dissociation constants K_a ([↷] [pKₐ]) onto linear additive scales:

$$\mathrm{pH} = -\log_{10}[H^+] \Leftrightarrow [H^+] = 10^{-\mathrm{pH}}, \quad pK_a = -\log_{10} K_a \Leftrightarrow K_a = 10^{-pK_a}, \quad \mathrm{pH} + \mathrm{pOH} = 14.00 \qquad \textbf{(Eq. 17.1)}$$

In these equations, $[H^+]$ is the molar hydrogen-ion concentration (or thermodynamic activity) in mol/L, and K_a is the equilibrium acid dissociation constant for a weak acid HA $\rightleftharpoons$ $H^+ + A^-$. Because the scale is logarithmic with base ten, each 1.00-unit decrease in pH represents a **tenfold increase** in free proton concentration $[H^+]$: a strong acid solution at $[H^+] = 0.0010$ M (10^{-3} M) has pH = **3.00**, while acetic acid with $K_a = 1.75 \times 10^{-5}$ has $pK_a = -\log_{10}(1.75 \times 10^{-5}) \approx$ **4.757**.

17.1.2 The Henderson-Hasselbalch Equation and Conjugate Species Partitioning

When a weak acid HA and its conjugate base salt A^- coexist in solution at equilibrium, the law of mass action requires $K_a = [H^+][A^-] / [HA]$. Solving for $[H^+] = K_a [HA] / [A^-]$ and taking $-\log_{10}$ of both sides yields the **Henderson–Hasselbalch Buffer Equation** (BUF):

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10}\left(\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}\right) \Leftrightarrow r = \frac{[\mathrm{A}^-]}{[\mathrm{HA}]} = 10^{\mathrm{pH}-\mathrm{p}K_a} \qquad \textbf{(Eq. 17.2)}$$

Here $[A^-]$ is the molar concentration (or total moles in the same vessel volume V) of the proton-accepting conjugate base, $[HA]$ is the molar concentration (or moles) of the proton-donating weak acid, and $r = [A^-] / [HA]$ is the dimensionless conjugate base-to-acid ratio. Notice the structural unity across physical science: just as the Nernst equation shifts electrochemical cell voltage by $(2.3026\ R\ T / n_e\ F) \log_{10}([Ox]/[Red])$ (59.16 mV per decade at 25°C) and decibels measure $10 \log_{10}(P_2/P_1)$, the Henderson–Hasselbalch equation shifts chemical proton potential by 1.00 pH unit per decade change in the conjugate ratio $[A^-]/[HA]$.

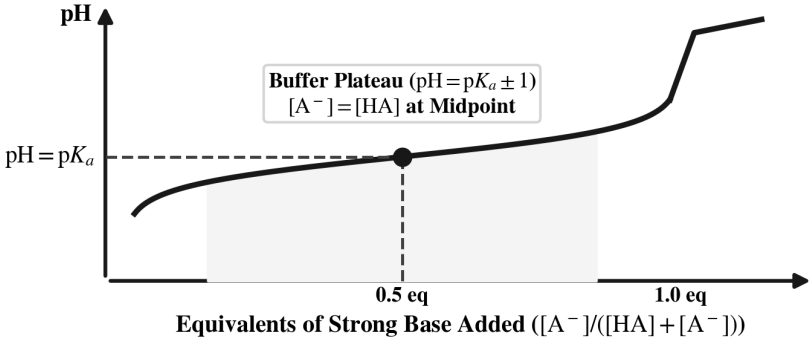


Figure 17.1: *Weak-acid titration curve showing the Henderson-Hasselbalch buffer plateau centered at pH = pKa.*

When a biochemist formulates a buffer of specified total concentration $C_{\mathrm{total}} = [HA] + [A^-]$ at a target pH using a weak acid of known pK_a, the base-to-acid ratio $r = 10^{\mathrm{pH} - \mathrm{p}K_a}$ partitions the total buffer molarity into exact weak-acid and conjugate-base concentrations:

$$[\mathrm{HA}] = \frac{C_{\mathrm{total}}}{1+r} = \frac{C_{\mathrm{total}}}{1+10^{\mathrm{pH}-\mathrm{p}K_a}}, \qquad [\mathrm{A}^-] = C_{\mathrm{total}} - [\mathrm{HA}] = \frac{r \cdot C_{\mathrm{total}}}{1+r} \qquad \textbf{(Eq. 17.3)}$$

For our kinase assay's $C_{\mathrm{total}} = 0.100$ M (100 mM) sodium phosphate buffer at pH = 7.40 using the standard thermodynamic $pK_{a2} = 7.20$, the required base-to-acid ratio is:

$$r = \frac{[\mathrm{HPO_4^{2-}}]}{[\mathrm{H_2PO_4^-}]} = 10^{7.40-7.20} = 10^{0.20} \approx \mathbf{1.5849} \qquad \textbf{(Eq. 17.4)}$$

Dividing C_{total} = 0.100 M by 1 + r = 2.5849 resolves the two individual species concentrations and their required gram masses for V = 1.00 L:

- **Weak Acid (NaH_2PO_4, M = 119.976 g/mol):** $[H_2PO_4^-]$ = 0.100 / 2.5849 ≈ **0.03869 M** (38.69 mM), requiring W_{acid} = 0.03869 × 119.976 ≈ **4.642 g**.
- **Conjugate Base (Na_2HPO_4, M = 141.958 g/mol):** $[HPO_4^{2-}]$ = 0.100 - 0.03869 ≈ **0.06131 M** (61.31 mM), requiring W_{base} = 0.06131 × 141.958 ≈ **8.703 g**.

When ionic-strength activity corrections at physiological salt concentration (I = 0.15 M) lower phosphate's apparent dissociation constant to $pK_a' = 6.86$, the required conjugate ratio at target pH = 7.40 rises to $r' = 10^{7.40 - 6.86} = 10^{0.54} \approx \mathbf{3.4674}$ ($[H_2PO_4^-]$ = 22.38 mM, $[HPO_4^{2-}]$ = 77.62 mM)—an inverse calculation that the BUF Solver Workspace handles directly.

17.2 Buffer Capacity Against Acid Spikes and Volumetric Titrations

Why does that mixture of 38.69 mM $H_2PO_4^-$ and 61.31 mM HPO_4^{2-} hold pH nearly invariant when strong acid is released into the assay? Because the conjugate base scavenges incoming free protons quantitatively before they can accumulate in the water!

17.2.1 Stoichiometric Proton Neutralization Inside a Conjugate Buffer

When a strong acid spike of molar concentration ΔC_{H^+} (such as our kinase reaction's ΔC_{H^+} = 5.0 mM = 0.0050 M proton release) enters a conjugate buffer, the strong acid reacts to completion with the conjugate base ($HPO_4^{2-} + H^+ \rightarrow H_2PO_4^-$). Each mole of added H^+ converts one mole of conjugate base A^- into one mole of weak acid HA, shifting the Henderson–Hasselbalch ratio to:

$$\text{pH}_{\text{after acid}} = pK_a + \log_{10}\left(\frac{[A^-]_0 - \Delta C_{H^+}}{[HA]_0 + \Delta C_{H^+}}\right), \qquad \text{pH}_{\text{after base}} = pK_a + \log_{10}\left(\frac{[A^-]_0 + \Delta C_{OH^-}}{[HA]_0 - \Delta C_{OH^-}}\right) \qquad \textbf{(Eq. 17.5)}$$

Substituting our initial phosphate concentrations ($[A^-]_0$ = 61.31 mM and $[HA]_0$ = 38.69 mM) and the ΔC_{H^+} = 5.0 mM acid spike gives the new concentrations $[A^-]$ = 61.31 - 5.0 = 56.31 mM (0.05631 M) and [HA] = 38.69 + 5.0 = 43.69 mM (0.04369 M). The new base-to-acid ratio is r_{new} = 56.31 / 43.69 ≈ **1.2889**, and the resulting solution pH is:

$$\text{pH}_{\text{after}} = 7.20 + \log_{10}(1.2889) = 7.20 + 0.1102 \approx \mathbf{7.310} \qquad \textbf{(Eq. 17.6)}$$

Instead of plunging 4.7 pH units from 7.00 to 2.30 as it would in unbuffered saline, the 100 mM phosphate buffer restricts the entire 5.0 mM acid spike to a gentle ΔpH = 7.310 - 7.400 = **-0.090 pH units**, keeping the kinase active throughout the thirty-minute assay!

17.2.2 Monoprotic and Diprotic Equivalence Point Stoichiometry

When an analytical chemist standardizes an acid or base by **Volumetric Titration** (↰ EQV), a burette dispenses a measured volume V_A of titrant of known molarity C_A (producing ν_A moles of H^+ or OH^- per formula unit) into a sample volume V_B of unknown molarity C_B (accepting or donating ν_B equivalents per formula unit). At the stoichiometric **equivalence point**, total acid equivalents equal total base equivalents:

$$\nu_A \cdot C_A \cdot V_A = \nu_B \cdot C_B \cdot V_B \Leftrightarrow C_B = \left(\frac{\nu_A}{\nu_B}\right)\frac{C_A \cdot V_A}{V_B} \qquad \textbf{(Eq. 17.7)}$$

For a **monoprotic** neutralization ($\nu_A = \nu_B = 1$, such as HCl + NaOH → NaCl + H2O), equivalence simplifies to $C_A\ V_A = C_B\ V_B$: if $V_A = 24.50$ mL of $C_A = 0.1000$ M standard NaOH neutralizes a $V_B = 25.00$ mL monoprotic acid sample, the acid molarity is $C_B = (0.1000 \times 24.50) / 25.00 =$ **0.0980 M**.

For a **diprotic** acid such as sulfuric acid (H2SO4 + 2 NaOH → Na2SO4 + 2 H2O), each mole of H2SO4 donates $\nu_{acid} = 2$ moles of protons while each mole of NaOH supplies $\nu_{base} = 1$ mole of hydroxide ions. Consequently, neutralizing $V_{acid} = 25.00$ mL of $C_{acid} = 0.0500$ M diprotic H2SO4 ($2 \times 0.0500 \times 25.00 = 2.500$ mmol of H^+ equivalents) with $C_{base} = 0.1000$ M NaOH requires an equivalence volume of $V_{base} = 2.500 / 0.1000 =$ **25.00 mL**. Conversely, if $V_{base} = 24.50$ mL of 0.1000 M NaOH (2.450 mmol OH^-) brings a 25.00 mL sample of diprotic H2SO4 to its second phenolphthalein endpoint, the monoprotic equivalent concentration is 0.0980 N (normal) and the molar concentration of H2SO4 is half that value: $C_{H_2SO_4} = 0.0980 / 2 =$ **0.0490 M**.

17.3 Formulating Conjugate Buffers and Standardizing Acids on the SC-60

On the **SC-60**, logarithmic acidity and dissociation keys ([pH], [→] [pKa], [+/−] [←] [10ˣ]), the Henderson–Hasselbalch buffer operator and workspace ([BUF]), formula gravimetric weighing ([FML], [MW]), and titration equivalence ([←] [EQV]) chain across the RPN stack so a researcher can design a conjugate buffer forward or backward and audit acid spikes or burette endpoints in seconds.

Key / Softkey	RPN Stack Action	Reg Y	Reg X	Solver Workspace ([→] [SOLVE] / [XEQ])
[pH]	Solution Acidity pH = -log10[H^+]	—	[H^+] (M)	Direct 1-key logarithmic evaluation
[→] [pKa]	Acid Dissociation Index pK_a = -log10 K_a	—	Constant K_a	Direct 1-key logarithmic evaluation
[+/−] [←] [10ˣ]	Antilog Concentration [H^+] = 10^{-X}	—	pH or pK_a	Inverts [pH] and [→] [pKa] back to molarity or K_a
[BUF]	Henderson–Hasselbalch pH = pK_a + log10(X)	Acid pK_a	Ratio [A^-] / [HA]	[BUF]: [P], [K], [A], [h] (**P** = **K** + **LOG(A/H)**)
[←] [EQV]	Titration Equivalence $C_A\ V_A = C_B\ V_B$ (Z=C_A, Y=V_A, X=V_B)	Volume V_A or $C_A\ V_A$	Sample V_B or C_B	[←] [EQV]: [A], [V], [B], [U] (**A** × **V** = **B** × **U**)

17.3.1 Partitioning Phosphate Buffer Masses for Physiological pH 7.40

To formulate 1.00 L of the 0.100 M sodium phosphate buffer at pH = 7.40 (pK_a = 7.20), the researcher confirms the base-to-acid ratio $r = 10^{7.40\ -\ 7.20} = 1.5849$ (**7.2** [ENTER] **1.5849** [BUF] → 7.400), partitions 0.100 mol into $n_{acid} = 0.03869$ mol of monobasic NaH2PO4 and $n_{base} = 0.06131$ mol of dibasic Na2HPO4, and converts both mole amounts directly into analytical balance gram weights via [FML] and [×] (or [M] in [MW]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.03869 [ENTER]	0	0	0.0387	0.0387	Load monobasic weak-acid moles $n_{acid} = 0.03869$ mol into Register Y
[MW] [FML] **NaH2PO4** [FML]	0	0.03869	0.0387	119.976	Parse NaH2PO4 molar mass M = 119.976 g/mol into Register X

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[×]	0	0	0.03869	4.642	Compute monobasic NaH_2PO_4 mass $W_{acid} = 4.642$ g
0.06131 [ENTER]	0.03869	4.642	0.0613	0.0613	Load dibasic conjugate-base moles $n_{base} = 0.06131$ mol into Register Y
[MW] [FML] **Na2HPO4** [FML]	4.6419	0.06131	0.0613	141.958	Parse Na_2HPO_4 molar mass $M = 141.958$ g/mol into Register X
[×]	4.6419	4.6419	0.06131	**8.703**	Compute dibasic Na_2HPO_4 mass $W_{base} = 8.703$ g; W_{acid} in Y

Weighing **4.642 g** of NaH_2PO_4 and **8.703 g** of anhydrous Na_2HPO_4 and bringing the solution to 1.00 L poises the assay buffer at pH 7.400.

17.3.2 Evaluating Acid-Spike pH Shifts and Back-Solving Buffer Ratios in the Solver Workspace

Next, the researcher verifies the buffer's pH after the 5.0 mM (0.0050 M) ATP-hydrolysis acid spike reduces $[A^-]$ from 61.31 mM to 56.31 mM and raises [HA] from 38.69 mM to 43.69 mM ([BUF]), and standardizes the 25.00 mL sulfuric acid sample neutralized by 24.50 mL of 0.1000 M NaOH ([↶] [EQV]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
7.2 [ENTER] **56.31** [ENTER] **43.69** [÷]	0	0	7.2	1.2889	Compute post-spike ratio $[A^-]/[HA] = 56.31 / 43.69 = 1.2889$; pK_a in Y
[BUF]	0	0	0	7.3102	Evaluate post-spike buffer pH = 7.20 + $\log_{10}(1.2889) = 7.310$ (ΔpH = -0.090)
0.1 [ENTER] **24.5** [ENTER] **25** [↶] [EQV]	7.3102	7.3102	7.3102	0.098	Evaluate acid normality $N_B = (0.1000 \times 24.50) / 25.00 = 0.0980$ N
2 [÷]	7.3102	7.3102	7.3102	**0.049**	Divide by $\nu = 2$ protons per H_2SO_4 for molarity $C_{H_2SO_4} = 0.0490$ M

What if the researcher is given the target physiological pH = 7.40 and phosphate's ionic-strength-corrected dissociation constant $pK_a' = 6.86$ (or thermodynamic $pK_a = 7.20$), and needs to **solve backward** for the required conjugate base-to-acid ratio A / H without exponentiating by hand? Pressing [BUF] opens the **BUF: P = K + LOG(A/H)** workspace:

Solver Action / Softkey	Workspace Binding	Register X	Physical Meaning
[BUF]	Opens **BUF: P = K + LOG(A/H)**	**0.0000**	Softkeys map to [P], [K], [A], [h] ([LFU0]–[LFU3])
7.40 → [P] ([LFU0])	pH = 7.40	**7.4000**	Bind target physiological buffer pH
6.86 → [K] ([LFU1])	$pK_a' = 6.86$	**6.8600**	Bind apparent phosphate pK_a' at 0.15 M ionic strength
1.00 → [h] ([LFU3])	[HA] = 1.00 (normalized)	**1.0000**	Set weak-acid reference to unity so [A] yields ratio A/H
[↷] [SOLVE] → [A] ([LFU2])	Solves $A = H \cdot 10^{P - K}$	**3.4674**	Required conjugate base-to-acid ratio $A/H =$ **3.4674** (or **1.5849** if $K = 7.20$)

17.3.3 Try It on the Stack

Strengthen your command of logarithmic acidity, Henderson–Hasselbalch buffers, and volumetric titrations with these three analytical exercises:

1. **Acetic Acid Dissociation and 10:1 Acetate Buffer:** Compute pK_a for acetic acid given $K_a = 1.75 \times 10^{-5}$ (**0.0000175** [R↓] [pKa]), and evaluate the pH of an acetate buffer (p$K_a = 4.75$) prepared with a 10 : 1 conjugate-base-to-acid ratio (**4.75** [ENTER] **10** [BUF]). *(Verified Register X results: p$K_a \approx$ **4.757** and pH = **5.750**.)*

2. **Back-Solving Tris Conjugate Base Molarity in the Buffer Solver:** Using the [BUF] solver (**P = K + LOG(A/H)**), find the conjugate base concentration $[A^-]$ required to poise a Tris buffer (p$K_a = 8.06$) with weak acid concentration [HA] = 0.050 M at pH = 8.36 (**8.36** [P] **8.06** [K] **0.05** [h] [R↓] [SOLVE] [A]). *(Verified Register X result: $[A^-] \approx$ **0.0998 M**.)*

3. **Diprotic Sulfuric Acid Equivalence Volume:** What volume V_{NaOH} (in mL) of 0.1250 M NaOH is required to completely neutralize 20.00 mL of 0.0750 M diprotic H_2SO_4 ($\nu = 2$)? (**2** [ENTER] **0.075** [×] **20** [ENTER] **0.125** [R↑] [EQV]). *(Verified Register X result: $V_{NaOH} =$ **24.00 mL**.)*

Having stabilized our enzyme in a pH 7.40 conjugate buffer, we carry the sample to the UV-Vis spectrophotometer, temperature bath, and high-speed centrifuge in Chapter 18 to measure protein concentration with light, model thermal reaction kinetics, and spin down the precipitate.

Chapter 18: Reading Protein Concentration Through a Quartz Cuvette

With the purified enzyme eluted in its pH 7.40 phosphate buffer, the biochemist pipettes a 1.0 mL sample into a fused-silica quartz cuvette of internal optical path length $b = 1.00$ cm and slides it into a UV-Vis spectrophotometer set to $\lambda = 280$ nm. At 280 nm, the conjugated indole rings of tryptophan and phenolic rings of tyrosine absorb ultraviolet photons strongly. Based on the enzyme's amino-acid sequence, its molar extinction coefficient is $\varepsilon_{280} = 42{,}500\ \mathrm{M^{-1}cm^{-1}}$ (or $45{,}000\ \mathrm{M^{-1}cm^{-1}}$ for the wild-type variant). When the monochromator shutter opens, the detector reports an optical absorbance of $A = 0.680$ AU, allowing the biochemist to determine the exact molar concentration c of the enzyme without destroying a single microgram of protein.

Next, the researcher runs the catalytic assay across a 10.0 K temperature step—comparing the baseline reaction rate at standard room temperature $T_1 = 298.15$ K (25.0°C) against body-temperature incubation at $T_2 = 308.15$ K (35.0°C)—for an enzymatic pathway with activation energy barrier $E_a = 52{,}000$ J/mol (52.0 kJ/mol). Finally, once the assay finishes and trichloroacetic acid precipitates the protein, the protocol calls for pelleting the precipitate at a Relative Centrifugal Force of $\mathrm{RCF} = 12{,}000 \times g$ in a benchtop microcentrifuge rotor of radius $r_{\mathrm{cm}} = 9.5$ cm (or 10.0 cm in the floor rotor).

This chapter weaves together three physical-chemistry themes at the analytical bench:

1. Measure dissolved biomolecule molarity and percent transmittance through a quartz cuvette using the Beer–Lambert law.
2. Evaluate head-space gas pressure, Gibbs free-energy spontaneity ($\Delta G = \Delta H - T\Delta S$), equilibrium constants, and Nernst redox potentials.
3. Model Boltzmann–Arrhenius thermal rate acceleration across temperature steps and convert Relative Centrifugal Force ($\times\ g$) into exact rotor Revolutions Per Minute (RPM).

18.1 Exponential Light Attenuation and the Beer-Lambert Law

Why does optical absorbance scale linearly with molar concentration even though light intensity decays exponentially through a coloured or ultraviolet-absorbing liquid? Following a beam of photons slice by slice through the quartz cuvette reveals the origin of August Beer's 1852 law.

18.1.1 Differential Photon Absorption and Logarithmic Optical Density

When a collimated beam of monochromatic light of incident intensity I_0 enters a transparent quartz cuvette of path length b (in centimeters, standardized at $b = 1.00$ cm) containing an absorbing solute at molar concentration c (in mol/L or M), each infinitesimal fluid slice of thickness dx absorbs a fraction of the local photon flux proportional to $c\ dx$.

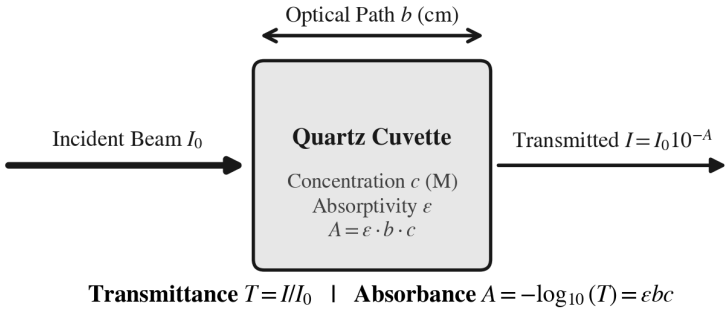


Figure 18.1: *Monochromatic light attenuation through an optical cuvette of pathlength b under the Beer-Lambert absorption law.*

Integrating $dI / I = -\alpha\ c\ dx$ from $x = 0$ to $x = b$ shows that the emergent intensity I and fractional **Transmittance** $T = I / I_0 \in (0, 1]$ decay exponentially with both path length b and concentration c. Defining **Optical Absorbance** A (↶ %T, in dimensionless Absorbance Units, AU) as the negative base-ten logarithm of transmittance linearizes that exponential decay into the **Beer–Lambert Law** (εbc):

$$T = \frac{I}{I_0} = 10^{-A}, \qquad A = -\log_{10}(T) = -\log_{10}\left(\frac{I}{I_0}\right) = \varepsilon \cdot b \cdot c \Leftrightarrow c = \frac{A}{\varepsilon \cdot b} \qquad \textbf{(Eq. 18.1)}$$

In this fundamental equation of analytical spectroscopy, T is the dimensionless fractional transmittance, A is the dimensionless optical absorbance, ε is the compound's **molar absorptivity** (molar extinction coefficient) at the measurement wavelength in $M^{-1}cm^{-1}$ ($L \cdot mol^{-1} \cdot cm^{-1}$), b is the optical path length in cm, and c is the molar concentration in M (mol/L). Notice that A = -log₁₀(I/I_0) is identical to the optical density D of a neutral-density camera filter in Unit IV ($I/I_0 = 10^{-D} = 2^{-\text{stops}}$) and one-tenth of the decibel attenuation dB = -10 log₁₀(P/P_0) along an RF transmission line in Unit V.

18.1.2 Ultraviolet Protein Extinction Coefficients and Stray-Light Linearity

Because A = -log₁₀(T), each +1.000 AU increase in absorbance corresponds to a tenfold reduction in transmitted light intensity:

Transmittance $T = I/I_0$	Percent Transmittance (%T)	Light Absorbed (%)	Absorbance A (↶ %T)	Analytical Instrument Regime
1.000	100.0%	0.0%	0.0000 AU	Pure blank buffer baseline
0.500	50.0%	50.0%	0.3010 AU	Optimal high-precision window
0.2089	20.89%	79.11%	0.6800 AU	Optimal protein assay reading
0.100	10.0%	90.0%	1.0000 AU	Upper linear standard range
0.010	1.0%	99.0%	2.0000 AU	High optical density; dilute aliquot
0.001	0.1%	99.9%	3.0000 AU	Detector stray-light saturation limit

For our purified enzyme reading $A = 0.680$ AU ($T = 10^{-0.680} \approx 0.2089$, meaning 79.11% of the 280 nm photons are absorbed across the $b = 1.00$ cm quartz cuvette):

- If the engineered enzyme has molar extinction coefficient $\varepsilon = 42{,}500\ \mathrm{M^{-1}cm^{-1}}$, its molar concentration is $c = 0.680 / (42{,}500 \times 1.00) = \mathbf{1.600 \times 10^{-5}\ M}$ (**16.00 μM**).
- If the wild-type variant with $\varepsilon = 45{,}000\ \mathrm{M^{-1}cm^{-1}}$ reads $A = 0.680$ AU (or $A = 0.650$ AU), its concentration is $c = 0.680 / 45{,}000 \approx \mathbf{1.5111 \times 10^{-5}\ M}$ (15.11 μM, or $\mathbf{1.4444 \times 10^{-5}\ M}$ at $A = 0.650$ AU).

18.2 Chemical Thermodynamics, Arrhenius Kinetics, and Centrifugal Fields

Once the enzyme concentration is quantified, physical chemistry governs how temperature accelerates its catalytic turnover, how free energy drives chemical and electrochemical equilibria, and how centrifugal fields sediment precipitated macromolecules.

18.2.1 Ideal Gases, Gibbs Free Energy, Nernst Potentials, and Arrhenius Barriers

Across physical and biophysical chemistry, the universal gas constant $R = 8.314462618$ J/(mol·K) ([⇗] [R]), equal to Avogadro's constant [↖] [Nₐ] times Boltzmann's constant [↖] [kB]) and Faraday's constant $F = 96{,}485.33$ C/mol ([↖] [F]) link thermal kinetic energy $R\ T$ (at absolute temperature T in Kelvin, K = °C + 273.15, converted via [⇗] [→K]) to five foundational state and rate equations:

1. **Ideal Gas Law ([PVT]):** A gas of chemical amount n (mol) in volume V (L) exerts pressure $P = n\ R\ T\ /\ V$ (where $n = 1.00$ mol at $T = 298.15$ K in $V = 24.79$ L exerts $P \approx$ **100.0 kPa** or 0.9869 atm, convertible between [↖] [→atm] and [⇗] [→kPa]).
2. **Gibbs Free Energy ([↖] [ΔG]) and Equilibrium Constant ([⇗] [K₌q]):** Combining enthalpy change ΔH and entropy change ΔS defines $\Delta G = \Delta H - T\ \Delta S$ (convertible between [↖] [→J] and [⇗] [→cal]), which determines the standard free-energy relation $\Delta G^\circ = -R\ T \ln K_{eq}$ ([⇗] [K₌q]) and equilibrium constant $K_{eq} = \exp(-\Delta G^\circ\ /\ R\ T)$ at 298.15 K ($R\ T \approx 2478.96$ J/mol).
3. **Nernst Electrochemical Potential ([⇗] [E°]):** For a redox cell with standard potential E°, electron transfer count n_e, and reaction quotient Q, the operating voltage at 298.15 K is $E = E^\circ - (0.05916\ /\ n_e) \log_{10} Q$.

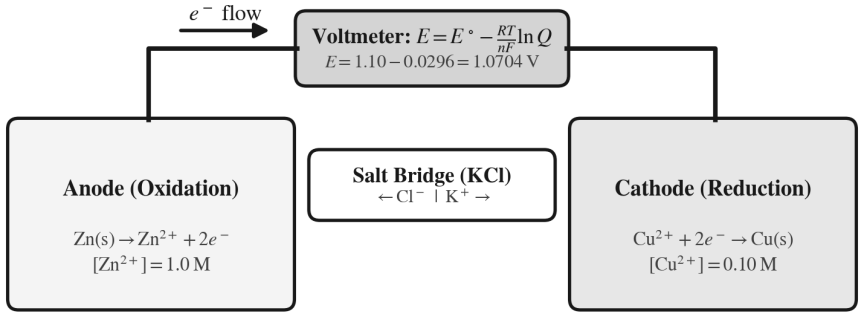


Figure 18.2: *Galvanic Zn-Cu Daniell cell schematic showing anode oxidation, cathode reduction, salt bridge, and Nernst potential.*

4. **Arrhenius Activation Rate Law (↗ Eₐ) and First-Order Half-Life (↗ t½):** In 1889, Swedish physicist and chemist Svante Arrhenius showed that reactants must surmount an activation energy barrier E_a (in J/mol) to reach the transition state, so the rate constant $k = A_{freq} \exp(-E_a / R\ T)$ (whose Boltzmann factor $\exp(-E_a / R\ T)$ is evaluated directly by ↗ Eₐ, paired with first-order half-life $t_{1/2} = \ln(2) / k$, ↗ t½) grows exponentially with temperature. Comparing reaction rates k_1 and k_2 between two absolute temperatures T_1 and T_2 yields both the forward **Arrhenius Acceleration Ratio** and the inverse **Temperature Ceiling** T_2 for a maximum allowable rate ratio k_2 / k_1:

$$\frac{k_2}{k_1} = \frac{\exp(-E_a/RT_2)}{\exp(-E_a/RT_1)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] \Leftrightarrow \frac{1}{T_2} = \frac{1}{T_1} - \frac{R}{E_a}\ln\left(\frac{k_2}{k_1}\right) \quad \textbf{(Eq. 18.2)}$$

For our enzyme assay with activation barrier E_a = 52,000 J/mol warmed across a 10.0 K interval from T_1 = 298.15 K (25.0°C) to T_2 = 308.15 K (35.0°C), the inverse-temperature factor is $(T_2 - T_1) / (T_1\ T_2) = 10.0 / (298.15 \times 308.15) \approx 1.08844 \times 10^{-4}\ K^{-1}$. Multiplying by E_a / R = 52,000 / 8.31446 ≈ 6254.16 K gives an exponent of 0.68073, so the catalytic turnover rate accelerates by:

$$\frac{k_2}{k_1} = \exp(0.68073) \approx \mathbf{1.9753\times} \quad \textbf{(Eq. 18.3)}$$

Conversely, if a chemical reactor or enzymatic bath must never exceed k_2 / k_1 = 3.00× its 25.0°C (298.15 K) baseline rate before thermal runaway or denaturation begins, solving the inverse Arrhenius equation gives $1/T_2 = 1/298.15 - (8.31446 / 52{,}000) \ln(3.00) = 0.00335402 - 0.00017565 = 0.00317837\ K^{-1}$, establishing a maximum safe temperature ceiling of T_2 = **314.63 K** (**41.48°C**).

18.2.2 Rotor Geometry and Relative Centrifugal Force Scaling

When a centrifuge spins a sample tube at angular velocity $\omega = 2\pi \cdot \text{RPM} / 60$ (in radians per second) at radial distance r from the central spindle axis, the outward centripetal/centrifugal acceleration is $a_c = \omega^2\ r$. Because different centrifuge rotors have different physical radii r_{cm} (in centimeters), quoting rotor speed in RPM alone is ambiguous. Dividing a_c by standard Earth gravity (g = 9.80665 m/s² = 980.665 cm/s²) defines the dimensionless **Relative Centrifugal Force** (↗ RCF, expressed in multiples of Earth gravity, × g):

$$\mathrm{RCF} = \frac{\omega^2 r}{g} = \left(\frac{2\pi}{60}\right)^2 \frac{r_{\mathrm{cm}}}{980.665} \cdot \mathrm{RPM}^2 \approx 1.118 \times 10^{-5} \cdot r_{\mathrm{cm}} \cdot \mathrm{RPM}^2 \Leftrightarrow \mathrm{RPM} = \sqrt{\frac{\mathrm{RCF}}{1.118 \times 10^{-5} \cdot r_{\mathrm{cm}}}} \qquad \textbf{(Eq. 18.4)}$$

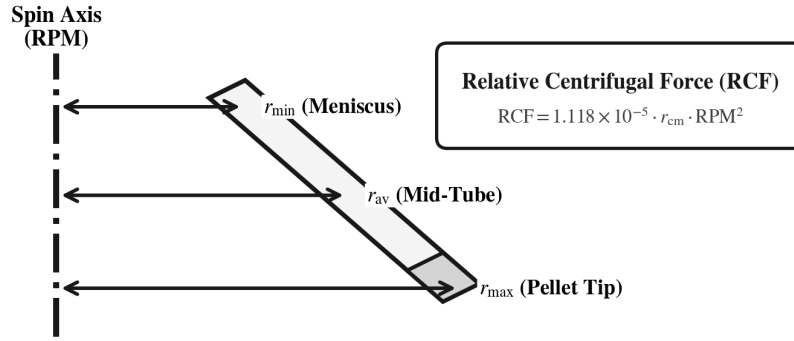


Figure 18.3: *Fixed-angle centrifuge rotor radial geometry showing r_min, r_av, and r_max relative to the spin axis.*

In a floor rotor of radius $r_{\mathrm{cm}} = 10.0$ cm spinning at 10,000 RPM, the sample experiences $\mathrm{RCF} = 1.118 \times 10^{-5} \times 10.0 \times 10^{8} = \mathbf{11{,}180} \times \boldsymbol{g}$. Conversely, when our protein precipitation protocol specifies a sedimentation field of RCF = 12,000 × *g* in a benchtop rotor of radius $r_{\mathrm{cm}} = 9.5$ cm (or $r_{\mathrm{cm}} = 10.0$ cm), solving the inverse square-root relation gives the required spindle speed:

- For $r_{\mathrm{cm}} = 9.5$ cm: $\mathrm{RPM} = \sqrt{(12{,}000 \,/\, (1.118 \times 10^{-5} \times 9.5))} = \sqrt{(12{,}000 \,/\, 1.0621 \times 10^{-4})} \approx \mathbf{10{,}629.4\ RPM}$.
- For $r_{\mathrm{cm}} = 10.0$ cm: $\mathrm{RPM} = \sqrt{(12{,}000 \,/\, (1.118 \times 10^{-5} \times 10.0))} = \sqrt{(12{,}000 \,/\, 1.118 \times 10^{-4})} \approx \mathbf{10{,}360.2\ RPM}$.

18.3 Quantifying Proteins, Reaction Rates, and Rotor Speeds on the SC-60

On the **SC-60**, optical transmittance-to-absorbance conversion ([↶] [%T]), the three-variable Beer–Lambert law and solver ([εbc]), ideal gas pressure ([PVT]), Gibbs free energy and equilibrium constants ([↶] [ΔG], [↷] [K₀q]), Nernst cell voltage ([↷] [E°]), Arrhenius rate kinetics ([↷] [Eₐ], [↷] [t½]), calorimetry ([↶] [CAL]), and centrifuge force conversion ([↷] [RCF]) operate both directly across the RPN stack ([4-LV] or [8-LV]) and as bidirectional Equation Solver workspaces ([↷] [SOLVE] / [XEQ]).

Key / Softkey	RPN Stack Action	Reg Z	Reg Y	Reg X	Solver Workspace ([↷] [SOLVE] / [XEQ])
[↶] [%T]	Optical Absorbance A = -log₁₀(T)	—	—	Transmittance T	Direct 1-key logarithmic evaluation
[εbc]	Beer–Lambert Law $A = \varepsilon \cdot b \cdot c$	Extinction ε	Path b (cm)	Conc c (M)	[εbc]: [A], [E], [B], [C/cm] ($\mathbf{A} = \mathbf{E} \times \mathbf{B} \times \mathbf{C}$)

Key / Softkey	RPN Stack Action	Reg Z	Reg Y	Reg X	Solver Workspace ([→] SOLVE / XEQ)
PVT	Ideal Gas Law $P = n\,R\,T\,/\,V$ (Z=n, Y=T, X=V; or Y=n, X=V at 298.15 K)	Moles n (mol)	Temp T (K) or n	Volume V (L)	PVT: P, V, N, T (**P × V = N × R × T**)
[→] Eₐ / [→] t½	Boltzmann Factor $\exp(-E_a\,/\,R\,T)$ / Half-life $\ln(2)/k$	—	Energy E_a (J/mol)	Temp T (K) or k	Evaluates Boltzmann factor or $t_{1/2}$
[←] ΔG / [→] K⇄q	Gibbs $\Delta G = \Delta H - T\,\Delta S$ / $\Delta G^\circ = -R\,T \ln K_{eq}$	Enthalpy ΔH	Temp T (K)	Entropy ΔS or K_{eq}	[←] ΔG: G, h, T, S (**G = H - T × S**)
[→] E°	Nernst Voltage $E = E^\circ - (0.05916/n_e)\,\log_{10} Q$	—	Standard E° (V)	Reaction Quotient Q	[→] E°: E, S, N, Q (**E = S - .059/N LOG Q**)
[→] RCF	Relative Centrifugal Force ($\times\,g$)	—	Radius r_{cm}	Speed (RPM)	Evaluates RCF = $1.118 \times 10^{-5} \cdot r_{cm} \cdot \text{RPM}^2$

18.3.1 Solving Cuvette Protein Concentration and Arrhenius Rate Ratios

At the spectrophotometer, the biochemist converts a 50% calibration filter ($T = 0.50$) to absorbance ([←] %T), resolves the purified enzyme concentration c from $A = 0.680$ AU in the $b = 1.0$ cm quartz cuvette for $\varepsilon = 42{,}500\ \text{M}^{-1}\text{cm}^{-1}$ (and checks the forward product via [εbc]), and computes the Arrhenius rate acceleration between $T_1 = 298.15$ K and $T_2 = 308.15$ K for $E_a = 52{,}000$ J/mol ([→] Eₐ):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.5 [←] %T	0	0	0	0.301	Verify 50% transmittance ($T = 0.50$) equals $A = 0.3010$ AU
0.68 ENTER **42500** ÷	0	0	0.301	0.000016	Solve enzyme concentration $c = A\,/\,(\varepsilon\,b) = 0.680\,/\,42{,}500 = 1.60 \times 10^{-5}$ M
42500 ENTER **1** x⇄y εbc	0.30103	0.30103	0.301	0.68	Forward [εbc] check: $\varepsilon(42{,}500) \times b(1.0) \times c(16\ \mu\text{M}) = 0.680$ AU
52000 ENTER **308.15** [→] Eₐ **52000** ENTER **298.15** [→] Eₐ ÷	0.68	0.68	0.68	**1.9753**	Evaluate Arrhenius rate acceleration $k_2\,/\,k_1 = 1.9753\times$ from 298.15 K to 308.15 K

18.3.2 Evaluating Free-Energy Equilibria, Cell Potentials, Rotor Speeds, and Inverse Solver Runs

Next, the researcher evaluates ΔG for an exothermic binding step ($\Delta H = -100$ kJ/mol, $T = 300$ K, $\Delta S = -0.10$ kJ/(mol·K)), verifies $\Delta G^\circ = -5708$ J/mol for an equilibrium constant $K_{eq} = 10.0$ at 298.15 K ([→] K⇄q), checks a redox cell Nernst voltage ($E^\circ = 1.10$ V, $n_e = 2$, $Q = 10$ via [→] E°), evaluates RCF at $r_{cm} = 10$ cm and 10,000 RPM, and solves the required RPM for RCF = 12,000 × g on the $r_{cm} = 9.5$ cm benchtop rotor:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
100 [+/−] [ENTER] **300** [ENTER] **0.1** [+/−] [↶] [ΔG]	0	0	0	-70	Evaluate ΔG = -100 - 300(-0.10) = -70.0 kJ/mol
298.15 [ENTER] **10** [→] [K⇄q]	0	0	-70	-5708.0	Standard free energy ΔG° = $-R(298.15)\ \ln(10)$ = -5708.0 J/mol
[→] [E°] **1.1** [S] **2** [N] **10** [Q] [→] [SOLVE] [E]	1.1	2	10	1.0704	Nernst cell voltage E = 1.10 - (0.05916/2) $\log_{10}(10)$ = 1.0704 V
10 [ENTER] **10000** [→] [RCF]	10	10	1.0704	11180	Relative Centrifugal Force RCF = 11,180 × g at r = 10 cm, 10,000 RPM
12000 [ENTER] **1.118** [E] **5** [+/−] [÷] **9.5** [÷] [√x]	1.0704	1.0704	11180	**10629.4**	Solve required rotor speed RPM = √(12,000 / (1.118 × 10^{-5} × 9.5)) = 10,629 RPM

When an analytical question runs **backward**—such as finding enzyme concentration c from absorbance A, finding the unknown reaction quotient Q from a measured Nernst electrode potential E = 1.0408 V, or finding the maximum safe reactor temperature T_2 before an Arrhenius rate triples (k_2/k_1 = 3.00)—the **Bidirectional Solver Workspaces** ([εbc] and [→] [E°]) and [→] [R] constant key eliminate algebraic rearrangement:

Solver / Stack Action	Workspace Binding / Sequence	Register X	Physical Meaning
[εbc]: **0.68** [A] **42500** [E] **1** [B] → [→] [SOLVE] [c/cm]	**A = E × B × C** solved for C	**1.6000e-5**	Purified enzyme concentration c = **16.00 μM** (**0.0160 mM**)
[→] [E°]: **1.0408** [E] **1.10** [S] **2** [N] → [→] [SOLVE] [Q]	**E = S - .059/N LOG Q** solved for Q	**100.3**	Unknown redox reaction quotient Q = **100** ($\log_{10} Q$ = 2.00)
1 [ENTER] **298.15** [÷] **3** [LN] [→] [R] [×] **52000** [÷] [-] **1** [x⇄y] [÷]	Inverse Arrhenius T_2 for k_2/k_1 = 3.0	**314.63**	Maximum safe reactor temperature T_2 = **314.63 K** (**41.48°C**)

18.3.3 Try It on the Stack

Consolidate your command of Beer–Lambert spectrophotometry, chemical thermodynamics, and centrifugation with these three biophysical chemistry exercises:

1. **Wild-Type Enzyme Concentration in the Beer-Lambert Solver:** A wild-type enzyme with ε_{280} = 45,000 $M^{-1}cm^{-1}$ in a b = 1.0 cm cuvette gives an absorbance reading of A = 0.650 AU. Open [εbc] (**EBC: A = E × B × C**) and solve for molar concentration c (**0.65** [A] **45000** [E] **1** [B] [→] [SOLVE] [c/cm]). *(Verified Register X result: $c \approx$ **1.4444 × 10^{-5} M**, or 14.44 μM.)*

2. **Ideal Gas Headspace Pressure and Redox Potential:** Find the pressure P (in kPa) exerted by n = 1.00 mol of ideal gas at T = 298.15 K in a V = 24.79 L vessel (**1** [ENTER] **24.79** [PVT]), and evaluate the Nernst potential for a half-cell with E° = 0.800 V, n_e = 2, and Q = 100 ([→] [E°] **0.8** [S] **2** [N] **100** [Q] [→] [SOLVE] [E]). *(Verified Register X results: $P \approx$ **100.0 kPa** and $E \approx$ **0.7408 V**.)*

3. **High-Speed Microcentrifuge RCF at 14,000 RPM:** A microcentrifuge with rotor radius r_{cm} = 8.5 cm spins at 14,000 RPM. Compute the Relative Centrifugal Force in multiples of Earth gravity (**8.5** [ENTER] **14000** [→] [RCF]). *(Verified Register X result: $RCF \approx$ **18,625.9 × g**.)*

From analytical balances, conjugate buffers, and quartz cuvettes in the wet lab, we turn in **Unit VII: Commercial Baking, Dough Thermodynamics & Brewing Science** to applied food and fermentation science—where flour-anchored baker's percentages, dough thermal balances, wort gravity scales, and hop isomerization kinetics govern the bakery and

brewhouse.

Unit VII: Commercial Baking, Dough Thermodynamics & Brewing Science

In domestic kitchens, recipes are traditionally recorded in volumetric scoops—two cups of flour, one cup of water, a teaspoon of salt—yet granular solids pack with variable bulk density. Depending on ambient humidity, milling particle size, and whether the grain was sifted or tamped, a standard United States cup of wheat flour (236.6 mL) weighs anywhere from 110 g to 160 g. That 45% mass variation disrupts gluten cross-linking, yeast osmotic pressure, and charcuterie curing safety. Even after replacing volumetric scoops with gram balances, expressing a dough formula in conventional weight percentages that sum to 100% (Flour 51.49% + Water 37.08% + Levain 10.30% + Salt 1.13% = 100.00%) couples every ingredient together. Raising water by 2% to compensate for a drier winter wheat harvest forces the numerical percentages of flour, levain, and salt to shrink so the ledger still totals 100%. Nineteenth-century European guild and commercial bakers resolved that coupling by anchoring every ingredient mass to total flour mass fixed at 100.0%. In this Baker's Percentage system, hydration, inoculation, and salt ratios remain independent invariants across any batch size.

Controlling biological fermentation in the bakery and brewhouse requires balancing mechanical work against sensible and latent heat. Baker's yeast (*Saccharomyces cerevisiae*) and sourdough lactic acid bacteria (*Fructilactobacillus sanfranciscensis*) roughly double their metabolic carbon-dioxide evolution for every 8°C (15°F) rise in dough temperature. When heavy electric spiral mixers entered twentieth-century bakeries, viscous shear between the rotating steel hook and the gluten matrix converted mechanical work into thermal energy, warming the dough during mixing. To hit a repeatable post-mix Desired Dough Temperature, bakers solve an inverse enthalpy balance for the strike-water temperature. Alternatively, they replace part of the tap water with flake ice, whose 80 kcal/kg latent heat of fusion absorbs excess summer heat. Meanwhile, all-grain brewers balance the specific heat of dry malted barley against hot strike liquor to land on the 64°C–69°C enzymatic saccharification rest.

Quantitative brewing science (*zymurgy*) traces its mathematical foundation to 1843 in Prague. There, Bohemian chemist Carl Joseph Napoleon Balling calibrated aqueous sucrose density tables so brewers could monitor malt starch conversion and yeast attenuation with a floating hydrometer measuring Specific Gravity. In 1854, Prussian civil engineer Adolf Ferdinand Wenceslaus Brix refined Balling's pycnometric tables for the beet-sugar and fermentation trades. In 1900, German chemist Fritz Plato standardized the extract scale (1°P = 1 g sucrose per 100 g wort) for the Imperial Normal Calibration Commission, linking the gravity drop from temperature-corrected Original Gravity to Final Gravity to Alcohol by Volume. Nearly a century later in 1995, Oregon brewing scientist Glenn Tinseth modeled the first-order thermal isomerization kinetics of hop alpha acids (*Humulus lupulus*) across boil time and wort gravity. Notice the cross-domain mathematical bridge between the brewhouse and the physics laboratory. Tinseth's boil-time utilization factor $f_t(t) = (1 - e^{-0.04t})/4.15$ obeys the identical first-order exponential saturation law $1 - e^{-t/\tau}$ that governs capacitor charging across an RC network in Unit III (**SC-16**). Similarly, the grain-water mash enthalpy balance mirrors the first-law calorimetry equation $Q = m\ c\ \Delta T$ in Unit VI (**SC-60**).

The **SC-78 Culinary, Baking, Brewing & Hospitality** calculator brings these gravimetric, thermodynamic, and fermentation equations to the production bench. Each

chapter in this Unit follows a single production narrative through three connected mathematical themes:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 19	Scaling 200 country sourdough loaves (176.2 kg dough) or a constrained 85.0 kg mixer bowl on a July morning and tempering 9°C spiral-mixer friction with flake ice	Flour-anchored Baker's Percentages, true pre-ferment hydration, inverse batch scaling, four-term Desired Dough Temperature balance, and latent-heat ice substitution	1. Flour-Anchored Baker's Percentages & Batch Scaling • 2. True Dough Hydration Across Pre-Ferments • 3. Spiral-Mixer Friction & Flake-Ice Latent Heat
Chapter 20	Formulating mash strike liquor for a 10 BBL pale ale, tracking attenuation from OG 1.056 to FG 1.012, and solving hop charges for 41–42 IBU	First-law grain-water mash calorimetry, Balling–Brix–Plato extract polynomials, apparent attenuation, Tinseth hop isomerization kinetics, and Morey color	1. Mash Strike-Water Enthalpy Balance • 2. Wort Extract Scales, Attenuation & Alcohol by Volume • 3. Tinseth Hop Alpha-Acid Isomerization & Malt Color
Chapter 21	Trimming 45 lb of whole beef tenderloin across multi-stage fabrication yields, pricing an $8.40 entrée at 28% food cost (72% gross margin), and auditing water activity and wine SO_2	Compounding multi-stage fabrication yields, hyperbolic asymmetry between Margin and Markup, Raoult water activity, and pH-dependent molecular SO_2 equilibrium	1. Compounding Multi-Stage Fabrication Yields • 2. Raoult Water Activity & Wine Sulfite Equilibria • 3. Yield-Adjusted Costing & Margin vs. Markup

Chapter 19: Scaling Two Hundred Country Loaves on a Hot July Morning

At 4:30 a.m. on a humid July morning, the ambient air inside a stone-hearth wholesale bakery already stands at $R = 28.0$°C, and the organic bread flour stored in the mezzanine silos measures $F = 26.0$°C. The head baker must mix enough naturally leavened country sourdough to divide $N = 200$ loaves scaled at $W = 881$ g (0.881 kg) each—a total deck-oven batch of $T = 176.2$ kg (176,200 g) of finished dough. The bakery's master formula specifies 72.0% water, 20.0% mature liquid levain (maintained at 100% internal hydration), and 2.2% sea salt anchored to 100.0% mixer flour. Simultaneously, the pastry baker on the adjacent bench has a smaller spiral mixer whose removable stainless bowl has a maximum mechanical dough capacity of $T = 85.0$ kg. That baker needs to know the exact anchor flour mass F required to fill the 85.0 kg bowl to capacity when a seeded bâtard formula sums to $P = 179.2\%$.

Scaling the ingredient masses solves only the gravimetric half of the morning's problem. Fifteen minutes of first- and second-speed kneading in the heavy spiral mixer injects $M = 9.0$°C of mechanical friction heat into the gluten network. If the baker draws municipal tap water at 21.0°C straight from the wall hose, the dough will leave the mixer above 30°C: lactobacilli will race ahead of yeast fermentation, proteolytic enzymes will slacken the gluten matrix, and the shaped boules will puddle flat on the loader peels before reaching the steam-injected hearth decks.

To hold the post-mix Desired Dough Temperature at an exact $D = 24.0$°C, the baker works through three connected calculations:

1. Scale the flour-anchored Baker's Percentage ledger forward for the 200-loaf deck-oven batch and backward for the constrained 85.0 kg spiral mixer bowl.
2. Partition the liquid levain into its underlying flour and water components to verify True Overall Dough Hydration.
3. Solve the four-term Desired Dough Temperature balance for the required strike-water temperature (9.0°C), and calculate the exact mass of sub-zero flake ice needed to cool the 21.0°C municipal tap supply.

19.1 Flour-Anchored Baker's Percentages and Pre-Ferment Hydration

Commercial dough formulation rests on decoupling ingredient adjustments from one another through a single invariant mass reference. By anchoring every solid and liquid component to total flour weight rather than total batch weight, a baker can modify hydration for a thirsty spring-wheat harvest or adjust salt for ambient humidity while leaving every other ratio in the formula unchanged.

19.1.1 Decoupling Ingredient Ratios Through the Flour Anchor and Total Batch Scaling

In the **Baker's Percentage** (BK%) coordinate system, the combined mass of all milled grain flours introduced into the mixer—high-protein bread flour, stoneground whole wheat, rye, or Khorasan—is defined as the reference flour mass F and assigned 100.0%:

$$F = \sum_{k=1}^{m} F_k \equiv 100.0\%, \qquad P_i = \left(\frac{I_i}{F}\right) \times 100, \qquad I_i = F \times \left(\frac{P_i}{100}\right) \tag{Eq. 19.1}$$

In these expressions, P_i is the Baker's Percentage ([BK%]) of the i-th ingredient of mass I_i (in grams or kilograms) relative to the anchor flour mass F.

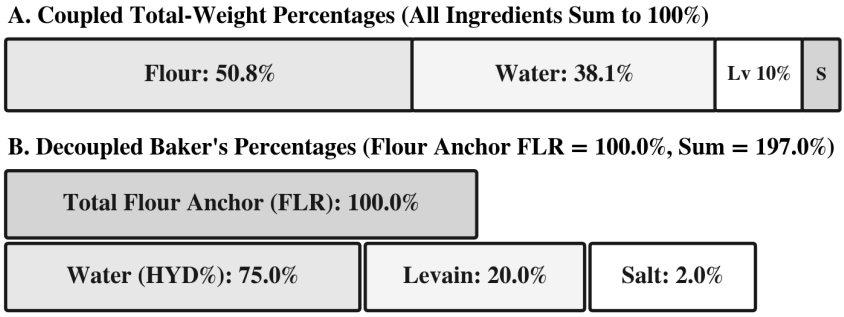


Figure 19.1: *Comparison of coupled total-weight percentages (sum = 100%) versus flour-anchored Baker's Percentages (FLR = 100%).*

Summing all individual Baker's Percentages including the 100.0% flour anchor defines the **Total Formula Percentage** $P = 100.0 + \sum_i P_i$. On the **SC-78**, entering the reference flour mass F in Register Y and the first ingredient mass I_1 in Register X and pressing [BK%] locks F as the active base flour weight, logs the ingredient into the live **@BK%** recipe ledger in [↷] [▦], and returns $P_1 = (I_1 \,/\, F) \times 100$. Subsequent presses of **I_k** [BK%] log each remaining ingredient against the locked flour anchor F. Pressing [ΣW] recalls the cumulative dough batch weight $T = F + \sum_i I_i$ from the **@BK%** ledger, pressing [↰] [SCL] scales the entire **@BK%** ledger to a new target batch weight T_{target} (or by a multiplier factor), and pressing [↰] [DIV] governs the **Dough Divider Scaling Law** linking total batch mass T, loaf count N, and unit portion weight W:

$$T = \frac{F \times P}{100} \Leftrightarrow F = \frac{100 \times T}{P}, \qquad W = \frac{T}{N} \Leftrightarrow N = \frac{T}{W} \tag{Eq. 19.2}$$

Furthermore, opening the [BK%] Equation Solver Workspace (**P = (I / F) × 100**) binds softkeys [P], [I], and [F] so the baker can solve **backward** via [XEQ] for the exact flour mass F required to yield a total batch mass $I = T$ when the total formula percentage is P.

In the country sourdough formula on our July production sheet, mixer flour (100.0%), water (72.0%), mature levain (20.0%), and sea salt (2.2%) sum to a Total Formula Percentage of $P = 100.0 + 72.0 + 20.0 + 2.2 = 194.2\%$. To produce $N = 200$ loaves at $W = 0.881$ kg each ([↰] [DIV]), the required total dough batch is $T = 200 \times 0.881 =$ **176.20 kg**. Scaling the ledger via [↰] [SCL] (or solving the [BK%] relation backward for the anchor flour mass) yields $F = (100 \times 176.20) \,/\, 194.2 =$ **90.7312 kg**. Scaling each percentage against $F = 90.7312$ kg produces $I_{\text{water}} = 90.7312 \times 0.72 =$ **65.3265 kg** of water, $I_{\text{levain}} = 90.7312 \times 0.20 =$ **18.1462 kg** of mature levain, and $I_{\text{salt}} = 90.7312 \times 0.022 =$ **1.9961 kg** of sea salt, which sum to **176.200 kg** of mixed dough. Meanwhile, for the adjacent spiral mixer capped at $T = 85.0$ kg running the seeded bâtard formula ($P = 179.2\%$), solving [BK%] backward for [F] gives $F = (100 \times 85.0) \,/\, 179.2 =$ **47.4330 kg** of flour.

19.1.2 Decomposing Pre-Ferment Mass and True Overall Hydration

Notice why bakers distinguish between **Mixer Formula Hydration** (72.0%) and **True Overall Dough Hydration** ($\mathrm{HYDR}_{\mathrm{true}}$, evaluated via [→] [HYD] or [BK%]). When a formula incorporates a pre-ferment of mass W_{pre}—such as a liquid levain ([→] [LV%], [LEV]), a 100% poolish, or a stiff 55% Italian biga—maintained at its own internal hydration percentage H_{pre}, that pre-ferment already carries pre-hydrated flour $W_{\mathrm{pre,flour}}$ and water $W_{\mathrm{pre,water}}$ into the mixing bowl:

$$W_{\mathrm{pre,\,flour}} = \frac{W_{\mathrm{pre}}}{1 + (H_{\mathrm{pre}}/100)}, \qquad W_{\mathrm{pre,\,water}} = W_{\mathrm{pre}} - W_{\mathrm{pre,\,flour}}, \qquad \mathrm{HYDR}_{\mathrm{true}} = \left(\frac{W_{\mathrm{water}} + W_{\mathrm{pre,\,water}}}{W_{\mathrm{flour}} + W_{\mathrm{pre,\,flour}}}\right) \times 100 \qquad \textbf{(Eq. 19.3)}$$

Here $W_{\mathrm{pre,flour}}$ and $W_{\mathrm{pre,water}}$ partition the pre-ferment mass W_{pre}, and $\mathrm{HYDR}_{\mathrm{true}}$ expresses the combined water from both the main mix and the pre-ferment as a percentage of total combined grain flour ($W_{\mathrm{flour}} + W_{\mathrm{pre,flour}}$).

In the 176.2 kg batch above, the W_{levain} = 18.1462 kg of liquid levain built at H_{pre} = 100.0% hydration contains equal masses of fermented flour and water: $W_{\mathrm{pre,flour}}$ = 18.1462 / 2 = **9.0731 kg** and $W_{\mathrm{pre,water}}$ = **9.0731 kg**. Adding those pre-fermented components to the main mixer additions reveals a combined grain flour mass of 90.7312 + 9.0731 = **99.8043 kg** and a combined water mass of 65.3265 + 9.0731 = **74.3996 kg**, giving a True Overall Dough Hydration ([→] [HYD]) of $\mathrm{HYDR}_{\mathrm{true}}$ = (74.3996 / 99.8043) × 100 = **74.545%** ((72 + 10) / (100 + 10) = (82/110) ≈ 74.55%).

19.2 Dough Mixing Thermodynamics and Latent-Heat Ice Substitution

Once the dry and liquid masses are weighed on the bench platform scale, the baker must control the thermal state of the dough leaving the spiral mixer. Because flour and ambient air temperatures cannot be altered rapidly on the production floor, water temperature serves as the bakery's primary thermal control actuator.

19.2.1 The Four-Variable Desired Dough Temperature Balance ([TW])

To maintain a repeatable four-hour bulk fermentation schedule, mixed sourdough must exit the bowl at a target **Desired Dough Temperature** (D, typically 24.0°C for country hearth loaves). Under the commercial Three-Factor Rule ([TW]), the three thermal reservoirs that govern final dough temperature—ambient bakery room air (R), stored flour (F), and strike water (W)—average together alongside the mechanical temperature rise M generated by viscous shear in the mixer bowl:

$$W = 3D - R - F - M \Leftrightarrow M = 3D_{\mathrm{actual}} - R - F - W \qquad \textbf{(Eq. 19.4)}$$

In this five-variable balance (**W = 3×D - R - F - M**), all temperatures are expressed in degrees Celsius (°C) or degrees Fahrenheit (°F). In forward production mode, the baker knows D, R, F, and M and solves for the required strike-water temperature W. Conversely, when commissioning a new spiral mixer or testing a stiffer bagel dough, the baker runs an instrumentation batch with measured room (R), flour (F), strike water (W), and post-mix dough temperature (D_{actual}) and solves the exact same [TW] workspace **backward** via [XEQ] for [M] to calibrate the mixer's mechanical friction factor.

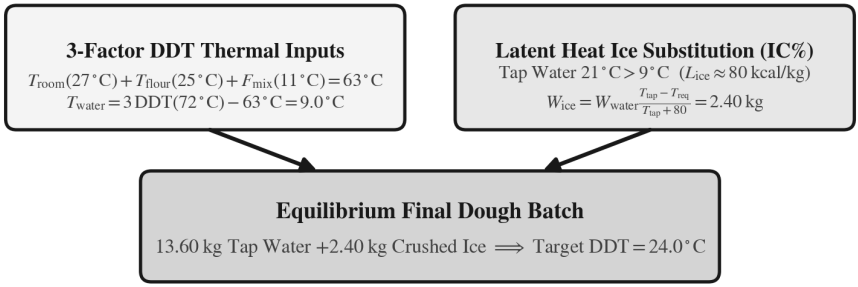


Figure 19.2: *Three-factor Desired Dough Temperature (DDT) thermal balance and latent-heat ice substitution (IC%).*

On our July morning, the baker targets $D = 24.0°C$ while facing a room temperature of $R = 28.0°C$, a flour temperature of $F = 26.0°C$, and a spiral-mixer friction factor of $M = 9.0°C$. Evaluating [Tw] yields a required strike-water temperature of $W = 3(24.0)$ - 28.0 - 26.0 - 9.0 = 72.0 - 63.0 = **9.0°C**.

19.2.2 Latent Heat of Fusion and Flake-Ice Weight Calculation ([↗] [ICE])

When the required strike-water temperature $T_{target} = W$ falls below the temperature of municipal tap water ($T_{tap} = 21.0°C$) and the bakery lacks a refrigerated water chiller, the baker substitutes crushed or flake ice ([↗] [ICE]) for part of the liquid water mass W_{water}. Liquid water has a specific heat capacity of $c_w = 1.0$ kcal/(kg·°C), whereas solid ice at 0°C absorbs a latent heat of fusion of $L_f = 79.72 \approx 80.0$ kcal/kg solely to transition from crystalline solid at 0°C into liquid water at 0°C. Dividing the sensible cooling deficit W_{water} c_w (T_{tap} - W) by the 80 kcal/kg latent heat constant ([↗] [ICE]) determines the required flake-ice mass W_{ice} and remaining tap-water mass W_{liquid}:

$$W_{ice} = \frac{W_{water} \cdot (T_{tap} - W)}{80}, \qquad W_{liquid} = W_{water} - W_{ice} \qquad \textbf{(Eq. 19.5)}$$

Applying [↗] [ICE] to our $W_{water} = 65.3265$ kg of mixer water with tap water at $T_{tap} = 21.0°C$ and target strike temperature $W = 9.0°C$ ($\Delta T = 21.0$ - 9.0 = 12.0°C in Register X) gives $W_{ice} = 65.3265 \times (12.0 / 80) = 65.3265 \times 0.15 =$ **9.7990 kg** of flake ice and W_{liquid} = 65.3265 - 9.7990 = **55.5275 kg** of 21.0°C tap water.

19.3 Dough Scaling and Thermal Balancing on the Stack and in the Solver Workspace

On the **SC-78**, entering the base flour weight in Register Y and the first ingredient weight in Register X before pressing [BK%] locks the base flour weight F and activates the live **@BK%** recipe ledger in [↗] [▦]. Subsequent presses of [BK%] compute each ingredient's Baker's Percentage and append it to the ledger, while [ΣW], [↖] [SCL], [↖] [DIV], [Tw], and [↗] [ICE] handle batch scaling, portion dividing, and thermal balancing.

Key / Solver	Mode	Inputs / Softkeys (LFU0–LFU4)	Governing Expression
BK%	Stack / Solver	Stack: $Y = F$, $X = I_1$ (then $X = I_k$) • Solver: P I F	$P = (I \,/\, F) \times 100$ (locks F and logs each ingredient row to **@BK%** in ↱ ▦)
ΣW	Immediate Ledger	Stack: recalls cumulative batch dough weight T from **@BK%**	$T = F + \sum I_k = F \times P \,/\, 100$
↰ SCL	Immediate Ledger	Register X: Target dough mass T_{target} (or multiplier factor)	Scales active flour F and all **@BK%** rows in ↱ ▦ by $T_{\text{target}} \,/\, T$
↰ DIV	Immediate Stack	Register Y: Total dough mass T, Register X: Count N (or unit mass W)	$W = T \,/\, N \Leftrightarrow N = T \,/\, W$
Tw	Stack / Solver	Stack: $Y = D$, $X = T_{\text{avg}}$ • Solver: W D R F M	$W = 3\,D$ - R - F - M (Strike Water / Mixer Friction Solver)
↱ ICE	Immediate Stack	Register Y: Water mass W_{water}, Register X: Cooling $\Delta T = T_{\text{tap}}$ - W	$W_{\text{ice}} = W_{\text{water}} \cdot (\Delta T \,/\, 80)$ (80 kcal/kg latent heat of fusion)

19.3.1 Immediate Stack Trace: Building and Scaling the @BK% Recipe Ledger

To build the country sourdough ledger from a 1000 g test formula (**720 g** water, **200 g** levain, **22 g** salt), recall the cumulative test-batch dough weight with ΣW, scale the entire recipe to 200 loaves at 0.881 kg (176.2 kg total dough) with ↰ SCL, and compute the flake-ice weight for $W = 9.0$°C strike water with ↱ ICE, the baker chains:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1000 ENTER **720** BK%	0	0	0	72.0000	Lock base flour $F = 1000$ g and log water (72.0%) into **@BK%** ledger in ↱ ▦
200 BK% **22** BK%	0	72.0000	20.0000	2.2000	Log levain (20.0%) and sea salt (2.2%) into **@BK%** (**P = 194.2%**)
ΣW	72.0000	20.0000	2.2000	1942.000	Recall cumulative test-batch dough mass $T = 1942$ g (194.2%) from **@BK%**
176.2 x⇄y ÷ **1000** ×	20	20	2.2	90.7312	Scale anchor flour to 176.2 kg batch: $F = 90.7312$ kg
0.72 × **12** ↱ ICE	20	20	2.2	**9.7990**	Compute mixer water (65.3265 kg) and flake ice $W_{\text{ice}} = 9.7990$ kg at $\Delta T = 12$°C

19.3.2 Bidirectional Solver Workspaces (BK% and Tw): Solving Backward for Flour Capacity and Strike Water

Now see how the Equation Workspaces eliminate manual algebra when working backward from physical equipment constraints. First, on a cleared stack (C ON), the pastry baker opens BK% (**P = I / F × 100**) to find the exact flour weight F that fills an 85.0 kg mixer bowl ($I = 85.0$ kg) when the seeded bâtard formula totals $P = 179.2\%$, and uses ↰ DIV (**85** ENTER **0.85** ↰ DIV) to find how many $W = 0.850$ kg loaves that 85.0 kg bowl yields. Second, the hearth baker opens Tw (**W = 3×D - R - F - M**) to solve forward for the 9.0°C strike water temperature and backward for the mixer friction factor M when a stiffer rye dough exits the mixer at $D = 25.5$°C:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar (LFU0–LFU4)	Register X Result & Physical Meaning
[C ON] [BK%] (open workspace)	**BK%: P = I / F × 100**	[P] [I] [F]	Opens Baker's Percentage / batch scaling workspace
85 → [I], **179.2** → [P], [XEQ] → [F]	**F = 47.4330**	**[P=179.2] [I=85] [F=47.43]**	Solves backward for required flour F = **47.433 kg** to fill 85 kg mixer
[C ON] **85** [ENTER] **0.85** [↰] [DIV]	*(Normal Stack)*	*(Standard Spine)*	Divides 85.0 kg batch by 0.85 kg portion: N = **100.00 loaves**
[C ON] [Tw] (open workspace)	**TW: W = 3×D - R - F - M**	[W] [D] [R] [F] [M]	Opens five-variable dough thermodynamics workspace
24 → [D], **28** → [R], **26** → [F], **9** → [M]	**TW: W = 3×D - R - F - M**	[W] **[D=24] [R=28] [F=26] [M=9]**	Stores target D=24°C, room 28°C, flour 26°C, friction 9°C
[XEQ] → [W]	**W = 9.0000**	**[W=9] [D=24] [R=28] [F=26] [M=9]**	Solves forward for required strike-water temperature W = **9.0°C**
25.5 → [D], **12** → [W], [XEQ] → [M]	**M = 10.5000**	**[W=12] [D=25.5] [R=28] [F=26] [M=10.5]**	Solves backward for empirical mixer friction rise M = **10.5°C**

19.3.3 Try It on the Stack

Practice scaling artisan dough formulas and balancing summer mixer heat across these three bakery problems:

1. **Ciabatta High-Hydration Inverse Scaling ([BK%] Solver):** A hearth ciabatta formula uses 100% bread flour, 82% water, 15% poolish, 2.5% olive oil, and 2.2% salt (P = 201.7%). Open the [BK%] Solver Workspace ([C ON] [BK%], **P = I / F × 100**), store **80.68** → [I] and **201.7** → [P], and press [XEQ] → [F] to solve backward for the flour weight needed to mix 80.68 kg of total dough, then multiply by 0.82 to find the water weight. *(Verified results: F = **40.00 kg** and I_{water} = **32.80 kg**.)*

2. **True Hydration with a Stiff Biga ([↗] [HYD] / [BK%]):** A panettone primo impasto combines 1,000 g main flour and 620 g water with 300 g of stiff biga maintained at 50% internal hydration (200 g flour and 100 g water). Compute True Overall Dough Hydration via **1000** [ENTER] **200** [+] **620** [ENTER] **100** [+] [↗] [HYD]. *(Verified Register X result: $HYDR_{true}$ = **60.00%**.)*

3. **Bagel Mixer Friction and Flake-Ice Cooling ([Tw] & [↗] [ICE]):** A New York bagel dough targets D = 22.0°C when R = 27.0°C, F = 25.0°C, and high-gluten spiral friction is M = 12.0°C. In the [Tw] Solver Workspace, solve via [XEQ] for strike water [W], then for 24.0 kg of recipe water and tap water at 20.0°C (ΔT = 18.0°C), compute the flake-ice weight (**24** [ENTER] **18** [↗] [ICE]). *(Verified results: W = **2.0°C** and W_{ice} = **5.40 kg**.)*

Having balanced flour-anchored dough proportions and spiral-mixer thermodynamics on the bakery floor, we step across the courtyard in Chapter 20 into the brewhouse—where grain-water calorimetry, hydrometer extract density, and hop alpha-acid isomerization govern a ten-barrel pale ale.

Chapter 20: Hitting Original Gravity and Bitterness in the Brewhouse Kettle

Next door to the bakery, the head brewer is mashing in a 10 BBL (310 U.S. gallon, or 1,173.5 L) batch of flagship West Coast pale ale. Seven hundred pounds of crushed two-row malted barley sits in the overhead grist case at an ambient grain-room temperature of $G = 19.0$°C. When the slide gate opens and the dry grist meets the hot liquor flowing through the mash hydrator at a standard thickness of $R = 1.5$ qt/lb, the combined mash must equilibrate at a single-infusion saccharification rest of $T = 66.5$°C—warm enough for α-amylase to dextrinize long amylopectin chains, yet cool enough to preserve β-amylase so the finished beer remains crisp and fermentable.

Ninety minutes later, after lautering the sweet wort into the steam-jacketed kettle, the brewer draws a sample to verify extract concentration before starting the boil. An optical refractometer reading a single drop of wort reports 13.80 Brix. Once the brewer confirms a 20.0°C temperature-corrected **Original Gravity** of $O = 1.0560$ and pitches *Saccharomyces cerevisiae* (Chico ale strain), ten days of active fermentation will lower the density to a **Final Gravity** of $F = 1.0120$, fixing both the apparent attenuation and the finished **Alcohol by Volume**.

During the sixty-minute kettle boil, the brewer works through three connected zymurgy themes:

1. Solve the first-law grain-water enthalpy balance in the mash tun for the required strike-water temperature.
2. Convert between Brix, Plato, and Specific Gravity across lautering and fermentation to determine Apparent Attenuation and Alcohol by Volume.
3. Model hop alpha-acid isomerization kinetics across boil time and wort gravity to hit 41.0 IBU, and solve backward for the exact hop charge mass needed when a new hop lot arrives with an alpha-acid rating of $A = 11.8\%$.

20.1 Mash Strike-Water Calorimetry and Wort Gravity Scales

All-grain brewing begins with an adiabatic enthalpy balance in the mash tun and continues by tracking the mass fraction of dissolved maltose, maltotriose, and dextrins across lautering and fermentation.

20.1.1 First-Law Enthalpy Balance in the Mash Tun (MSH)

When dry crushed malt of mass W_{grain} and specific heat capacity $c_g \approx 0.40$ kcal/(kg·°C) (or 0.40 BTU/(lb·°F)) at initial grain temperature G mixes with hot strike water of mass W_{water} and specific heat $c_w = 1.00$ kcal/(kg·°C) at **Mash Strike Temperature** (MSH, S), conservation of thermal energy requires the sensible heat lost by the cooling strike liquor to match the sensible heat gained by the warming grain bed at equilibrium rest temperature T:

$$W_{\text{water}} c_w (S - T) = W_{\text{grain}} c_g (T - G) \Rightarrow S = \frac{0.2}{R}(T - G) + T \tag{Eq. 20.1}$$

In the standard American brewing convention where mash thickness R is expressed in U.S. quarts of water per pound of grain (qt/lb), 1 qt of water weighs 2.085 lb, so the thermal heat-capacity ratio c_g / (2.085 c_w) ≈ 0.40 / 2.085 ≈ 0.20 yields the thermodynamic constant 0.2 / R (and when R is expressed in metric L/kg, 0.2/R is replaced by $0.41/R_{\mathrm{L/kg}}$, corresponding to an equivalent $R = 0.488\ R_{\mathrm{L/kg}}$).

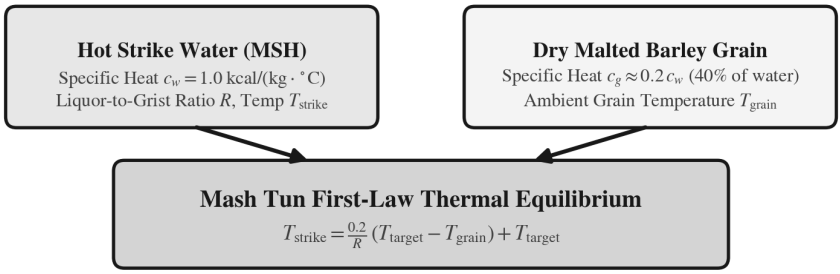


Figure 20.1: *First-law enthalpy balance between hot strike water and dry malted grain reaching mash rest equilibrium (MSH).*

For our 10 BBL pale ale mashed at a standard liquor-to-grist ratio of R = 1.50 qt/lb with malt stored at G = 19.0°C and a target saccharification rest of T = 66.5°C, the temperature differential is T - G = 66.5 - 19.0 = 47.5°C. Multiplying by 0.2 / 1.5 = 0.13333 gives a thermal drop of 6.333°C during dough-in, requiring the hot liquor tank to deliver strike water at S = 66.5 + 6.333 = **72.833°C**.

20.1.2 Extract Polynomials ([→] [→SG], [↶] [→Brx], [↶] [→°P]), Hydrometer Calibration ([→] [SGT]), and Alcohol by Volume ([ABV])

Brewers, winemakers, and distillers quantify dissolved sugar concentration using three interconvertible density scales: **Specific Gravity** ([→] [→SG] and [↶] [BRX], the density ratio of wort or must to pure water at 20.0°C), **Degrees Brix** ([↶] [→Brx], B), and **Degrees Plato** ([↶] [→°P], grams of sucrose-equivalent extract per 100 g of solution). When a glass hydrometer is read in warm kettle wort at temperature $T_{°\mathrm{C}}$ rather than its 20.0°C calibration reference, [→] [SGT] applies thermal expansion correction ($\mathrm{SG}_{20°\mathrm{C}} = \mathrm{SG}_{\mathrm{meas}} + 0.0013 \cdot (T_{°\mathrm{C}} - 20)/10$). The American Society of Brewing Chemists (ASBC) rational and cubic polynomials convert directly between extract B and Specific Gravity SG:

$$\mathrm{SG} = 1 + \frac{B}{258.6 - \left(\frac{B}{258.2}\right) \times 227.1}, \qquad B \approx {}^{\circ}\mathrm{P} = ((182.4601 \cdot \mathrm{SG} - 775.6821) \cdot \mathrm{SG} + 1262.7794) \cdot \mathrm{SG} - 669.5622 \qquad \textbf{(Eq. 20.2)}$$

Once yeast ferments the wort from **Original Gravity** (O) down to **Final Gravity** (F), the fraction of gravity points consumed defines **Apparent Attenuation** ($\mathrm{ATT}_{\%}$), and the density drop scaled by the stoichiometric ethanol-to-CO_2 conversion factor 131.25 determines **Alcohol by Volume** ([ABV], A) and U.S. distilled spirit proof ([→] [→PRF], Proof = 2 × $\mathrm{ABV}_{\%}$):

$$\mathrm{ATT}_{\%} = \left(\frac{O - F}{O - 1.000}\right) \times 100, \qquad A = (O - F) \times 131.25 \qquad \textbf{(Eq. 20.3)}$$

In the 10 BBL brewhouse, the optical refractometer reading of B = 13.80 Brix converts via [→] [→SG] (or [↶] [BRX]) to a calibrated Original Gravity of O = 1 + 13.80 / (258.6 - 12.138) = **1.0560**. Fermenting from O = 1.0560 down to F = 1.0120 removes 44.0 of the initial 56.0 gravity points, yielding an Apparent Attenuation of $\mathrm{ATT}_{\%}$ = (0.044 / 0.056) ×

100 = **78.571%** and a finished ethanol strength of A = (1.0560 - 1.0120) × 131.25 = **5.775% ABV**.

20.2 Hop Alpha-Acid Isomerization, Malt Colorimetry, and Carbonation

During the kettle boil, insoluble humulones (alpha acids) inside the lupulin glands of hop cones undergo a thermal acyloin ring contraction into soluble, bitter *cis*- and *trans*-iso-alpha-acids.

20.2.1 Glenn Tinseth's Gravity-Time Utilization Model ([IBU])

One **International Bitterness Unit** ([IBU], I) is defined analytically as 1.0 mg of isomerized alpha acids per liter of beer (1 mg/L). Depending on whether the brewhouse weighs hops and wort in metric units (W_g in grams, decimal alpha-acid fraction $A \in (0, 1)$, and batch volume V_L in liters, where 1000 mg/g converts grams per liter to mg/L) or in U.S. customary units (W_{oz} in ounces, percentage alpha acid $A_{\%}$, and volume V_{gal} in gallons, where 74.89 = 28,349.5 / (3.78541 × 100)), kettle bitterness I and Glenn Tinseth's (1995) decimal utilization $U(t, \mathrm{SG})$ are governed by:

$$I = \frac{W_g \cdot A \cdot U \cdot 1000}{V_L} = \frac{W_{oz} \cdot A_{\%} \cdot U \cdot 74.89}{V_{gal}}, \qquad U(t, \mathrm{SG}) = \underbrace{\left(\left[1.65 \times 0.000125^{(\mathrm{SG}-1)}\right]\right)}_{[f_G(\mathrm{SG})]} \times \underbrace{\left(\left[\frac{1 - e^{-0.04t}}{4.15}\right]\right)}_{[f_t(t)]} \qquad \textbf{(Eq. 20.4)}$$

Here $f_G(\mathrm{SG})$ captures the physical inhibition of alpha-acid solubility and break-material coagulation as wort gravity SG rises, while $f_t(t)$ models first-order isomerization approaching an asymptotic ceiling of 1 / 4.15 ≈ 24.1% as boil duration t exceeds 60 to 90 minutes.

Evaluating Tinseth's equation for our 10 BBL (V_{gal} = 310.0 gal = 1173.5 L) pale ale at SG = 1.0560 yields a bigness factor of f_G = 1.65 × $(0.000125)^{0.056}$ = **0.9985**. For the t = 60 min Magnum addition (W_{oz} = 52.0 oz, $A_{\%}$ = 12.0%), $f_t(60)$ = (1 - $e^{-2.40}$) / 4.15 = 0.21910, giving U_{60} = 0.9985 × 0.21910 = **0.2188** (21.88%) and contributing I_{60} = (52.0 × 12.0 × 0.2188 × 74.89) / 310.0 = **32.98 IBU**. For the t = 15 min Cascade flavor addition (W_{oz} = 30.0 oz, $A_{\%}$ = 10.0%), U_{15} = **0.1086** (10.86%), contributing I_{15} = (30.0 × 10.0 × 0.1086 × 74.89) / 310.0 = **7.87 IBU**, so the two additions total I_{total} = 32.98 + 7.87 = **40.85 ≈ 41.0 IBU**.

Conversely, suppose the brewery receives a new lot of bittering hops with alpha-acid fraction A = 0.118 ($A_{\%}$ = 11.8%) and wants a single 60-minute charge (U = 0.2188) to deliver I = 42.0 IBU in the V_{gal} = 310.0 gal (V_L = 1173.5 L) kettle. Solving the [IBU] equation backward via [XEQ] for the hop mass W gives:

$$W_{oz} = \frac{I \cdot V_{gal}}{A_{\%} \cdot U \cdot 74.89} = \frac{42.0 \times 310.0}{11.8 \times 0.2188 \times 74.89} = \mathbf{67.338}\ \mathrm{oz}, \qquad W_g = \frac{I \cdot V_L}{A \cdot U \cdot 1000} = \frac{42.0 \times 1173.5}{0.118 \times 0.2188 \times 1000} = \mathbf{1908.99}\ \mathrm{g} \qquad \textbf{(Eq. 20.5)}$$

20.2.2 Malt Colorimetry ([→] [SRM]), Bright-Tank Carbonation ([←] [CO₂]), and Coffee Extraction ([→] [TDS])

Alongside hop bitterness, brewers and beverage directors track Dan Morey's power-law beer color in **Standard Reference Method** ([→] [SRM]) units from **Malt Color Units** (MCU = $\sum$ ($W_{grain,lb} \cdot L$) / V_{gal}), bright-tank **Carbonation** ([←] [CO₂]), and refractometric coffee **Extraction Yield** ([→] [TDS], $EY_{\%}$) from liquid beverage weight W_{brew}, **Total Dissolved Solids** ($TDS_{\%}$), and dry dose W_{dose} (18.0 g standard espresso dose on the stack):

$$\mathrm{SRM} = 1.4922 \cdot (\mathrm{MCU})^{0.6859}, \qquad \mathrm{EY}_{\%} = \frac{W_{\mathrm{brew}} \cdot \mathrm{TDS}_{\%}}{W_{\mathrm{dose}}} \qquad \textbf{(Eq. 20.6)}$$

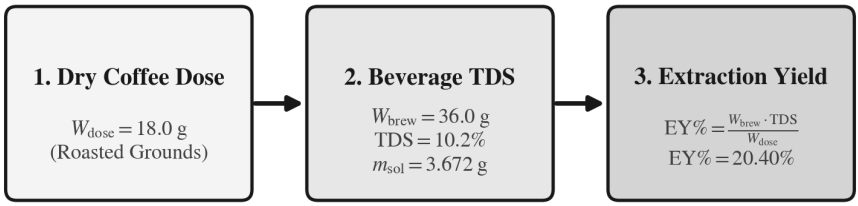


Figure 20.2: *Coffee extraction mass balance relating dry dose mass, beverage weight, Total Dissolved Solids (TDS), and Extraction Yield (EY%).*

In our 10 BBL (V_{gal} = 310 gal) pale ale mashed with 650 lb of two-row pale malt (2.0°L) and 50 lb of Caramel 40 malt (40.0°L), total grain color units are 650 × 2.0 + 50 × 40.0 = 3300 lb·°L, giving MCU = 3300 / 310.0 = 10.645 and producing a golden-copper hue of SRM = 1.4922 × $(10.645)^{0.6859}$ = **7.560 SRM**.

20.3 Brewhouse Gravity, Attenuation, and Bitterness on the Stack and in the Solver Workspace

On the **SC-78**, mash infusion thermodynamics ([MSH]), ASBC extract conversions ([↱] [→SG], [↰] [→Brx], [↰] [→°P], [↰] [BRX], [↱] [SGT]), fermentation alcohol ([ABV], [↱] [→PRF]), hop isomerization ([IBU]), malt color ([↱] [SRM]), carbonation ([↰] [CO₂]), and coffee extraction ([↱] [TDS]) execute both as immediate two-register RPN stack operators and as multi-variable Equation Solver Workspaces ([MSH], [ABV], [IBU]).

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU4])	Governing Expression
[MSH]	Stack / Solver	Stack: $Y = T$, $X = G$ • Solver: [S] **[R=1.5]** [T] [G]	$S = (0.2 \,/\, R) \cdot (T - G) + T$ (R = 1.5 qt/lb default)
[↱] [→SG] / [↰] [→Brx] / [↰] [→°P]	Immediate Stack	Register X: Extract B ([↱] [→SG], [↰] [BRX]) or Specific Gravity ([↰] [→Brx], [↰] [→°P])	ASBC rational (Brix → SG) and cubic (SG → Brix/**°P**) polynomials
[↱] [SGT]	Immediate Stack	Register Y: Measured SG, Register X: Sample temp $T_{^\circ\mathrm{C}}$	Hydrometer thermal expansion correction $\mathrm{SG}_{20^\circ\mathrm{C}} = Y + 0.0013(X - 20)/10$
[ABV] / [↱] [→PRF]	Stack / Solver	Stack: $Y = O$, $X = F$ • Solver: [A] [O] [F]	$A = (O - F) \times 131.25$; [↱] [→PRF] converts $\mathrm{ABV}_{\%} \to 2 \times \mathrm{ABV}_{\%}$
[IBU]	Stack / Solver	Stack: $Y = W_{\mathrm{oz}}$, $X = A_{\%}$ • Solver: [I] [W] [A] [U] [V]	$I = W \cdot A \cdot U \cdot 74.89 \,/\, V$ (5 gal, U=0.25 stack shortcut)
[↱] [SRM] / [↱] [TDS]	Immediate Stack	[↱] [SRM]: $Y = \sum(W_{\mathrm{lb}} \cdot L)$, $X = V_{\mathrm{gal}}$ • [↱] [TDS]: $Y = W_{\mathrm{brew}}$, $X = \mathrm{TDS}_{\%}$	Morey power law $\mathrm{SRM} = 1.4922 \cdot (Y/X)^{0.6859}$; Coffee $\mathrm{EY}_{\%} = Y \cdot X \,/\, 18$

20.3.1 Immediate Stack Trace: Mash Strike Liquor, Refractometer Conversion, and Attenuation

Starting at the mash tun with target rest $T = 66.5$°C and grain temperature $G = 19.0$°C, the brewer computes the strike-water temperature ([MSH]), converts the 13.80 Brix refractometer reading into calibrated Specific Gravity ([↗] [→SG]), and evaluates Alcohol by Volume ([ABV]), Apparent Attenuation, and Morey color ([↗] [SRM]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
66.5 [ENTER] **19** [MSH]	0	0	0	72.8333	Compute strike-water temperature $S = 72.833$°C at $R = 1.5$ qt/lb
13.8 [↗] [→SG]	0	0	72.8333	1.0560	Convert 13.80 Brix wort reading to Original Gravity $O = 1.0560$
[ENTER] [ENTER] **1.012** [ABV]	72.8333	72.8333	1.0560	5.7750	Duplicate O and compute finished ethanol $A = 5.775\%$ ABV at $F = 1.0120$
[x⇄y] **1.012** [−] **0.056** [÷] **100** [×]	72.8333	72.8333	5.7750	78.5714	Evaluate Apparent Attenuation $\text{ATT}_{\%} = (0.044 \,/\, 0.056) \times 100 = 78.57\%$
3300 [ENTER] **310** [↗] [SRM]	5.774	5.7750	78.5714	**7.5599**	Evaluate Morey beer color $\text{SRM} = 1.4922 \times (3300/310)^{0.6859} = 7.56$ SRM

20.3.2 Bidirectional Solver Workspaces ([MSH] and [IBU]): Solving Backward for Hop Charge Mass and Rest Temperature

When a new lot of bittering hops arrives with alpha-acid rating $A = 11.8\%$ and the brewer needs to know the exact hop mass W required to hit $I = 42.0$ IBU in the $V = 310.0$ gal (1173.5 L) kettle at 60-minute Tinseth utilization $U = 0.2188$—or wants to check what mash rest temperature T will result if thick strike liquor ($R = 1.25$ qt/lb) enters at $S = 74.0$°C into $G = 18.0$°C malt—the [IBU] and [MSH] Equation Workspaces solve both inverse problems directly via [XEQ]:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU4])	Register X Result & Physical Meaning
[C ON] [IBU] (open workspace)	**IBU: I = W×A×U×74.89/V**	[I] [W] [A] [U] [V]	Opens five-variable hop isomerization workspace
42 → [I], **11.8** → [A]	**IBU: I = W×A×U×74.89/V**	**[I=42]** [W] **[A=11.8]** [U] [V]	Stores target bitterness $I = 42.0$ IBU and alpha acid $A = 11.8\%$
0.2188 → [U], **310** → [V]	**IBU: I = W×A×U×74.89/V**	**[I=42]** [W] **[A=11.8]** **[U=.219]** **[V=310]**	Stores 60-min Tinseth utilization $U = 0.2188$ and volume $V = 310$ gal
[XEQ] → [W]	**W = 67.3381**	**[I=42]** **[W=67.34]** **[A=11.8]** **[U=.219]** **[V=310]**	Solves backward for required hop charge $W =$ **67.34 oz** ($\times$ 28.3495 = **1909.0 g**)
[C ON] [MSH] (open workspace)	**MSH: S = .2/R×(T-G)+T**	[S] **[R=1.5]** [T] [G]	Opens mash strike-water enthalpy workspace
74 → [S], **1.25** → [R], **18** → [G], [XEQ] → [T]	**T = 66.2759**	**[S=74]** **[R=1.25]** **[T=66.28]** **[G=18]**	Solves backward for equilibrium mash rest temperature $T =$ **66.28°C**

20.3.3 Try It on the Stack

Test your brewhouse stack and Solver Workspace fluency on these three zymurgy problems:

1. **Imperial Stout Mash Strike and Final Gravity Target ([MSH] & [ABV]):** A brewer mashes dark roasted grain at $G = 17.5$°C to hit a full-bodied rest of $T = 68.0$°C (**68** [ENTER] **17.5** [MSH]), then converts 22.0 Brix wort to Specific Gravity (**22** [↗] [→SG]) and uses the [ABV] Solver Workspace ([C ON] [ABV], **A = (O - F) × 131.25**) to solve backward via [XEQ] for the Final Gravity [F] required to reach $A = 9.166\%$ ABV. *(Verified results: S = **74.733°C**, O = **1.0918**, and F = **1.0220**.)*

2. **Pilot Batch Centennial Hop Bitterness and Morey Color ([IBU] & [↗] [SRM]):** In a 5.0 gal pilot batch ($U = 0.25$), compute kettle bitterness for 1.75 oz of Centennial hops at $A_{\%} = 10.5\%$ (**1.75** [ENTER] **10.5** [IBU]) and beer color in [↗] [SRM] for a 5.0 gal grain bill contributing 90.0 lb·°L (**90** [ENTER] **5** [↗] [SRM], or MCU = 18.0). *(Verified Register X results: I = **68.805 IBU** and SRM = **10.828 SRM**.)*

3. **Inverse Hop Charge for a 15-Barrel IPA ([IBU] Solver):** A 15 BBL ($V = 465$ gal = 1760.2 L) West Coast IPA requires $I = 55.0$ IBU from a single 60-minute Citra hop addition with $A_{\%} = 13.4\%$ ($A = 0.134$) and utilization $U = 0.225$. Solve the [IBU] Workspace backward via [XEQ] for [W] in ounces and convert to grams. *(Verified results: W = **113.065 oz** = **3205.3 g**.)*

From the brewhouse kettle and fermentation cellar, we move in Chapter 21 to the butcher block, the charcuterie curing room, and the executive chef's desk—where multi-stage fabrication yields, the hyperbolic geometry of margin versus markup, water activity, and sulfite equilibrium govern culinary profitability and preservation.

Chapter 21: Trimming the Tenderloin and Pricing the Tasting Menu

In the butcher room of a hotel banquet kitchen, the garde-manger chef is breaking down $W_{\text{AP}} = 45.0$ lb (20.41 kg) of whole vacuum-packed beef tenderloin (*psoas major*, purchased as PSMO—peeled, side muscle on) for a 120-guest Saturday evening tasting menu. Before a single medallion can hit the porcelain plate, the raw commodity passes through a sequence of mass-reducing stages: peeling away the tough elastin silverskin membrane and detaching the fatty chain muscle on the butcher block (80.0% raw fabrication yield), followed by dry-aging and plancha searing purge (90.0% thermal cooking retention), compounding into an overall net **Edible Portion Yield** of $Y_{\%} = 0.80 \times 0.90 = 72.0\%$.

The executive chef faces two coupled mathematical challenges at the butcher table. First, how many pounds and ounces of finished **Edible Portion** (EP) beef remain from the 45.0 lb delivery (32.40 lb, providing a 4.32 oz tasting-menu medallion for each of the 120 guests)? Conversely, if a wedding contract stipulates a full 6.0 oz (0.375 lb) cooked filet per guest (45.0 lb of finished EP meat on the pass), how many pounds of untrimmed AP tenderloin must the purchasing steward order from the purveyor?

Second, because 28.0% of the raw tenderloin mass disappears across fabrication and cooking, the purveyor's invoice price of \$16.20/lb does not represent the true cost of the meat sitting on the plate. Dividing the invoice cost by the 0.72 cumulative yield factor uncovers a true edible cost of \$22.50/lb (0.27 lb × \$22.50 = \$6.075 per tasting portion). Adding \$2.325 for truffled pomme purée, glazed cipollini onions, and marrow bordelaise brings the plated entrée cost to $C = \$8.40$. In this chapter, we follow three culinary-management themes:

1. Compound multi-stage fabrication and cooking yields to convert between As-Purchased and Edible Portion weights.
2. Audit Raoult water activity for charcuterie preservation and pH-dependent molecular SO_2 for cellar wine stability.
3. Navigate the hyperbolic asymmetry between Gross Margin and Markup on Cost to price the tasting menu at a 28.0% food cost target (72.0% gross margin).

21.1 Multi-Stage Fabrication Yields and Preservation Equilibria

Every raw agricultural commodity arriving at a restaurant loading dock—whole fish, bone-in subprimals, unpeeled root vegetables, or fresh berries—loses mass across multiple sequential preparation stages before reaching the guest's fork.

21.1.1 Compounding Multi-Stage Fabrication Yields (YLD)

In culinary cost accounting, the gross weight of an untrimmed commodity as delivered by the supplier is the **As-Purchased Mass** (W_{AP}), whereas the net weight of fabricated, cooked food ready for service is the **Edible Portion Mass** (W_{EP}). When a product undergoes m sequential processing stages with individual decimal stage yields $y_1, y_2, \ldots, y_m \in (0, 1]$ (where $y_k = 1 - \ell_k$ for stage fractional loss ℓ_k), mass conservation dictates that the stage yields **multiply geometrically** rather than subtracting linearly:

$$y_{\text{cum}} = \prod_{k=1}^{m} y_k = \prod_{k=1}^{m} (1 - \ell_k), \qquad Y_{\%} = 100 \cdot y_{\text{cum}} = \left(\frac{W_{\text{EP}}}{W_{\text{AP}}}\right) \times 100, \qquad W_{\text{AP}} = \frac{W_{\text{EP}}}{Y_{\%}/100} \qquad \textbf{(Eq. 21.1)}$$

Notice why additive mental math misleads kitchen managers: if whole tenderloin loses ℓ_1 = 20.0% on the butcher table (y_1 = 0.80) and the trimmed steaks lose ℓ_2 = 10.0% moisture during searing and resting (y_2 = 0.90), the cumulative yield ([YLD]) is not 100% - (20% + 10%) = 70.0%, but rather y_{cum} = 0.80 × 0.90 = **0.720** ($Y_{\%}$ = **72.0%**). Conversely, if a three-stage fish fabrication involves 55% filleting yield (y_1 = 0.55), 92% skinning yield (y_2 = 0.92), and 88% pan-roasting retention (y_3 = 0.88), the cumulative yield plunges to y_{cum} = 0.55 × 0.92 × 0.88 = **0.44528** (44.53%).

When the butcher processes our W_{AP} = 45.0 lb delivery of whole PSMO beef tenderloin at $Y_{\%}$ = 72.0% cumulative yield, the Edible Portion equation gives W_{EP} = 45.0 × 0.72 = **32.40 lb** (32.40 × 16 = **518.4 oz**), providing an exact 518.4 / 120 = **4.32 oz** (0.270 lb) medallion per guest. Conversely, when a wedding contract requires W_{EP} = 45.0 lb of finished beef on the pass (120 guests × 6.0 oz), evaluating the inverse purchasing formula shows that the steward must order W_{AP} = 45.0 / 0.72 = **62.50 lb** of untrimmed tenderloin.

21.1.2 Raoult Water Activity ([→] [Aw]) and Henderson–Hasselbalch Sulfite Equilibrium ([→] [SO₂])

Preserving the charcuterie, pastry confections, and cellar wines that accompany the banquet menu relies on two physical-chemistry equilibria built directly into the **SC-78** keypad: **Water Activity** ([→] [Aw], a_w) and **Sulfite Addition** ([→] [SO₂]). Unlike bulk moisture percentage, water activity a_w measures the thermodynamic chemical potential (escaping tendency) of unbound water molecules via Raoult's law—the ratio of the food's equilibrium water-vapor partial pressure P_v to pure water's saturation vapor pressure P_0 at the same temperature, which equals the Equilibrium Relative Humidity ($\text{ERH}_{\%}$) divided by 100:

$$a_w = \frac{P_v}{P_0} = \frac{\text{ERH}_{\%}}{100} \approx \frac{n_{\text{water}}}{n_{\text{water}} + \nu\, n_{\text{solute}}} \qquad \textbf{(Eq. 21.2)}$$

Pathogenic bacteria (*Clostridium botulinum*, *Staphylococcus aureus*) cannot proliferate when osmotic solutes (sodium chloride in cured meats via [↶] [SAL] or sucrose in syrups via [→] [SYR]) depress a_w below 0.85–0.91. In the wine cellar, potassium metabisulfite ($K_2S_2O_5$, which yields 57.6% mass fraction of SO_2 gas in acidic must) protects wine against oxidation and *Brettanomyces* spoilage. Because only the un-ionized **molecular SO₂** species ($SO_2 \cdot H_2O$, first acid dissociation pK_{a1} = 1.81) crosses microbial cell membranes, the required **Free SO₂** concentration (in mg/L, or ppm) to achieve a target antimicrobial molecular level $[SO_2]_{\text{mol}}$ (typically 0.80 mg/L for white wine or 0.50 mg/L for red wine) follows the Henderson–Hasselbalch power law ([→] [SO₂], matching the buffer physics of Unit VI):

$$\text{Free SO}_2 = [\text{SO}_2]_{\text{mol}} \cdot (1 + 10^{\text{pH} - 1.81}), \qquad W_{\text{SO}_2,\text{mg}} = \Delta\text{ppm} \times V_{\text{L}}, \qquad W_{\text{K}_2\text{S}_2\text{O}_5,\text{g}} = \frac{\Delta\text{ppm} \times V_{\text{L}}}{576} \qquad \textbf{(Eq. 21.3)}$$

At a wine pH of 3.41, $10^{3.41 - 1.81} = 10^{1.60}$ = 39.81, so achieving 0.80 mg/L of active molecular SO_2 requires Free SO_2 = 0.80 × 40.81 = **32.65 ppm**. Raising a V_{L} = 225 L Bordeaux barrique by Δppm = 25 mg/L requires 25 × 225 = **5625 mg** (5.625 g) of pure SO_2, or 5625 / 576 = **9.766 g** of $K_2S_2O_5$ ([→] [SO₂]).

21.2 The Hyperbolic Geometry of Margin, Markup, and Yield-Adjusted Costing

A restaurant that confuses **Markup on Cost** with **Gross Margin on Revenue**—or that prices its menu using raw supplier invoice prices rather than yield-adjusted edible costs—suffers an immediate, mathematically predictable collapse in operating profit.

21.2.1 Yield-Adjusted Edible Cost Across Compounding Fabrication Stages

When an untrimmed commodity is purchased at an **As-Purchased Unit Cost** of C_{AP} (in dollars per pound or kilogram) and yields a cumulative decimal fraction $y_{\text{cum}} = Y_{\%} \,/\, 100 = \prod_{k=1}^{m} y_k$ of edible product after fabrication and cooking, the entire invoice expenditure must be recovered across the smaller edible mass W_{EP}. Dividing C_{AP} by y_{cum} gives the true **Edible Portion Unit Cost** (C_{EP}) and the portion cost C_{portion} for an edible serving of weight w_{portion}:

$$C_{\text{EP}} = \frac{C_{\text{AP}}}{Y_{\%}/100} = \frac{C_{\text{AP}}}{\prod_{k=1}^{m} y_k}, \qquad C_{\text{portion}} = w_{\text{portion}} \cdot C_{\text{EP}} \tag{Eq. 21.4}$$

Because C_{EP} scales with the **reciprocal** of y_{cum}, low fabrication yields penalize plate cost hyperbolically: while our 72.0% beef tenderloin yield multiplies the \$16.20/lb invoice price by 1 / 0.72 = 1.3889× to C_{EP} = **\$22.50/lb** (making each guest's 0.270 lb / 4.32 oz medallion cost $C_{\text{portion}} = 0.270 \times \$22.50 =$ **\$6.075**, and bringing the complete plated entrée with \$2.325 of garnish to $C =$ **\$8.40**), a flatfish with a 44.53% three-stage yield multiplies the dock price by 1 / 0.4453 = 2.2458×.

21.2.2 The Hyperbolic Asymmetry Between Food Cost ([CST] / [↖ PRC]), Gross Margin, and Markup

Let C denote the total yield-adjusted plate cost and P denote the menu selling price. Commercial accounting defines three coupled dimensionless ratios—decimal **Food Cost Fraction** $f = F \,/\, 100$ ([CST] and [↖ PRC]), decimal **Gross Margin** $m = M_{\text{margin}} \,/\, 100 = 1 - f$, and decimal **Markup on Cost** $u = M_{\text{markup}} \,/\, 100$:

$$f = \frac{C}{P}, \qquad m = \frac{P - C}{P} = 1 - f, \qquad u = \frac{P - C}{C} = \frac{P}{C} - 1 \tag{Eq. 21.5}$$

Although gross margin m and markup u share the exact same numerator—the unit gross profit dollar contribution $\Delta = P - C$—they normalize that profit against different denominators: **Margin divides by selling price *P***, whereas **Markup divides by kitchen cost *C***. Eliminating $P \,/\, C = 1 \,/\, (1 - m) = 1 + u$ between the two definitions reveals a **Möbius (fractional linear) hyperbolic transformation** with a vertical asymptote at $m \to 1^{-}$, matching the structure of the Standing Wave Ratio $S = (1 + \Gamma)/(1 - \Gamma)$ in Unit V (**SC-50**):

$$P = \frac{C}{f} = \frac{C}{1 - m} = C(1 + u), \qquad u = \frac{m}{1 - m} = \frac{1 - f}{f}, \qquad m = \frac{u}{1 + u} \tag{Eq. 21.6}$$

Consider why linear intuition bankrupts kitchens when a chef mistakes markup for margin:

- To achieve our hotel controller's target **Food Cost Percentage** of $F = 28.0\%$ ($f = 0.28$), the required **Gross Margin** is $m = 1 - 0.28 =$ **0.720** ($M_{\text{margin}} =$ **72.0%**).

- What **Markup on Cost** u is required to achieve that 72.0% gross margin? Evaluating the hyperbolic mapping gives $u = 0.72 \text{ / } (1 - 0.72) = 0.72/0.28 = (18/7) \approx \mathbf{2.57143}$, or a **257.14% markup** above plate cost ($P = \$8.40 \times (1 + 2.57143) = \$8.40 \text{ / } 0.28 = \mathbf{\$30.00}$ via [↖] [PRC], leaving $M_{\text{gross}} = \$30.00 - \$8.40 = \mathbf{\$21.60}$ per plate).
- If an untrained chef hears "we need a 72% return on this \$8.40 plate" and applies a **72.0% markup** ($u = 0.72$) instead of a **72.0% gross margin** ($m = 0.72$), the menu price is set to $P_{\text{wrong}} = \$8.40 \times 1.72 = \mathbf{\$14.45}$. At \$14.45, evaluating [CST] shows that the food cost jumps to $F = \mathbf{58.14\%}$ and the actual gross margin collapses to $m = 0.72 \text{ / } (1 + 0.72) = \mathbf{41.86\%}$, wiping out \$15.55 of contribution margin on every single plate served.

21.3 Butcher Yield, Menu Costing, and Preservation on the Stack and in the Solver Workspace

On the **SC-78**, fabrication yield ([YLD]), food-cost percentage and pricing ([CST], [↖] [PRC]), water activity ([↗] [Aw]), sulfite dosing ([↗] [SO₂]), and utensil-snapping volume conversions ([↖] [→cup], [↗] [→pt], [↖] [→qt], [↗] [→dL], [↗] [→tb], [↖] [→tsp]) operate across both the RPN stack and the bidirectional [CST] Solver Workspace.

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU2])	Governing Expression
[YLD]	Immediate Stack	Register Y: As-Purchased W_{AP}, Register X: Edible W_{EP}	Trim yield percentage $Y_{\%} = (X \text{ / } Y) \times 100$
[CST]	Stack / Solver	Stack: $Y = C$, $X = P$ • Solver: [P] [C ON] **[F=30]**	Stack: Food cost $F_{\%} = (C \text{ / } P) \times 100$; Solver: $P = C \text{ / } (F \text{ / } 100)$
[↖] [PRC]	Immediate Stack	Register Y: Plate cost C, Register X: Target food cost $F_{\%}$	Menu selling price $P = C \text{ / } (F_{\%} \text{ / } 100)$ (Gross margin $M_{\%} = 100 - F_{\%}$)
[↗] [Aw] / [↗] [SO₂]	Immediate Stack	[↗] [Aw]: $X = \text{ERH}_{\%}$ • [↗] [SO₂]: $Y = W_{\text{SO2,mg}}$, $X = 576$ (or V_{L})	Water activity $a_w = X \text{ / } 100$; Sulfite ratio $Y \text{ / } X$
[↖] [→cup] [↗] [→tb] [↖] [→tsp]	Immediate Stack	Register X: Volume in mL	Converts volume to cups, tablespoons, or teaspoons with fraction snap

21.3.1 Immediate Stack Trace: Tenderloin Yield, Plate Costing, and Margin Audit

At the butcher table, the chef verifies the 72.0% fabrication yield on 45.0 lb of tenderloin ([YLD]), calculates the \$8.40 yield-adjusted plate cost, prices the entrée at a 28.0% food-cost target ([↖] [PRC]), and audits the achieved food cost percentage ([CST]) and gross margin ($100 - F_{\%}$) at a \$32.00 menu tariff:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
45 [ENTER] **32.4** [YLD]	0	0	0	72.0000	Verify cumulative trim yield $Y_{\%} = (32.4 \text{ / } 45.0) \times 100 = 72.0\%$
100 [÷] **16.20** [x⇄y] [÷]	0	0	0	22.5000	Compute true Edible Portion cost $C_{\text{EP}} = \$16.20 \text{ / } 0.72 = \$22.50/\text{lb}$
0.27 [×] **2.325** [+]	0	0	0	8.4000	Multiply by 0.27 lb (4.32 oz) and add \$2.325 garnish: $C = \$8.40$
[ENTER] [ENTER] **28** [↖] [PRC]	0	0	8.4000	30.0000	Price \$8.40 entrée at $F = 28.0\%$ food cost: $P = \$8.40 \text{ / } 0.28 = \$30.00$
[x⇄y] **32** [CST] **100** [x⇄y] [−]	0	0	30.0000	**73.7500**	Compute food cost at $P = \$32.00$ (26.25%) and Gross Margin $M_{\%} = 73.75\%$

21.3.2 Bidirectional Solver Workspace ([CST]): Solving Backward for Maximum Plate Cost

Suppose the banquet sales director locks the printed tasting-menu price for the beef course at $P = \$29.00$ per guest while the controller enforces a strict $F = 26.0\%$ food-cost ceiling ($M = 74.0\%$ gross margin). What is the maximum allowable kitchen plate cost C that the chef can spend on meat and garnish? Opening the [CST] (**P = C / (F / 100)**) Equation Workspace on a cleared stack ([C ON]) solves backward via [XEQ] for [C ON] in one step:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU2])	Register X Result & Physical Meaning
[C ON] [CST] (open workspace)	**CST: P = C / (F / 100)**	[P] [C ON] **[F=30]**	Opens food-cost pricing workspace with $F = 30.0\%$ default
29 → [P], **26** → [F]	**CST: P = C / (F / 100)**	**[P=29]** [C ON] **[F=26]**	Stores locked menu price $P = \$29.00$ and target food cost $F = 26.0\%$
[XEQ] → [C ON]	**C = 7.5400**	**[P=29] [C=7.54] [F=26]**	Solves backward for maximum allowable plate cost $C =$ **$7.54**
8.40 → [C ON], **29** → [P], [XEQ] → [F]	**F = 28.9655**	**[P=29] [C=8.4] [F=28.97]**	Solves backward for actual food cost F = **28.97%** if $C = \$8.40$ is kept

With the maximum allowable plate cost $C = \$7.54$ in Register X, subtracting the \$2.325 garnish leaves \$5.215 for the tenderloin medallion; dividing by $C_{\text{EP}} = \$22.50$/lb and multiplying by 16 oz/lb tells the butcher to portion the medallions at **3.708 oz** instead of 4.32 oz to hit the 26.0% cost target.

21.3.3 Try It on the Stack

Consolidate your command of multi-stage fabrication yields, margin-versus-markup geometry, and preservation equilibria with these three culinary exercises:

1. **Two-Stage Atlantic Halibut Fabrication and Portion Cost ([YLD] & [←] [PRC]):** A 35.0 lb whole dressed Atlantic halibut invoiced at $C_{\text{AP}} = \$14.30$/lb yields 19.25 lb of skinless center-cut fillet (**35** [ENTER] **19.25** [YLD]). Find the true edible cost per pound C_{EP} (**100** [÷] **14.30** [x⇄y] [÷]), the protein cost of a 6.0 oz (0.375 lb) fillet (**0.375** [×]), and the menu price required for a 25.0% food cost (**25** [←] [PRC]). *(Verified Register X results: $Y_{\%} =$ **55.00%**, $C_{EP} =$ **$26.00/lb**, $C_{portion} =$ **$9.75**, and $P =$ **$39.00**.)*

2. **Hyperbolic Margin vs. Markup Audit ([CST]):** A pastry chef plates a chocolate soufflé costing $C = \$3.15$ and prices it on the dessert menu at $P = \$14.00$. Compute the Food Cost percentage (**3.15** [ENTER] **14** [CST]), subtract from 100 (**100** [x⇄y] [-]) to find the Gross Margin percentage, and compute the Markup percentage on cost (**14** [ENTER] **3.15** [→] [Δ%]). *(Verified Register X results: $F_{\%} =$ **22.50%**, $M_{margin} =$ **77.50%**, and $M_{markup} =$ **344.44%**.)*

3. **Barrique Sulfite Addition ([→] [SO₂]) and Water Activity ([→] [Aw]):** Compute the milligrams of pure SO_2 required to raise a 225 L wine barrel by 30 ppm (**30** [ENTER] **225** [×]), press **576** [→] [SO₂] to find grams of $K_2S_2O_5$, and check the water activity of a cured sausage whose Equilibrium Relative Humidity is $\text{ERH}_{\%} = 85.0\%$ (**85** [→] [Aw], or **19.89** [ENTER] **23.40** [÷]). *(Verified Register X results: $W_{SO_2} =$ **6750 mg**, $W_{K_2S_2O_5} =$ **11.719 g**, and $a_w =$ **0.8500**.)*

Having mastered flour-anchored dough scaling, brewhouse zymurgy, and culinary yield costing on the **SC-78**, we turn in **Unit VIII: Clinical Dosing, Renal Clearance &**

Pharmacokinetics to the hospital ward and intensive care unit—where human body surface area, glomerular filtration, and continuous intravenous titrations govern patient dosing on the **SC-80**.

Unit VIII: Clinical Dosing, Renal Clearance & Pharmacokinetics

In 1916, Delafield Du Bois and Eugene F. Du Bois encased hospitalized patients at the Russell Sage Institute of Pathology in New York in tight-fitting paper-and-paraffin molds to measure human skin area directly. They demonstrated that basal metabolic rate, extracellular fluid volume, cardiac output, and renal blood flow scale with two-dimensional surface area rather than raw body mass. Seven decades later, in a 1987 letter to the *New England Journal of Medicine*, biostatistician Robert D. Mosteller replaced their multi-exponent nomogram with a square-root geometric mean of stature and weight. In outpatient primary care, oral medications are routinely prescribed in fixed adult unit doses because common pharmaceuticals possess a wide therapeutic index—a generous physiological margin between the minimum effective concentration required to treat disease and the toxic threshold at which organ damage begins.

In hospital oncology wards, intensive care units, emergency resuscitation bays, and neonatal nurseries, that margin vanishes. Cytotoxic antineoplastic agents, aminoglycoside and glycopeptide antibiotics, antiarrhythmics, and vasoactive catecholamines operate within a narrow therapeutic window. Under-dosing by 15% allows a systemic infection or malignancy to progress, while over-dosing by 15% can induce irreversible nephrotoxicity, ototoxicity, or ventricular arrhythmias. Because two patients who both weigh 88 kg on a bed scale can differ substantially in skeletal stature and adipose tissue fraction, clinical pharmacology partitions total weight into Ideal Body Weight (formulated by Ben Devine in 1974), Adjusted Body Weight, Body Mass Index, and Body Surface Area, alongside Malcolm Holliday and William Segar's 1957 three-tier pediatric fluid maintenance law.

Administering an accurate initial loading dose solves only the first half of the clinical dosing problem. Recurring maintenance doses must match the speed at which the kidneys and liver clear the compound from blood plasma. In 1953, German pediatrician and pharmacologist Friedrich Hartmut Dost published *Der Blutspiegel* (*The Blood Level*), coining the term *pharmacokinetics* and modeling the human body as a mathematical compartment governed by exponential first-order elimination. Notice the cross-domain mathematical identity linking clinical pharmacology to electrical engineering (Unit III, **SC-16**) and chemical kinetics (Unit VI, **SC-60**): a drug distributing across an apparent volume of distribution V_d and cleared by the kidneys at volumetric flow rate CL decays as $C(t) = C_0\, e^{-(\mathrm{CL}/V_d)t}$ with biological time constant $\tau_{\mathrm{bio}} = V_d \,/\, \mathrm{CL}$ and half-life $t_{1/2} = \ln(2)\, V_d \,/\, \mathrm{CL}$. That is the exact differential equation governing a capacitor C_{cap} discharging through a conductance $G = 1/R$ with time constant $\tau = R\, C_{\mathrm{cap}}$. Two decades after Dost, in 1976, Donald Cockcroft and Henry Gault analyzed 24-hour urine creatinine excretions from 249 patients aged 18 to 92 at Queen Mary Veterans Hospital in Montreal. They proved that steady-state serum creatinine must be scaled by chronological age, dosing weight, and biological sex to estimate glomerular filtration as Creatinine Clearance.

The **SC-80 Clinical, Nursing, Pharmacy & Critical Care** calculator anchors every clinical equation to a persistent Patient Profile (PAT) so clinicians can verify anthropometric compartments, infusion rates, and pharmacokinetic decay without manual transcription errors. Each chapter in this Unit follows a single hospital case through three connected clinical themes:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 22	Admitting a 68-year-old, 175 cm, 88 kg patient with serum creatinine 1.6 mg/dL and adjusting renal antibiotic dosing and 4-2-1 maintenance fluids	Persistent Patient Profile, Mosteller Body Surface Area, Devine Ideal and 40% Adjusted Body Weight, Cockcroft–Gault clearance, and Holliday–Segar fluid scaling	1. Body Surface Area & Lean Mass Partitioning • 2. Cockcroft–Gault Glomerular Filtration Kinetics • 3. Pediatric 4-2-1 Fluid & Caloric Maintenance
Chapter 23	Auditing a 5 µg/kg/min vasopressor order for an 80 kg ICU patient (400 mg in 250 mL), back-solving an inherited pump rate, and calibrating a 15 gtt/mL transport drip	Weight-based microgram-to-milligram mass flux, inverse pump-to-dose verification, gravity tubing drop factors, Mean Arterial Pressure, and Parkland burn resuscitation	1. Weight-Based Vasoactive Drip Titrations • 2. Inverse Infusion Pump Audits at Shift Change • 3. Gravity Drip Calibration, MAP & Burn Resuscitation
Chapter 24	Modeling a 500 mg IV antibiotic ($V_d = 25$ L, $t_{1/2} = 6.0$ hr), solving hold time for a toxic trough, and auditing an ICU blood gas, QTc, and helicopter O_2 tank	One- and two-compartment exponential decay, steady-state accumulation, Alveolar–arterial O_2 gradient, anion gap, corrected Na^+ and Ca^{2+}, Bazett QT_c, and O_2 tank duration	1. First-Order Elimination & 5-Half-Life Steady State • 2. Arterial Blood Gases, Anion Gap & Corrected Electrolytes • 3. Bazett QT_c Normalization & Transport Oxygen

Chapter 22: Adjusting the Vancomycin Regimen for Declining Renal Function

During morning infectious-disease rounds on the medical-surgical ward, a 68-year-old male patient standing $H = 175.0$ cm (5 ft 8.9 in) tall and weighing $W = 88.0$ kg is admitted with methicillin-resistant *Staphylococcus aureus* (MRSA) bacteremia. His morning basic metabolic panel shows that his steady-state serum creatinine has climbed from a baseline of 1.0 mg/dL to $S = 1.60$ mg/dL. Because intravenous vancomycin and aminoglycosides are eliminated almost entirely by glomerular filtration in the kidneys—and accumulate to nephrotoxic and ototoxic plasma concentrations when filtration slows—the clinical pharmacist must store the patient's biometrics in the **Patient Profile** and audit his anthropometric compartments and renal clearance before signing off on the maintenance dosing interval.

Evaluating the patient's chart requires resolving a fundamental tension in clinical anthropometrics. At 175.0 cm and 88.0 kg (BMI $= 28.73$ kg/m^2), the patient carries significant excess adipose mass above his lean skeletal frame. If the pharmacist plugs his raw 88.0 kg bed-scale weight into the Cockcroft–Gault clearance numerator, the equation treats every kilogram of poorly perfused fat tissue as creatinine-producing skeletal muscle, overestimating his kidney function at 55.0 mL/min. Conversely, partitioning his mass into Devine Ideal Body Weight (70.46 kg) and 40% Adjusted Body Weight (77.48 kg) reveals that his true glomerular filtration has fallen below the 50.0 mL/min clinical threshold, requiring an extension of his dosing interval.

In this chapter, we work through three connected clinical admission calculations:

1. Partition total bed-scale weight into Body Surface Area, Devine Ideal Body Weight, and 40% Adjusted Body Weight.
2. Evaluate Cockcroft–Gault creatinine clearance across weight compartments, and solve backward for the exact serum creatinine threshold S where clearance drops below 30.0 mL/min.
3. Scale pediatric weight-based medication doses and Holliday–Segar 4-2-1 hourly fluid maintenance for a 26.0 kg child in the step-down unit.

22.1 Anthropometric Mass Partitioning and Body Surface Area

Every clinical dosing protocol on the **SC-80** is anchored by the persistent **Patient Profile** (PAT), accessed from the right-hand softkey inside BSA, GFR, DOS, or DRP. The PAT profile stores patient weight W (in kg, W), height H (in cm, h), chronological age A (in years, A), steady-state serum creatinine S (in mg/dL, S), and biological sex (MALE or FEM), automatically pre-populating those biometric variables across every clinical Equation Workspace (BSA, GFR, DOS, DRP, → BRN).

22.1.1 Mosteller Body Surface Area and Allometric Scaling (BSA)

Two patients who each weigh 88.0 kg can possess markedly different internal drug distribution volumes. A tall, muscular patient has an expansive vascular bed, high cardiac output, and large hepatic and renal parenchymal mass, whereas a shorter, sedentary patient of the same 88.0 kg weight carries a substantial fraction of mass as poorly perfused adipose tissue. Because extracellular fluid volume, glomerular filtration, and basal metabolic rate correlate closely with external **Body Surface Area** (BSA, B) in square meters (m^2, averaging 1.73 m^2 in a reference adult), Robert D. Mosteller (1987) formulated BSA from patient height H (in cm) and total body weight W (in kg) using the dimensional normalization constant $3600 = 60^2$ cm·kg/m^4:

$$B = \sqrt{\frac{H \cdot W}{3600}} \Leftrightarrow W = \frac{3600\,B^2}{H} \qquad \textbf{(Eq. 22.1)}$$

Here $H \cdot W$ forms a two-dimensional product of linear stature and mass, and taking the square root yields a geometric mean whose behavior closely tracks the classical 1916 Du Bois & Du Bois power law ($B = 0.007184\ W^{0.425}\ H^{0.725}$) across infant, pediatric, and adult populations.

For our 68-year-old male admission measuring $H = 175.0$ cm and weighing $W = 88.0$ kg, the product of height and weight is $175.0 \times 88.0 = 15{,}400$ cm·kg. Dividing by 3600 gives 4.27778 m^4, and extracting the square root yields a Body Surface Area of $B = \sqrt{(4.27778)} =$ **2.0683 m^2**.

22.1.2 Devine Ideal Body Weight (↰ IBW) and Forty-Percent Adjusted Body Weight (↱ ABW)

When dosing hydrophilic medications that remain inside lean tissue and extracellular water, using raw total body weight (TBW = W) in an overweight patient causes systematic over-dosing. To estimate lean metabolically active mass from skeletal stature alone, Ben Devine (1974) established the **Ideal Body Weight** (↰ IBW, in kg) equations for adults at least 60 inches (152.4 cm, or 5 feet) tall, where $H_{\text{in}} = H\ /\ 2.54$ is height in inches, 50.0 kg (male) and 45.5 kg (female) are the baseline lean weights at 60 inches, and 2.3 kg/in (0.9055 kg/cm) is the linear lean-mass increment per inch above five feet:

$$\text{IBW}_{\text{male}} = 50.0 + 2.3\left(\frac{H}{2.54} - 60\right), \qquad \text{IBW}_{\text{female}} = 45.5 + 2.3\left(\frac{H}{2.54} - 60\right) \qquad \textbf{(Eq. 22.2)}$$

When a patient's total body weight TBW exceeds IBW by more than 20%, polar antibiotics such as aminoglycosides penetrate roughly 40% into excess adipose extracellular water, which clinical pharmacists capture via **Adjusted Body Weight** (↱ ABW, in kg) using the empirical adipose distribution factor 0.40:

$$\text{ABW} = \text{IBW} + 0.40(\text{TBW} - \text{IBW}) \qquad \textbf{(Eq. 22.3)}$$

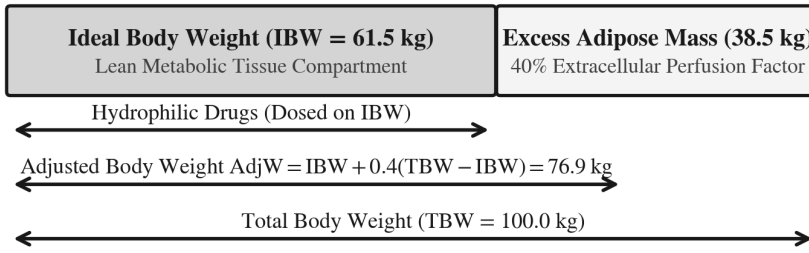


Figure 22.1: *Anthropometric weight compartments (IBW, AdjW, and Total Body Weight) governing hydrophilic and lipophilic drug distribution.*

Evaluating Devine's formula for our $H = 175.0$ cm ($H_{\text{in}} = 175.0 / 2.54 = 68.8976$ in) male patient via [↰] [IBW] gives $\text{IBW}_{\text{male}} = 50.0 + 2.3(68.8976 - 60.0) = 50.0 + 20.4646 =$ **70.4646 kg** (whereas a female patient of the same 175.0 cm stature has $\text{IBW}_{\text{female}} = 45.5 + 20.4646 =$ **65.9646 kg**). Because his actual weight of TBW = 88.0 kg is $88.0 / 70.4646 = 1.2489\times$ his ideal weight (24.9% above [↰] [IBW]), pressing [↱] [ABW] adds 40% of his 17.5354 kg excess adipose mass to yield an Adjusted Body Weight of ABW $= 70.4646 + 0.40(88.0 - 70.4646) =$ **77.4787 kg**.

22.2 Glomerular Filtration Kinetics and Pediatric Maintenance Scaling

With the patient's lean and adjusted mass compartments established, the pharmacist can translate his morning serum creatinine level into a physiological volumetric clearance rate.

22.2.1 The Cockcroft–Gault Creatinine Clearance Equation ([GFR])

Creatinine is a nitrogenous breakdown product of creatine phosphate in skeletal muscle, released into blood plasma at a steady daily rate proportional to muscle mass and filtered freely by the renal glomeruli into the urine. At steady state, urinary creatinine excretion matches muscle creatinine production, making **Creatinine Clearance** ([GFR], C)—the volume of plasma cleared of creatinine per minute (mL/min)—inversely proportional to steady-state serum creatinine (S, in mg/dL). Because skeletal muscle mass and nephron count decline linearly with adult age (A, in years), Donald Cockcroft and Henry Gault (1976) formulated bedside [GFR] from dosing weight W, using the empirical constant 72 in the denominator and a 0.85 factor in biological females ([FEM] in [PAT]) to reflect lower average skeletal muscle mass per kilogram:

$$C_{\text{male}} = \frac{(140 - A) \cdot W}{72 \cdot S}, \qquad C_{\text{female}} = 0.85 \cdot C_{\text{male}} \tag{Eq. 22.4}$$

Hospital pharmacy protocols enforce a three-tier weight selection rule for W inside the Cockcroft–Gault numerator: underweight patients (TBW $<$ IBW) use actual TBW. Normal-weight patients (IBW $\le$ TBW $\le$ 1.20 IBW) use Devine [↰] [IBW]. And overweight or obese patients (TBW $>$ 1.20 IBW) use 40% Adjusted Body Weight ([↱] [ABW]).

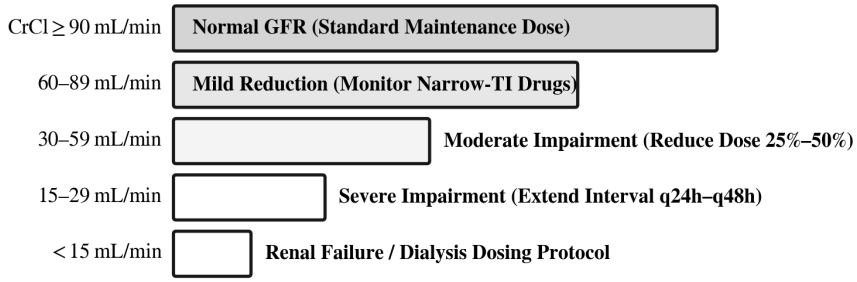


Figure 22.2: *Clinical Cockcroft-Gault creatinine clearance (CrCl) staging tiers for renal dose adjustment.*

In our 68-year-old male patient with $S = 1.60$ mg/dL, notice how the age factor (140 - 68) = 72 equals the denominator constant 72, reducing C_{male} to W / S. If a clinician uses unadjusted Total Body Weight ($W = 88.0$ kg), estimated clearance appears to be $C_{\text{TBW}} = 88.0 / 1.60 =$ **55.00 mL/min**. Substituting his obesity-corrected Adjusted Body Weight (W = ABW = 77.4787 kg) lowers estimated glomerular filtration to $C_{\text{ABW}} = 77.4787 / 1.60 =$ **48.424 mL/min** (and using strict lean [↰] [IBW] of 70.4646 kg gives $C_{\text{IBW}} = 70.4646 / 1.60 =$ **44.040 mL/min**), confirming that his renal clearance has dropped into the < 50 mL/min moderate impairment tier where vancomycin maintenance dosing must be slowed. Conversely, solving the Cockcroft–Gault equation backward via [XEQ] for [S] at $C = 30.0$ mL/min ($S = (140 - A)W / (72\ C) = 77.4787 / 30.0 =$ **2.583 mg/dL**) tells the medical team the exact serum creatinine threshold that marks severe renal impairment for this patient.

22.2.2 Holliday–Segar Pediatric Fluid Maintenance ([↱] [421]), Resting Energy ([↰] [REE]), and Patient-Bound Dosing ([DOS])

To replace daily insensible (respiratory and cutaneous) and urinary water losses in hospitalized children and adults, Malcolm Holliday and William Segar (1957) tied basal water requirements to metabolic caloric expenditure across three body-mass tiers, establishing the **4-2-1 Maintenance Fluid Rule** ([↱] [421]), while across clinical nutrition ([↰] [REE]) and veterinary medicine ([↱] [VET]), basal caloric demand follows Max Kleiber's 3/4-power allometric **Resting Energy Expenditure** (in kcal/day):

$$R_{\text{maint, mL/hr}} = 4 \cdot W \text{if } W \le 10,\ 40 + 2(W - 10) \text{if } 10 < W \le 20,\ 60 + 1(W - 20) \text{if } W > 20, \qquad \text{REE}_{\text{kcal/day}} = 70 \cdot W^{0.75} \tag{Eq. 22.5}$$

Here the first 10 kg of body weight demands 4 mL/kg/hr (up to 40 mL/hr), the next 10 kg between 10 kg and 20 kg adds 2 mL/kg/hr (up to +20 mL/hr), and each kilogram above 20 kg adds 1 mL/kg/hr. For the $W = 26.0$ kg pediatric patient on the step-down unit, pressing **26** [↱] [421] evaluates the hourly isotonic maintenance fluid rate as $R_{\text{maint}} = 40 + 20 + 1(26.0 - 20.0) =$ **66.0 mL/hr** (1,584 mL/day). When prescribing weight-based medications via [DOS] (**T** = **W** × **D**), the **SC-80** binds [W] directly to the active Patient Profile ([PAT]) and logs the verified total dose T to the **@MAR** Medication Administration Record in [↱] [▦]—and in the [DOS] Solver Workspace, allows the clinician to solve backward via [XEQ] for the delivered mg/kg dose intensity $D = T / W$ when auditing a fixed tablet or vial strength T.

22.3 Anthropometric Sizing and Renal Clearance on the Stack and in the Solver Workspace

On the **SC-80**, storing a patient's admission parameters into the persistent **Patient Profile** (PAT: W, h, A, S, MALE/FEM) automatically pre-populates W, h, A, and S into the BSA, GFR, DOS, DRP, and →BRN Equation Solver Workspaces while updating the live biometric header in →▦.

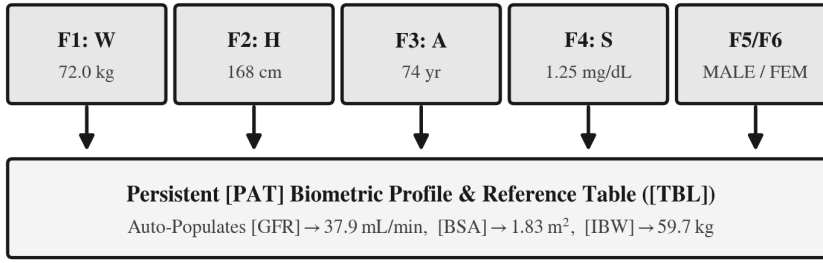


Figure 22.3: *Patient Profile (@PAT) softkey architecture linking bedside biometric inputs to automated clinical clearance indices.*

Key / Solver	Mode	Inputs / Softkeys (LFU0–LFU4)	Governing Expression
PAT	Profile Bar	Softkeys: W h A S MALE/FEM PAT	Stores persistent patient biometrics (W, H, A, S, Sex) and updates →▦
BSA	Stack / Solver	Stack: $Y = H$, $X = W$ • Solver: B h W PAT	$B = \sqrt{(H \cdot W / 3600)}$ (**H** and **W** pre-populated from PAT)
←IBW / →ABW	Immediate Stack	←IBW: $X = H$ (cm) • →ABW: Y = IBW, X = TBW	Devine IBW = 50 + 2.3(H/2.54 - 60); ABW = Y + 0.40(X - Y)
GFR	Stack / Solver	Stack: $Z = A$, $Y = W$, $X = S$ • Solver: C_{CN} A W S PAT	$C = (140 - A) \cdot W / (72 \cdot S)$ (**A**, **W**, **S** pre-populated from PAT)
→421 / ←REE	Immediate Stack	Register X: Patient weight W (kg, or uses PAT W)	Holliday–Segar 4-2-1 mL/hr (→421); Kleiber 70 · $W^{0.75}$ kcal/day (←REE)
DOS	Stack / Solver	Stack: $Y = W$, $X = D$ (or $X = D$ with PAT) • Solver: T W D PAT	$T = W \times D$ (**W** pre-populated from PAT; logs to **@MAR** in →▦)

22.3.1 Immediate Stack Trace: Partitioning Mass Compartments and Auditing Renal Clearance

At the bedside computer cart, the pharmacist evaluates Mosteller BSA for the 68-year-old male patient ($H = 175$ cm, $W = 88$ kg), chains Devine ←IBW and 40% Adjusted Body Weight (→ABW), and compares →ABW-adjusted versus unadjusted GFR at $S = 1.6$ mg/dL:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
175 [ENTER] **88** [BSA]	0	0	0	2.0683	Compute Mosteller surface area $B = \sqrt{(175 \times 88 / 3600)} = 2.0683$ m²
175 [↶] [IBW]	0	0	2.0683	70.4646	Lift [BSA] to Register Y; evaluate Devine $\text{IBW}_{\text{male}} = 70.4646$ kg
88 [↷] [ABW]	0	0	2.0683	77.4787	Blend IBW (Y) and TBW = 88 kg (X) into ABW = 77.4787 kg
68 [x⇄y] **1.6** [GFR]	2.0683	2.0683	2.0683	48.4242	Evaluate [↷] [ABW]-adjusted clearance $C_{\text{ABW}} = 48.42$ mL/min (Z=68, Y=77.48, X=1.6)
68 [ENTER] **88** [ENTER] **1.6** [GFR]	48.4242	48.4242	48.4242	**55.0000**	Compare unadjusted total-weight clearance $C_{\text{TBW}} = 55.00$ mL/min

22.3.2 Bidirectional Solver Workspaces ([GFR] and [DOS]): Pre-Populated Patient Biometrics and Inverse Clearance

Now see how the persistent Patient Profile ([PAT]) accelerates clinical what-if analysis in the [GFR] (**C = (140-A)W/(72×S)**) and [DOS] (**T = W × D**) Equation Workspaces. When the pharmacist opens [GFR] on a cleared stack ([C ON] [GFR]) after setting the patient profile via [PAT], the patient's age **A = 68**, weight **W = 77.48** (updated to [↷] [ABW]), and serum creatinine **S = 1.6** are already bound on the softkey bar. The pharmacist first solves forward via [XEQ] for [C ON], then stores **30** into [C ON] and solves **backward** via [XEQ] on [S] to find the exact serum creatinine level at which this patient would enter severe renal impairment ($C = 30.0$ mL/min), and finally opens [DOS] ([C ON] [DOS]) to back-calculate the exact mg/kg dose intensity [D] delivered by a standard $T = 1250$ mg vancomycin minibag:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU4])	Register X Result & Physical Meaning
[C ON] [GFR] (open workspace)	**GFR: C = (140-A)W/(72×S)**	[C ON] **[A=68] [W=77.48] [S=1.6]** [PAT]	Opens [GFR] workspace pre-populated from Patient Profile [PAT]
[XEQ] → [C ON]	**C = 48.4242**	**[C=48.42] [A=68] [W=77.48] [S=1.6]** [PAT]	Solves forward for [↷] [ABW] Creatinine Clearance C = **48.42 mL/min**
30 → [C ON], [XEQ] → [S]	**S = 2.5826**	**[C=30] [A=68] [W=77.48] [S=2.58]** [PAT]	Solves backward for critical creatinine threshold S = **2.58 mg/dL**
[C ON] [DOS] (open workspace)	**DOS: T = W × D**	[T] **[W=77.48]** [D] [PAT]	Opens weight-based dosing workspace with **W = 77.48 kg** pre-bound
1250 → [T], [XEQ] → [D]	**D = 16.1335**	**[T=1250] [W=77.48] [D=16.13]** [PAT]	Solves backward for delivered dose intensity D = **16.13 mg/kg** (within 15–20 mg/kg)

22.3.3 Try It on the Stack

Practice anthropometric partitioning, [PAT]-coupled renal clearance, and inverse dosing audits across these three clinical admissions:

1. **Oncology Carboplatin & BSA Sizing ([BSA], [↶] [IBW] & [↷] [ABW]):** A female oncology patient measures $H = 162.56$ cm (64.0 in) and weighs TBW = 84.0 kg. Compute her Mosteller [BSA], her female Devine [↶] [IBW] (IBW_{male} - 4.5 kg), and her 40% Adjusted Body Weight (**162.56** [ENTER] **84** [BSA] **162.56** [↶] [IBW] **4.5** [-] **84** [↷] [ABW]). *(Verified Register X results: B = **1.9476 m²**, IBW_{female} = **54.70 kg**, and ABW = **66.42 kg**.)*

2. **Geriatric Female Renal Clearance Comparison ([GFR]):** An 80-year-old female patient weighs $W = 54.0$ kg with a seemingly normal serum creatinine of $S = 1.20$

mg/dL. In the [GFR] Solver Workspace ([C ON] [GFR]), store **80** in [A], **54** in [W], and **1.2** in [S], press [XEQ] → [C ON], and multiply by the 0.85 female factor (or toggle [FEM] inside [PAT]). *(Verified result: C_{female} = **31.875 mL/min**.)*

3. **Pediatric 4-2-1 Fluid Maintenance ([↗] [421]) and Amoxicillin Dose Audit ([DOS]):** Compute the hourly Holliday–Segar maintenance fluid rate for an 18.5 kg toddler (**18.5** [↗] [421]), and in the [DOS] Solver Workspace ([C ON] [DOS]) with **W = 18.5 kg**, store **832.5** in [T] and press [XEQ] → [D] to solve backward for mg/kg/dose when a clinician orders a fixed $T = 832.5$ mg amoxicillin dose. *(Verified results: R_{maint} = **57.00 mL/hr** and D = **45.00 mg/kg/dose**.)*

From admission anthropometrics and renal clearance staging on the ward, we move in Chapter 23 to the intensive care unit at shift change—where weight-based microgram titrations, inverse infusion-pump audits, and gravity tubing drop factors govern continuous vasopressor support.

Chapter 23: Titrating the ICU Norepinephrine Pump at Shift Change

At 7:00 p.m. shift change in the medical intensive care unit, an oncoming critical-care nurse takes handover on an $W = 80.0$ kg patient in distributive septic shock. The bedside peristaltic infusion pump is actively infusing a high-alert central-venous vasopressor at a volumetric flow rate of $R = 15.0$ mL/hr from a pharmacy bag labeled 400 mg dissolved in 250 mL of 5% dextrose in water (D_5W, concentration $C = 1600$ μg/mL). Before co-signing the shift-change medication administration record (**@MAR**), the oncoming nurse must verify the pump in **both directions**: first, given the physician's active order of $D = 5.0$ μg/kg/min, confirming the forward pump rate $R = 15.0$ mL/hr. And second, when inheriting a running infusion pump in the neighboring trauma bay set to $R = 27.0$ mL/hr on a $W = 72.0$ kg patient from a standard norepinephrine bag (4.0 mg in 250 mL, so $C = 16.0$ μg/mL), solving **backward** to verify the exact delivered dose rate in μg/kg/min.

Twenty minutes into the shift, the 80.0 kg patient requires an urgent computed-tomography (CT) angiogram in the basement radiology suite. To prepare a backup manual drip count in case the transport infusion pump battery fails in the elevator, the nurse checks a macro-drip intravenous administration set rated at a Drop Factor of $DF = 15$ gtt/mL alongside a pediatric micro-drip chamber rated at 60 gtt/mL, converting the hourly volumetric flow rate into visible drops per minute.

In this chapter, we work through three critical-care infusion and hemodynamic themes:

1. Convert weight-based vasoactive dose orders (μg/kg/min) into hourly volumetric infusion pump rates (mL/hr).
2. Solve running infusion pumps backward at shift change to audit the exact delivered microgram-per-kilogram-per-minute dose rate.
3. Calibrate manual gravity tubing drop rates (gtt/min), monitor Mean Arterial Pressure and Cardiac Output, and size a biphasic 24-hour Parkland burn resuscitation protocol.

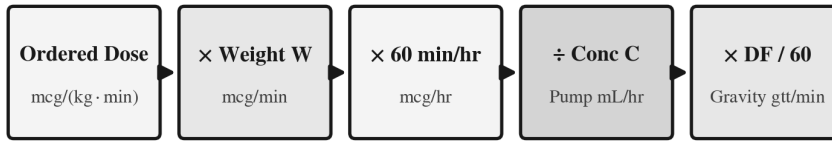
23.1 Weight-Based Vasoactive Titrations and Inverse Pump Verification

Critical-care catecholamines and inotropes (norepinephrine, epinephrine, dopamine, dobutamine, and milrinone) have plasma half-lives of only one to two minutes, so they must be infused continuously and titrated to the patient's real-time arterial pressure. Bridging the physician's physiological order (D, in μg/kg/min) and the mechanical pump's volumetric displacement rate (R, in mL/hr) requires a dimensional chain through mass flux and solution concentration.

23.1.1 The Four-Variable Vasoactive Drip Law (DRP)

When a physician orders a continuous weight-based dose D (in $\mu g \cdot kg^{-1} \cdot min^{-1}$) for a patient of body mass W (in kg, pre-populated from PAT), multiplying $D \cdot W$ gives the microgram delivery rate per minute ($\mu g/min$), and multiplying by 60 min/hr yields the hourly microgram flux $\dot{m}_{mcg/hr} = 60\ D\ W$ (or dividing by 1000 $\mu g/mg$ via the immediate DRP stack operator gives the hourly milligram flux $\dot{m}_{mg/hr} = 0.060\ D\ W$). Dividing the hourly microgram flux by the pharmacy bag's microgram concentration $C = 1000 \cdot (M_{bag,mg} / V_{bag,mL})$ (in $\mu g/mL$) yields the **Volumetric Infusion Pump Rate** R (in mL/hr), while algebraic inversion in the DRP Solver Workspace solves backward via XEQ for the **Delivered Dose Rate** D from an observed pump rate R:

$$R = \frac{D \cdot W \cdot 60}{C} \Leftrightarrow D = \frac{R \cdot C}{60 \cdot W}, \qquad C = \frac{1000 \cdot M_{bag,mg}}{V_{bag,mL}} \tag{Eq. 23.1}$$



Master Vasoactive Identity: $R_{mL/hr} = (D \times W \times 60) / C_{mcg/mL}$

Figure 23.1: *Dimensional unit-cancellation chain transforming weight-based vasoactive dose orders (mcg/kg/min) into volumetric pump rates (mL/hr).*

In our first ICU shift-change audit ($W = 80.0$ kg, $D = 5.0$ $\mu g/kg/min$, and 400 mg in 250 mL D_5W so $C = 1600$ $\mu g/mL = 1.60$ mg/mL), the hourly drug delivery is $\dot{m} = 5.0 \times 80.0 \times 60 = 24{,}000$ $\mu g/hr$ = **24.00 mg/hr**, and dividing by $C = 1600$ $\mu g/mL$ sets the volumetric pump to $R = 24{,}000 / 1600 =$ **15.00 mL/hr**. Conversely, when the nurse inherits the running norepinephrine pump in the next bay set to $R = 27.0$ mL/hr for a $W = 72.0$ kg patient with a standard 4.0 mg / 250 mL bag ($C = 16.0$ $\mu g/mL$), solving DRP backward via XEQ for D gives $D = (27.0 \times 16.0) / (60 \times 72.0) = 432.0 / 4320.0 =$ **0.100 µg/kg/min**.

23.2 Gravity Drip Calibration, Arterial Perfusion, and Burn Resuscitation

When transporting unstable patients or operating in field and disaster settings without electronic peristaltic pumps, nurses regulate intravenous flow by counting liquid drops falling through a calibrated drip chamber against the sweep second hand of a watch.

23.2.1 Orifice Drop Factors and Gravity Drip Rate Calibration (gtt)

Every manual intravenous administration set manufactures its drip-chamber orifice to form drops of a fixed volumetric size, specified by the tubing **Drop Factor** ($DF_{\text{gtt/mL}}$, in drops per milliliter, gtt/mL). Standard adult **macro-drip** sets deliver 10, 15, or 20 gtt/mL, whereas pediatric and critical-care **micro-drip** sets use a fine stainless-steel needle orifice that delivers 60 gtt/mL. Multiplying the hourly volumetric flow rate $R_{\text{mL/hr}}$ by $DF_{\text{gtt/mL}}$ gives drops per hour, and dividing by 60 min/hr yields the **Gravity Drip Rate** ([gtt], $R_{\text{gtt/min}}$, in drops per minute):

$$R_{\text{gtt/min}} = \frac{R_{\text{mL/hr}} \cdot DF_{\text{gtt/mL}}}{60} \qquad \textbf{(Eq. 23.2)}$$

Notice the geometric elegance of the 60 gtt/mL micro-drip chamber: when $DF_{\text{gtt/mL}} = 60$, the ratio 60 / 60 equals unity, so $R_{\text{gtt/min}} = R_{\text{mL/hr}}$ numerically. For our ICU transport running at $R_{\text{mL/hr}} = 15.00$ mL/hr, a macro-drip set with $DF = 15$ gtt/mL produces a drip rate via [gtt] of $R_{\text{gtt/min}} = (15.00 \times 15) / 60 =$ **3.75 gtt/min**, whereas switching to a 60 gtt/mL micro-drip chamber raises the drip frequency to **15.00 gtt/min** (one drop every 4.0 seconds).

23.2.2 Mean Arterial Pressure ([↰] [MAP]), Cardiac Output ([ICU]), and Parkland Burn Crystalloid Resuscitation ([↱] [BRN])

Vasoactive infusions are titrated to restore organ perfusion pressure and cardiac output, while severe thermal burns require high-volume crystalloid infusions to replace capillary plasma leakage. At resting heart rates, the cardiac ventricles spend roughly one-third of each cardiac cycle in systolic ejection (SBP, in mmHg) and two-thirds in diastolic relaxation (DBP, in mmHg), defining **Mean Arterial Pressure** ([↰] [MAP], in mmHg), while multiplying heart rate H (bpm) by stroke volume S (mL/beat) and dividing by 1000 mL/L gives **Cardiac Output** ([ICU], $Q = H \cdot S$ / 1000 L/min). Simultaneously, Charles Baxter's 1968 **Parkland Formula** ([↱] [BRN]) sizes the total 24-hour Lactated Ringer's crystalloid resuscitation volume V (in mL) from patient weight W (pre-populated from [PAT]) and the percentage of Total Body Surface Area burned (B = % TBSA, entered as a whole number such as 35 for 35%):

$$\text{MAP} = \text{DBP} + \tfrac{1}{3}(\text{SBP} - \text{DBP}) = \frac{\text{SBP} + 2 \cdot \text{DBP}}{3}, \qquad V = 4\ \text{mL} \times W \times B \qquad \textbf{(Eq. 23.3)}$$

Parkland protocol infuses the first half of V over the **first 8 hours** from the time of burn injury ($R_{\text{1st 8h}} = V$ / 16 mL/hr) and the second half over the **subsequent 16 hours** ($R_{\text{next 16h}} = V$ / 32 mL/hr).

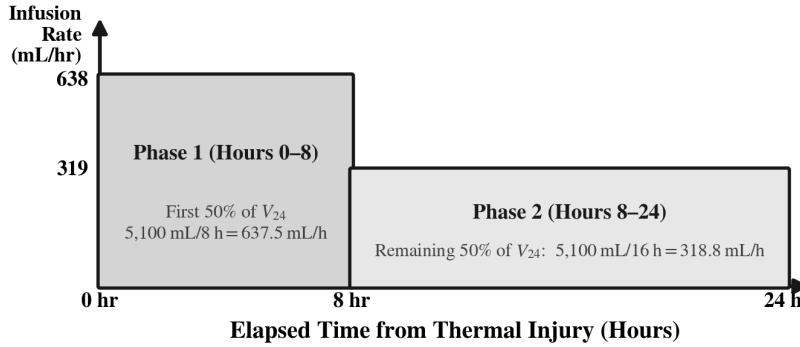


Figure 23.2: *Parkland 24-hour biphasic crystalloid resuscitation infusion profile (50% in the first 8 hours, 50% over the next 16 hours).*

When the 80.0 kg ICU patient's arterial line reads 106 / 52 mmHg on 5.0 µg/kg/min of vasopressor, pressing [↶] [MAP] shows that his perfusion pressure is MAP = 52 + 1/3(106 - 52) = 52 + 18 = **70.0 mmHg**, safely above the 65 mmHg autoregulatory threshold. In the adjacent trauma bay, an $W = 80.0$ kg firefighter with second- and third-degree burns across $B = 35\%$ TBSA requires a 24-hour Parkland volume ([↷] [BRN]) of $V = 4 \times 80.0 \times 35 =$ **11,200 mL**, setting the volumetric infusion pump ([IV]: **R** = **V** / **T**) to 5,600 / 8 = **700.0 mL/hr** for the first 8 hours and 5,600 / 16 = **350.0 mL/hr** for the following 16 hours.

23.3 Vasopressor Titrations and Inverse Pump Audits on the Stack and in the Solver Workspace

On the **SC-80**, vasoactive drip titration ([DRP]), volumetric pump rate ([IV]), gravity drip calibration ([gtt]), Mean Arterial Pressure ([↶] [MAP]), Parkland burn resuscitation ([↷] [BRN]), and hemodynamic cardiac output ([ICU]) operate both on the RPN stack and in [PAT]-linked Equation Solver Workspaces.

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU4])	Governing Expression
[DRP]	Stack / Solver	Stack: $Y = D$, $X = W$ • Solver: [R] [D] [W] [ON/C] [PAT]	Stack: $\dot{m}_{\mathrm{mg/hr}} = D \cdot W \cdot 60 \,/\, 1000$; Solver: $R = D \cdot W \cdot 60 \,/\, C$
[IV]	Stack / Solver	Stack: $Y = V_{\mathrm{mL}}$, $X = T_{\mathrm{hr}}$ • Solver: [R] [V] [T]	$R = V \,/\, T \Leftrightarrow T = V \,/\, R$ (Volumetric pump rate & bag duration)
[gtt]	Immediate Stack	Register Y: Flow rate $R_{\mathrm{mL/hr}}$, Register X: Drop factor DF	Gravity drip rate $R_{\mathrm{gtt/min}} = (Y \cdot X) \,/\, 60$
[↶] [MAP]	Immediate Stack	Register Y: Systolic SBP, Register X: Diastolic DBP	MAP $= X + (Y - X) \,/\, 3 = (Y + 2X) \,/\, 3$ (mmHg)
[↷] [BRN]	Stack / Solver	Stack: $Y = W$, $X = B$ • Solver: [V] [W] [B]	Parkland 24-hr volume $V = 4 \cdot W \cdot B$ (**W** pre-populated from [PAT])
[ICU]	Stack / Solver	Stack: $Y = H_{\mathrm{bpm}}$, $X = S_{\mathrm{mL}}$ • Solver: [Q] [h] [S]	Cardiac output $Q = H \cdot S \,/\, 1000$ (L/min)

23.3.1 Immediate Stack Trace: Auditing the Vasopressor Pump, Gravity Drip, and Burn Resuscitation

During the shift-change double check, the nurse evaluates the hourly mass flux ([DRP]), divides by the bag concentration (400 mg / 250 mL = 1.60 mg/mL) to obtain the volumetric pump rate (**15.00 mL/hr**), calibrates the 15 gtt/mL macro-drip set ([gtt]), verifies [↶] [MAP] at 106 / 52 mmHg, and computes the Parkland burn infusion rates ([↷] [BRN], [IV]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
5 [ENTER] 80 [DRP]	0	0	0	24.0000	Compute vasopressor mass delivery $\dot{m}$ = 24.00 mg/hr (24,000 µg/hr)
400 [ENTER] 250 [÷] [÷]	0	0	0	15.0000	Divide by bag concentration 1.60 mg/mL: pump rate R = 15.00 mL/hr
[ENTER] [ENTER] 15 [gtt]	0	0	15.0000	3.7500	Preserve R = 15.00 mL/hr in Y and compute 15 gtt/mL drip: 3.75 gtt/min
106 [ENTER] 52 [↶] [MAP]	15	15.0000	3.7500	70.0000	Evaluate Mean Arterial Pressure MAP = 52 + (106 - 52)/3 = 70.0 mmHg
80 [ENTER] 35 [↷] [BRN] 16 [IV]	3.75	3.7500	70.0000	**700.000**	Compute 24-hr Parkland volume (11,200 mL) and first-8-hr rate 700.0 mL/hr

23.3.2 Bidirectional Solver Workspaces ([DRP] and [IV]): Back-Calculating Delivered Dose from an Inherited Pump

Now see how the [DRP] Equation Workspace (**R = D × W × 60 / C**) protects patient safety at shift change when a nurse inherits an infusion pump already running at R mL/hr. With patient weight [W] automatically pre-populated from [PAT], the nurse opens [DRP] on a cleared stack ([C ON] [DRP]), confirms the forward pump rate R = 15.0 mL/hr for the 80.0 kg patient (D = 5.0 µg/kg/min, C = 1600 µg/mL), and then audits the inherited norepinephrine pump running at R = 27.0 mL/hr on the W = 72.0 kg patient (4.0 mg / 250 mL ⇒ C = 16.0 µg/mL) by pressing [XEQ] on [D]:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU4])	Register X Result & Physical Meaning
[C ON] [DRP] (open workspace)	**DRP: R = D × W × 60 / C**	[R] [D] **[W=80]** [C ON] [PAT]	Opens vasoactive drip solver with **W = 80 kg** pre-bound from [PAT]
5 → [D], **1600** → [C ON], [XEQ] → [R]	**R = 15.0000**	**[R=15] [D=5] [W=80] [C=1600]** [PAT]	Solves forward for volumetric pump rate R = **15.00 mL/hr**
27 → [R], **72** → [W], **16** → [C ON]	**DRP: R = D × W × 60 / C**	**[R=27]** [D] **[W=72] [C=16]** [PAT]	Stores inherited pump rate R = 27 mL/hr, W = 72 kg, C = 16 µg/mL
[XEQ] → [D]	**D = 0.1000**	**[R=27] [D=0.1] [W=72] [C=16]** [PAT]	Solves backward for delivered norepinephrine dose D = **0.100 µg/kg/min**
[C ON] [IV], **250** → [V], **27** → [R], [XEQ] → [T]	**T = 9.2593**	**[R=27] [V=250] [T=9.26]**	Solves [IV] (**R = V / T**) backward for time until bag empty T = **9.26 hr**

23.3.3 Try It on the Stack

Verify these three critical-care infusion and hemodynamic calculations on the stack and in the [DRP] / [IV] Solver Workspaces:

1. **Inverse Epinephrine Pump Audit ([DRP] Solver):** An oncoming ICU nurse inherits an epinephrine infusion running at $R = 21.0$ mL/hr on a $W = 70.0$ kg patient from a standard bag of 4.0 mg in 250 mL ($C = 16.0$ µg/mL). In the [DRP] Solver Workspace ([C ON] [DRP]), store **21** in [R], **70** in [W], and **16** in [C ON], and press [XEQ] → [D] to solve backward for the delivered dose rate. *(Verified result: D = **0.080 µg/kg/min**.)*

2. **Dobutamine Inotropic Infusion and Macro-Drip Count ([DRP] & [gtt]):** A $W = 90.0$ kg heart-failure patient receives dobutamine at $D = 7.5$ µg/kg/min from a 500 mg in 250 mL bag ($C = 2000$ µg/mL $= 2.0$ mg/mL). Compute the pump rate in mL/hr and the gravity drop rate through a 20 gtt/mL tubing set (**7.5** [ENTER] **90** [DRP] **2** [÷] **20** [gtt]). *(Verified Register X results: R = **20.25 mL/hr** and R_{gtt} = **6.75 gtt/min**.)*

3. **Vancomycin Minibag Duration ([IV]) and Mean Arterial Pressure ([↶] [MAP]):** In the [IV] Solver Workspace ([C ON] [IV], **R** = **V** / **T**), store minibag volume **250** → [V] (mL) and maximum safe vancomycin infusion rate **166.67** → [R] (mL/hr), press [XEQ] → [T] to solve backward for minimum infusion time in hours, and check [↶] [MAP] for an arterial pressure of 92 / 50 mmHg (**92** [ENTER] **50** [↶] [MAP]). *(Verified results: T = **1.50 hr** (90 min) and MAP = **64.00 mmHg**.)*

Having mastered continuous intravenous pump titrations and hemodynamic monitoring in Chapter 23, we turn in Chapter 24 to therapeutic drug monitoring, arterial blood gas and electrolyte interpretation, and aeromedical transport oxygen budgeting.

Chapter 24: Tracking the Trough Across Five Half-Lives

Twelve hours after receiving an initial 500 mg intravenous bolus of a renally cleared antibiotic regimen, an ICU patient's blood is drawn for therapeutic drug monitoring (TDM) alongside an arterial blood gas (ABG), a 12-lead electrocardiogram (ECG), and a comprehensive metabolic panel. Pharmacokinetic population modeling for this patient estimates an **Apparent Volume of Distribution** of $V = 25.0$ L and a biological **Elimination Half-Life** of $T_{1/2} = 6.0$ hours. Before ordering the next 500 mg maintenance dose on an every-12-hour ($\tau = 12.0$ hr) schedule, the clinical pharmacist must calculate the initial post-bolus peak concentration C_0, the first-order elimination rate constant K, the 12-hour pre-dose trough concentration $C(12)$, and the steady-state peak and trough levels that will be reached after five half-lives (30.0 hours) of repeated dosing. When a neighboring renal-failure patient returns a supratherapeutic vancomycin trough of $C_0 = 32.0$ mg/L ($K = 0.05776$ hr^{-1}, $T_{1/2} = 12.0$ hr), the pharmacist also solves the exponential decay equation **backward** for the exact hold time T required for the serum level to decay safely to $C = 15.0$ mg/L.

Simultaneously, the patient's chemistry panel, arterial blood gas, and ECG return from the laboratory with $[Na^+] = 140$ mEq/L, $[Cl^-] = 102$ mEq/L, $[HCO_3^-] = 14$ mEq/L, measured total calcium $Ca_{meas} = 7.6$ mg/dL with severe hypoalbuminemia of Albumin $= 2.0$ g/dL, blood glucose 350 mg/dL, inspired oxygen $F_iO_2 = 0.40$ with arterial $P_aO_2 = 110$ mmHg and $P_aCO_2 = 28$ mmHg, and a measured ECG QT interval of 420 ms at heart rate 96 bpm. Finally, before transferring the patient by helicopter for a $T = 70.0$-minute flight using an E-cylinder pressurized to $P = 1950$ psi, the respiratory therapist must solve for the maximum safe oxygen flow rate F in L/min.

This chapter weaves together three critical-care diagnostic themes:

1. Model first-order exponential drug elimination, steady-state accumulation across five half-lives, and inverse hold-time calculations for supratherapeutic troughs.
2. Evaluate pulmonary oxygenation (P/F ratio and Alveolar–arterial O_2 gradient), metabolic acid-base status (anion gap), and corrected serum calcium, sodium, and osmolality.
3. Normalize cardiac repolarization (QT_c) across heart rate and budget compressed oxygen cylinder duration for aeromedical transport.

24.1 Multi-Compartment Elimination and Steady-State Superposition

In linear pharmacokinetics, the body is modeled either as a single well-mixed fluid volume V (one-compartment model) or as a central vascular compartment V_1 exchanging drug reversibly with a peripheral tissue compartment V_2 before renal and hepatic elimination:

$$C_{2-\text{comp}}(t) = A\,e^{-\alpha t} + B\,e^{-\beta t} \xrightarrow{t \gg 1/\alpha} B\,e^{-\beta t}, \qquad C(t) = C_0\,e^{-Kt} \tag{Eq. 24.1}$$

For glycopeptides and aminoglycosides sampled after the rapid α-distribution phase (30–60 minutes post-infusion) has equilibrated, plasma concentrations follow a single terminal exponential governed by the elimination rate constant $K = \beta$ and apparent volume of

distribution V (PK).

24.1.1 Exponential Decay, Rate Constants, and Elimination Half-Life (PK)

When an intravenous bolus dose of mass D (in mg) mixes across the **Apparent Volume of Distribution** (V, in liters L) cleared at volumetric **Systemic Clearance** rate C_{CL} (in L/hr), it establishes an initial peak concentration $C_0 = D \,/\, V$ (in mg/L). Because the rate of drug elimination $-dM/dt = C_{\text{CL}} \cdot C(t)$ is proportional to instantaneous concentration $C(t)$, plasma concentration decays exponentially according to the first-order **Elimination Rate Constant** (K, in hr^{-1}) and **Elimination Half-Life** ($T_{1/2}$, in hours), where $\ln(2) \approx 0.693147$ (PK):

$$T_{1/2} = \frac{\ln(2) \cdot V}{C_{\text{CL}}} = \frac{\ln(2)}{K}, \qquad C(T) = C_0\, e^{-KT} \Leftrightarrow T = \frac{\ln(C_0/C)}{K} \qquad \textbf{(Eq. 24.2)}$$

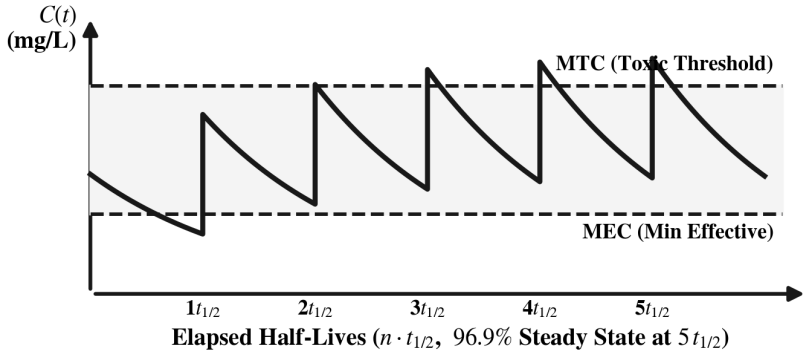


Figure 24.1: *Multi-dose pharmacokinetic plasma accumulation profile approaching steady state within the narrow therapeutic window between MEC and MTC.*

In our ICU patient receiving a $D = 500$ mg intravenous antibiotic bolus into $V = 25.0$ L with $T_{1/2} = 6.0$ hr, the initial peak plasma concentration is $C_0 = 500 \,/\, 25.0 = \mathbf{20.00\ mg/L}$. Pressing **6** (PK) divides $\ln(2) = 0.693147$ by $T_{1/2} = 6.0$ hr to give a first-order elimination rate constant of $K = 0.693147 \,/\, 6.0 = \mathbf{0.11552\ hr^{-1}}$ and a systemic clearance of $C_{\text{CL}} = K \cdot V = 0.11552 \times 25.0 = \mathbf{2.8881\ L/hr}$. At $T = 12.0$ hr (2.0 half-lives), the plasma concentration decays to $C(12) = 20.00 \times e^{-0.11552 \times 12.0} = 20.00 \times (0.5)^2 = \mathbf{5.00\ mg/L}$. Conversely, for the neighboring patient whose vancomycin trough returned supratherapeutic at $C_0 = 32.0$ mg/L with $T_{1/2} = 12.0$ hr ($K = \ln(2)/12 = 0.057762\ \text{hr}^{-1}$), solving the decay law backward for the hold time T needed to reach a safe target trough of $C = 15.0$ mg/L gives $T = \ln(32.0 \,/\, 15.0) \,/\, 0.057762 = 0.75769 \,/\, 0.057762 = \mathbf{13.117\ hr}$ (or at $T_{1/2} = 6.0$ hr, $K = 0.11552\ \text{hr}^{-1} \Rightarrow T = \mathbf{6.559\ hr}$).

24.1.2 Superposition and Steady-State Accumulation Across Five Half-Lives

When the same maintenance dose D is repeated at a fixed dosing interval τ (in hours), each new dose adds $C_0 = D / V$ on top of the residual trough remaining from all prior doses. After n doses, the finite geometric series $\sum_{j=0}^{n-1} r^j = (1 - r^n)/(1 - r)$ with per-interval survival ratio $r = e^{-K\tau} = 2^{-\tau / T_{1/2}}$ converges as $n \to \infty$ to the **Steady-State Accumulation Factor** R_{acc}, determining the **Steady-State Peak** ($C_{ss,max}$), **Steady-State Trough** ($C_{ss,min}$), and **Average Steady-State Concentration** ($C_{ss,avg}$, which equals $AUC_{0\to\tau} / \tau$):

$$R_{acc} = \sum_{j=0}^{\infty} e^{-jK\tau} = \frac{1}{1 - e^{-K\tau}}, \qquad C_{ss,max} = \frac{C_0}{1 - e^{-K\tau}}, \qquad C_{ss,min} = \frac{C_0 e^{-K\tau}}{1 - e^{-K\tau}}, \qquad C_{ss,avg} = \frac{D}{CL \cdot \tau} \qquad \textbf{(Eq. 24.3)}$$

During repeated maintenance dosing or a constant-rate infusion, plasma levels climb to $1 - 2^{-n}$ of steady state: 50% after 1 $T_{1/2}$, 75% after 2 $T_{1/2}$, 87.5% after 3 $T_{1/2}$, 93.75% after 4 $T_{1/2}$, and 96.875% (clinical steady state) after **5 $T_{1/2}$** ($5 \times 6.0 = 30.0$ hours). For our $D = 500$ mg regimen given every $\tau = 12.0$ hr (2.0 half-lives, so $e^{-K\tau} = 0.25$), the accumulation factor is $R_{acc} = 1 / (1 - 0.25) = \mathbf{1.3333}$, stabilizing at $C_{ss,max} = 20.00 \times 1.3333 = \mathbf{26.67}$ **mg/L** and $C_{ss,min} = 26.67 \times 0.25 = \mathbf{6.67}$ **mg/L**.

24.2 Serum Electroneutrality, Pulmonary Gas Exchange, and Oxygen Transport

Beyond drug kinetics, critical-care physiology rests on four conservation and equilibrium laws mapped directly to the **SC-80** keypad: serum electroneutrality ([↗] [AG]), macromolecular and osmotic ion corrections ([↖] [Ca²⁺], [↗] [Na⁺], [↗] [OSM]), alveolar-arterial gas exchange and cardiac repolarization ([↗] [ABG], [↖] [QTc]), and compressed-gas cylinder depletion ([↖] [O₂]).

24.2.1 Anion Gap ([↗] [AG]), Corrected Calcium ([↖] [Ca²⁺]), and Hyperglycemic Sodium ([↗] [Na⁺])

Subtracting measured extracellular anions ($[Cl^-]$ + $[HCO_3^-]$) from sodium ($[Na^+]$) defines the **Serum Anion Gap** ([↗] [AG], normally 8–12 mEq/L), which James Figge, Thomas Fenichel, and Vladimir Jabor (1998) corrected for hypoalbuminemia by adding 2.5 mEq/L per 1.0 g/dL albumin deficit below 4.0 g/dL, alongside Albert Winter's (1967) respiratory compensation law:

$$AG = [Na^+] - ([Cl^-] + [HCO_3^-]), \qquad AG_{corr} = AG + 2.5(4.0 - Alb), \qquad P_aCO_2 = 1.5[HCO_3^-] + 8 \pm 2 \qquad \textbf{(Eq. 24.4)}$$

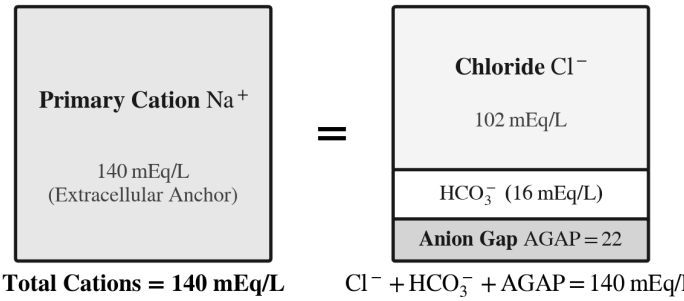


Figure 24.2: *Serum electrolyte Gamblegram illustrating macroscopic cation-anion electroneutrality and the unmeasured Anion Gap (AGAP).*

Simultaneously, Robert Payne's (1973) **Albumin-Corrected Calcium** ([↖] [Ca²⁺]) restores protein-bound calcium in hypoalbuminemia, while Marvin Katz's (1973) **Hyperglycemic Sodium Correction** ([↗] [Na⁺]) accounts for osmotic water translocated from the intracellular space by elevated extracellular glucose (Glc, in mg/dL), and [↗] [OSM] evaluates effective serum osmolality ($2[Na^+] + Glc/18$):

$$Ca_{corr} = Ca_{meas} + 0.8(4.0 - Alb), \qquad Na_{corr} = Na_{meas} + 0.016(Glc - 100) \qquad \textbf{(Eq. 24.5)}$$

For our ICU patient ($Na^+ = 140$, $Cl^- = 102$, $HCO_3^- = 14$ mEq/L, $Ca_{meas} = 7.6$ mg/dL, Alb = 2.0 g/dL, and Glc = 350 mg/dL), the uncorrected anion gap ([↗] [AG]) is AG = 140 - 116 = **24.0 mEq/L** and the albumin-corrected gap is $AG_{corr} = 24.0 + 2.5(2.0) =$ **29.0 mEq/L**; Payne's calcium ([↖] [Ca²⁺]) is $Ca_{corr} = 7.6 + 0.8(2.0) =$ **9.20 mg/dL**. And Katz's corrected sodium ([↗] [Na⁺]) is $Na_{corr} = 140 + 0.016(250) =$ **144.0 mEq/L**.

24.2.2 Arterial Blood Gas Oxygenation ([↗] [ABG]), Bazett [↖] [QTc], and Transport Oxygen ([↖] [O₂])

In mechanical ventilation and ARDS staging, [↗] [ABG] computes the **Horowitz Oxygenation Index (P_aO_2 / F_iO_2 ratio)** from arterial oxygen tension P_aO_2 in Register Y and inspired oxygen fraction F_iO_2 in Register X (where $P/F < 300$ mmHg indicates acute lung injury), while the **Alveolar Gas Equation** computes alveolar oxygen tension P_AO_2 at barometric pressure $P_{atm} = 760$ mmHg, 37°C water-vapor pressure $P_{H_2O} = 47$ mmHg (P_{atm} - $P_{H_2O} = 713$ mmHg), and respiratory quotient $R = 0.80$, and subtracts measured arterial P_aO_2 to obtain the **A–a Gradient** ($\Delta P_{A\text{-}a}O_2$). Alongside blood gas analysis, Henry Cuthbert Bazett's (1920) **Corrected QT Interval** ([↖] [QTc], in ms) normalizes ventricular repolarization time QT_{ms} to a 60 bpm ($RR = 1.0$ s) reference via $\sqrt{RR} = \sqrt{(60 / HR)}$, and the **Compressed Oxygen Cylinder Law** ([↖] [O₂]) relates remaining tank pressure P_{psi} (above the 200 psi safe residual cushion) and cylinder conversion factor k_{tank} (0.28 L/psi for a portable E-cylinder; 3.14 L/psi for a bulk H-cylinder) to flow rate F (L/min) and transport duration T (min):

$$\frac{P_aO_2}{F_iO_2} = \frac{Y}{X}, \qquad \Delta P_{A-a}O_2 = \left[F_iO_2(713) - \frac{P_aCO_2}{0.8}\right] - P_aO_2, \qquad QT_c = \frac{QT_{ms}}{\sqrt{60/HR}}, \qquad T = \frac{(P_{psi} - 200) \times 0.28}{F} \qquad \textbf{(Eq. 24.6)}$$

At $F_iO_2 = 0.40$, $P_aCO_2 = 28$ mmHg, and $P_aO_2 = 110$ mmHg, pressing **110** [ENTER] **0.40** [↗] [ABG] gives a Horowitz P/F ratio of **275.0 mmHg**, while alveolar oxygen is $P_AO_2 =$ 0.40(713) - 28/0.8 = 285.2 - 35.0 = 250.2 mmHg, giving $\Delta P_{A\text{-}a}O_2 = 250.2 - 110 =$ **140.2**

mmHg. On the ECG monitor (QT = 420 ms at HR = 96 bpm, so $RR = 60/96 = 0.625$ s), pressing **420** [ENTER] **96** [←] [QTc] gives $\mathrm{QT}_c = 420 \,/\, \sqrt{(0.625)} = \mathbf{531.26\ ms}$ (alerting the team to pause QT-prolonging antiarrhythmics). Finally, because T and F are reciprocally symmetric in $T \cdot F = (P_{\mathrm{psi}} - 200) \times 0.28$, for the $T = 70.0$-minute helicopter transfer with an E-cylinder at $P_{\mathrm{psi}} = 1950$ psi, pressing **1950** [ENTER] **70** [←] [O₂] directly yields the maximum safe oxygen flow rate $F = (1950 - 200) \times 0.28 \,/\, 70.0 = 490.0 \,/\, 70.0 = \mathbf{7.00\ L/min}$.

24.3 Compartment Kinetics, Blood Gases, and Transport Oxygen on the Stack and in the Solver Workspace

On the **SC-80**, pharmacokinetic elimination ([PK]), acid-base and electrolyte corrections ([→] [AG], [←] [Ca²⁺], [→] [Na⁺], [→] [OSM]), cardiopulmonary indices ([→] [ABG], [←] [QTc], [ICU]), and oxygen tank budgeting ([←] [O₂]) execute both on the RPN stack and in bidirectional Equation Solver Workspaces.

Key / Solver	Mode	Inputs / Softkeys ([LFU0]–[LFU2])	Governing Expression
[PK]	Stack / Solver	Stack: $X = T_{1/2}$ or K • Solver: [T] [V] [c/cM]	Stack: $K = \ln(2) \,/\, T_{1/2} \Leftrightarrow T_{1/2} = \ln(2) \,/\, K$; Solver: $T_{1/2} = 0.693147 \cdot V \,/\, C_{\mathrm{CL}}$
[→] [AG] / [←] [Ca²⁺] / [→] [Na⁺]	Immediate Stack	[→] [AG]: $Y = \mathrm{Na^+}$, $X = \mathrm{Cl^-} + \mathrm{HCO_3^-}$ • [←] [Ca²⁺]: $Y = \mathrm{Ca}$, $X = \mathrm{Alb}$ • [→] [Na⁺]: $Y = \mathrm{Na}$, $X = \mathrm{Glc}$	$\mathrm{AG} = Y - X$; $\mathrm{Ca}_{\mathrm{corr}} = Y + 0.8(4 - X)$; $\mathrm{Na}_{\mathrm{corr}} = Y + 0.016(X - 100)$
[→] [ABG] / [←] [QTc]	Immediate Stack	[→] [ABG]: $Y = P_a\mathrm{O_2}$, $X = F_i\mathrm{O_2}$ • [←] [QTc]: $Y = \mathrm{QT}_{\mathrm{ms}}$, $X = \mathrm{HR}_{\mathrm{bpm}}$	Horowitz $P/F = Y \,/\, X$; Bazett $\mathrm{QT}_c = Y \,/\, \sqrt{(60/X)}$
[←] [O₂]	Immediate Stack	Register Y: Pressure P_{psi}, Register X: Flow $F_{\mathrm{L/min}}$ (or duration T_{min})	E-cylinder $T = (P - 200) \times 0.28 \,/\, F \Leftrightarrow F = (P - 200) \times 0.28 \,/\, T$

24.3.1 Immediate Stack Trace: Antibiotic Trough Decay, Electrolyte Corrections, and [←] [QTc]

To evaluate the 500 mg IV bolus in $V = 25.0$ L with $T_{1/2} = 6.0$ hr at $T = 12.0$ hr and at steady state ($\tau = 12.0$ hr), followed by the Figge–Fenichel Anion Gap ([→] [AG]), Payne Corrected Calcium ([←] [Ca²⁺]), Katz Corrected Sodium ([→] [Na⁺]), and Bazett [←] [QTc], the clinician chains:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
500 [ENTER] **25** [÷] **6** [PK]	0	0	20.0000	0.1155	Peak $C_0 = 20.0$ mg/L in Y; rate constant $K = \ln(2)/6 = 0.11552$ hr⁻¹ in X
12 [×] [+/−] [→] [e^x] [×] **0.75** [÷]	0	0	0	6.6667	First trough $C(12) = 5.0$ mg/L scaled by $1/(1\text{-}0.25) \rightarrow C_{\mathrm{ss,min}} = 6.67$ mg/L
140 [ENTER] **102** [ENTER] **14** [+] [→] [AG] **5** [+]	6.6667	6.6667	6.6667	29.0000	Uncorrected AG = 24.0 plus +5.0 albumin correction → $\mathrm{AG}_{\mathrm{corr}} = 29.0$ mEq/L
7.6 [ENTER] **2.0** [←] [Ca²⁺]	6.6667	6.6667	29.0000	9.2000	Payne Albumin-Corrected Calcium $\mathrm{Ca}_{\mathrm{corr}} = 7.6 + 0.8(4 - 2) = 9.20$ mg/dL
420 [ENTER] **96** [←] [QTc]	29	29.0000	9.2000	**531.2627**	Bazett corrected interval $\mathrm{QT}_c = 420 \,/\, \sqrt{(60/96)} = 531.26$ ms

24.3.2 Bidirectional Solver Workspaces ([PK] and [ICU]) and Inverse Oxygen Budgeting ([↖] [O₂])

Now see how the [PK] and [ICU] Equation Workspaces and [↖] [O₂] solve critical clinical constraints backward. First, when a patient with $V = 25.0$ L and $T_{1/2} = 6.0$ hr ($K = 0.11552$ hr^{-1}) has a supratherapeutic trough of $C_0 = 32.0$ mg/L, the pharmacist opens [PK] ([C ON] [PK]) to solve via [XEQ] for systemic clearance $C_{CL} = 2.888$ L/hr and determines the exact hold time $T = \ln(32.0/15.0)/0.11552 =$ **6.559 hr** required before resuming dosing. Second, before loading the patient into the helicopter for a $T = 70.0$-minute flight with an E-cylinder at $P = 1950$ psi, the respiratory therapist evaluates **1950 [ENTER] 70 [↖] [O₂]** to find the maximum safe oxygen flow rate $F =$ **7.00 L/min**, and uses the [ICU] Solver Workspace (**Q = H × S / 1000**) to solve backward for stroke volume [S] given cardiac output $Q = 5.28$ L/min at heart rate $H = 96$ bpm:

Step & Action	Line 1 Workspace Formula	Softkey Legend Bar ([LFU0]–[LFU2])	Register X Result & Physical Meaning
[C ON] [PK] (open workspace)	**PK: T = .69315 × V / C**	[T] [V] [C ON]	Opens compartment pharmacokinetic workspace
6 → [T], **25** → [V], [XEQ] → [C ON]	**C = 2.8881**	**[T=6] [V=25] [C=2.89]**	Solves backward for systemic clearance $C_{CL} =$ **2.888 L/hr**
[C ON] **32** [ENTER] **15** [÷] [↖] [LN] **6** [PK] [÷]	*(Normal Stack)*	*(Standard Spine)*	Solves exponential decay backward for hold time $T =$ **6.559 hr**
1950 [ENTER] **70** [↖] [O₂]	*(Normal Stack)*	*(Standard Spine)*	Solves E-cylinder law for maximum safe oxygen flow rate $F =$ **7.00 L/min**
[C ON] [ICU], **5.28** → [Q], **96** → [h], [XEQ] → [S]	**S = 55.0000**	**[Q=5.28] [H=96] [S=55]**	Solves [ICU] (**Q = H × S / 1000**) backward for stroke volume $S =$ **55.0 mL/beat**

24.3.3 Try It on the Stack

Reinforce your command of compartment pharmacokinetics, blood-gas homeostasis, and transport oxygen budgeting with these three clinical problems:

1. **Gentamicin Half-Life ([PK]) and Supratherapeutic Hold Time:** A septic patient has $V = 18.0$ L and renal clearance $C_{CL} = 3.00$ L/hr. In the [PK] Solver Workspace ([C ON] [PK]), store **18** in [V] and **3** in [C ON] and press [XEQ] → [T] to solve for half-life $T_{1/2}$, then find the hold time required for a toxic serum concentration of $C_0 = 14.0$ mg/L to decay to $C = 2.0$ mg/L (**14 [ENTER] 2 [÷] [↖] [LN] 4.1589 [PK] [÷]**). *(Verified results: $T_{1/2} =$ **4.1589 hr**, $K =$ **0.1667 hr^{-1}**, and $T_{hold} =$ **11.676 hr**.)*

2. **Diabetic Ketoacidosis Anion Gap ([↗] [AG]) and Hyperglycemic Sodium ([↗] [Na⁺]):** A patient in diabetic ketoacidosis presents with $[Na^+] = 131$ mEq/L, $[Cl^-] = 95$ mEq/L, $[HCO_3^-] = 8$ mEq/L, and blood glucose Glc = 600 mg/dL. Compute the Serum Anion Gap (**131 [ENTER] 95 [ENTER] 8 [+] [↗] [AG]**) and Katz Corrected Sodium (**131 [ENTER] 600 [↗] [Na⁺]**). *(Verified Register X results: $AG =$ **28.00 mEq/L** and $Na_{corr} =$ **139.0 mEq/L**.)*

3. **Blood Gas Oxygenation ([↗] [ABG]) and Transport Cylinder Reserve ([↖] [O₂]):** Evaluate the Horowitz P/F oxygenation index (**85 [ENTER] 0.50 [↗] [ABG]**) and A–a oxygen gradient for a ventilated patient on $F_iO_2 = 0.50$ with $P_aCO_2 = 40$ mmHg and $P_aO_2 = 85$ mmHg (**0.50 [ENTER] 713 [×] 40 [ENTER] 0.8 [÷] [-] 85 [-]**), and compute the E-cylinder duration at $P = 1700$ psi and $F = 10.0$ L/min (**1700 [ENTER] 10 [↖] [O₂]**). *(Verified results: $P/F =$ **170.0 mmHg**, $\Delta P_{A\text{-}a}O_2 =$ **221.5 mmHg**, and $T =$*

42.00 min.)

With clinical anthropometrics, vasoactive infusion titrations, and compartment pharmacokinetics established on the **SC-80**, we turn in **Unit IX: Quantitative Finance, Fixed-Income Valuation & Six Sigma Quality** to our ninth and final domain instrument—the **SC-95**—where compound discounting, derivative pricing, and statistical process capability govern capital and manufacturing quality.

Unit IX: Quantitative Finance, Fixed-Income Valuation & Six Sigma Quality

In 1202, Leonardo of Pisa—known to posterity as Fibonacci—posed a foundational question in *Liber Abaci*: if a merchant is promised a series of future payments while capital earns compound interest each year, what single lump sum paid today is its fair equivalent? In physics, dimensional analysis forbids adding quantities that do not share the same physical units. In quantitative finance, a parallel conservation law forbids adding or subtracting cash flows that occur at different points in time until each flow is transported to a common reference date. At a 6.0% annual compound return, an investment of \$10,000 today grows over ten years to $V_{10} = 10{,}000(1.06)^{10} = \$17{,}908.48$, whereas a promise of \$10,000 delivered ten years in the future discounts back to a present purchasing power of $V_0 = 10{,}000(1.06)^{-10} = \$5{,}583.95$.

Seven centuries after Fibonacci, Frederick R. Macaulay's 1938 treatise for the National Bureau of Economic Research generalized multi-period discounting to fixed-income bond pricing, loan amortization timelines, corporate capital budgeting, and the weighted-average maturity of cash flows now known as Macaulay Duration. In 1973, Fischer Black, Myron Scholes, and Robert C. Merton extended continuous-time discounting into stochastic calculus. They proved that a dynamically rebalanced portfolio of shares and risk-free debt eliminates diffusion risk and prices European options via the cumulative standard normal integral. Notice how the continuous-time discount factor e^{-rT} that transports option strike prices backward in time is the exact mathematical twin of exponential physical relaxation across our earlier field instruments: RC capacitor discharge $e^{-t/\tau}$ in **Unit III: Digital Systems, Bitwise Logic & Embedded Timing**, Beer–Lambert optical attenuation $10^{-\varepsilon \ell c}$ in **Unit VI: Stoichiometry, Chemical Equilibrium & Wet-Lab Analytics**, and first-order drug elimination $e^{-k_e t}$ in **Unit VIII: Clinical Dosing, Renal Clearance & Pharmacokinetics**.

While financial economists formalized the time value of capital and derivative risk, industrial engineers and statisticians developed the mathematics of commercial unit economics and statistical quality control. In 1908, William Sealy Gosset—publishing under the pseudonym "Student" while working as Head Experimental Brewer at Arthur Guinness & Son in Dublin—derived the heavy-tailed small-sample t-distribution, complementing Karl Pearson's 1900 chi-square goodness-of-fit law and George W. Snedecor's variance-ratio F-distribution. On May 16, 1924, Walter A. Shewhart drafted a one-page memorandum at Bell Telephone Laboratories introducing the statistical control chart to separate random common-cause variation from assignable mechanical drift. Six decades later, Motorola engineers Bill Smith and Mikel Harry synthesized Shewhart's dispersion limits with blueprint engineering tolerances into the Six Sigma process capability indices C_p and C_{pk}.

Across the three chapters of this unit, we follow a corporate finance and operations team as they structure a plant mortgage, hedge fixed-income and derivative exposure, and audit a high-precision manufacturing line on the **SC-95**:

Chapter	Narrative Problem	Mathematical Concepts	Three Chapter Themes
Chapter 25	Structuring a $450,000, 20-year monthly facility mortgage at 6.0%, decomposing Year 1 amortization, and valuing a 4-year plant expansion	Five-variable compound Time Value of Money balance, geometric principal/interest amortization, Net Present Value, Internal Rate of Return, and gross margin versus markup	1. Commercial Mortgage Payment & Year 1 Amortization • 2. Plant Automation NPV & Internal Rate of Return • 3. Gross Margin, Markup & Economic Order Quantity
Chapter 26	Pricing a 10-year, 5% semiannual corporate bond yielding 6.0%, hedging a +50 bps rate hike, and valuing a 6-month European option (S=100, K=100)	Semiannual bond discounting, Yield to Maturity root-finding, Macaulay and Modified Duration, Black–Scholes–Merton continuous hedging, and Put–Call Parity	1. Semiannual Coupon Bond Valuation & Yield to Maturity • 2. Macaulay Duration, Modified Duration & Rate Convexity • 3. Black–Scholes–Merton Option Pricing & Put–Call Parity
Chapter 27	Auditing a precision shaft grinder (LSL 9.95 mm, USL 10.05 mm) whose mean has drifted to $\mu = 10.01$ mm ($\sigma = 0.01$ mm)	Bivariate linear regression trend, Gaussian cumulative tail integrals, Student's t confidence intervals, sample size n , Six Sigma capability (C_p, C_{pk}), and DPMO	1. Hourly Spindle Regression & Drift Hypothesis Testing • 2. Six Sigma Potential (C_p) vs. Centered Capability (C_{pk}) • 3. Audit Sample Sizing & Defects Per Million (DPMO)

Chapter 25: Structuring the Equipment Lease and Valuing the Expansion Plant

The chief financial officer of a precision valve manufacturer is finalizing the capital structure for a new regional assembly plant. To acquire the building, the company is taking out a \$450,000 commercial facility mortgage amortized over 20 years with monthly payments (N = 240 months) at a fixed 6.00% nominal annual interest rate (0.50% per month). Before signing the loan covenant, the CFO must determine the exact monthly debt-service payment and provide the corporate tax accountants with the first-year amortization breakdown—separating the twelve monthly payments of Year 1 (Months 1 through 12) into tax-deductible interest expense and balance-sheet principal reduction.

Inside the newly acquired building, the VP of Operations proposes installing a robotic valve-machining cell requiring an upfront capital outlay of I_0 = \$120,000 at t = 0. Engineering time studies project that the automated cell will generate uniform net operating cash savings of C = \$42,000 per year across a 4-year planning horizon. Discounting those future cash inflows at the company's 10.0% weighted-average cost of capital and solving for the project's break-even yield (IRR) determines whether the automation investment creates net shareholder value.

Finally, to set wholesale catalog pricing for the new valve line—which costs C = \$72.00 per unit to manufacture and is targeted to sell at P = \$120.00—the commercial team must reconcile the accounting department's **Gross Margin** requirement against the sales force's **Markup on Cost** schedule.

25.1 Compound Discounting, Loan Amortization, and the Five-Variable Balance

Every level-payment commercial mortgage, equipment lease, sinking fund, and retirement annuity rests on transporting a schedule of signed periodic cash flows to a common reference epoch and enforcing conservation of value.

25.1.1 The Five-Variable Time Value of Money Recurrence (TVM)

Across any level-payment financial contract in the TVM workspace, five parameters are coupled together on softkeys N, I%, PV, PMT, and FV (alongside payment timing END/BEG): the total number of compounding or payment periods N (N), the periodic interest rate I% expressed as a percentage per period where the decimal rate is i = (I%)/100 (obtained from the nominal annual rate by dividing by M compounding periods per year), the initial **Present Value** PV (PV) at t = 0, the level periodic **Payment** PMT (PMT) at the end of each period t = 1, 2, ..., N, and the terminal **Future Value** or balloon balance FV (FV) at horizon $t = N$. Discounting each cash flow back to t = 0 and summing the finite geometric series $\sum_{t=1}^{N} (1+i)^{-t} = (1 - (1+i)^{-N})/i$ yields the **Time Value of Money Balance Equation**:

$$\mathrm{PV} + \mathrm{PMT}\left(\frac{1-(1+i)^{-N}}{i}\right) + \mathrm{FV}(1+i)^{-N} = 0, \qquad \mathrm{PV}(1+i)^{N} + \mathrm{PMT}\left(\frac{(1+i)^{N}-1}{i}\right) + \mathrm{FV} = 0 \qquad \textbf{(Eq. 25.1)}$$

Because this equation enforces cash conservation across a two-party transaction, cash inflows received by the decision-maker carry a **positive (+)** sign, whereas cash outflows paid out carry a **negative (-)** sign.

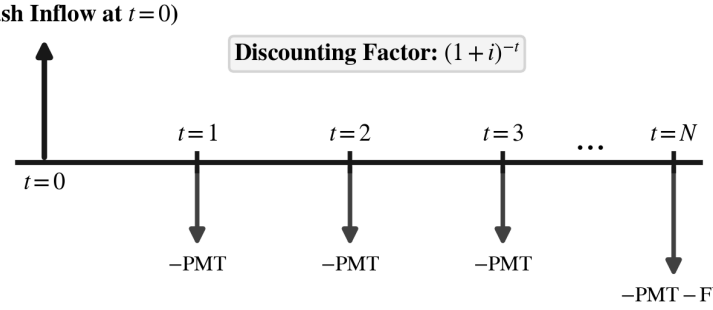


Figure 25.1: *Cash-flow timeline convention showing positive inflows, negative outflows, and exponential discounting across N compounding periods.*

For the valve manufacturer's $450,000 commercial mortgage (PV = +450,000) over 20 years paid monthly ($N = 20 \times 12 = 240$ months) at a 6.00% nominal annual rate (i = 6.00% / 12 = 0.50% = 0.005 per month) fully amortized to FV = 0, the 240-month present-value annuity factor is $a_{240|0.005} = (1 - 1.005^{-240}) / 0.005 = 0.6979039 / 0.005 = 139.58077$. Dividing -PV by $a_{240|0.005}$ gives a monthly debt-service outflow of PMT = -450,000 / 139.58077 = **-$3,223.94/month** (-$3,223.9398).

25.1.2 Geometric Principal and Interest Decomposition Across Amortization Periods ([↰] [AMR])

Although the total monthly payment |PMT| = $3,223.94 remains constant across all 240 months, its internal composition between interest expense I_t and principal reduction P_t shifts every month ([↰] [AMR]). In Month 1, interest on the initial PV = $450,000 balance (**450000** [ENTER] **0.5** [↰] [AMR]) is $I_1 = \text{PV} \cdot i = 450{,}000 \times 0.005 =$ **$2,250.00**, leaving $P_1 =$ |PMT| - I_1 = 3,223.9398 - 2,250.00 = **$973.94** to reduce principal. Because each month's principal reduction lowers the outstanding loan balance for the following month, the principal portion grows geometrically as $P_t = P_1 (1 + i)^{t-1}$, so cumulative principal paid (ΔPrin_{1-k}), cumulative interest paid (ΔInt_{1-k}), and remaining loan balance B_k after k payments follow closed-form geometric sums:

$$\Delta\text{Prin}_{1-k} = P_1\left(\frac{(1+i)^k - 1}{i}\right), \qquad \Delta\text{Int}_{1-k} = k \cdot |\text{PMT}| - \Delta\text{Prin}_{1-k}, \qquad B_k = \text{PV} - \Delta\text{Prin}_{1-k} \qquad \textbf{(Eq. 25.2)}$$

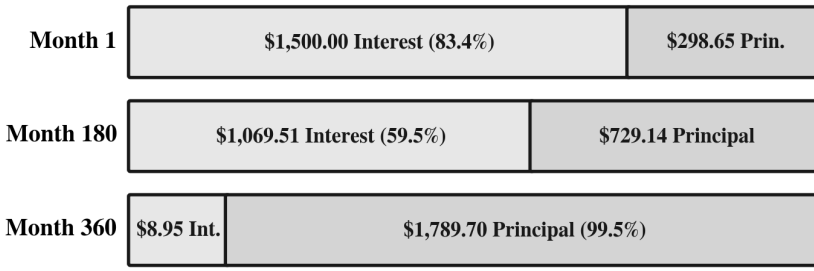


Figure 25.2: *Evolution of monthly interest versus principal reduction across a 360-month level-payment mortgage amortization schedule.*

Evaluating these sums across the first twelve months ($k = 12$) of the \$450,000 facility mortgage shows that total cash outflow in Year 1 is 12 × \$3,223.9398 = **\$38,687.28**. The cumulative Year 1 principal reduction is $\Delta\text{Prin}_{1–12}$ = \$973.9398 × [(1.005^{12} - 1)/0.005] = \$973.9398 × 12.335562 = **\$12,014.09**, which reduces the ending Month 12 loan balance to B_{12} = \$450,000 - \$12,014.09 = **\$437,985.91** and leaves a Year 1 tax-deductible interest expense of $\Delta\text{Int}_{1–12}$ = \$38,687.28 - \$12,014.09 = **\$26,673.19**.

25.2 Capital Budgeting and Commercial Unit Economics

Once long-term facility financing is in place, corporate capital allocation compares upfront equipment outlays against discounted operating cash inflows and links unit manufacturing cost to wholesale selling price.

25.2.1 Net Present Value, Internal Rate of Return ([↗] [CF]), and Effective Annual Rates ([↗] [EAR])

When a capital project requires an initial investment I_0 at $t = 0$ and generates a uniform net operating cash inflow C across N periods discounted at the firm's opportunity cost of capital (hurdle rate) $r = R$ / 100, **Net Present Value** ([↗] [CF], opening the [↗] [CF] Solver Workspace) subtracts I_0 from the present value of the cash-flow annuity. A positive NPV > 0 certifies that the asset earns more than the hurdle rate r. The break-even discount rate r^* at which $\text{NPV}(r^*) = 0$ defines the **Internal Rate of Return** (IRR), while a nominal Annual Percentage Rate ($\text{APR}_{\%}$) compounded M times per year converts to the **Effective Annual Rate** ([↗] [EAR]), and multi-year lump-sum growth follows **Compound Annual Growth Rate** ([↗] [CGR]) and **Return on Investment** ([↗] [ROI]):

$$\text{NPV} = C\left(\frac{1-(1+r)^{-N}}{r}\right) - I_0, \qquad \text{NPV}(r^*) = 0 \Leftrightarrow \frac{1-(1+r^*)^{-N}}{r^*} = \frac{I_0}{C}, \qquad \text{EAR}_{\%} = \left[\left(1+\frac{\text{APR}_{\%}}{100\cdot M}\right)^{M} - 1\right] \times 100 \qquad \textbf{(Eq. 25.3)}$$

For a single-period cash outlay CF_0 and return CF_1, the rate of return ([↗] [ROI]) simplifies directly to $\text{ROI}_{\%} = [(CF_1 - |CF_0|) / |CF_0|] \times 100$.

In the valve manufacturer's automation proposal (I_0 = \$120,000, C = \$42,000/yr over N = 4 years at hurdle rate R = 10.0%, so r = 0.10), the 4-year annuity factor is (1 - 1.10^{-4}) / 0.10 = 3.169865. Multiplying by \$42,000/yr gives a discounted inflow of \$133,134.35, and subtracting the \$120,000 outlay yields a positive Net Present Value of NPV = \$133,134.35 - \$120,000 = +**\$13,134.35**. Setting $V = \text{NPV}(r^*) = 0$ in the [↗] [CF] ([↗] [CF]) Solver Workspace requires $a_{4|r^*}$ = 120,000 / 42,000 = 2.857143, which resolves via [↗] [SOLVE] →

[R] to an Internal Rate of Return of IRR = **14.96%/yr**—exceeding the 10.0% hurdle rate by nearly 500 basis points.

25.2.2 Gross Margin ([↖] [MGN]), Markup on Cost ([↗] [MKP]), Breakeven ([↖] [BEP]), and Economic Order Quantity ([↗] [EOQ])

For a manufactured product with unit cost C and wholesale selling price P, the unit gross profit (P - C) is measured in financial statements as **Gross Margin** ([↖] [MGN], profit as a percentage of selling price P) and in cost-plus pricing sheets as **Markup on Cost** ([↗] [MKP], profit as a percentage of unit cost C):

$$\mathrm{MGN}_{\%} = \left(\frac{P-C}{P}\right)\times 100, \qquad \mathrm{MKP}_{\%} = \left(\frac{P-C}{C}\right)\times 100, \qquad P = \frac{C}{1-(\mathrm{MGN}_{\%}/100)} = C\left(1+\frac{\mathrm{MKP}_{\%}}{100}\right) \qquad \textbf{(Eq. 25.4)}$$

Simultaneously, a plant with fixed overhead F_{fixed} and unit contribution margin $M_c = P$ - V reaches its **Breakeven Volume** ([↖] [BEP]) at $Q_{\mathrm{BEP}}^{*} = F_{\mathrm{fixed}} / (P - V)$, depreciates an asset of net cost (C_{cost} - S_{salvage}) over L years via **Straight-Line Depreciation** ([↗] [DEP], $D_{\mathrm{annual}} = (C_{\mathrm{cost}} - S_{\mathrm{salvage}})/L$), and minimizes combined annual ordering and holding cost for annual demand D, setup cost S, and holding cost H at Ford W. Harris and R. H. Wilson's **Economic Order Quantity** ([↗] [EOQ]):

$$Q_{\mathrm{BEP}}^{*} = \frac{F_{\mathrm{fixed}}}{P-V}, \qquad D_{\mathrm{annual}} = \frac{C_{\mathrm{cost}} - S_{\mathrm{salvage}}}{L}, \qquad Q_{\mathrm{EOQ}}^{*} = \sqrt{\frac{2\cdot D\cdot S}{H}} \qquad \textbf{(Eq. 25.5)}$$

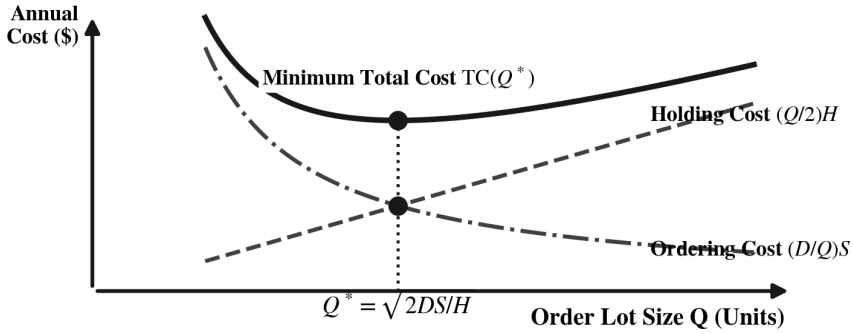


Figure 25.3: *Wilson Economic Order Quantity (EOQ) optimization balancing annual ordering cost against linear inventory holding cost.*

For the valve assembly with unit cost C = \$72.00 and selling price P = \$120.00 (unit gross profit \$48.00), the Gross Margin ([↖] [MGN]) is $\mathrm{MGN}_{\%}$ = (48.00 / 120.00) × 100 = **40.00%**, whereas the Markup on Cost ([↗] [MKP]) is $\mathrm{MKP}_{\%}$ = (48.00 / 72.00) × 100 = **66.67%**. Covering \$36,000 of monthly fixed plant overhead at P - V = \$48.00/unit ([↖] [BEP]) requires a breakeven volume of Q_{BEP}^{*} = 36,000 / 48.00 = **750 valves/month**, and ordering raw brass forgings with annual demand D = 12,000 units/yr, setup cost S = \$90/order, and holding cost H = \$6.00/unit/yr ([↗] [EOQ]) yields an optimal batch size of $Q_{\mathrm{EOQ}}^{*} = \sqrt{((2 \times 12{,}000 \times 90)/6.00)} = \sqrt{(360{,}000)}$ = **600 units/order**.

25.3 Mortgage Structuring and Capital Budgeting on the Stack and in the Solver Workspace

The **SC-95** integrates the five-variable Time Value of Money workspace ([TVM]: softkeys [N], [I%], [PV], [PMT], [FV], [END]/[BEG]), loan interest amortization ([↖] [AMR]), discounted cash-flow analysis ([↗] [CF], opening the [↗] [CF] Solver Workspace), and commercial unit-economics keys ([↖] [MGN], [↗] [MKP], [↗] [DEP], [↖] [BEP], [↗] [EOQ], [↗] [EAR], [↗] [CGR], [↗] [ROI]) across the RPN stack and **Bidirectional Equation Solver Workspaces** ([↗] [SOLVE] / [XEQ]):

Key / Solver	Operation	Stack Inputs (Y, X)	Governing Expression & Solver Softkeys ([LFU0]–[LFU5])
[TVM]	Five-Variable TVM Workspace	Value in $X \rightarrow$ tap softkey	$\mathrm{PV} + \mathrm{PMT}\cdot a_{\overline{N}\vert i} + \mathrm{FV}(1+i)^{-N} = 0$ ([N] [I%] [PV] [PMT] [FV] [END])
[↖] [AMR]	Loan Amortization Interest	Y = Balance B, X = $i_{\%}$	Period interest $I_k = B_k \cdot (i_{\%}/100)$; Geometric $\Delta\mathrm{Prin}_{1-k} = P_1((1+i)^{k} - 1) / i$
[↗] [CF]	Cash Flow / [↗] [CF] & [↗] [CF] Solver	$Y = CF_1$, $X = R_{\%}$	Stack: $Y / (1+X/100)$; Solver ([↗] [CF]): **V = C×(1-(1+R/100)⁻ᴺ)/(R/100) - I** ([V] [C0] [R] [N] [I])
[↗] [ROI] / [↗] [CGR]	Return on Investment & CAGR	$Y = P_0$, $X = F_N$	Stack: $(X - Y) / Y \times 100$; Solver ([↗] [CGR]): **F = P × (1 + R/100)^N** ([F] [P] [R] [N])
[↖] [MGN] / [↗] [MKP]	Gross Margin & Markup	Y = Cost C, X = Price P	Stack & Solver: $\mathrm{MGN}_{\%} = (P - C) / P \times 100$ ([M] [P] [C0]); $\mathrm{MKP}_{\%} = (P - C) / C \times 100$ ([U] [P] [C0])
[↖] [BEP] / [↗] [EOQ]	Breakeven & Wilson EOQ	[↗] [EOQ]: $Y = D \cdot S$, $X = H$	[↖] [BEP]: **Q = F / (P - V)** ([Q] [F] [P] [V]); [↗] [EOQ]: **Q = √(2×D×S/H)** ([Q] [D] [S] [h])

25.3.1 Solving the Commercial Mortgage and Year 1 Amortization on the Stack

To structure the \$450,000, 20-year monthly mortgage (PV = 450,000, FV = 0) without pre-dividing the 6.0% annual interest rate or 20-year term in his head, the CFO presses [TVM], lets the RPN stack compute $N = 20 \times 12 = 240$ months ([N]) and periodic rate I% = 6.0 / 12 = 0.50%/month ([I%]), solves for [PMT], and evaluates the Month 1–12 principal reduction and interest expense ([↖] [AMR]) on the stack:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[TVM] 20 [ENTER] 12 [×] [N] 6 [ENTER] 12 [÷] [I%]	0	0	240	0.5	Open [TVM] and store $N = 20 \times 12 = 240$ mo and $i = 6 / 12 = 0.50\%$/mo
450000 [PV] 0 [FV] [↗] [SOLVE] [PMT]	240	0.5	450000	**-3223.94**	Store PV = 450,000, FV = 0 and solve PMT = -\$3,223.94/mo
[+/−] [ENTER] [ENTER] 450000 [ENTER] 0.5 [↖] [AMR] [−]	3223.94	3223.94	3223.94	**973.9398**	Subtract Month 1 interest (\$2,250.00 via [↖] [AMR]) from |PMT|: P_1 = \$973.94
1.005 [ENTER] 12 [↗] [y^x] 1 [−] 0.005 [÷] [×]	3223.94	3223.94	3223.94	**12014.09**	Accumulate Year 1 principal reduction $\Delta\mathrm{Prin}_{1-12}$ = \$12,014.09
[x⇄y] 12 [×] [x⇄y] [−]	3223.94	3223.94	3223.94	**26673.19**	Subtract $\Delta\mathrm{Prin}_{1-12}$ from 12|PMT|: Year 1 interest $\Delta\mathrm{Int}_{1-12}$ = \$26,673.19

Subtracting Year 1 principal **12014.09** in Register Y from **450000** in Register Z confirms the Month 12 ending balance B_{12} = **$437,985.91**.

25.3.2 Evaluating Plant Automation NPV, IRR, and Valve Margin in the Solver Workspaces

Next, the CFO opens the [→] [CF] (**NPV: V = C×(1-(1+R/100)^-N)/(R/100) - I**) Solver Workspace to evaluate the 4-year, $42,000/yr automation cash-flow stream at the R = 10.0% hurdle rate net of the I = $120,000 outlay, solves backward via Secant–Brent root-finding for the multi-period **Internal Rate of Return** (IRR) where $V = 0$ and the break-even annual savings C, and computes both Gross Margin ([←] [MGN]) and Markup on Cost ([→] [MKP]) for the $72.00 / $120.00 valve assembly:

Keystroke	Solver Display / Register X	What Happens in the [→] [CF], [←] [MGN], and [→] [MKP] Workspaces
[C ON] [→] [CF]	**NPV: V = C×(1-(1+R)^-N)/R - I**	Open the [→] [CF] Solver Workspace with softkeys [V], [C ON], [R], [N], [I]
42000 [C ON] **10** [R] **4** [N] **120000** [I] [→] [SOLVE] [V]	**13,134.3515**	Solve forward for Net Present Value V = **+$13,134.35** at R = 10.0% hurdle rate
0 [V] [→] [SOLVE] [R]	**14.9631**	Store break-even $V = 0$ and solve backward via root-finding for IRR = **14.9631%/yr**
10 [R] [→] [SOLVE] [C ON]	**37,856.4965**	Restore R = 10.0% and solve backward for break-even annual savings C = **$37,856.50/yr**
72 [ENTER] **120** [←] [MGN]	**40.0000**	Compute valve Gross Margin $MGN_{\%}$ = (120 - 72)/120 = **40.00%** (opens [←] [MGN] workspace)
72 [ENTER] **120** [→] [MKP]	**66.6667**	Compute valve Markup on Cost $MKP_{\%}$ = (120 - 72)/72 = **66.67%** (opens [→] [MKP] workspace)

25.3.3 Try It on the Stack

Apply the five-key [TVM] solver, discounted cash-flow [→] [CF] ([→] [CF]) workspace, and unit-economics operators to these three corporate finance problems:

1. **Equipment Lease with Residual Buyout Balloon ([TVM]):** A 5-year monthly CNC equipment lease (N = 60 months) finances PV = $180,000 at 7.20% APR with a $30,000 residual purchase obligation at maturity (FV = -30,000). Press [TVM], key **5** [ENTER] **12** [×] [N], **7.2** [ENTER] **12** [÷] [I%], **180000** [PV], and **30000** [+/-] [FV], then press [→] [SOLVE] [PMT]. *(Verified Register X result: PMT = **-$3,177.26/month**.)*

2. **Packaging Line NPV and Multi-Period IRR ([→] [CF] & [→] [ROI]):** In the [→] [CF] ([→] [CF]) Solver Workspace ([C ON] [→] [CF]), evaluate the Net Present Value of a packaging line costing I = $85,000 that saves C = $26,000/yr over N = 5 years at an R = 8.0% discount rate (**26000** [C ON] **8** [R] **5** [N] **85000** [I] [→] [SOLVE] [V]), then store **0** [V] and press [→] [SOLVE] [R] to find the multi-year IRR (and compare against the single-year return on a $50,000 note returning $56,250 via **50000** [ENTER] **56250** [→] [ROI]). *(Verified Register X results: NPV = **+$18,810.46**, 5-yr IRR = **16.09%**, and 1-yr ROI = **12.50%**.)*

3. **Gross Margin ([←] [MGN]), Breakeven Volume ([←] [BEP]), and Wilson EOQ ([→] [EOQ]):** For an industrial pump costing C = $210 and selling for P = $350, evaluate [←] [MGN] and [→] [MKP] (**210** [ENTER] **350** [←] [MGN] **210** [ENTER] **350** [→] [MKP]), open the [←] [BEP] Solver Workspace ([C ON] [←] [BEP]) to find the monthly breakeven volume [Q] for F = $49,000, P = $350, V = $210 (**49000** [F] **350** [P] **210** [V] [→] [SOLVE] [Q]), and compute [→] [EOQ] for annual demand D = 8,000, order cost S = $100, and holding cost H = $4.00 (**8000** [ENTER] **100** [×] **4** [→] [EOQ]). *(Verified Register X results: $MGN_{\%}$ =*

$\boldsymbol{40.00\%}$, $MKP_{\%} = \boldsymbol{66.67\%}$, $Q_{BEP}^{*} = \boldsymbol{350.0\ units}$, and $Q_{EOQ}^{*} = \boldsymbol{632.46\ units}$.)

From corporate mortgage amortization and plant capital budgeting in Chapter 25, we move in Chapter 26 to the fixed-income and derivatives trading desk—pricing a ten-year corporate bond, measuring its duration risk against a fifty-basis-point rate spike, and valuing a six-month European option via Black–Scholes–Merton.

Chapter 26: Hedging the Semiannual Corporate Bond Against a Rate Spike

On the institutional fixed-income desk, a portfolio manager holds a 10-year senior corporate bond with a par redemption value of $F = 100.00$ that pays a 5.00% nominal annual coupon in semiannual installments ($C_{\text{semi}} = 2.50$ every six months across $N = 20$ half-year periods, or $C_{\text{ann}} = 5.00$ across 10 annual periods). Credit spreads and Treasury yields have risen since issuance, pushing the bond's market **Yield to Maturity** to 6.00% ($Y = 3.00\%$ per half-year). Before the Federal Open Market Committee announcement at 2:00 p.m., the manager must price the discount bond and calculate its **Macaulay Duration** and **Modified Duration** to quantify how many dollars the position will lose if yields spike by another +50 basis points ($\Delta y = +0.50\%$).

Across the trading floor on the equity derivatives desk, a strategist is structuring a 6-month ($T = 0.50$ yr) European collar on an industrial index currently trading at a spot price of $S_0 = 100.00$ with an at-the-money strike price of $K = 100.00$, a continuous risk-free interest rate of $r = 5.00\%$ (0.05), and an implied annualized volatility of $\sigma = 20.00\%$ (0.20).

To price both the European **Call Option** and **Put Option**—and determine the exact share delta ($\Delta = \Phi(d_1)$) required to hedge the position dynamically—the strategist evaluates Fischer Black, Myron Scholes, and Robert Merton's partial differential equation solution using the cumulative standard normal distribution.

26.1 Fixed-Income Bond Valuation and Duration Interest-Rate Sensitivity

A fixed-coupon bond is a portfolio of two distinct cash-flow promises: a level annuity of periodic coupon payments C and a single zero-coupon lump-sum redemption of par value F at maturity.

26.1.1 Discounting Coupon Annuities and Par Redemption Across Market Yields (BND)

For a bond with par redemption value $F = 100$, periodic coupon payment C, and N remaining periods to maturity discounted at periodic **Yield to Maturity** ($y = Y/100$), the fair market **Bond Price** (BND, P_{bond}) is the sum of the present value of the coupon stream and the discounted par redemption:

$$P_{\text{bond}} = C\left(\frac{1-(1+y)^{-N}}{y}\right) + F(1+y)^{-N} \qquad \textbf{(Eq. 26.1)}$$

In the United States corporate and Treasury bond markets, a 10-year bond paying a 5.00% annual coupon rate distributes $C_{\text{semi}} = 2.50$ every six months across $N_{\text{semi}} = 20$ semiannual periods at a periodic yield of $Y_{\text{semi}} = Y_{\text{nom}} / 2 = 3.00\%$ ($y = 0.030$), whereas Eurobonds pay $C_{\text{ann}} = 5.00$ once per year across $N_{\text{ann}} = 10$ annual periods at $Y_{\text{ann}} = 6.00\%$ ($y = 0.060$). When the market yield equals the coupon rate ($y = C/F$), $P_{\text{bond}} = 100.00$ (**par**). When $y > C/F$, $P_{\text{bond}} < 100.00$ (**discount**). And when $y < C/F$, $P_{\text{bond}} > 100.00$ (**premium**).

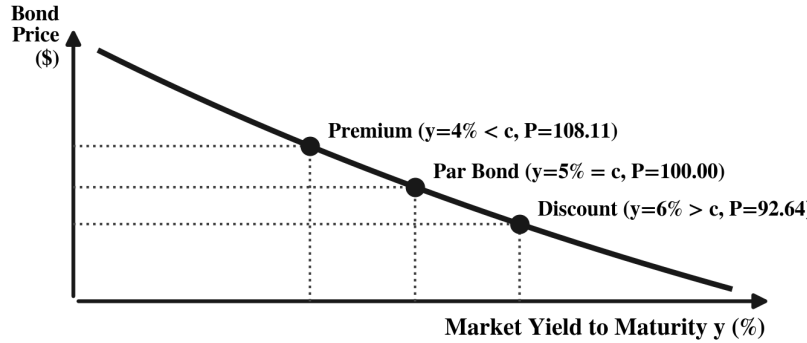


Figure 26.1: *Convex price-yield relationship of a fixed-coupon bond across premium, par, and discount market yields.*

Evaluating the 10-year, 5.00% corporate bond at a 6.00% market yield under exact semiannual compounding ($N = 20$, $C = 2.50$, $Y = 3.00\%$) in the [↗] [BND] workspace gives a coupon annuity present value of $2.50 \times [(1 - 1.03^{-20}) / 0.03] = 2.50 \times 14.877475 = 37.19369$ and a discounted par value of $100(1.03)^{-20} = 55.36758$, summing to a discount price of $P_{\text{semi}} = \mathbf{92.5613\ USD}$. Under annual compounding ($N = 10$, $C = 5.00$, $Y = 6.00\%$), the coupon stream is $5.00 \times 7.360087 = 36.80044$ and discounted par is $100(1.06)^{-10} = 55.83948$, yielding $P_{\text{ann}} = \mathbf{92.6400\ USD}$.

26.1.2 Macaulay Duration, Modified Duration, and Rate-Hike Convexity

In 1938, Frederick Macaulay defined the effective time horizon of a bond—**Macaulay Duration** (D_{Mac})—as the present-value-weighted average arrival time of its cash flows, which admits a closed-form algebraic expression in terms of periodic coupon rate $c = C/F$, periodic yield y, and period count N. Dividing Macaulay Duration (in years) by $(1 + y)$ yields the first-derivative price sensitivity with respect to yield, known as **Modified Duration** (D_{Mod}):

$$D_{\text{Mac, periods}} = \frac{1+y}{y} - \frac{1+y+N(c-y)}{c[(1+y)^N - 1] + y}, \qquad D_{\text{Mod}} = -\frac{1}{P_{\text{bond}}}\frac{dP_{\text{bond}}}{dy_{\text{nom}}} = \frac{D_{\text{Mac, yr}}}{1+y}, \qquad \frac{\Delta P_{\text{bond}}}{P_{\text{bond}}} \approx -D_{\text{Mod}} \cdot \Delta y_{\text{nom}} \qquad \textbf{(Eq. 26.2)}$$

Here D_{Mod} measures the percentage price change per 100 percentage-point (1.00) change in yield. Because the true price-yield curve $P_{\text{bond}}(y)$ is strictly convex ($d^2P / dy^2 > 0$), the linear tangent line $-D_{\text{Mod}} \cdot \Delta y$ slightly overstates price drops when yields rise and slightly understates price gains when yields fall.

For our semiannual 10-year, 5.00% corporate bond yielding 6.00% ($c = 0.025$, $y = 0.030$, $N = 20$ half-years), the closed-form formula gives $D_{\text{Mac,periods}} = (1.03 / 0.03) - [1.03 + 20(-0.005)] / [0.025(1.03^{20} - 1) + 0.03] = 34.3333 - (0.93 / 0.0501528) = \mathbf{15.7900\ half\text{-}years}$, or $D_{\text{Mac,yr}} = 15.7900 / 2 = \mathbf{7.8950\ years}$. Dividing by $(1 + 0.03)$ yields a Modified Duration of $D_{\text{Mod}} = 7.8950 / 1.03 = \mathbf{7.6650\ years}$ (and for the annual-coupon bond, $D_{\text{Mac}} = \mathbf{8.0225\ years}$ and $D_{\text{Mod}} = 8.0225 / 1.06 = \mathbf{7.5684\ years}$). Against a sudden $+50$ bps ($\Delta y = +0.0050$) rate spike to 6.50%, Modified Duration predicts a price decline of $\Delta P / P \approx -7.6650 \times 0.0050 = \mathbf{-3.8325\%}$, or $\Delta P \approx -92.5613 \times 0.038325 = \mathbf{-\$3.5474}$ per \$100 par (to \$89.0139, compared to the exact repriced value of \$89.1085 at $Y_{\text{semi}} = 3.25\%$).

26.2 Black–Scholes–Merton Continuous-Time Option Valuation

While a bond promises deterministic cash flows, an equity option confers the right—without the obligation—to buy (Call) or sell (Put) an underlying asset at a fixed strike price K on expiration date T only if the terminal spot price S_T finishes favorably.

26.2.1 Risk-Neutral Log-Normal Diffusion and the Call Option Formula (pN)

Assuming the underlying asset price S_t follows geometric Brownian motion $dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t$ with annualized volatility σ and continuous risk-free interest rate r, Fischer Black, Myron Scholes, and Robert Merton proved in 1973 that the drift rate μ cancels out of a continuously delta-hedged portfolio. The arbitrage-free price of a European **Call Option** (C_{call}) on a non-dividend-paying asset of spot price S_0 with strike price K and time to expiration T (in years) is governed by the standard normal cumulative distribution function $\Phi(z) = \text{pnorm}(z)$ (pN):

$$C_{\text{call}} = S_0\,\Phi(d_1) - K e^{-rT}\Phi(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r + \tfrac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \qquad \textbf{(Eq. 26.3)}$$

In this PDE solution, $\Phi(d_1)$ is the option's **Delta** ($\Delta_{\text{call}} = \partial C_{\text{call}} / \partial S_0$, the number of underlying shares held per option to remain delta-neutral), $K\, e^{-rT}$ is the continuously discounted present value of the strike price, and $\Phi(d_2)$ is the risk-neutral probability that the option expires in the money ($S_T > K$).

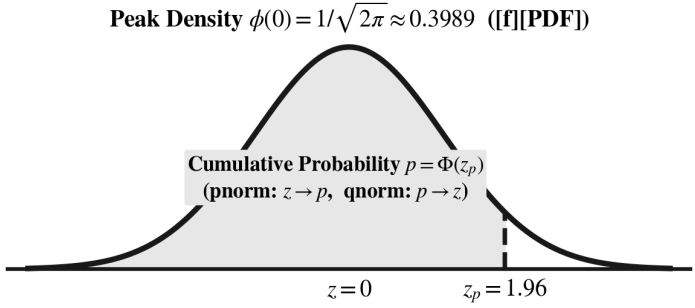


Figure 26.2: *Standard normal density phi(z) illustrating cumulative probability integration (pnorm) and inverse quantile lookup (qnorm).*

For our 6-month ($T = 0.50$ yr) European call option with $S_0 = 100.00$, $K = 100.00$, $r = 0.05$, and $\sigma = 0.20$, the volatility-time product is $\sigma\sqrt{T} = 0.20\sqrt{(0.50)} = \mathbf{0.141421}$. Because the option is at the money ($\ln(100/100) = 0$), the numerator of d_1 is $(0.05 + 1/2(0.04))(0.50) = 0.070 \times 0.50 = 0.0350$, giving $d_1 = 0.0350 / 0.141421 = +\mathbf{0.247487}$ and $d_2 = 0.247487 - 0.141421 = +\mathbf{0.106066}$. Integrating the standard normal curve at d_1 and d_2 via pN yields $\Phi(d_1) = \mathbf{0.597734}$ ($\Delta_{\text{call}} \approx 59.77$ shares per 100-share contract) and $\Phi(d_2) = \mathbf{0.542235}$. With discounted strike $K\, e^{-rT} = 100\, e^{-0.025} = \mathbf{97.53099}$, the fair European call premium is $C_{\text{call}} = 100(0.597734) - 97.53099(0.542235) = 59.7734 - 52.8847 = \mathbf{\$6.8887 \approx \$6.89}$.

26.2.2 Put–Call Parity and European Put Option Valuation

To prevent static arbitrage between a protective put portfolio ($S_0 + P_{\text{put}}$) and a fiduciary call portfolio ($C_{\text{call}} + K\,e^{-rT}$)—both of which deliver an identical payoff of $\max(S_T, K)$ at expiration T—the European **Put Option** (P_{put}) must satisfy Hans Stoll's (1969) **Put–Call Parity** identity:

$$P_{\text{put}} = C_{\text{call}} - S_0 + Ke^{-rT} = Ke^{-rT}\Phi(-d_2) - S_0\,\Phi(-d_1) \qquad \textbf{(Eq. 26.4)}$$

Here $\Phi(-d_1) = 1 - \Phi(d_1)$ gives the magnitude of the put option's negative delta ($\Delta_{\text{put}} = \Phi(d_1) - 1 = -0.402266$).

Applying Put–Call Parity to our 6-month at-the-money option ($C_{\text{call}} = \$6.8887$, $S_0 = \$100.00$, $K\,e^{-rT} = \$97.5310$) yields a European put price of $P_{\text{put}} = 6.8887 - 100.00 + 97.5310 = \mathbf{\$4.4197 \approx \$4.42}$. Notice why an at-the-money call ($S_0 = K = 100$) trades \$2.47 higher than the corresponding put: positive risk-free interest rates ($r = 5\%$) reduce the present value of the strike price paid by the call holder by $100 - 97.5310 = \$2.4690$.

26.3 Bond Duration and Black–Scholes Option Pricing on the Stack and in the Solver Workspace

The **SC-95** combines the fixed-income **Bidirectional** [→] [BND] **Solver Workspace** ([→] [BND]: $\mathbf{P = C\times(1\text{-}(1+Y/100)^{-N})/(Y/100) + F\times(1+Y/100)^{-N}}$, softkeys [P], [C ON], [Y], [N], [F]), calendar day-count operators ([←] [→YR], [→] [→YMD], [←] [DT+], [→] [ΔD]), and direct Row 2 normal distribution keys ([→] [pN], [←] [qN], [rN]) to solve forward for clean bond price P, solve backward via Secant–Brent / Newton–Raphson root-finding for periodic Yield to Maturity Y, and price European options on the RPN stack:

Key / Solver	Operation	Stack Inputs / Softkeys	Governing Expression & Solver Schema
[→] [BND]	Bond Price & YTM Solver	Stack: $Y = C$, $X = P$ • Solver: [P] [C ON] [Y] [N] [F]	Stack: Current yield $(C/P)\times 100$; Solver: $P = C\,a_{\overline{N}\vert\,y} + F(1+y)^{-N}$
[→] [pN] / [←] [qN]	Normal CDF & Quantile	Register X: z-score ([→] [pN]) or area p ([←] [qN])	$\Phi(z) = \text{pnorm}(X)$ ([→] [pN] workspace: [P] [X] [M] [S])
[→] [EAR]	Effective Annual Rate	Register Y: $\text{APR}_{\%}$, Register X: Periods M	$\text{EAR}_{\%} = \left[(1 + Y/(100X))^{X} - 1\right]\times 100$ ([E] [A] [M])
[←] [→YR] / [→] [→YMD] / [→] [ΔD]	Calendar & Day-Count Accrual	Y = Start Date, X = End Date	Converts day counts and ISO dates for accrued bond interest

26.3.1 Pricing the Corporate Bond and Solving Yield to Maturity in the [→] [BND] Solver Workspace

On the fixed-income desk, the portfolio manager opens the [→] [BND] ($\mathbf{P = C\times f(Y,N) + F(1+Y)^{-N}}$) Solver Workspace ([C ON] [→] [BND]) to price the 10-year, 5.0% corporate bond under both annual compounding ($C = 5.0$, $Y = 6.0\%$, $N = 10$, $F = 100$) and exact semiannual compounding ($C = 2.50$, $Y = 3.00\%$, $N = 20$, $F = 100$), reprices the bond at a +50 bps (6.50% nominal, $Y = 3.25\%$ semiannual) yield spike, and solves backward via [→] [SOLVE] → [Y] to verify the implied Yield to Maturity from the screen price:

Keystroke	Solver Display / Register X	What Happens in the [→] [BND] Solver Workspace
[C ON] [→] [BND]	**BND: P = C×f(Y,N)+F(1+Y)⁻ᴺ**	Open the [→] [BND] Solver Workspace with softkeys [P], [C ON], [Y], [N], [F]
5 [C ON] **6** [Y] **10** [N] **100** [F] [→] [SOLVE] [P]	**92.6399**	Price 10-yr, 5% annual-coupon bond at 6.0% yield: P_{ann} = **$92.6400**
2.5 [C ON] **3** [Y] **20** [N] **100** [F] [→] [SOLVE] [P]	**92.5613**	Price exact semiannual bond (C=2.5, Y=3%, N=20): P_{semi} = **$92.5613**
3.25 [Y] [→] [SOLVE] [P]	**89.1085**	Reprice semiannual bond after +50 bps hike (Y = 3.25%): P_{exact} = **$89.1085**
92.5613 [P] [→] [SOLVE] [Y] **2** [×]	**6.0000**	Solve backward via root-finding for semiannual Y = 3.0000% and double for **6.0000%** nominal YTM

Comparing the linear Modified Duration estimate (92.5613 × (1 - 0.038325) = **$89.0139**) against the exact repriced bond (P_{exact} = **$89.1085**) displays the +$0.0946 convexity cushion earned by the bondholder during the +50 bps rate shock.

26.3.2 Valuing the Black–Scholes Call and Put Options on the Stack

Switching to the equity derivatives desk (S_0 = 100, K = 100, r = 0.05, σ = 0.20, T = 0.50 yr, so $\sigma\sqrt{T}$ = 0.141421 and d_1 = 0.247487), the strategist presses [→] [pN] (on Row 2 Col 2) to integrate $\Phi(d_1)$ and $\Phi(d_2)$ and evaluate both the European Call and Put prices:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[C ON] **0.247487** [→] [pN] **100** [×]	0	0	0	**59.7734**	Integrate $\Phi(d_1)$ = 0.597734 ([→] [pN]) and scale by S_0 = 100: $S_0\Phi(d_1)$ = $59.7734
0.247487 [ENTER] **0.141421** [-] [→] [pN]	0	0	59.7734	**0.5422**	Compute d_2 = 0.106066 and integrate $\Phi(d_2)$ = 0.542235 ([→] [pN])
0.025 [+/−] [e^x] **100** [×]	59.7734	59.7734	0.5422	**97.5310**	Compute continuously discounted strike $K\,e^{-rT} = 100\,e^{-0.025}$ = $97.5310
[×] [-]	59.7734	59.7734	59.7734	**6.8887**	Multiply $K\,e^{-rT}\Phi(d_2)$ = $52.8847 and subtract from $S_0\Phi(d_1)$: C_{call} = $6.89
100 [-] **97.5310** [+]	59.7734	59.7734	59.7734	**4.4197**	Apply Put–Call Parity $P_{put} = C_{call} - S_0 + K\,e^{-rT}$ = $4.42

In five stack steps, both the call premium ($6.89) and put premium ($4.42) are resolved to four decimal places.

26.3.3 Try It on the Stack

Test your fixed-income and option-pricing stack fluency across these three trading-desk scenarios:

1. **Premium Bond Valuation and Backward YTM Verification ([→] [BND]):** A 10-year corporate bond (F = 100) carries a 7.0% annual coupon and trades in a market where yields have fallen to y = 5.0%. In the [→] [BND] Solver Workspace ([C ON] [→] [BND]), price the premium bond under annual compounding (**7** [C ON] **5** [Y] **10** [N] **100** [F] [→] [SOLVE] [P]) and semiannual compounding (**3.5** [C ON] **2.5** [Y] **20** [N] **100** [F] [→] [SOLVE] [P]), then store **115.5892** [P] and press [→] [SOLVE] [Y] to verify the semiannual yield. *(Verified Register X results: P_{ann} = **$115.4435**, P_{semi} = **$115.5892**, and Y = **2.5000%**.)*

2. **Effective Annual Rate of Daily vs. Monthly Compounding ([→] [EAR]):** Convert a nominal APR = 9.0% into its Effective Annual Rate when compounded monthly ($M = 12$: **9** [ENTER] **12** [→] [EAR]) and when compounded daily ($M = 365$: **9** [ENTER] **365** [→] [EAR]). *(Verified Register X results: EAR_{12} = **9.3807%** and EAR_{365} = **9.4162%**.)*

3. **One-Year Out-of-the-Money Black–Scholes Call ([→] [pN]):** For a 1-year ($T = 1.0$) European call with $S_0 = 100$, $K = 110$, $r = 0.05$, and $\sigma = 0.25$, verify that d_1 = -0.09724, d_2 = -0.34724, and $K\,e^{-rT}$ = 104.6353, then evaluate $C_{\text{call}} = 100\Phi(d_1)$ - $104.6353\Phi(d_2)$ (**0.09724** [+/-] [→] [pN] **100** [×] **0.34724** [+/-] [→] [pN] **104.6353** [×] [-]). *(Verified Register X result: C_{call} = **$8.0263**.)*

From bond convexity and Black–Scholes option hedging in Chapter 26, we return to the factory floor in Chapter 27—applying linear regression, normal tail integrals, and Six Sigma process capability indices to catch a drifting grinding spindle before scrap occurs.

Chapter 27: Catching the Drifting Grinding Spindle Before Scrap Happens

On a high-volume automotive powertrain line, a cylindrical cubic-boron-nitride grinding machine finishes hardened steel transmission output shafts to a blueprint nominal diameter of 10.000 mm with bilateral specification limits of LSL = 9.950 mm and USL = 10.050 mm (a total engineering tolerance window of USL - LSL = 0.100 mm, or 100 μm). Under short-term steady-state operation, the grinder exhibits a natural within-lot standard deviation of σ = 0.010 mm (10 μm). When the process mean is centered at μ₀ = 10.000 mm, each specification limit lies five standard deviations (± 5.0σ) away from the center line.

By 1:00 p.m., however, gradual thermal expansion of the hydrostatic wheelhead bearing and microscopic dressing-diamond wear have caused the process mean across the latest five hourly inspection sub-lots (t = 1, 2, 3, 4, 5 hr, with sub-lot means 9.990, 10.000, 10.010, 10.020, 10.030 mm) to average μ = 10.010 mm—a +0.010 mm (+1.0σ) upward drift toward the Upper Specification Limit USL = 10.050 mm. Every single shaft inspected by the operator still measures inside the [9.950, 10.050] mm blueprint tolerance band, so a naive go/no-go snap gauge reports zero defects.

To catch the drifting spindle *before* the wheelhead crosses the scrap threshold, the quality engineer uses statistical process control on the stack. She accumulates the hourly inspection series, extracts the sample mean, standard deviation, and linear regression drift rate, compares potential and centered capability, sizes the inspection sample, and integrates the upper Gaussian tail to quantify expected **Defects Per Million Opportunities** (DPMO).

27.1 Sample Regression, Sampling Distributions, and Statistical Inference

Statistical process control begins by summarizing empirical measurements into sufficient sample statistics and modeling how a process parameter drifts over time or varies across finite samples.

27.1.1 Sample Mean, Dispersion, and Ordinary Least-Squares Linear Regression (Σ+, x̄,ȳ, →, s,σ, →, L.R.)

Given n paired observations (x_i, y_i) accumulated via the statistical summation operator (Σ+, with relative share ↖ %Σ)—where x_i represents elapsed production hours and y_i represents measured shaft diameter—the sample means ($\bar{x}$, $\bar{y}$) (x̄,ȳ) and Bessel-corrected sample standard deviations (s_x, s_y) (→ s,σ, with n - 1 degrees of freedom in the denominator) pair with the sample covariance s_{xy} to determine the ordinary least-squares **Linear Regression** (→ L.R., binding softkeys ŷ, x̂, m, b, r) slope m, intercept b, and Pearson correlation coefficient r:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s_y = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}, \qquad m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad \hat{y}(x) = b + mx \qquad \textbf{(Eq. 27.1)}$$

Here $b = \bar{y} - m\bar{x}$ is the baseline dimension at $x = 0$, m is the thermal drift velocity (in mm/hr), and $r = m(s_x / s_y) \in [-1, +1]$ measures linear trend strength.

Across the five hourly grinding sub-lots (1, 9.990), (2, 10.000), (3, 10.010), (4, 10.020), and (5, 10.030), the mean time is $\bar{x} = 3.0$ hr and the overall sample mean diameter is $\bar{y} = \mu = \mathbf{10.010\ mm}$ (with sub-lot dispersion $s_y = \mathbf{0.01581\ mm}$ across the five hours and within-hour natural machine standard deviation $\sigma = \mathbf{0.010\ mm}$). Least-squares regression ([→] [L.R.]) extracts a drift slope of $m = +\mathbf{0.010\ mm/hr}$ from an initial intercept of $b = 9.980$ mm ($r = \mathbf{1.0000}$), predicting via [ŷ] that by Hour 6 the spindle mean will reach $\hat{y}(6) = 9.980 + 0.010(6) = \mathbf{10.040\ mm}$—only 1.0σ below USL.

27.1.2 Normal Tail Integrals, Student's t Intervals, and Hypothesis Testing

When a continuous dimension $X \sim N(\mu, \sigma^2)$ is standardized via $z = (x - \mu)/\sigma$, the lower-tail probability $P(X \le x) = \Phi(z)$ is given by the **Standard Normal Cumulative Integral** ([→] [pN]), inverted by [←] [qN]), while the upper-tail exceedance probability above a threshold x is $P(X > x) = 1 - \Phi(z) = \Phi(-z)$:

$$z = \frac{x-\mu}{\sigma}, \qquad \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\,du, \qquad E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}, \qquad n^{*} = \left(\frac{z_{\alpha/2} \cdot s}{E}\right)^2, \qquad Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \textbf{(Eq. 27.2)}$$

In these inference equations, E ([→] [CI]) is the confidence interval margin of error around $\bar{x}$, n^{*} ([←] [n*]) is the required sample size to achieve a target margin of error E at critical quantile $z_{\alpha/2}$ (1.96 for 95% confidence), and Z ([←] [z₀] / [←] [z₀], or two-sample [←] [t₂] / [←] [t₂] and proportion test [→] [A/B] / [→] [A/B], alongside William Sealy Gosset's Student's t-distribution [→] [pt]/[←] [qt]/[rt], chi-square [→] [pχ²]/[←] [qχ²]/[rχ²], Snedecor's F [→] [pF]/[←] [qF]/[rF], and binomial [→] [pB]/[←] [qB]/[rB]) tests whether the observed shift $\bar{x} - \mu_0$ is statistically distinguishable from random sampling noise.

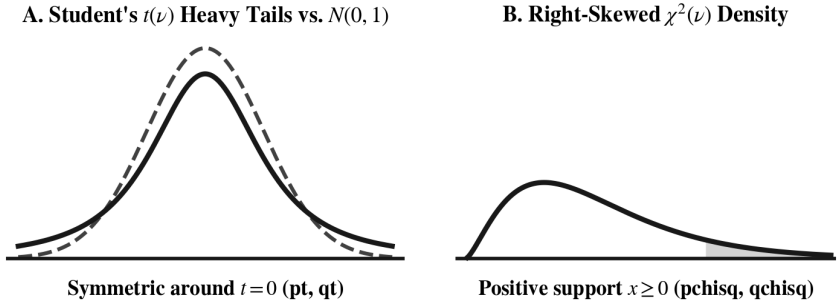


Figure 27.1: *Comparison of heavy-tailed Student's t sampling density against N(0,1) and right-skewed Chi-Square critical region.*

If the quality engineer inspects a sample of $n = 25$ shafts ($\sqrt{n} = 5$) at 1:00 p.m. and observes $\bar{x} = 10.010$ mm against the blueprint target $\mu_0 = 10.000$ mm with $\sigma = 0.010$ mm, the standard error of the mean is $\sigma / \sqrt{n} = 0.010 / 5 = 0.0020$ mm. The 95% margin of error ([→] [CI]) is $E = 1.96 \times 0.0020 = \mathbf{0.00392\ mm}$, and the one-sample test statistic ([←] [z₀]) is $Z = (10.010 - 10.000) / 0.0020 = +\mathbf{5.0000}$—confirming at better than 99.9999% confidence that an assignable mechanical drift has occurred.

27.2 Six Sigma Process Capability and Defect Rate Prediction

Walter Shewhart's control charts ([↗] [SPC]) track process stability over time; Motorola's **Six Sigma Capability Indices** ([↗] [Cpk]) compare the natural width and centering of that process against the customer's engineering blueprint limits (LSL and USL).

27.2.1 Process Potential Versus Centered Process Capability ([↗] [Cpk])

For a manufacturing process with mean μ and short-term standard deviation σ operating between **Lower Specification Limit** LSL and **Upper Specification Limit** USL, **Process Potential** (C_p) divides the blueprint tolerance width (USL - LSL) by the natural 6σ process spread (± 3σ). Because C_p ignores where the mean μ is located, **Centered Process Capability** (C_{pk}) evaluates the upper capability C_{pu} and lower capability C_{pl} separately and takes the minimum—measuring the distance from μ to the *nearer* specification wall in units of 3σ:

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}, \qquad C_{pu} = \frac{\text{USL} - \mu}{3\sigma}, \qquad C_{pl} = \frac{\mu - \text{LSL}}{3\sigma}, \qquad C_{pk} = \min(C_{pu},\, C_{pl}) = C_p(1 - k_{\text{shift}}) \qquad \textbf{(Eq. 27.3)}$$

Here $k_{\text{shift}} = |\mu - T| \,/\, [(\text{USL} - \text{LSL})/2]$ is the fractional off-center shift relative to the half-tolerance (USL - LSL)/2, and the process Sigma Quality Level is $Z_{\min} = 3 \cdot C_{pk}$.

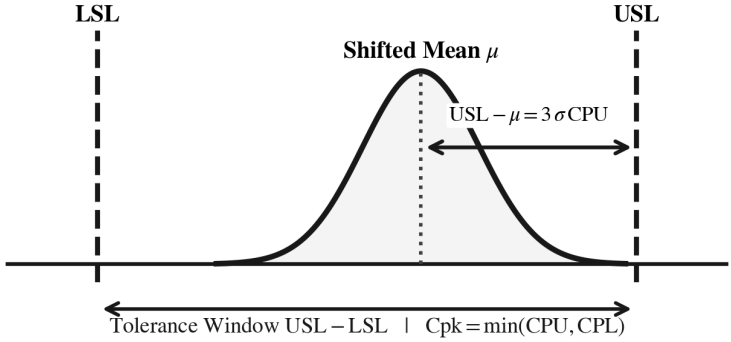


Figure 27.2: *Six Sigma process capability geometry (Cp and Cpk) comparing a shifted 6-sigma process bell curve against LSL and USL specification limits.*

Evaluating our automotive shaft grinder (LSL = 9.950 mm, USL = 10.050 mm, σ = 0.010 mm) via [↗] [Cpk] reveals a Process Potential of C_p = (10.050 - 9.950) / (6 × 0.010) = 0.100 / 0.060 = **1.6667** (5.0σ potential). Once the mean drifts by +0.010 mm to μ = 10.010 mm, however, the lower margin expands to C_{pl} = (10.010 - 9.950) / 0.030 = **2.0000** (6.0σ), while the upper margin shrinks to C_{pu} = (10.050 - 10.010) / 0.030 = 0.040 / 0.030 = **1.3333** (4.0σ). Consequently, Centered Process Capability drops from 1.6667 to C_{pk} = min(1.3333, 2.0000) = **1.3333**—landing right on the 4.0σ automotive quality action limit and triggering a tool-offset compensation before capability erodes further.

27.2.2 Gaussian Tail Integration and Defects Per Million Opportunities

To translate C_p and C_{pk} into expected scrap volume in **Defects Per Million Opportunities** (DPMO), the engineer integrates both normal tails outside [LSL, USL] via [→] [pN] and multiplies by 1,000,000:

$$z_{USL} = \frac{USL - \mu}{\sigma} = 3 \cdot C_{pu}, \qquad z_{LSL} = \frac{LSL - \mu}{\sigma} = -3 \cdot C_{pl}, \qquad DPMO = 10^6 \times [\Phi(z_{LSL}) + \Phi(-z_{USL})] \qquad \textbf{(Eq. 27.4)}$$

In Six Sigma quality reporting, a process is audited both at its current short-term mean μ (unshifted DPMO) and under Motorola's standard 1.5σ long-term drift allowance ($Z_{LT} = Z_{ST}$ - 1.5).

At the grinder's drifted mean of μ = 10.010 mm, the upper specification limit lies at z_{USL} = (10.050 - 10.010) / 0.010 = **+4.00σ** and the lower limit lies at z_{LSL} = (9.950 - 10.010) / 0.010 = **-6.00σ**. Integrating the upper normal tail at +4.00σ via **4** [+/-] [→] [pN] gives Φ(-4.00) = 1 - 0.999968329 = **3.1671 × 10⁻⁵** (while the lower tail at -6.00σ is 9.87 × 10⁻¹⁰). Multiplying by 10^6 yields an instantaneous defect rate of DPMO = **31.67 DPMO** (compared to 0.57 DPMO when centered at μ = 10.000 mm, a **55-fold increase in scrap risk** caused by a 10 μm drift).

27.3 Statistical Process Control and Six Sigma Capability on the Stack and in the Solver Workspace

The **SC-95** unites the dedicated faceplate R-style statistical distribution keys ([→] [pN], [←] [qN], [rN], [→] [pt], [←] [qt], [rt], [→] [pχ²], [←] [qχ²], [rχ²], [→] [pF], [←] [qF], [rF], [→] [pβ], [←] [qβ], [rβ], plus [←] [PDF]), bivariate regression accumulators ([Σ+], [x̄,ȳ], [→] [ŝ,σ̂], [→] [L.R.], [←] [%Σ]), and the Six Sigma capability operator ([→] [Cpk]) on the RPN stack with the **Bidirectional Equation Solver Workspaces** ([→] [SOLVE] / [XEQ]) for [→] [Cpk] (**C = (U - L) / (6×S)**, softkeys [C ON], [U], [L], [S]), [→] [CI] (**E = Z × S / √N**), [←] [z₀] (**Z = (X - M) / (S / √N)**), [←] [t₂] (**T = (X - Y) / (S × √(2/N))**), and [←] [n*] (**N = (Z × S / E)²**, softkeys [N], [Z], [S], [E]):

Key / Solver	Operation	Stack Inputs (*Z*, *Y*, *X*)	Governing Expression & Solver Softkeys ([LFU0]–[LFU4])
[Σ+], [x̄,ȳ], [→] [ŝ,σ̂], [→] [L.R.]	Sample Stats & Regression	$Y = y_i$, $X = x_i$	[→] [L.R.] softkeys: [ŷ] [x̂] [m] [b] [r]
[→] [CI] / [←] [z₀]	Margin of Error & *Z*-Test	[→] [CI]: $Y = s$, $X = n$ • [←] [z₀]: $T = \bar{x}$, $Z = \mu_0$, $Y = \sigma$, $X = n$	[→] [CI]: **E = Z×S/√N** ([E] [Z] [S] [N]); [←] [z₀]: **Z = (X-M)/(S/√N)** ([Z] [X] [m] [S] [N])
[←] [n*]	Audit Sample Size ([←] [n*])	Stack: $Y = s$, $X = E$	Stack: (1.96 *Y* / *X*)²; Solver: **N = (Z × S / E)²** ([N] [Z] [S] [E])
[→] [pN] / [←] [qN] / [rN]	Normal CDF, Quantile & RNG	Register *X*: *z*-score ([→] [pN]) or area *p* ([←] [qN])	$\Phi(z)$ = pnorm(*X*) and $\Phi^{-1}(p)$ = qnorm(*X*) ([→] [pN]: [P] [X] [m] [S])
[→] [pt] [←] [qt] [→] [pχ²] [←] [qχ²] [→] [pF] [←] [qF] [→] [pβ] [←] [qβ]	Student's *t*, χ^2, *F*, and Binomial	Y = stat or *p*, $X = \nu$ (or Z=*k*, Y=*n*, X=*p*)	Exact cumulative tail probabilities, inverse quantiles, and [→] [pt]/[→] [pχ²]/[→] [pF]/[→] [pβ] solvers
[→] [Cpk] / [→] [SPC]	Six Sigma Capability & Control Limit	[→] [Cpk]: Y = USL-LSL, $X = \sigma$ • [→] [SPC]: $Y = \mu$, $X = \sigma$	Stack: $C_p = Y / 6\ X$; Solver: **C = (U - L) / (6×S)** ([C ON] [U] [L] [S]); [→] [SPC]: μ + 3σ

27.3.1 Auditing Spindle Drift Significance and Process Capability on the Stack

At the grinding cell, the quality engineer first verifies the 95% confidence interval Margin of Error ([→] [CI]) and one-sample Z-test statistic ([←] [z_0]) for the $n = 25$ shaft sample ($\bar{x} = 10.010$ mm, $\mu_0 = 10.000$ mm, $\sigma = 0.010$ mm), then evaluates Process Potential C_p ([→] [Cpk]) and Centered Process Capability C_{pk} across LSL = 9.950 mm, USL = 10.050 mm, and $\mu = 10.010$ mm:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.010 [ENTER] 25 [→] [CI]	0	0	0	**0.0039**	Compute 95% Margin of Error $E = 1.96 \times 0.010 / \sqrt{25} = 0.00392$ mm ([→] [CI])
10.01 [ENTER] 10.00 [ENTER] 0.010 [ENTER] 25 [←] [z_0]	10.01	10.01	10.01	**5.0000**	Compute drift test statistic $Z = (10.01 - 10.00) / (0.010/\sqrt{25}) = +5.0000$ ([←] [z_0])
10.05 [ENTER] 9.95 [-] 0.010 [→] [Cpk]	10.01	10.01	5.0000	**1.6667**	Evaluate centered Process Potential $C_p = 0.100 / (6 \times 0.010) = 1.6667$ ([→] [Cpk])
10.05 [ENTER] 10.01 [-] 0.03 [÷]	5	5.0000	1.6667	**1.3333**	Evaluate drifted Centered Capability $C_{pk} = (10.05 - 10.01)/0.03 = 1.3333$

Comparing Register Y ($C_p = 1.6667$) and Register X ($C_{pk} = 1.3333$) quantifies the exact capability loss (1.6667 - 1.3333 = 0.3333) caused by the +10 μm thermal wheelhead shift.

27.3.2 Sizing Audit Samples Forward and Backward in the [←] [n] Solver Workspace

Suppose the automotive customer tightens the quality audit protocol, requiring the 95% confidence interval margin of error around the hourly sample mean not to exceed $E = \pm 0.0025$ mm (2.5 μm) at $Z = 1.96$ and $\sigma = 0.010$ mm. Opening the [←] [n*] (**NSIZE: N = (Z × S / E)²**) Solver Workspace ([C ON] [←] [n*]) solves forward via [→] [SOLVE] → [n] for the required sample size N, and if metrology cycle time caps the hourly sample at $N = 25$ shafts, solving backward on [E] reveals the tightest margin of error achievable:

Keystroke	Solver Display / Register X	What Happens in the [←] [n*] Solver Workspace
[C ON] [←] [n*]	**NSIZE: N = (Z × S / E)²**	Open the Sample Size ([←] [n*]) Solver Workspace with softkeys [n], [z], [s], [E]
1.96 [z] 0.010 [s] 0.0025 [E]	0.0025	Store 95% critical quantile $Z = 1.96$, std dev $S = 0.010$ mm, target $E = 0.0025$ mm
[→] [SOLVE] [n]	**61.4656**	Solve forward for required sample size $N = (1.96 \times 0.010 / 0.0025)^2 =$ **61.47 → 62 shafts**
25 [n] [→] [SOLVE] [E]	**0.00392**	Store budget cap $N = 25$ and solve backward for achievable margin $E =$ **0.00392 mm**

27.3.3 Computing Gaussian Tail Probability and DPMO on the Stack

Continuing with $C_{pk} = 1.3333$ on the stack, the engineer multiplies by 3 to obtain the upper specification z-score ($z_{USL} = +4.00\sigma$), integrates the upper Gaussian tail via **4** [+/-] [→] [pN], scales by 10^6 to obtain DPMO, and compares the result against a 1.5σ-shifted process ($z = +2.50\sigma$):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1.333333 ENTER **3** × +/− [→] pN	0	0	0	**0.0000317**	Convert C_{pk}=1.3333 to -z_{USL} = -4.00 and integrate tail $\Phi(-4.00) = 3.167 \times 10^{-5}$ ([→] pN)
1000000 ×	0	0	0	**31.6712**	Multiply tail probability by 10^6: unshifted DPMO = 31.67 ppm
2.5 +/− [→] pN **1000000** ×	0	0	31.6712	**6209.67**	Compare 4σ process after 1.5σ shift ($z = 2.5\sigma$): $\text{DPMO}_{1.5\sigma}$ = 6,209.7 ppm

Seeing DPMO jump from 31.67 ppm in Register Y to 6,209.7 ppm in Register X if the spindle drifts another 1.5σ explains why Six Sigma control loops re-center the grinding wheel as soon as C_{pk} touches 1.33.

27.3.4 Try It on the Stack

Master statistical inference, non-normal sampling distributions, and Six Sigma capability on the stack with these three quality-engineering exercises:

1. **Aerospace Fastener Six Sigma Audit ([→] Cpk & [→] pN):** A titanium bolt thread-rolling cell operates with blueprint tolerance width USL - LSL = 0.072 mm and standard deviation σ = 0.006 mm. Compute [→] Cpk (**0.072** ENTER **0.006** [→] Cpk), and find the centered two-tailed defect rate in DPMO at ± 6.0σ (**6** +/− [→] pN **2** × **1000000** ×). *(Verified Register X results: C_p = **2.0000** and $DPMO_{centered}$ = **0.00197 ppm**.)*

2. **Small-Sample Student's *t* Quantile and Test Statistic ([←] z₀ & [←] qt):** A laboratory tests $n = 16$ concrete cylinders ($\nu = 15$ degrees of freedom) with sample mean $\bar{x} = 42.10$ MPa against target $\mu_0 = 40.00$ MPa ($\bar{x}$ - μ_0 = 2.10 MPa) and sample standard deviation $s = 3.50$ MPa. Compute the one-sample test statistic (**42.10** ENTER **40.00** ENTER **3.50** ENTER **16** [←] z₀) and compare it against the 95% two-tailed Student's t critical quantile $t_{0.975,\ 15}$ (**0.975** ENTER **15** [←] qt). *(Verified Register X results: t = **2.4000** and $t_{0.975,\ 15}$ = **2.1314**, rejecting H_0.)*

3. **Chi-Square Variance Threshold ([←] qx²) and Binomial Lot Acceptance ([→] pB):** Find the 95% chi-square critical value for $\nu = 10$ degrees of freedom (**0.95** ENTER **10** [←] qx²). Then open the [→] pB ([→] pB) Solver Workspace ([SOLVE] [→] pB) for a binomial lot sample of $N = 20$ boards with defect rate $Q = 0.05$ and find the cumulative probability P of observing $K \leq 2$ defects (**2** K **20** n **0.05** Q [→] SOLVE P). *(Verified Register X results: $\chi^2_{0.95,\ 10}$ = **18.3070** and $P(X \leq 2)$ = **0.9245**.)*

This concludes our twenty-seven-chapter journey across the nine instruments of **Volume III: The Stack in the Field**. For a unified mathematical and operational reference across all nine calculators (**SC-35** through **SC-95**), consult **Appendix A: Cross-Domain Formula & Equation Compendium** and **Appendix B: Domain Instrument Keycap & Workspace Directory**.

Appendix A: Cross-Domain Formula & Equation Compendium

This compendium consolidates the governing mathematical identities, physical constants, four-register RPN stack conventions, and bidirectional Equation Solver Workspace schemas across all nine specialized instruments in the StackCalc Domain Series (**SC-35**, **SC-14**, **SC-16**, **SC-24**, **SC-50**, **SC-60**, **SC-78**, **SC-80**, and **SC-95**). Each section summarizes the first-principles equations derived across Units I through IX (Chapters 1 through 27) so engineers, navigators, clinicians, and analysts can verify input dimensions, immediate stack arguments, and closed-form or numerical solver relationships at a glance.

A.1 Universal Three-Pillar Stack, Precision, and Bidirectional Solver Conventions

Every instrument in the StackCalc Domain Series shares a common **three-pillar computational architecture** backed by IEEE 754 double-precision floating-point arithmetic (64-bit word, 53-bit binary significand, providing 15 to 17 significant decimal digits) and selectable fixed-width integer arithmetic on the **SC-16**:

1. **Configurable RPN Stack Depth ([4-LV], [8-LV], or [INF] via [→] [FLAGS]):** Practitioners can operate with a classic four-register (X, Y, Z, T) operational stack ([4-LV]), expand to an eight-register stack ([8-LV]), or enable unbounded stack capacity ([INF]) for multi-branch field derivations. Across all stack configurations, the active four-register window at the bottom of the stack (**Register T**, **Register Z**, **Register Y**, **Register X**) obeys deterministic RPN lift and drop semantics:

Register	Operational Role	Behavior on Binary Operation ([+], [−], [×], [÷], [y^x])	Behavior on [ENTER]
Register *T*	Top / 4th Visible Level	Replicates downward into Z ($T' = T$) in [4-LV] mode (or drops from **R5** in [8-LV]/[INF])	Receives prior Z ($T' = Z$)
Register *Z*	3rd Stack Level	Drops into Y ($Z = T$)	Receives prior Y ($Z = Y$)
Register *Y*	2nd Operand Level	Consumed as left operand ($Y \circ X$); receives prior Z	Replicates X ($Y' = X$)
Register *X*	Active Display Line	Consumed as right operand; receives result $f(Y, X)$	Remains in X (lift disabled for next digit)

2. **Dual-Mode Domain Keys (Immediate RPN Stack Shortcuts + Bidirectional Equation Solver Workspaces):**

   - **Immediate Stack Mode:** Pressing a domain key when arguments are staged in Registers Z, Y, and X evaluates the forward physical transformation in a single keystroke while automatically injecting embedded physical constants ([→] [C ON], [←] [kB], [→] [r], [←] [F], [→] [g₀], [→] [PW]) and regulatory lookup tables (NEC Table 8 wire gauges, IUPAC atomic masses).
   - **Bidirectional Equation Solver Workspace ([→] [SOLVE] / [XEQ] + Softkeys [LFU0]–[LFU5]):** Opening a domain key's Equation Workspace renders the governing equation across Line 1 of the LCD and binds the six top-row softkeys ([LFU0]–[LFU5]) to its symbolic variables. Storing known values in any order and pressing [→] [SOLVE] (or [XEQ]) followed by the target variable softkey solves the governing law **forward or**

backward for any unknown parameter via exact algebraic isolation or Secant–Brent numerical root-finding.

3. **Domain Modes, Native Input Parsers, and Live Reference Ledgers ([↷] [▦]):** Native domain entry modes ([↷] [▸HMS] via [↶] [▸HR]/[↷] [▸HMS], [↶] [f-i] via [↶] [f-i], [HEX]/[↶] [DEC]/[↶] [OCT]/[↷] [BIN] + [↷] [NSZ] + [↶] [NEG], [TC] + [FPS], chemical formula entry via [Z]) and the non-destructive Reference Table ([↷] [▦]) maintain live domain ledgers (**@COGO**, **@BK%**, **@PAT**/**@MAR**, and bivariate statistical accumulators).

A.1.1 Cross-Domain Mathematical Isomorphisms

Across the nine physical trades, identical mathematical structures recur under domain-specific physical variables:

Mathematical Structure	Governing Law Form	Domain Instantiations Across Units I–IX
First-Order Exponential Decay & Discounting	$y(t) = y_0\, e^{-\lambda\, t}$	**Unit III (SC-16):** RC settling $v_C(t) = V_{in}(1 - e^{-t/\tau})$ ([↶] [τ], [PWM], [↷] [BPS]) • **Unit VI (SC-60):** Beer–Lambert transmittance $I/I_0 = 10^{-\varepsilon\, \ell\, c}$ ([εbc], [εbc], [C₁V₁]) & Arrhenius kinetics $k = A\, e^{-E_a/(RT)}$ ([↷] [Eₐ], [↷] [t½], [↷] [Kₑq], [↶] [ΔG], [↷] [E°]) • **Unit VIII (SC-80):** One-compartment drug elimination $C(t) = C_0\, e^{-k_e\, t}$ ([PK], [GFR], [BSA], [↶] [IBW], [↷] [ABW], [DOS], [DRP], [IV]) • **Unit IX (SC-95):** Multi-period and continuous discounting ([TVM], [↶] [AMR], [↷] [CF], [↷] [BND], [↷] [EAR], [↷] [CGR], [↷] [ROI], [↷] [pN], [↶] [qN])
Logarithmic Ratio Compression	$L = \varkappa \log_{10}(X\ /\ X_{ref})$	**Unit IV (SC-24):** Acoustic sound pressure level SPL = 20 $\log_{10}(p/p_0)$ ([↶] [SPL], [dB], [↶] [SPL], [EV], [DOF], [FOV], [FOV], [TC], [↷] [+TC], [↷] [-TC], [FPS]) • **Unit V (SC-50):** RF power ([dBm], [↶] [→dBm], [dBm]), Friis path loss ([FSL], [↶] [LNK], [↶] [LNK], [↷] [F₁]), and noise cascade ([NF], [↷] [Pₙ], [↶] [IP3], [↷] [FDR]) • **Unit VI (SC-60):** Sørensen acidity pH = -$\log_{10}[H^+]$ ([pH], [↷] [pKₐ], [BUF], [↶] [EQV], [MW], [MW], [↶] [w/w], [PVT], [↷] [RCF])
Shannon Information & Quantization Capacity	$C = B\, \log_2(1 + \mathrm{SNR})$	**Unit III (SC-16):** N-bit ADC dynamic range ([↶] [ADC], [↷] [DAC], [↶] [CRC], [NET], [FLT], [END], [↶] [•B], [↷] [CLZ], [↶] [MSK], [↶] [SB], [↶] [CB], [↷] [B?]) • **Unit V (SC-50):** Shannon–Hartley channel capacity ([↷] [C/B], [↷] [Δf], [↶] [RCS], [↶] [f₀], [↶] [f₀], [↶] [λ/L])
Euclidean & Stoichiometric Decomposition	$r = \sqrt{(x^2 + y^2)}$	**Unit I (SC-35):** Survey inverse ([INV], [TRV], [BRG], [SLP], [↷] [GR%], [↷] [CUT], [↷] [△], [FLW]) & wind triangle ([WND], [↷] [XWC], [WND], [↶] [ALT], [TSD]) • **Unit II (SC-14):** Conduit & rafter geometry ([OFS], [↶] [MIT], [↷] [FIT], [↷] [BC], [↶] [SFM], [↶] [TAP]), three-phase power ([pF], [pF], [pF], [pF], [AWG], [↷] [FLA], [↶] [MTR]), and process hydraulics ([PIP], [↷] [Cᵥ], [PMP], [↷] [NPH], [mA], [↶] [RTD]) • **Unit V (SC-50):** Complex impedance & VSWR ([↶] [CMPLX], [↷] [Z₀], [Γ], [SWR], [RL], [RL], [↷] [Z₀]) • **Unit VII (SC-78):** Culinary & zymurgy scaling ([BK%], [BK%], [ΣW], [↶] [DIV], [Tw], [↷] [ICE], [MSH], [↷] [→SG], [ABV], [IBU], [↷] [SRM], [↷] [TDS], [YLD], [CST], [↶] [PRC], [↷] [Aw], [↷] [SO₂], [↶] [CO₂]) • **Unit IX (SC-95):** Unit economics & Six Sigma ([↶] [MGN], [↷] [MKP], [↶] [BEP], [↷] [EOQ], [↷] [Cpk], [↷] [SPC], [↷] [CI], [↶] [Z₀], [↶] [t₂], [↷] [A/B], [↶] [n*])

A.2 Geomatics, Land Surveying, and Flight Navigation

Covered in **Unit I, Chapters 1–3**, the **SC-35** geomatics and flight-navigation instrument links sexagesimal angular encoding (**DDD.MMSS** via [↶] [▸HR] and [↷] [▸HMS]), plane coordinate geometry ([TRV], [INV], [↶] [PT+], [↷] [CLS]), horizontal and vertical highway curves ([CRV], [↶] [VCV]), differential leveling ([↶] [LVL]), and atmospheric flight mechanics ([TSD], [WND], [↷] [XWC], [↶] [ALT], [↶] [CG]).

A.2.1 Sexagesimal Angle Conversion and Horizontal/Vertical Curves

Packed sexagesimal format (**DDD.MMSS**) encodes integer degrees D, arc-minutes $M \in [0, 59]$, and arc-seconds $S \in [0, 59.99]$ as $D + M/100 + S/10{,}000$, converting to decimal degrees ϑ_{dec} ([↶] [▸HR], with inverse [↷] [▸HMS]), while the [CRV] ([CRV]) Equation Workspace (**L = R × D × π / 180** on [LFU0]–[LFU2]: [L], [r], [D]) relates circular arc length L, radius R, tangent $T = R\tan(D/2)$, and central deflection angle D:

$$\theta_{\text{dec}} = D + \frac{M}{60} + \frac{S}{3600}, \qquad L = R \cdot D_{\text{deg}} \cdot \frac{\pi}{180^\circ}, \qquad T = R\tan\left(\frac{D}{2}\right) \tag{Eq. A.1}$$

A.2.2 Forward and Inverse Coordinate Geometry

Given a known survey station A at coordinates (N_1, E_1), horizontal leg distance D, and clockwise North azimuth α, forward traverse ([TRV]) computes the target station (N_2, E_2), whereas inverse traverse ([INV]) recovers D and α from coordinate differences $\Delta N = N_2 - N_1$ and $\Delta E = E_2 - E_1$:

$$N_2 = N_1 + D\cos\alpha, \qquad E_2 = E_1 + D\sin\alpha, \qquad D = \sqrt{(\Delta N)^2 + (\Delta E)^2}, \qquad \alpha = \operatorname{atan2}(\Delta E, \Delta N) \pmod{360^\circ} \tag{Eq. A.2}$$

A.2.3 Bowditch Compass Rule Traverse Adjustment

For a closed polygon traverse ([↶] [PT+], [↷] [CLS], and the **@COGO** ledger in [↷] [▦]) of total perimeter $P = \sum D_i$ with linear misclosure $E_{\text{mis}} = \sqrt{((\sum \Delta N)^2 + (\sum \Delta E)^2)}$, the Bowditch compass rule distributes closure corrections (δN_i, δE_i) in proportion to leg length D_i:

$$\delta N_i = -\Sigma\Delta N \cdot \frac{D_i}{P}, \qquad \delta E_i = -\Sigma\Delta E \cdot \frac{D_i}{P}, \qquad \text{Precision Ratio} = 1 : \frac{P}{E_{\text{mis}}} \tag{Eq. A.3}$$

A.2.4 Aviation Wind Triangle, Crosswind, and Density Altitude

With runway or course heading ψ, wind direction ϑ_w ($A = \vartheta_w - \psi$), wind speed V_w, and true airspeed V_{TAS}, the wind-triangle solver ([WND], opening the [WND] workspace **X = V × SIN(A)** on [X], [V], [A]) and [↷] [XWC] decompose crosswind V_{XW}, headwind V_{HW}, and wind correction angle WCA, while the atmospheric solver ([↶] [ALT], opening the [↶] [ALT] workspace **D = P + 120 × (T - 15)** on [D], [P], [T]) computes Pressure Altitude P (in feet) and Density Altitude D from field elevation Elev, altimeter setting Alt_{inHg}, outside air temperature T (°C), and standard temperature $T_{\text{ISA}} = 15°\text{C} - 2°\text{C} \times (P / 1000\ \text{ft})$:

$$V_{\text{XW}} = V_w \sin(\theta_w - \psi), \qquad V_{\text{HW}} = V_w \cos(\theta_w - \psi), \qquad \text{WCA} = \arcsin\left(\frac{V_{\text{XW}}}{V_{\text{TAS}}}\right) \tag{Eq. A.4}$$

$$P = \text{Elev} + 1000 \times (29.92 - \text{Alt}_{\text{inHg}}), \qquad D \approx P + 120 \times (T - T_{\text{ISA}}) \tag{Eq. A.5}$$

A.3 Industrial Trades, Electrical, and Mechanical

Covered in **Unit II, Chapters 4–6**, the **SC-14** industrial trades calculator combines feet-inch-fraction ([↶] [f-i]) carpentry layout, electrical conduit offset geometry ([OFS], [↶] [MIT], [↷] [FIT]), National Electrical Code (NEC) power and conductor drop formulas ([↶] [Ω], [pF], [AWG], [↷] [FLA], [↷] [kVA]), and industrial process flow ([PIP], [PMP], [↷] [Cv], [↶] [SFM], [↷] [WLD], [mA]).

A.3.1 Framing Trigonometry and Rafter Layout

For a pitched roof with unit rise U (inches of vertical rise per 12 in of horizontal run) and building span S (where half-span run is $R = S/2$), the framing and diagonal keys ([↖] [MIT], [↗] [FIT], [↖] [f-i]) evaluate roof pitch angle ϑ, common rafter length L_{common}, and 45° hip or valley rafter length L_{hip}:

$$\theta = \arctan\left(\frac{U}{12}\right), \qquad L_{\text{common}} = R\sqrt{1 + \left(\frac{U}{12}\right)^2}, \qquad L_{\text{hip}} = R\sqrt{2 + \left(\frac{U}{12}\right)^2} \tag{Eq. A.6}$$

A.3.2 Electrical Conduit Offset Bends and Shrink

To clear a mechanical obstruction of height O using two equal conduit bends of angle $A = \vartheta$ ([OFS], with [OFS] Equation Workspace **T = O / SIN(A)** binding [T], [O], [A] on [LFU0]–[LFU2]), the center-to-center travel distance $T = L_{\text{travel}}$ along the pipe and the axial conduit pull-back shrink S_{shrink} follow cosecant and half-angle tangent scaling:

$$T = O \cdot \csc\theta = \frac{O}{\sin\theta}, \qquad S_{\text{shrink}} = O \cdot \tan\left(\frac{\theta}{2}\right) \tag{Eq. A.7}$$

Bend Angle (ϑ)	Travel Multiplier ($\csc\vartheta$)	Shrink per Inch of Offset ($\tan(\vartheta/2)$)
10°	5.759 × O	0.087 in/in (≈ 1/16 in)
22.5°	2.613 × O	0.199 in/in (≈ 3/16 in)
30°	2.000 × O	0.268 in/in (≈ 1/4 in)
45°	1.414 × O	0.414 in/in (≈ 3/8 in)
60°	1.155 × O	0.577 in/in (≈ 9/16 in)

A.3.3 Three-Phase AC Power, Conductor Voltage Drop, and Control Valve Flow

For a balanced load with line-to-line voltage V_{LL}, line current I_L, power factor $F = \cos\varphi$, one-way conduit run length L (in feet), conductor cross-section CM (in circular mils), and resistivity constant K (12.9 Ω·CM/ft for copper or 21.2 Ω·CM/ft for aluminum), active power ([PF] Equation Workspace **P = V × I × F**, alongside [↗] [FLA] and [↗] [kVA]) and conductor voltage drop ([AWG] Equation Workspace **D = 2 × R × L × I / 1000** on [LFU0]–[LFU4]: [D], [R], [L], [I], [G], plus control valve flow **Q = C × √(P / G)** under [↗] [Cv]) are:

$$P_{3\phi} = \sqrt{3} \cdot V_{LL} \cdot I_L \cdot \cos\phi, \qquad V_{\text{drop},3\phi} = \frac{\sqrt{3} \cdot K \cdot I_L \cdot L}{\text{CM}}, \qquad D = \frac{2 \cdot K \cdot I_L \cdot L}{\text{CM}} = \frac{2 \cdot R_{\Omega/\text{kft}} \cdot L \cdot I_L}{1000} \tag{Eq. A.8}$$

A.4 Systems Programming, Embedded, and Cyber Operations

Covered in **Unit III, Chapters 7–9**, the **SC-16** systems and cybersecurity instrument operates on selectable fixed-width integer words (8-bit, 16-bit, 32-bit, 64-bit via [↗] [WSZ] and unsigned mode [↗] [UNS]), radix bases ([HEX], [↖] [DEC], [↖] [OCT], [↗] [BIN]), IEEE 754 binary floating-point encodings ([FLT], [↖] [F16], [↗] [F64]), embedded timer/ADC peripherals ([PWM], [↗] [BPS], [↖] [ADC], [↗] [DAC], [↖] [τ]), and network addressing ([NET], [↖] [CRC], [↗] [MOD]).

A.4.1 Two's Complement Signed Integer Representation and Bitwise Operators

In an N-bit word ($N \in (8, 16, 32, 64)$, cycled via [↷] [WSZ]), a binary bit vector $b_{N-1}b_{N-2}\ldots b_1 b_0$ represents the signed integer value V_{signed}, where two's complement negation ([↶] [NEG] / [+/−]) adds 1 to the bitwise one's complement ([↷] [~]) modulo 2^N, and shift/mask operators ([≪], [≫], [↶] [ASR], [↷] [ROR], [↶] [SB], [↶] [CB], [↷] [B?], [↶] [MSK], [&], [|], [⊕], [↶] [~&], [↶] [~|], [↶] [*B], [↷] [CLZ]) manipulate individual register fields:

$$V_{\text{signed}} = -b_{N-1}2^{N-1} + \sum_{k=0}^{N-2} b_k 2^k, \qquad -x \equiv (\sim x + 1) \pmod{2^N} \tag{Eq. A.9}$$

A.4.2 IEEE 754 Single-Precision Floating-Point Layout and Embedded Timer/PWM Synthesis

For a 32-bit **float32** word ([FLT], paired with byte-order swap [↷] [SWP]) partitioned into sign bit $s \in (0, 1)$ (bit 31), 8-bit biased exponent $E \in [1, 254]$ (bits 30–23), and 23-bit significand mantissa M (bits 22–0), the real value x and the hardware timer PWM frequency ([PWM] Equation Workspace **F = C / (P × A)** binding [F], [C ON], [P], [A] on [LFU0]–[LFU3], alongside [↷] [BPS] divisor **D = C / (16 × B)**) are:

$$x = (-1)^s \times \left(1 + \frac{M}{2^{23}}\right) \times 2^{E-127}, \qquad F = \frac{C}{P \cdot A}, \qquad \text{BRR} = \frac{f_{\text{clk}}}{16 \times f_{\text{baud}}} \tag{Eq. A.10}$$

A.4.3 Mixed-Signal ADC Quantization, RC Settling, Modular Cryptography, and CIDR

An N-bit successive-approximation ADC ([↶] [ADC] Equation Workspace **D = V / R × 2^N** binding [D], [V], [R], [N] on [LFU0]–[LFU3]) maps analog input V against reference R with ideal signal-to-noise ratio $\text{SNR}_{\text{dB}} = 6.0206\ N + 1.7609$ dB and Effective Number of Bits ENOB = (SINAD - 1.7609)/6.0206, while the RC anti-aliasing filter ([↶] [τ] [↶] [τ] Equation Workspace **T = R × C** binding [T], [R], [C ON] on [LFU0]–[LFU2]) requires $t_{\text{acq}} \geq (N + 1)\ \ln(2)\ \tau$ to settle within (1/2) LSB. In network and cryptographic analysis ([↷] [MOD], [↶] [CRC], [NET], [↶] [→KiB], [↶] [→MHz], [↷] [→ns]), modular arithmetic, Shannon entropy $H(X)$, and IPv4 subnet host capacity follow:

$$\boldsymbol{D = \lfloor\ V/R \cdot 2^N\ \rfloor,\ t_{\text{acq}} \geq (N+1)\ \ln(2)\ R\ C,\ C = M^e \pmod{n},} \\ \boldsymbol{H(X) = -\sum_{i=1}^{m} p_i \log_2 p_i,\ N_{\text{hosts}} = 2^{32-P} - 2} \tag{Eq. A.11}$$

A.5 Cinema Optics, Broadcast Timecode, and Sound

Covered in **Unit IV, Chapters 10–12**, the **SC-24** motion-picture and broadcast calculator unites SMPTE frame-accurate timecode arithmetic ([TC], [FPS], [↷] [DF], [↷] [+TC], [↷] [-TC]), geometrical camera optics ([FOV], [DOF], [↶] [HFD]), rotary shutter exposure ([EV], [↶] [SHT], [LUX], [↷] [GEL]), and acoustic and data propagation ([dB], [↶] [SPL], [↷] [DLY], [BIT]).

A.5.1 SMPTE Timecode and NTSC Drop-Frame Compensation

Color NTSC broadcast video runs at $f_{\text{NTSC}} = 30 \times 1000 / 1001 \approx 29.97002997$ fps, causing a nominal 30 fps frame counter to drift by +108 frames (+3.6 s) per hour. SMPTE **29.97 DF** Drop-Frame timecode ([TC], [FPS], [↷] [DF], with [TC] Equation Workspace **F = S × R** on [F], [S], [R]) reconciles wall-clock time by skipping frame labels **:00** and **:01** at the start

of every minute except minutes divisible by 10:

$$N_{\text{dropped/hr}} = 60 \times 2 - 6 \times 2 = 108 \text{ frames/hr}, \qquad F = S \times R \qquad \textbf{(Eq. A.12)}$$

A.5.2 Hyperfocal Distance, Depth of Field, and Rotary Shutter Exposure

For lens focal length F, sensor width W, aperture N (f-stop or photometric T-stop), sensor circle of confusion C, focus distance d, frame rate R, and rotary shutter angle $\vartheta_{\text{shutter}}$ (in degrees), the optical solvers ([FOV] **A = 2 × ATAN(W / (2 × F))**, [DOF] **H = F² / (N × C)**, [EV] **N² = T × 2^E**, [↶] [SHT], and [LUX] **E = I / D²**) evaluate field of view A, hyperfocal distance H, near and far depth-of-field limits D_{near} and D_{far}, and frame exposure time T:

$$A = 2\arctan\left(\frac{W}{2F}\right), \quad H = \frac{F^2}{N \cdot C}, \quad D_{\text{near}} = \frac{H \cdot d}{H + (d - F)}, \quad D_{\text{far}} = \frac{H \cdot d}{H - (d - F)} \ (d < H), \quad T = \frac{\vartheta_{\text{shutter}}}{360^\circ \times R} \qquad \textbf{(Eq. A.13)}$$

A.6 RF Spectrum, Transmission Lines, and Microwave

Covered in **Unit V, Chapters 13–15**, the **SC-50** radio-frequency and microwave engineering instrument evaluates logarithmic power ratios ([dBm], [↶] [→dBm], [dBm], [↷] [→dBW], [↶] [→dBV], [↷] [→µV]), resonant LC and waveguide structures ([↶] [f₀], [↷] [fc], [↶] [λ/L]), transmission-line impedance mismatches ([Γ], [SWR], [↷] [TLN]), free-space link budgets ([FSL], [↶] [LNK], [↷] [F₁]), and receiver noise and channel capacity ([↷] [Pₙ], [NF], [↷] [C/B], [↶] [RCS], [↷] [Δf], [↶] [IP3], [↷] [FDR]).

A.6.1 Decibel Power Ratios and Resonant LC Tank Circuits

Logarithmic gain G_{dB} relates power ratio P_2 / P_1 and equal-impedance voltage ratio V_2 / V_1, while absolute power P_{dBm} ([dBm], [↶] [→dBm], with inverse [dBm] and watt-referenced [↷] [→dBW]) references linear power to 1 mW (0.001 W), and the resonant tank solver ([↶] [f₀] [↶] [f₀] Equation Workspace **F = 1 / (2π√(L×C))** binding [F], [L], [C 0N] on [LFU0]–[LFU2]) relates resonant frequency F, inductance L, and capacitance C:

$$G_{\text{dB}} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 20\log_{10}\left(\frac{V_2}{V_1}\right), \quad P_{\text{dBm}} = 10\log_{10}\left(\frac{P_{\text{mW}}}{1\text{ mW}}\right) = P_{\text{dBW}} + 30, \quad F = \frac{1}{2\pi\sqrt{LC}} \qquad \textbf{(Eq. A.14)}$$

A.6.2 Reflection Coefficient, Standing Wave Ratio, and Quarter-Wave Matching

For a transmission line of characteristic impedance Z_0 ([↷] [Z₀], defaulting to 50 Ω) terminated in load impedance Z_L, the reflection coefficient $G = |\Gamma|$ ([Γ]) governs Voltage Standing Wave Ratio S = VSWR ([SWR] Equation Workspace **S = (1 + G) / (1 - G)** binding [S], [G] on [LFU0]–[LFU1]), quarter-wave transformer impedance $Z = \sqrt{(S \cdot L)}$ ([↷] [TLN] [↷] [Z₀] workspace on [Z], [S], [L]), Return Loss RL_{dB}, and Mismatch Loss ML_{dB}:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad S = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad \text{RL}_{\text{dB}} = -20\log_{10}|\Gamma|, \quad \text{ML}_{\text{dB}} = -10\log_{10}(1 - |\Gamma|^2) \qquad \textbf{(Eq. A.15)}$$

A.6.3 Friis Free-Space Path Loss, Thermal Noise Floor, Shannon Capacity, and Doppler Shift

Across a line-of-sight propagation link of distance D at carrier frequency F (with speed of light [→] [c ON] $c = 2.99792458 \times 10^8$ m/s), [FSL] (Equation Workspace **L = 20LOG(D×F) + 32.4** binding [L], [D], [F] on [LFU0]–[LFU2]) and [↖] [LNK] (**R = P + G - L** binding [R], [P], [G], [L] on [LFU0]–[LFU3]) compute Free-Space Path Loss L and received power R; [→] [Pₙ] and [NF] (**F = A + (B - 1) / G** on [F], [A], [b], [G]) evaluate the thermal noise floor P_n at [→] [T₀] $T_0 = 290$ K ($k_B T_0 = -174$ dBm/Hz via Boltzmann constant [↖] [kB]) and Friis cascaded noise factor. And [→] [C/B] (**C = B × LOG2(1 + S)** on [LFU0]–[LFU2]: [c ON], [b], [S]) and [→] [Δf] solve Shannon–Hartley channel capacity and monostatic radar Doppler shift:

$$L_{\mathrm{FSPL,\,dB}} = 20\log_{10}(D_{\mathrm{km}}) + 20\log_{10}(F_{\mathrm{MHz}}) + 32.44, \qquad P_{n,\mathrm{dBm}} = -174\ \mathrm{dBm/Hz} + 10\log_{10}(B_{\mathrm{Hz}}) + \mathrm{NF}_{\mathrm{dB}} \qquad \textbf{(Eq. A.16)}$$

$$C = B\log_2(1+S), \qquad \Delta f_{\mathrm{radar}} = \frac{2\,v_r f_0}{c}, \qquad r_{F_1} = 17.32\sqrt{\frac{d_{\mathrm{km}}}{4 f_{\mathrm{GHz}}}} \qquad \textbf{(Eq. A.17)}$$

A.7 Wet-Lab Chemistry, Stoichiometry, and Spectroscopy

Covered in **Unit VI, Chapters 16–18**, the **SC-60** analytical chemistry and biochemistry instrument connects gravimetric stoichiometry ([MW], [Z], [↖] [RXN], [↖] [w/w]), serial dilutions and titrations ([C₁V₁], [↖] [EQV]), Beer–Lambert spectrophotometry ([εbc], [↖] [%T]), acid-base buffer equilibria ([pH], [→] [pKₐ], [BUF]), and chemical thermodynamics and kinetics ([PVT], [↖] [ΔG], [→] [E°], [↖] [CAL], [→] [Eₐ], [→] [RCF]).

A.7.1 Molarity, Dilutions, and Beer–Lambert Spectrophotometry

Solution molarity C ([↖] [RXN] Equation Workspace **N = C × V** on [n], [c ON], [V] and [MW] **W = N × M** on [W], [n], [m]) relates solute mass W, molar mass M ([MW] / [Z]), moles N, and solution volume V. Stock dilution ([C₁V₁] **C × V = D × U** on [c ON], [V], [D], [U]) conserves solute moles across concentrations C, D and volumes V, U. And optical absorbance A ([εbc] [εbc] Equation Workspace **A = E × B × C** binding [A], [E], [b], [c ON] on [LFU0]–[LFU3], alongside [↖] [%T]) links transmitted intensity I / I_0 to molar absorptivity $E = \varepsilon$, path length $B = b$, and concentration $C = c$:

$$N = \frac{W}{M} = C \cdot V, \qquad C \cdot V = D \cdot U, \qquad A = -\log_{10}\!\left(\frac{I}{I_0}\right) = E \cdot B \cdot C \qquad \textbf{(Eq. A.18)}$$

A.7.2 Acid-Base Buffers, Thermodynamics, and Arrhenius Activation Kinetics

Aqueous acidity ([pH]) and weak-acid buffer equilibria ([BUF], with Equation Workspace **P = K + LOG(A/H)** binding [P], [K], [A], [h] on [LFU0]–[LFU3]) follow Sørensen's logarithmic scale and the Henderson–Hasselbalch equation for conjugate pair [A⁻] / [HA] around pK_a, while [PVT] (**P × V = N × R × T**), [↖] [ΔG] (**G = H - T × S**), [→] [E°] (**E = S - (.0592/N)LOG(Q)**), [↖] [CAL] (**Q = M × C × T**), and [→] [Eₐ] relate thermodynamic state functions via universal gas constant [→] [R] $R = 8.3144626$ J·mol^{-1}·K^{-1}, Faraday's constant [↖] [F] $F = 96{,}485.33$ C/mol, and Avogadro's constant [↖] [Nₐ] $N_A = 6.02214076 \times 10^{23}$ mol^{-1}:

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+], \qquad P = K + \log_{10}\!\left(\frac{A}{H}\right), \qquad \ln\!\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \qquad \textbf{(Eq. A.19)}$$

A.8 Commercial Bakery, Fermentation Science, and Brewing

Covered in **Unit VII, Chapters 19–21**, the **SC-78** culinary and fermentation instrument standardizes flour-referenced baker's percentages ([BK%], [↷] [HYD], [↷] [LV%], [ΣW], [↶] [SCL], [↶] [DIV]), dough thermal balance ([Tw], [↷] [ICE]), mash strike temperatures ([MSH]), wort attenuation ([ABV], [↷] [→SG], [↶] [→Brx], [↶] [→°P], [↷] [SGT]), hop bitterness ([IBU], [↷] [SRM], [↶] [CO₂]), and butcher yield costing ([YLD], [CST], [↶] [PRC], [↷] [Aw], [↷] [SO₂]).

A.8.1 Baker's Percentage, Dough Thermodynamics, and Culinary Yield Costing

Baker's percentage ([BK%] Equation Workspace **P = I / F × 100** on [P], [I], [F], paired with [ΣW], [↶] [SCL], and [↶] [DIV]) expresses each ingredient weight I as a percentage P of total flour mass F. The five-variable Desired Dough Temperature balance ([Tw] [Tw] Workspace **W = 3×D - R - F - M** on [W], [D], [R], [F], [M]) and flake-ice substitution ([↷] [ICE]) govern mixer calorimetry. And butcher trim yield $Y_{\%}$ ([YLD]) and food cost ([CST] **P = C / (F / 100)** on [P], [C ON], [F], paired with [↶] [PRC]) govern menu pricing:

$$P = \frac{I}{F} \times 100\%, \qquad W = 3D - R - F - M, \qquad W_{\text{ice}} = \frac{W_{\text{water}}(T_{\text{tap}} - W)}{80} \tag{Eq. A.20}$$

$$Y_{\%} = \frac{W_{\text{EP}}}{W_{\text{AP}}} \times 100\%, \qquad C_{\text{EP}} = \frac{C_{\text{AP}}}{Y_{\%}/100}, \qquad P = \frac{C}{F/100} \tag{Eq. A.21}$$

A.8.2 Brewing Gravity, Strike Water, and Hop Bitterness

During wort mashing and fermentation, [ABV] (**A = (O - F) × 131.25** on [A], [O], [F]) evaluates Alcohol by Volume (A) from Original Gravity (O) and Final Gravity (F), [MSH] (**S = .2/R×(T-G)+T** on [S], [R], [T], [G]) solves mash strike water temperature S from water-to-grist ratio R, dry grain temperature G, and target rest temperature T, and [IBU] (**I = W×A×U×74.89/V** on [I], [W], [A], [U], [V]) evaluates Glenn Tinseth's International Bitterness Units from hop weight W, alpha-acid rating A, decimal utilization U, and batch volume V:

$$A = (O - F) \times 131.25, \qquad S = \frac{0.2}{R}(T - G) + T, \qquad I = \frac{W \cdot A \cdot U \cdot 74.89}{V} \tag{Eq. A.22}$$

A.9 Clinical Nursing, Hospital Pharmacy, and Infusions

Covered in **Unit VIII, Chapters 22–24**, the **SC-80** clinical nursing and hospital pharmacy calculator implements persistent Patient Profile biometrics ([PAT]: [W], [h], [A], [S], [MALE]/[FEM]), anthropometric dosing-weight corrections ([BSA], [↷] [BMI], [↶] [IBW], [↷] [ABW]), glomerular filtration estimation ([GFR]), first-order compartment kinetics ([PK]), intravenous pump titrations ([DRP], [IV], [DOS], [gtt], [↷] [BRN], [ICU]), and acid-base electrolyte homeostasis ([↷] [AG], [↶] [Ca²⁺], [↷] [Na⁺], [↷] [OSM], [↷] [ABG], [↶] [QTc], [↶] [O₂]).

A.9.1 Body Surface Area and Dosing Weight Corrections

For a patient of height H (in cm, where $H_{\text{in}} = H / 2.54$) and total body weight W (in kg), [BSA] (**B = √(H × W / 3600)** on [b], [h], [W], [PAT]) evaluates Robert Mosteller's surface area (m²), [↶] [IBW] evaluates Ben Devine's lean Ideal Body Weight ($\text{IBW}_{\text{male}} = 50 + 2.3(H_{\text{in}} - 60)$ and $\text{IBW}_{\text{female}} = 45.5 + 2.3(H_{\text{in}} - 60)$), and [↷] [ABW] blends 40% of excess adipose mass into Adjusted Body Weight ABW:

$$B = \sqrt{\frac{H \times W}{3600}}, \qquad \text{ABW} = \text{IBW} + 0.40(W - \text{IBW}) \qquad \textbf{(Eq. A.23)}$$

A.9.2 Renal Clearance, Intravenous Infusions, Compartment Kinetics, and Anion Gap

Given chronological age A, dosing weight W, and steady-state serum creatinine S, [GFR] (**C = (140-A)W/(72×S)** on [c ON], [A], [W], [S], [PAT]) evaluates Donald Cockcroft and Henry Gault's glomerular filtration estimate (mL/min); [DRP] (**R = D × W × 60 / C** on [r], [D], [W], [c ON], [PAT]), [IV] (**R = V / T** on [r], [V], [T]), [DOS] (**T = W × D** on [T], [W], [D], [PAT]), and [gtt] convert weight-based vasoactive orders (D) and bag concentrations (C) into volumetric pump rates (R) and gravity drop rates (gtt/min). And [PK] (**T = .69315 × V / C** on [T], [V], [c ON]), [→] [AG], and [←] [Ca²⁺] govern one-compartment elimination and serum electroneutrality:

$$C = \frac{(140 - A) \times W}{72 \times S} \times (0.85 \text{ if female}) \qquad \textbf{(Eq. A.24)}$$

$$R = \frac{D \times W \times 60}{C}, \qquad R_{\text{gtt/min}} = \frac{R_{\text{mL/hr}} \times DF_{\text{gtt/mL}}}{60} \qquad \textbf{(Eq. A.25)}$$

$$T_{1/2} = \frac{\ln(2) \cdot V}{C_{\text{CL}}} = \frac{\ln(2)}{K}, \qquad \text{AG} = [\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-]), \qquad \text{Ca}_{\text{corr}} = \text{Ca}_{\text{meas}} + 0.8(4.0 - \text{Alb}) \qquad \textbf{(Eq. A.26)}$$

A.10 Quantitative Finance, Actuarial Science, and Six Sigma

Covered in **Unit IX, Chapters 25–27**, the **SC-95** quantitative finance and statistical inference instrument unites multi-period cash-flow discounting ([TVM], [←] [AMR], [→] [CF], [→] [CGR], [→] [ROI]), fixed-income bond pricing and yield ([→] [BND]), unit economics ([←] [MGN], [→] [MKP], [←] [BEP], [→] [EOQ], [→] [DEP], [→] [EAR], [→] [TAX]), direct faceplate R-style probability distributions ([→] [pN], [←] [qN], [rN], [→] [pt], [←] [qt], [rt], [→] [px²], [←] [qx²], [rx²], [→] [pF], [←] [qF], [rF], [→] [pB], [←] [qB], [rB], [←] [PDF]), and Six Sigma process capability ([→] [Cpk], [→] [SPC], [←] [Z₀], [←] [t₂], [→] [A/B], [→] [CI], [←] [n*]).

A.10.1 Compound Time Value of Money, Bond Valuation, and Net Present Value

Across N compounding periods at periodic interest rate i = I%/100 with payment timing indicator mode ∈ (0, 1) ([END] = 0, [BEG] = 1), the five softkeys ([n], [I%], [PV], [PMT], [FV] under [TVM]) enforce cash-flow conservation between Present Value PV, periodic payment PMT, and Future Value FV. In the [→] [CF] ([→] [CF]) Equation Workspace (**V = C×(1-(1+R/100)⁻ᴺ)/(R/100) - I** binding [V], [c ON], [r], [n], [I] on [LFU0]–[LFU4]) and the [→] [BND] Equation Workspace (**P = C×(1-(1+Y/100)⁻ᴺ)/(Y/100) + F×(1+Y/100)⁻ᴺ** binding [P], [c ON], [Y], [n], [F] on [LFU0]–[LFU4]), Secant–Brent numerical root-finding solves forward for price/value or backward for internal yield:

$$\text{PV}(1+i)^N + \text{PMT} \cdot \frac{(1+i)^N - 1}{i} \cdot (1 + i \cdot \text{mode}) + \text{FV} = 0, \qquad V = C \cdot \frac{1 - (1+r)^{-N}}{r} - I \qquad \textbf{(Eq. A.27)}$$

$$P = C \cdot \frac{1 - (1+y)^{-N}}{y} + \frac{F}{(1+y)^N}, \qquad D_{\text{Mac}} = \frac{1}{P} \sum_{t=1}^{N} \frac{t \cdot CF_t}{(1+y)^t}, \qquad D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1+y} \qquad \textbf{(Eq. A.28)}$$

A.10.2 Six Sigma Process Capability, Sample Size Planning, and Black–Scholes Option Valuation

For a manufacturing process with mean μ and standard deviation $S = \hat{\sigma}$ operating between specification limits L = LSL and U = USL, [→] [Cpk] (**C = (U - L) / (6×S)** on [ON/C], [U], [L], [S]) evaluates potential capability C_p and centered capability C_{pk}, while the sample-size solver ([←] [n*] [←] [n*] Equation Workspace **N = (Z × S / E)²** binding [n], [Z], [S], [E] on [LFU0]–[LFU3]) relates sample size N, critical quantile $Z = z_{\alpha/2}$ ([←] [qN] / [←] [qt]), dispersion S, and margin of error E ([→] [CI]). In continuous-time finance, the Black–Scholes–Merton equations price European call and put options from spot price S_0, strike K, risk-free rate r, volatility σ, expiration T, and the standard normal cumulative integral $\Phi(z)$ ([→] [pN]):

$$C_p = \frac{U-L}{6S}, \qquad C_{pk} = \min\left(\frac{U-\mu}{3S}, \frac{\mu-L}{3S}\right), \qquad N = \left(\frac{Z\cdot S}{E}\right)^2 \tag{Eq. A.29}$$

$$C_{\text{call}} = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad P_{\text{put}} = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1), \quad d_1 = \frac{\ln(S_0/K) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \tag{Eq. A.30}$$

Appendix B: Domain Instrument Keycap & Workspace Directory

This directory catalogs the physical keycaps, Row 0 contextual/LFU softkeys ([LFU0]–[LFU5]), yellow ([↖]) and blue ([↗]) shift layers, bidirectional Equation Solver Workspaces ([↗] [SOLVE] / [XEQ]), and register input-output signatures across all nine instruments in the StackCalc Domain Series (**SC-35** through **SC-95**, covered in Units I through IX, Chapters 1 through 27). Each table lists the pre-execution stack arguments consumed from **Register T**, **Register Z**, **Register Y**, and **Register X** alongside the post-execution results returned to **Register X'** and **Register Y'** and the softkey variable bindings inside each instrument's Equation Solver Workspace.

B.1 Shared Three-Pillar Core Keys Across All Nine Instruments

Every calculator in the StackCalc Domain Series is built on the same **three-pillar architecture**: configurable RPN stack depth ([4-LV], [8-LV], or [INF] cycled on Flag 5 **5:DEP** inside [↗] [FLAGS]), dual-mode domain keys that combine one-key RPN stack evaluation with a bidirectional Equation Solver Workspace ([↗] [SOLVE] / [XEQ] + [LFU0]–[LFU5]), and non-destructive Live Reference Ledgers ([↗] [▦]) paired with ten general-purpose memory registers (**R**0–**R**9):

Keycap / Sequence	Function Name	Pre-Execution Stack (*Y*, *X*)	Post-Execution Stack (*Y′*, *X′*) & Architecture Role
[ENTER]	Stack Lift & Duplicate	**Y, X**	**X, X** (prior $Y \to Z$, $Z \to T$; lifts across [4-LV], [8-LV], or [INF] via Flag 5 **5:DEP** in [↗] [FLAGS])
[x⇄y]	Swap Registers *X* and *Y*	**Y, X**	**X, Y** (*Z* and *T* unchanged)
[R↓] / [↗] [R↑]	Roll Stack Down / Up	**T, Z, Y, X**	**X, T, Z, Y** or **Z, Y, X, T** (cycles active stack window)
[⌫] / [C CLX]	Backspace / Clear *X*	**Y, X**	**Y, 0.0000** (disables stack lift)
[+/−]	Change Sign / Negate	**Y, X**	**Y, -X** (or two's complement negation [↖] [NEG] on **SC-16**)
[STO] [0]–[9]	Store *X* in Register R_n	**Y, X**	**Y, X** ($R_n \leftarrow X$)
[RCL] [0]–[9]	Recall Register R_n to *X*	**Y, X**	**X, R**$_n$ (lifts stack first)
[↗] [SOLVE] / [XEQ]	Bidirectional Solver Arm	**Y, X**	Arms Equation Workspace to solve forward or backward for the next [LFU0]–[LFU5] softkey pressed
[LFU0]–[LFU5]	Row 0 Softkeys (Cols 0–5)	**Y, X**	Populates dynamically with recently/frequently used operations (LFU) when idle, or stores/solves symbolic variables in an active Equation Workspace
[↗] [▦]	Live Reference Ledger	**Y, X**	Opens non-destructive domain reference lookup and ledger (**@COGO**, **@BOM**, **@BK%**, **@PAT**, cash-flow / stats table)
[↖]	Yellow Primary Shift	—	Arms yellow shifted legend for next key

Keycap / Sequence	Function Name	Pre-Execution Stack (Y, X)	Post-Execution Stack (Y, X) & Architecture Role
[↗]	Blue Secondary Shift	—	Arms blue shifted legend for next key
[↗] [SHOW]	Full-Precision Mantissa Peek	**Y, X**	Temporarily reveals all 15–17 significant IEEE 754 digits of X

B.1.1 Master Directory of Bidirectional Equation Solver Workspaces

Opening a multi-parameter domain workspace (by pressing the key with $X = 0$, $Y = 0$ or after [↗] [SOLVE]) renders its governing law on Line 1 of the display and binds the Row 0 softkeys ([LFU0]–[LFU5]) to its symbolic variables so any parameter can be solved forward or backward via [XEQ] <**SOFTKEY**> (or [↗] [SOLVE] <**SOFTKEY**>):

Instrument	Workspace Key	Governing Equation on Line 1	Bound Softkeys ([LFU0]–[LFU5])
SC-35	[CRV] ([CRV])	**L = R × D × π / 180**	[L] arc length • [r] radius • [D] central angle Δ
SC-35	[↖] [LVL] ([↖] [LVL])	**E = H + B - F**	[E] elevation • [h] benchmark / HI • [b] backsight • [F] foresight
SC-35	[WND] ([WND])	**X = V × SIN(A)**	[X] crosswind • [V] wind speed • [A] wind angle γ
SC-35	[↖] [ALT] ([↖] [ALT])	**D = P + 120 × (T - 15)**	[D] density altitude • [P] pressure altitude • [T] OAT (°C)
SC-14	[AWG] ([AWG])	**D = 2 × R × L × I / 1000**	[D] drop (V) • [r] Ω/kft • [L] one-way run (ft) • [I] load (A) • [G] AWG gauge
SC-14	[OFS] ([OFS])	**T = O / SIN(A)**	[T] center-to-center travel • [O] obstruction offset • [A] bend angle ϑ
SC-14	[↗] [C_v] ([↗] [C_v])	**Q = C × √(P / G)**	[Q] flow (GPM) • [C_v] valve C_v • [P] ΔP (psi) • [G] specific gravity
SC-14	[PMP] ([PMP])	**P = Q × H / 3960**	[P] hydraulic/brake HP • [Q] flow (GPM) • [h] total dynamic head (ft)
SC-16	[PWM] ([PWM])	**F = C / (P × A)**	[F] PWM freq (Hz) • [C_{ON}] clock f_{clk} • [P] prescaler $P_{PSC}+1$ • [A] period $A_{ARR}+1$
SC-16	[↖] [ADC] ([↖] [ADC])	**D = (V / R) × 2^N**	[D] digital code • [V] input V_{in} • [r] reference V_{ref} • [n] bit depth
SC-16	[↖] [τ] ([↖] [τ])	**T = R × C**	[T] time constant τ (s) • [r] series resistance (Ω) • [C_{ON}] capacitance (F)
SC-24	[FOV] ([FOV])	**A = 2 × ATAN(W / (2 × F))**	[A] angle of view (°) • [W] sensor gate width (mm) • [F] focal length (mm)
SC-24	[EV] ([EV])	**N^2 = T × 2^E**	[n] aperture f-stop • [T] exposure time (s) • [E] exposure value EV_{100}
SC-24	[DOF] ([DOF])	**H = F^2 / (N × C)**	[h] hyperfocal • [F] focal length (mm) • [n] T-stop • [C_{ON}] circle of confusion (mm)
SC-24	[TC] ([TC])	**F = S × R**	[F] total frame count • [S] elapsed seconds • [r] frame rate (fps)

Instrument	Workspace Key	Governing Equation on Line 1	Bound Softkeys (LFU0–LFU5)
SC-50	↖ f₀ (↖ f₀)	F = 1 / (2π√(L×C))	F resonant f_0 (Hz) • L inductance (H) • C ON capacitance (F)
SC-50	SWR (SWR)	S = (1 + G) / (1 - G)	S standing wave ratio VSWR • G reflection magnitude $\lvert\Gamma\rvert$
SC-50	FSL (FSL)	L = 20LOG(D) + 20LOG(F) + 32.44	L path loss $\mathrm{FSPL_{dB}}$ • D link distance (km) • F carrier (MHz)
SC-50	↖ LNK (↖ LNK)	R = P + G - L	r received P_{rx} (dBm) • P transmit P_{tx} • G net antenna gain • L total loss
SC-50	↗ C/B (↗ C/B)	C = B × LN(1 + S) / LN(2)	C ON capacity (bps) • b bandwidth (Hz) • S linear SNR
SC-50	Γ (↗ Z₀)	Z = √(S × L)	Z quarter-wave transformer Z_q • S source impedance Z_S • L load impedance Z_L
SC-60	εbc (εbc)	A = E × B × C	A absorbance • E molar absorptivity ε • b path length (cm) • C ON molarity (M)
SC-60	BUF (BUF)	P = K + LOG(A / H)	P target pH • K acid pK_a • A base $[A^-]$ • h acid [HA]
SC-60	PVT (PVT)	P × V = N × 8.3145 × T	P pressure • V volume • n moles n • T absolute temperature (K)
SC-60	C₁V₁ (C₁V₁)	C × V = D × U	C ON stock C_1 • V aliquot V_1 • D target C_2 • U final volume V_2
SC-78	BK% (BK%)	P = (I / F) × 100	P baker's % • I ingredient weight • F total flour weight
SC-78	Tw (Tw)	W = 3 × D - R - F - M	W water temp T_w • D target DDT • r room T_r • F flour T_f • m friction
SC-78	MSH (MSH)	S = (0.2 / R) × (T - G) + T	S strike water temp • r water-grist ratio • T mash target • G dry grain temp
SC-78	IBU (IBU)	I = W × A × U × 74.89 / V	I bitterness IBU • W hop mass (oz) • A alpha acid % • U utilization • V volume (gal)
SC-78	CST (CST)	P = C / (F / 100)	P menu price • C ON portion cost • F target food cost %
SC-80	BSA (BSA)	B = √(H × W / 3600)	b BSA (m²) • h height (cm) • W weight (kg) • PAT patient profile
SC-80	GFR (GFR)	C = (140-A)×W / (72×S)	C ON clearance (mL/min) • A age (yr) • W weight (kg) • S SCr (mg/dL) • PAT
SC-80	DRP (DRP)	R = D × W × 60 / C	r pump rate (mL/hr) • D dose (mcg/kg/min) • W weight (kg) • C ON concentration (mcg/mL) • PAT
SC-80	PK (PK)	T = 0.693 × V / C	T half-life $t_{1/2}$ (hr) • V volume V_d (L) • C ON clearance CL (L/hr)
SC-95	TVM (TVM)	PV(1+I)^N + PMT... + FV = 0	n periods • I% rate • PV present value • PMT payment • FV future value • END/BEG

Instrument	Workspace Key	Governing Equation on Line 1	Bound Softkeys (LFU0–LFU5)
SC-95	[→] [CF] ([→] [CF])	$\mathbf{V = C\times(1-(1+R/100)^{-N})/(R/100) - I}$	[V] net present value • [$\frac{\$}{CPN}$] uniform cash flow • [I%] discount rate / IRR (%) • [n] periods • [I] outlay
SC-95	[→] [BND] ([→] [BND])	$\mathbf{P = C\times(1-(1+Y/100)^{-N})/(Y/100) + F/(1+Y/100)^{N}}$	[P] bond price • [$\frac{\$}{CPN}$] periodic coupon • [Y] periodic yield (%) • [n] periods • [F] par face value
SC-95	[↶] [n*] ([↶] [n*])	$\mathbf{N = (Z \times S \,/\, E)^2}$	[n] sample size n^* • [Z] critical quantile $z_{\alpha/2}$ • [S] dispersion s • [E] margin of error

B.2 Geomatics, Land Surveying, and Flight Navigation Keycaps

On the **SC-35** (**Unit I, Chapters 1–3**), the Row 1 and Row 2 geomatics keys and shifted trigonometric/navigation operators execute plane coordinate geometry (COGO), horizontal and vertical highway curves, Bowditch traverse closure, sexagesimal angle conversions, and E6B aviation wind-triangle and density-altitude solutions:

Keycap / Softkey	Layer	Operation & Stack Signature
[TRV] / [↶] [PT+]	Row 1 Primary / Yellow Shift	**Forward Traverse & Station Accumulator:** Consumes distance D and azimuth α to resolve (ΔN, ΔE) ([↶] [N↔E] swaps Northing/Easting), and [↶] [PT+] accumulates station coordinates.
[INV] / [→] [CLS]	Row 1 Primary / Blue Shift	**Coordinate Inverse & Bowditch Closure:** Consumes ΔN (Y) and ΔE (X) to return azimuth α and distance D; [→] [CLS] computes traverse closure error and precision ratio.
[BRG] / [↶] [ISC]	Row 1 Primary / Yellow Shift	**Quadrant Bearing & Intersection:** Formats azimuth in X as surveyor quadrant bearing (NE, SE, SW, NW), and [↶] [ISC] evaluates bearing-bearing intersections.
[CRV] / [→] [STA]	Row 1 Primary / Blue Shift	**Horizontal Curve Workspace ([CRV]) & Stationing:** Solves circular arc $\mathbf{L = R \times D \times \pi \,/\, 180}$ ([L], [r], [D]) and formats highway stationing **XX+XX.XX**.
[SLP] / [↶] [VCV] / [→] [GR%]	Row 1 Col 4 (Primary / Yellow / Blue)	**Slope Reduction, Vertical Curve & Grade %:** Reduces slope distance/angle ([SLP]), solves parabolic vertical curves ([↶] [VCV]), and computes percent grade ([→] [GR%]).
[↶] [LVL] / [FLW] / [→] [△]	Yellow / Primary / Blue Shift	**Differential Leveling ([↶] [LVL]), Open-Channel Flow ([FLW]) & Triangle ([→] [SOLVE]):** Solves $\mathbf{E = H + B - F}$ ([E], [h], [b], [F]), continuity $\mathbf{Q = A \times V}$, and triangle area $\mathbf{A = 0.5 \times B \times H}$.
[WND] / [↶] [ALT] / [→] [XWC]	Row 7 Col 3 (Primary / Yellow / Blue)	**E6B Wind Triangle ([WND]), Density Altitude ([↶] [ALT]) & Crosswind:** Solves $\mathbf{X = V \times SIN(A)}$ ([X], [V], [A]), density altitude $\mathbf{D = P + 120 \times (T - 15)}$ ([D], [P], [T]), and crosswind component ([→] [XWC]).
[TSD] / [↶] [CG] / [→] [TID]	Row 2 Col 2 (Primary / Yellow / Blue)	**Time-Speed-Distance ([TSD]), Weight & Balance ([↶] [CG]) & Tidal Height:** Solves $\mathbf{D = S \times T}$ ([D], [S], [T]), moment $\mathbf{M = W \times A}$ ([m], [W], [A]), and rule-of-twelfths tidal stage ([→] [TID]).
[↶] [▸HR] / [→] [▸HMS] / [↶] [▸DEG] / [→] [▸RAD] / [→] [→GRD]	Angular & Sexagesimal Shifts	Converts packed sexagesimal **DDD.MMSS** to decimal degrees/hours ([↶] [▸HR]) and back ([→] [▸HMS]), or converts between degrees, radians, and gradians.
[↶] [▸θ,r] / [→] [▸y,x] / [↶] [f-i]	Coordinate & Feet-Inch Shifts	Converts between Rectangular (Y, X) and Polar (r, ϑ) coordinates and enters/formats feet-inch-fractions ([↶] [f-i]).

B.3 Industrial Trades, Electrical, and Mechanical Keycaps

On the **SC-14** (**Unit II, Chapters 4–6**), the Row 1 and Row 2 electrical, conduit, piping, and machining keys pair with dimensional shift converters to resolve National Electrical Code (NEC) conductor sizing, three-phase power and motor full-load amps, conduit offset bends, industrial process flow, and spindle machining speeds:

Keycap / Softkey	Layer	Operation & Stack Signature
AWG / ↰ MTR / ↱ FLA	Row 1 Col 1 (Primary / Yellow / Blue)	**NEC Voltage Drop (AWG), Motor HP (↰ MTR) & Full-Load Amps (↱ FLA):** Looks up NEC Table 8 resistance R (Ω/kft) and solves $\mathbf{D = 2 \times R \times L \times I \,/\, 1000}$ (D, R, L, I, G), motor torque $\mathbf{H = T \times N \,/\, 5252}$, and three-phase ↱ FLA.
OFS / ↰ MIT / ↱ FIT	Row 1 Col 2 (Primary / Yellow / Blue)	**Conduit Offset (OFS), Miter Angle & Pipe Fitting Take-Off:** Solves $\mathbf{T = O \,/\, SIN(A)}$ (T, O, A) on the stack or bidirectionally, miter cut angle (↰ MIT), and elbow fitting take-off (↱ FIT).
pF / ↰ Ω / ↱ kVA / x ‖ y	Electrical Power & Impedance Keys	**Power Factor (pF), Ohm's Law (XEQ), Transformer (XEQ) & Parallel (x‖y):** Solves real power $\mathbf{P = V \times I \times F}$ (P, V, I, F), $\mathbf{P = V \times I}$, three-phase $\mathbf{S = V \times I \times 1.732 \,/\, 1000}$, and parallel impedance $X \parallel Y = XY/(X+Y)$.
PIP / ↰ SCH / ↱ WAL	Row 1 Col 4 (Primary / Yellow / Blue)	**Pipe Continuity (PIP), Schedule Lookup & Barlow Wall Thickness:** Solves $\mathbf{Q = A \times V}$ (Q, A, V), pipe internal diameter (↰ SCH), and pressure wall thickness (↱ WAL).
PMP / ↰ TNK / ↱ C_v / ↰ ORF / ↱ NPH	Hydraulics & Process Keys	**Pump Power (PMP), Tank Volume, Valve C_v (↱ C_v), Orifice & NPSH:** Solves $\mathbf{P = Q \times H \,/\, 3960}$ (P, Q, h), $\mathbf{Q = C \times \sqrt{(P \,/\, G)}}$ (Q, C_{ON}, P, G),partially filled tank volume (↰ TNK), orifice flow (↰ ORF), and net positive suction head (↱ NPH).
mA / ↰ RTD / ↱ STM	Row 2 Col 2 (Primary / Yellow / Blue)	**4–20 mA Loop Scaling, Callendar–Van Dusen RTD & Saturated Steam:** Converts between 4–20 mA current and engineering span (mA), Pt100 RTD resistance/temperature (↰ RTD), and steam pressure/temperature (↱ STM).
↰ SFM / ↱ WLD / ↰ TAP / ↱ RPM / ↱ BC	Machining, Welding & Millwright Keys	**Surface Speed (↰ SFM), Weld Heat (↱ WLD), Tap Drill, Belt Ratio & Bolt Circle (↱ BC):** Solves spindle speed $\mathbf{N = 3.82 \times S \,/\, D}$ (n, S, D), weld heat input $\mathbf{H = 60 \times V \times I \,/\, (1000 \times S)}$ (h, V, I, S), tap drill allowance (↰ TAP), pulley ratio (↱ RPM), and bolt-circle chord $\mathbf{C = D \times SIN(180 \,/\, N)}$ (C_{ON}, D, n).
↰ f-i / ↱ →mm / ↱ ▶in / ↰ ▶CM / ↰ →kW / ↱ →HP / ↰ →PSI / ↱ →kPa / ↰ →Nm / ↱ →lbf / ↰ →GPM / ↱ →MGD	Dimensional & Trade Converters	Enters/formats feet-inch-fractions (↰ f-i) and converts between metric and US customary length, power, pressure, torque, and flow units.

B.4 Systems Programming, Embedded, and Cyber Operations Keycaps

On the **SC-16** (**Unit III, Chapters 7–9**), radix and word-architecture keys pair with bitwise logical operators, bit shifts, circular rotations, embedded timer/ADC workspaces, and modular/network keys (while Row 0 softkeys LFU0–LFU5 automatically become hexadecimal digits A–F whenever HEX mode is active):

Keycap / Softkey	Layer	Operation & Stack Signature
HEX / ↶ DEC / ↶ OCT / ↷ BIN	Row 1 Col 1 & Col 5	Switches display and keypad entry across base-16 hexadecimal (binding LFU0–LFU5 to A–F), base-10 decimal, base-8 octal, and base-2 binary view.
↷ WSZ / ↷ UNS / ↶ NEG / ↷ OVF	Word Architecture & Sign Keys	Cycles hardware word width across 8-bit, 16-bit, 32-bit, and 64-bit registers (↷ WSZ), toggles unsigned interpretation (↷ UNS), applies two's complement negation $-X \equiv \sim X + 1 \pmod{2^W}$ (↶ NEG), and inspects carry/overflow status (↷ OVF).
FLT / ↶ F16 / ↷ F64 / ↶ Q.n / ↷ FXP	Row 1 Col 2 & Row 7 Col 2	Encodes/decodes IEEE 754 **float32** (FLT), **float16** (↶ F16), and **float64** (↷ F64) bit patterns, and converts real coefficients to/from $Q_{m.n}$ fixed-point integers (↶ Q.n, ↷ FXP).
& / ↶ ~& / ↷ ~ / | / ↶ ~| / ⊕	Row 2 Cols 3–5 (Primary / Yellow / Blue)	Bitwise logical & (&), ↶ ~& (↶ ~&), ↷ ~ (↷ ~), | (|), ↶ ~| (↶ ~|), and ⊕ (⊕) masked to active ↷ WSZ.
≪ / ≫ / ↶ ASR / ↷ ROR	Row 1 Cols 3–4 (Primary / Yellow / Blue)	Logical shift left (≪, $Y \ll X$), logical shift right (≫, $Y \gg X$), sign-preserving arithmetic shift right (↶ ASR, $Y \gg X$), and circular bit rotation (↷ ROR).
↶ SB / ↶ CB / ↷ B? / ↶ MSK	Bitfield Manipulation Keys	Sets (↶ SB), clears (↶ CB), or tests (↷ B?) bit index X in word Y, or generates a contiguous X-bit mask $2^X - 1$ (↶ MSK).
↶ *B / ↷ CLZ / ↷ 2ˣ / lg	Bit Counting & Base-2 Keys	Counts set **1**-bits (Hamming weight ↶ *B) or leading zeros (↷ CLZ) in X, and evaluates exact base-2 powers 2^X (↷ 2ˣ) and $\log_2(X)$ (lg).
PWM / ↶ τ / ↷ ALN / ↷ BPS	Row 7 Col 3 & Row 2 Col 1	**Timer (PWM), RC Settling (↶ τ), Memory Alignment & UART Baud:** Solves $\mathbf{F} = \mathbf{C} / (\mathbf{P} \times \mathbf{A})$ (F, &/⇅, P, A), RC time constant $\mathbf{T} = \mathbf{R} \times \mathbf{C}$ (τ, ∠, &/⇅), power-of-two address alignment (↷ ALN), and UART baud divisor (↷ BPS).
↶ ADC / ↷ DAC	Mixed-Signal Keys	**ADC Quantization Workspace (↶ ADC) & DAC Output:** Solves $\mathbf{D} = (\mathbf{V} / \mathbf{R}) \times \mathbf{2\hat{}N}$ (D, V, ∠, n) and reconstructs analog DAC output voltage (↷ DAC).
NET / ↶ CRC / ↷ SWP / ↶ ÷R / ↷ MOD / ↷ GCD	Row 2 Col 2 & Modular Keys	Computes IPv4 NET usable host count 2^{32-P}-2 (NET), IEEE 802.3 CRC-32 (↶ CRC), byte-order endian swap (↷ SWP), integer quotient (↶ ÷R), remainder $Y \bmod X$ (↷ MOD), and greatest common divisor (↷ GCD).
↶ →B / ↷ →bit / ↶ →KiB / ↷ →MiB / ↶ →GiB / ↷ →TiB / ↶ →ms / ↷ →µs / ↶ →MHz / ↷ →GHz / ↶ →cyc / ↷ →ns	Storage & Clock Converters	Converts between IEC binary storage prefixes (2^{10}–2^{40}), bytes/bits (× 8), and clock frequency, period, and cycle counts.

B.5 Cinema Optics, Broadcast Timecode, and Sound Keycaps

On the **SC-24** (**Unit IV, Chapters 10–12**), the Row 1 and Row 2 camera, lighting, audio, and DIT keys pair with Row 7 SMPTE timecode solvers to evaluate angle of view, rotary shutter exposure, hyperfocal and depth-of-field lens geometry, mired color-temperature shifts, and location sound acoustics:

Keycap / Softkey	Layer	Operation & Stack Signature
FOV / ↰ →35 / ↱ →16 / ↱ W:H	Optics & Format Keys	**Angle of View (FOV), Crop Factors & Aspect Ratio:** Solves $\mathbf{A} = \mathbf{2} \times \mathbf{ATAN(W} \,/\, \mathbf{(2} \times \mathbf{F))}$ (A, W, F), applies Super-35 (↰ →35) and 16mm (↱ →16) crop equivalents, and evaluates frame aspect ratio (↱ W:H).
EV / ↰ SHT / ↱ GEL	Row 1 Col 1 (Primary / Yellow / Blue)	**Exposure Value (EV), Rotary Shutter & CTO/CTB Gel Mired Shift:** Solves $\mathbf{N^2} = \mathbf{T} \times \mathbf{2\hat{}E}$ (n, T, E), shutter exposure time $t = \vartheta_{\text{shutter}} / (360 \cdot \text{FPS})$ (↰ SHT), and mired color shift $10^6/T_2$ - $10^6/T_1$ (↱ GEL).
LUX / ↰ BM° / ↰ →fc / ↱ →lux / ↰ →MRD / ↱ →K	Photometrics & Color Keys	**Inverse-Square Illuminance (LUX), Beam Spread & Photometric Units:** Solves $\mathbf{E} = \mathbf{I} \,/\, \mathbf{D^2}$ (E, I, D), luminaire beam diameter (↰ BM°), footcandles/lux (↰ →fc, ↱ →lux), and mired/Kelvin conversions (↰ →MRD, ↱ →K).
DOF / ↰ HFD / ↰ DPT / ↰ →EV / ↱ →ND	Depth of Field & Filter Keys	**Depth of Field (DOF), Hyperfocal, Diopter & ND Stops:** Solves $\mathbf{H} = \mathbf{F^2} \,/\, \mathbf{(N \times C)}$ (h, F, n, c/CN), direct hyperfocal distance (↰ HFD), close-up diopter power (↰ DPT), and optical density / EV stops (↰ →EV, ↱ →ND).
dB / ↰ SPL / λ / ↱ Vs / DLY / ↱ SYN	Acoustic & Audio Keys	Evaluates decibel voltage/power ratios (dB), inverse-square SPL attenuation (↰ SPL), acoustic wavelength (↱ λ) using speed of sound ↱ Vs, delay tower alignment (↱ DLY), and audio-video lip-sync offset (↱ SYN).
BIT / ↰ TB / ↰ DMX / ↱ PWR / ↱ W/A	DIT Data & Gaffer Power Keys	**Uncompressed Data Rate (BIT), RAID Capacity, DMX512 & Generator Load:** Solves $\mathbf{R} = \mathbf{B} \times \mathbf{F} \,/\, \mathbf{8}$ (Ƒ, b, F), storage footprint (↰ TB), DMX universe/address (↰ DMX), and stage tie-in load (↱ PWR, ↱ W/A).
TC / FPS / ↱ DF / ↱ +TC / ↱ -TC / ↰ →FRM / ↱ →TC	Timecode & Frame-Rate Keys	**SMPTE Timecode Workspace (TC) & Frame Arithmetic:** Solves $\mathbf{F} = \mathbf{S} \times \mathbf{R}$ (F, S, Ƒ), sets frame rate (FPS), computes SMPTE 29.97 drop-frame drift (↱ DF), adds/subtracts timecode (↱ +TC, ↱ -TC), and converts between frames and **HH:MM:SS:FF** (↰ →FRM, ↱ →TC).

B.6 RF Spectrum, Transmission Lines, and Microwave Keycaps

On the **SC-50** (**Unit V, Chapters 13–15**), the Row 1 and Row 2 RF propagation, impedance matching, and dynamic range keys pair with bidirectional Equation Workspaces and decibel converters to solve transmission-line standing waves, resonant LC networks, free-space link budgets, cascaded noise figures, Shannon capacity, and radar Doppler shifts:

Keycap / Softkey	Layer	Operation & Stack Signature
dBm / ↰ →dBm / ↱ →mW / ↱ →dBW / ↰ →dBV / ↱ →µV	Power & Voltage Decibel Keys	Converts linear power in milliwatts/Watts in X to dBm ($10\ \log_{10}(P_{\text{mW}})$), inverse mW ($10^{X/10}$), Watt-referenced dBW ($P_{\text{dBm}}$ - 30), or RMS voltage across active impedance ↱ Z₀ (50 Ω default).
SWR / Γ / ↱ TLN / ↱ COX	Impedance & Transmission-Line Keys	**Standing Wave (SWR) & Quarter-Wave (↱ Z₀) Workspaces:** Solves $\mathbf{S} = \mathbf{(1 + G)} \,/\, \mathbf{(1 - G)}$ (S, G) and quarter-wave transformer $\mathbf{Z} = \sqrt{\mathbf{(S \times L)}}$ (Z, S, L), transmission-line electrical length (↱ TLN), and coaxial attenuation (↱ COX).
FSL / ↰ LNK / ↱ F₁ / ↰ λ/L	Propagation & Antenna Keys	**Friis Path Loss (FSL), Link Budget (↰ LNK), Fresnel Zone & Whip Length:** Solves $\mathbf{L} = \mathbf{20LOG(D)} + \mathbf{20LOG(F)} + \mathbf{32.44}$ (L, D, F), link budget $\mathbf{R} = \mathbf{P} + \mathbf{G} - \mathbf{L}$ (Ƒ, P, G, L), first Fresnel zone radius (↱ F₁), and quarter-wave whip length (↰ λ/L).

Keycap / Softkey	Layer	Operation & Stack Signature
[NF] / [↶] [IP3] / [↷] [FDR] / [↷] [Pₙ] / [↷] [dBc]	Receiver Noise & Dynamic Range	**Friis Cascaded Noise ([NF]), Cascaded IIP3, SFDR & Thermal Floor:** Solves $\mathbf{F} = \mathbf{A} + (\mathbf{B} - 1) / \mathbf{G}$ ([F], [A], [b], [G]), cascaded third-order intercept ([↶] [IP3]), spurious-free dynamic range ([↷] [FDR]), thermal noise floor P_n = -174 + 10 log₁₀(*B*) ([↷] [Pₙ]), and oscillator phase noise ([↷] [dBc]).
[↶] [f₀] / [↶] [XLC] / [↶] [PAD] / [↷] [fc] / [↷] [EIA]	Resonance, Reactance & Filter Keys	**LC Tank ([↶] [f₀]), Reactance, Attenuator Pad & Cutoff:** Solves $\mathbf{F} = 1 / (2\pi\sqrt{\mathbf{L}\times\mathbf{C}})$ ([F], [L], [C/ON]), inductive/capacitive reactance ([↶] [XLC]), resistive π/T-pad resistors ([↶] [PAD]), RC/RL -3 dB cutoff ([↷] [fc]), and EIA SMD codes ([↷] [EIA]).
[↷] [C/B] / [↶] [RCS] / [↷] [Δf]	Spectrum & Radar Solvers	**Shannon Capacity ([↷] [C/B]), Radar Range & Doppler Shift:** Solves $\mathbf{C} = \mathbf{B} \times \mathbf{LN}(\mathbf{1} + \mathbf{S}) / \mathbf{LN}(\mathbf{2})$ ([C/ON], [b], [S]), monostatic radar range/RCS ([↶] [RCS]), and two-way radar Doppler shift $\Delta f = 2\ v\ f_0\ /\ c$ ([↷] [Δf]).
[↷] [Z₀] / [↷] [T₀] / [↷] [C/ON] / [↶] [kB]	RF Reference Constants	Recalls or configures characteristic system impedance Z_0 (50 Ω), IEEE reference noise temperature T_0 (290 K), speed of light c, and Boltzmann constant k_B.

B.7 Wet-Lab Chemistry, Stoichiometry, and Spectroscopy Keycaps

On the **SC-60** (**Unit VI, Chapters 16–18**), the Row 1 and Row 2 analytical chemistry keys and bidirectional Equation Workspaces evaluate formula molecular weights, gravimetric molarity, stock aliquot dilutions, Sørensen pH, Henderson–Hasselbalch buffer pairs, Beer–Lambert optical absorbance, chemical thermodynamics, and Nernstian electrochemistry:

Keycap / Softkey	Layer	Operation & Stack Signature
[Z] / [MW] / [↶] [w/w] / [↶] [ISO] / [↶] [ION]	Atomic & Gravimetric Keys	**Element Lookup (Z), Molar Mass ([MW]) & Mass Fraction:** Recalls 118-element IUPAC atomic mass ([Z] with $X > 0$) or opens Formula Mode ([Z] with $X = 0$; softkeys [ALP], [NUM], [FML]), solves $\mathbf{W} = \mathbf{N} \times \mathbf{M}$ ([W], [n], [m]), mass percent ([↶] [w/w]), isotope decay ([↶] [ISO]), and polyatomic ion ledger ([↶] [ION]).
[C₁V₁] / [↶] [RXN] / [↷] [w/v]	Row 2 Col 2 (Primary / Yellow / Blue)	**Dilution ([C₁V₁]), Stoichiometry ([↶] [RXN]) & Weight/Volume %:** Solves $\mathbf{C} \times \mathbf{V} = \mathbf{D} \times \mathbf{U}$ ([C/ON], [V], [D], [U]), molar stoichiometry $\mathbf{N} = \mathbf{C} \times \mathbf{V}$ ([n], [C/ON], [V]), and %(w/v) concentration ([↷] [w/v]).
[εbc] / [↶] [%T]	Row 2 Col 3 (Primary / Yellow)	**Beer–Lambert ([εbc]) & Optical Transmittance:** Solves $\mathbf{A} = \mathbf{E} \times \mathbf{B} \times \mathbf{C}$ ([A], [E], [b], [C/ON]) and converts percent transmittance $\%T$ to absorbance $A = 2$ - log₁₀($\%T$) ([↶] [%T]).
[pH] / [↷] [pKₐ] / [↶] [EQV] / [BUF] / [↶] [%Y]	Row 2 Cols 4–5 & Row 1 Col 2	**Sørensen pH, Titration Equivalence ([↶] [EQV]), Buffer ([BUF]) & Yield:** Converts [H⁺] or K_a to -log₁₀(X) ([pH], [↷] [pKₐ]), solves titration equivalence $\mathbf{A} \times \mathbf{V} = \mathbf{B} \times \mathbf{U}$ ([A], [V], [b], [U]), Henderson–Hasselbalch $\mathbf{P} = \mathbf{K} + \mathbf{LOG}(\mathbf{A} / \mathbf{H})$ ([P], [K], [A], [h]), and percent yield ([↶] [%Y]).
[PVT] / [↶] [CAL] / [↷] [RCF]	Row 7 Col 3 (Primary / Yellow / Blue)	**Ideal Gas ([PVT]), Calorimetry ([↶] [CAL]) & Centrifuge RCF:** Solves $\mathbf{P} \times \mathbf{V} = \mathbf{N} \times \mathbf{R} \times \mathbf{T}$ ([P], [V], [n], [T]), heat transfer $\mathbf{Q} = \mathbf{M} \times \mathbf{C} \times \mathbf{T}$ ([Q], [m], [C/ON], [T]), and relative centrifugal force RCF = $1.118 \times 10^{-5}\ r\ \mathrm{RPM}^2$ ([↷] [RCF]).

Keycap / Softkey	Layer	Operation & Stack Signature
[→] [Kₑq] / [→] [Eₐ] / [←] [ΔG] / [→] [E°] / [→] [t½]	Kinetics, Thermodynamics & Electrochemistry	Evaluates equilibrium constant ([→] [Kₑq]), Arrhenius rate factor ([→] [Eₐ]), Gibbs free energy $\mathbf{G = H - T \times S}$ ([←] [ΔG]: [G], [h], [T], [S]), Nernst cell potential $\mathbf{E = S - (0.05916/N)LOG(Q)}$ ([→] [E°]: [E], [S], [n], [Q]), and first-order half-life $t_{1/2} = \ln(2)/k$ ([→] [t½]).
[→] [R] / [→] [h] / [←] [F] / [←] [Nₐ] / [←] [kB] / [→] [c/λ]	Fundamental Physical Constants	Recalls CODATA molar gas constant R, Planck constant h, Faraday constant F, Avogadro constant N_A, Boltzmann constant k_B, and speed of light c.

B.8 Commercial Bakery, Fermentation Science, and Brewing Keycaps

On the **SC-78** (**Unit VII, Chapters 19–21**), the Row 1 bakery and culinary yield keys and Row 2 brewing and enology keys pair with bidirectional Equation Workspaces to anchor batch scaling to total flour mass, compute mix water and flake-ice mass from friction and ambient heat, evaluate brewing gravity attenuation, strike water temperature, and hop bitterness, and audit butcher trim yields and menu food cost:

Keycap / Softkey	Layer	Operation & Stack Signature
[BK%] / [→] [HYD] / [←] [SCL] / [PAN]	Baker's Percentage & Scaling Keys	**Baker's % ([BK%]), Hydration, Batch Scale & Pan Scaling:** Consumes flour mass (Y) and ingredient mass (X) or opens $\mathbf{P = (I \,/\, F) \times 100}$ ([P], [I], [F]), computes dough hydration % ([→] [HYD]), scales ingredient weights ([←] [SCL]), and scales across pan volumes ([PAN]).
[LEV] / [←] [PRF] / [→] [LV%]	Row 1 Col 1 (Primary / Yellow / Blue)	**Sourdough Starter ([LEV]), Proofing Time & inoculation ([→] [LV%]):** Splits levain build mass into flour/water components ([LEV]), adjusts fermentation proof time for ambient temperature Q_{10} ([←] [PRF]), and computes pre-ferment inoculation % ([→] [LV%]).
[Tw] / [ΣW] / [←] [DIV] / [→] [ICE]	Row 1 Cols 3–4 (Primary / Yellow / Blue)	**Desired Dough Temp ([Tw]), Total Weight, Portion Divide & Flake Ice:** Solves $\mathbf{W = 3 \times D - R - F - M}$ ([W], [D], [R], [F], [m]), computes total dough mass from flour and total baker's % ([ΣW]), divides batch into equal loaves ([←] [DIV]), and computes flake-ice replacement mass ([→] [ICE]).
[ABV] / [←] [CHP] / [→] [SRM] / [←] [BRX] / [→] [SGT]	Gravity, Alcohol & Color Keys	**Alcohol by Volume ([ABV]), Chaptalization, Morey SRM & Hydrometer:** Solves $\mathbf{A = (O - F) \times 131.25}$ ([A], [O], [F]), sugar addition ([←] [CHP]), beer color SRM $= 1.4922 \cdot \mathrm{MCU}^{0.6859}$ ([→] [SRM]), alcohol-corrected refractometer FG ([←] [BRX]), and hydrometer temperature correction ([→] [SGT]).
[MSH] / [←] [SPG] / [IBU] / [←] [CO₂]	Row 2 Cols 4–5 (Primary / Yellow)	**Mash Strike ([MSH]), Sparge Volume, Hop Bitterness ([IBU]) & Carbonation:** Solves $\mathbf{S = (0.2 \,/\, R) \times (T - G) + T}$ ([S], [R], [T], [G]), boil/sparge volume ([←] [SPG]), hop bitterness $\mathbf{I = W \times A \times U \times 74.89 \,/\, V}$ ([I], [W], [A], [U], [V]), and keg head pressure ([←] [CO₂]).
[C₁V₁] / [←] [TA] / [→] [WTR] / [→] [SO₂] / [←] [ACD]	Enology, Acid & Water Keys	Solves Pearson dilution $\mathbf{C \times V = D \times U}$ ([C₁V₁]), titratable acidity ([←] [TA]), brewing salt ppm ([→] [WTR]), potassium metabisulfite dosage ([→] [SO₂]), and acid blend adjustment ([←] [ACD]).

Keycap / Softkey	Layer	Operation & Stack Signature
YLD / ↰ MAC / CST / ↰ PRC / ↱ PU	Yield, Costing & Food Safety Keys	**Butcher Yield (YLD) & Plate Costing (CST):** Computes edible yield EP/AP× 100 (YLD), Atwater caloric macros (↰ MAC), solves **P = C / (F / 100)** (P, $\frac{¢}{ON}$, F) or menu price (↰ PRC), and evaluates thermal pasteurization units (↱ PU).
↰ →Brx / ↱ →SG / ↰ →°P / ↱ →PRF / ↱ TDS / ↰ DTR / ↰ SAL / ↱ SV / ↱ SYR / ↱ Aw	Culinary R&D & Sugar Scale Keys	Converts between Brix, Specific Gravity, Plato, and US Proof; evaluates coffee extraction yield (↱ TDS) and roast development time ratio (↰ DTR), equilibrium brine salinity (↰ SAL), sous-vide thermal diffusion (↱ SV), syrup Brix (↱ SYR), and water activity (↱ Aw).

B.9 Clinical Nursing, Hospital Pharmacy, and Infusions Keycaps

On the **SC-80** (**Unit VIII, Chapters 22–24**), the Row 1 and Row 2 clinical dosing, clearance, and pharmacy compounding keys pair with the Patient Profile softkey bar (PAT: W, h, A, S, MALE/FEM) to evaluate Mosteller body surface area, Devine ideal and adjusted dosing weights, Cockcroft–Gault renal clearance, pediatric maintenance rules, vasoactive pump and gravity drip rates, and one-compartment pharmacokinetics:

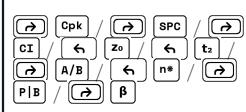
Keycap / Softkey	Layer	Operation & Stack Signature
BSA / ↰ IBW / ↱ ABW / ↱ BMI	Anthropometric & Dosing Weight Keys	**Mosteller BSA (BSA), Ideal/Adjusted Weight & BMI:** Consumes height in cm (Y) and weight in kg (X) or solves $\mathbf{B} = \sqrt{(\mathbf{H} \times \mathbf{W} / \mathbf{3600})}$ (b, h, W, PAT), Devine ideal body weight (↰ IBW), 40% obese-adjusted dosing weight (↱ ABW), and Quetelet BMI (↱ BMI).
GFR / PAT	Row 1 Col 2 & Patient Softkey	**Cockcroft–Gault Renal Clearance (GFR):** Solves $\mathbf{C} = ((140 - \mathbf{A}) \times \mathbf{W}) / (72 \times \mathbf{S})$ ($\frac{¢}{ON}$, A, W, S, PAT), automatically applying the × 0.85 female sex factor when FEM is active in the PAT profile.
DRP / IV / gtt / DOS / ↱ MGH	Critical-Care Infusion & Titration Keys	**Vasoactive Drip (DRP), Volumetric Pump (IV), Gravity (gtt) & Weight Dose (DOS):** Solves $\mathbf{R} = \mathbf{D} \times \mathbf{W} \times \mathbf{60} / \mathbf{C}$ (r, D, W, $\frac{¢}{ON}$, PAT), pump rate $\mathbf{R} = \mathbf{V} / \mathbf{T}$ (r, V, T), gravity roller clamp rate $R \times F_{\text{drop}} / 60$ (gtt), weight-based dose $\mathbf{T} = \mathbf{W} \times \mathbf{D}$ (T, W, D, PAT), and mg/hr mass rate (↱ MGH).
Rx / ↰ ALG / ↱ IST / ↰ VIL / TPN / ↰ GIR	Pharmacy Compounding & Nutrition Keys	Evaluates tablets/volume per dose (Rx), two-stock alligation alternate ratio (↰ ALG), sodium chloride isotonicity equivalent (↱ IST), powder vial reconstitution volume (↰ VIL), parenteral nutrition non-protein kcal (TPN), and neonatal glucose infusion rate (↰ GIR).
↰ PED / ↱ 421 / ↰ REE / ↱ VET	Pediatric, Metabolic & Veterinary Keys	Computes Clark/Young pediatric dose (↰ PED), Holliday–Segar 4-2-1 hourly maintenance fluid rate (↱ 421), Mifflin–St Jeor resting energy expenditure (↰ REE), and Kleiber allometric veterinary metabolic energy $70 \cdot W_{\text{kg}}^{0.75}$ (↱ VET).
PK / EBM / ↰ NNT / ↱ OR	Pharmacokinetics & Biostatistics Keys	**One-Compartment Elimination (PK) & Clinical Epidemiology:** Converts between k_e and $t_{1/2} = \ln(2)/k_e$ on the stack or solves $\mathbf{T} = \mathbf{0.693} \times \mathbf{V} / \mathbf{C}$ (T, V, $\frac{¢}{ON}$), and evaluates specificity/NPV (EBM), number needed to treat (↰ NNT), and odds ratio (↱ OR).

Keycap / Softkey	Layer	Operation & Stack Signature
[→] [AG] / [←] [Ca²⁺] / [→] [Na⁺] / [→] [OSM]	Serum Electrolyte & Osmolality Keys	Evaluates serum anion gap AG = Na^+ - (Cl^- + HCO_3^-) ([→] [AG]), Payne albumin-corrected calcium ([←] [Ca²⁺]), Katz hyperglycemia-corrected sodium ([→] [Na⁺]), and calculated serum osmolality ([→] [OSM]).
[ICU] / [←] [MAP] / [→] [BRN] / [→] [ABG] / [←] [QTc] / [←] [O₂]	Hemodynamics, Burn & Emergency Keys	**Cardiac Output ([ICU]), MAP, Parkland Burn ([→] [BRN]) & Respiratory Keys:** Solves **Q** = **H** × **S** / **1000** ([Q], [h], [S]), mean arterial pressure DBP + (SBP-DBP)/3 ([←] [MAP]), Parkland burn fluid **V** = **4** × **W** × **B** ([V], [W], [b]), Berlin ARDS P_aO_2/F_iO_2 ratio ([→] [ABG]), Bazett QTc ([←] [QTc]), and oxygen cylinder duration ([←] [O₂]).

B.10 Quantitative Finance, Actuarial Science, and Six Sigma Keycaps

On the **SC-95** (**Unit IX, Chapters 25–27**), the Row 7 Time Value of Money workspace key ([TVM]), Row 1 and Row 2 R-style probability distribution keys, and shifted financial and statistical solvers combine five-variable compound interest, discounted cash-flow and bond valuation, unit-economic accounting, and Six Sigma statistical inference:

Keycap / Softkey	Layer	Operation & Stack Signature
[TVM] → [n] [I%] [PV] [PMT] [FV] [END]/[BEG]	Row 7 Col 3 Primary & Softkeys	**Five-Variable TVM Workspace:** Pressing [TVM] binds [LFU0]–[LFU5] to [n], [I%], [PV], [PMT], [FV], and payment timing [END]/[BEG]; pressing <**VALUE**> <**SOFTKEY**> stores a known parameter, and [→] [SOLVE] <**SOFTKEY**> (or [XEQ] <**SOFTKEY**>) solves for the unknown variable.
[←] [AMR] / [→] [EAR]	Row 7 Col 3 (Yellow / Blue)	**Loan Amortization & Effective Annual Rate ([→] [EAR]):** Computes principal/interest reduction from the active [TVM] registers ([←] [AMR]), or solves **E** = **((1 + A / (100 × M))^M - 1) × 100** ([E], [A], [m]) via [→] [EAR].
[→] [CF] ([→] [CF] Workspace) / [→] [ROI] / [→] [CGR] / [←] [WAC] / [→] [CAP]	Capital Budgeting & Return Keys	**Discounted Cash-Flow Workspace ([→] [CF]) & Growth Solvers:** [→] [CF] solves **V** = **C×(1-(1+R/100)^−N)/(R/100) - I** ([V] for NPV, [r] for IRR at V=0, [C/Y], [n], [I]), return on investment ([→] [ROI]), compound annual growth **F** = **P** × **(1 + R/100)^N** ([→] [CGR]: [F], [P], [r], [n]), WACC ([←] [WAC]), and perpetuity capitalization ([→] [CAP]).
[←] [MGN] / [→] [MKP] / [←] [BEP] / [→] [EOQ] / [→] [DEP] / [→] [TAX] / [←] [%Σ]	Commercial & CPA Accounting Keys	Solves gross margin **M** = **(P - C) / P × 100** ([←] [MGN]: [m], [P], [C/Y]), markup **U** = **(P - C) / C × 100** ([→] [MKP]: [U], [P], [C/Y]), break-even units **Q** = **F / (P - V)** ([←] [BEP]: [Q], [F], [P], [V]), Wilson **Q** = **√(2×D×S/H)** ([→] [EOQ]: [Q], [D], [S], [h]), straight-line depreciation **D** = **(C - S) / L** ([→] [DEP]: [D], [C/Y], [S], [L]), sales tax **T** = **P** × **(1 + R/100)** ([→] [TAX]: [T], [P], [r]), and share of total ([←] [%Σ]).
[→] [BND] ([→] [BND] Workspace)	Row 2 Col 1 Blue Shift	**Fixed-Income Bond Workspace ([→] [BND]):** Solves **P** = **C×(1-(1+Y/100)^−N)/(Y/100) + F/(1+Y/100)^N** forward for clean price [P] or backward for periodic yield to maturity [Y], coupon [C/Y], remaining periods [n], or par redemption [F].
[→] [pN] / [←] [qN] / [rN] / [→] [pt] / [←] [qt] / [rt] / [→] [pχ²] / [←] [qχ²] / [rχ²] / [→] [pF] / [←] [qF] / [rF] / [→] [pB] / [←] [qB] / [rB] / [←] [PDF]	Row 1 Col 3 & Row 2 Cols 2–5	**R-Style CDF (p*), Inverse Quantile (q*), Random Variate (r*) & Density (PDF):** Evaluates cumulative probabilities (blue shift), critical quantiles (yellow shift), and random variates (unshifted primary) for Binomial ([→] [pB]/[←] [qB]/[rB]), Normal ([→] [pN]/[←] [qN]/[rN]), Student's t ([→] [pt]/[←] [qt]/[rt]), Chi-square ([→] [pχ²]/[←] [qχ²]/[rχ²]), and Fisher F ([→] [pF]/[←] [qF]/[rF]) distributions, or prefixes with [←] [PDF] for point density/mass.
[Σ+] / [←] [Σ−] / [x̄,ȳ] / [←] [MED] / [→] [s,σ] / [→] [L.R.]	Summary Statistics & Regression Keys	Accumulates/removes (x_i, y_i) pairs ([Σ+], [←] [Σ−]), computes sample means ([x̄,ȳ]), median/IQR ([←] [MED]), sample/population standard deviations ([→] [s,σ]), and opens the least-squares regression menu ([→] [L.R.] → [ŷ], [x̂], [m], [b], [r]).

Keycap / Softkey	Layer	Operation & Stack Signature
	Six Sigma, Inference & Bayesian Solvers	Solves process capability $\mathbf{C = (U - L) / (6 \times S)}$ (Cpk : & , U , L , S), 3σ Shewhart control limits (SPC), confidence margin $\mathbf{E = Z \times S / \sqrt{N}}$ (CI : E , Z , S , n), one-sample z-test (z_0), two-sample t-test (t_2), two-proportion test (A/B), sample size $\mathbf{N = (Z \times S / E)^2}$ (n* : n , Z , S , E), Bayes posterior (P|B), and three-point Beta-PERT mean $\mathbf{E = (O + 4{\times}M + P) / 6}$ (β : E , O , M , P).
	Calendar & Day-Count Keys	Converts day counts to annualized years (→YR), formats ISO **YYYY.MMDD** dates (→YMD), projects future maturity dates (DT+), and computes day counts between dates (ΔD).

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